Theory and Methods in Causal Inference

Lecture Notes

Jae Kwang Kim

Department of Statistics

Iowa State University

STAT 5900B

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Contents

Preface

These lecture notes accompany the graduate course **STAT 5900B: Theory and Methods in Causal Inference** at the Department of Statistics, Iowa State University (Version 2.0).

The course targets first-year PhD students in statistics. It assumes familiarity with probability theory, mathematical statistics, and linear models at the level of a graduate sequence, but no prior exposure to causal inference.

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Organization

The notes are organized in three parts, with a fourth part on policy learning deferred to a later release.

Part I — Foundations (Chapters 1–3) builds the graphical and conceptual foundation of causal inference. Chapter 1 introduces the three languages of causality — structural equation models (SEMs), directed acyclic graphs (DAGs), and potential outcomes — and explains how they relate. Chapter 2 develops DAGs and d-separation as tools for reading off conditional independence from causal structure. Chapter 3 introduces the do-calculus and the main identification criteria (back-door, front-door, do-calculus rules) that translate causal queries into statistical ones.

Part II — Identification (Chapters 4–9) bridges the graphical framework and observed data. Chapter 4 connects the potential outcomes framework to the do-calculus and establishes the consistency, exchangeability, and positivity assumptions. Chapter 5 covers randomization and the back-door adjustment formula. Chapter 6 develops propensity score methods — weighting, matching, and the balancing property. Chapter 7 develops instrumental variables: the Wald estimand, the local average treatment effect (LATE) for compliers, and the three IV assumptions. Chapter 8 covers mediation analysis and the front-door criterion, distinguishing the controlled direct effect from the natural direct and indirect effects. Chapter 9 introduces the estimation framework — estimating equations, asymptotic linearity, influence functions, and the semiparametric efficiency bound — that underlies the methods of Part III.

Part III — Estimation (Chapters 10–13) covers semiparametric efficiency theory and its modern applications. Chapter 10 develops the augmented inverse probability weighted (AIPW) estimator from three complementary perspectives — bias correction, optimal augmentation, and the efficient influence function — and establishes double robustness and semiparametric efficiency. Chapter 11 addresses flexible nuisance estimation with orthogonal scores and cross-fitting, culminating in the double machine learning (DML) estimator. Chapters 12 and 13 cover estimation under instrumental variables and further extensions (to be added).

Part IV — Policy Learning and Sequential Decision Making (Chapters 14–16, deferred) will cover policy evaluation, optimal policy learning, and dynamic treatment regimes.

Appendices

Appendix A provides graphical intuition for conditional independence and d-separation. **Appendix B** introduces Single World Intervention Graphs (SWIGs). **Appendix C** develops the formal foundations of pathwise differentiability and efficient influence functions for readers who want the semiparametric geometry underlying Chapter 9.

Notation

Throughout these notes:

- $\mathrm{do}(T = t)$ denotes the do-operator (intervention).
- $Y(t)$ denotes the potential outcome under $T = t$.
- $\mathbb{E}[\,\cdot\,]$ denotes expectation; $\perp\!\!\!\perp$ denotes conditional independence.
- $\mathcal{G}$ denotes a DAG; $\mathcal{G}_{\overline{T}}$ denotes the mutilated graph after intervening on T.
- $\pi(x) = P(T = 1 \mid X = x)$ denotes the propensity score.
- $\mu_t(x) = \mathbb{E}(Y \mid T = t, X = x)$ denotes the outcome regression.

Part I

Foundations

Chapter 1

Introduction

Learning Objectives

By the end of this chapter, students should be able to:

1. Articulate the distinction between an interventional distribution $P(y \mid \mathrm{do}(T{=}t))$ and an observational conditional distribution $P(y \mid T{=}t)$, and explain why they differ whenever T is endogenous.
2. Describe the same causal question in all three languages — SEM, DAG graph surgery, and potential outcomes — and begin to translate between them.
3. Derive the two densities explicitly in the Gaussian linear confounded model, and identify the endogeneity bias $\alpha\gamma\sigma_U^2/\sigma_T^2$ as the gap between them.
4. State the two-step paradigm (identification then estimation) and locate each step in the causal inference pipeline.

This chapter is an orientation, not a complete treatment. It introduces four ideas — the interventional/observational distinction, the causal trinity, the Gaussian confounded model as a running example, and the identification-versus-estimation paradigm — that will be developed rigorously in the chapters that follow.

1.1 Motivation: Why Causal Inference Is Hard

1.1.1 The Central Question

Causal inference is concerned with a deceptively simple question: what would happen to Y if we were to *set* $T = t$? This is fundamentally different from the question that standard statistical procedures answer: what is the distribution of Y among units *observed* to have $T = t$?

Definition: Interventional vs. Observational Distribution

- The *interventional distribution* $P(y \mid \mathrm{do}(T{=}t))$ is the distribution of Y in a hypothetical world where T has been *externally set* to t, regardless of the usual data-generating mechanism for T.
- The *observational conditional distribution* $P(y \mid T{=}t)$ is the distribution of Y among the subpopulation of units for whom $T = t$ was *observed* to hold.

These two distributions coincide when T is *exogenous* — that is, when T shares no common causes with Y. Whenever T is endogenous, the two distributions differ, and the difference is *endogeneity bias*.

Example: Drug Trial with Unmeasured Severity

Does a new drug ($T = 1$) reduce blood pressure (Y)? Suppose sicker patients are more likely to take the drug, so unmeasured severity U is a common cause of T and Y. Then:

- *Observational study*: $\mathbb{E}[Y \mid T{=}1] > \mathbb{E}[Y \mid T{=}0]$. Drug users have higher blood pressure on average. A naïve analyst concludes the drug is harmful.

- *Randomized trial*: $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] < \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$. Forcing everyone to take the drug reduces blood pressure. The drug is beneficial.

Same data. Different questions. Opposite answers. The entire discrepancy is caused by the back-door path $T \leftarrow U \rightarrow Y$.

The Fundamental Problem of Causal Inference

We observe $P(y \mid T{=}t)$ directly from data. We want $P(y \mid \mathrm{do}(T{=}t))$. The goal of *identification theory* is to find conditions under which the former can be used to recover the latter.

1.2 The Causal Trinity: Three Languages for One Idea

The same causal question can be expressed in three closely related languages: the *structural equation model* (SEM), the *directed acyclic graph* (DAG) with graph surgery, and the *potential outcomes* framework. Collectively, we call these the **causal trinity**. Each language illuminates a different facet of the same underlying idea, and fluency in all three is essential for modern causal reasoning.

```
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\definecolor{accent}{RGB}{46,117,182}
\definecolor{defbg}{RGB}{238,244,251}
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\begin{tikzpicture}[
  lang/.style={rectangle,rounded corners=6pt,draw,very thick,text width=3.0cm,align=center,minimum he
  sem/.style={lang,draw=semcolor,fill=semcolor!10,text=semcolor},
  dag/.style={lang,draw=dagcolor,fill=dagcolor!10,text=dagcolor},
  po/.style={lang,draw=pocolor,fill=pocolor!10,text=pocolor},
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  trans/.style={<->,thick,color=darkgrey,shorten >=4pt,shorten <=4pt},
  tlabel/.style={font=\footnotesize\itshape,color=darkgrey,fill=white,inner sep=2pt}
]
\node[po]  (PO)  at (90:3.8cm)   {Potential\\Outcomes\\$Y(t)$};
\node[sem] (SEM) at (210:3.8cm)  {SEM\\$Y=f_Y(T,U,\varepsilon)$};
\node[dag] (DAG) at (330:3.8cm)  {DAG \&\\Graph Surgery\\$\Gcal,\;\Gcal_{\overline{T}}$};
\node[center] (C) at (0,0) {Set $T=t$?};
\draw[-{Stealth[length=5pt]},thick,color=pocolor!70,shorten >=4pt,shorten <=4pt] (PO)--(C);
\draw[-{Stealth[length=5pt]},thick,color=semcolor!70,shorten >=4pt,shorten <=4pt] (SEM)--(C);
\draw[-{Stealth[length=5pt]},thick,color=dagcolor!70,shorten >=4pt,shorten <=4pt] (DAG)--(C);
\draw[trans] (PO)--(SEM) node[tlabel,midway,left=2pt]{\parbox{2.2cm}{\centering consistency\\+ no int
\draw[trans] (SEM)--(DAG) node[tlabel,midway,below=4pt]{\parbox{2.2cm}{\centering equation\\$\leftrig
\draw[trans] (DAG)--(PO) node[tlabel,midway,right=2pt]{\parbox{2.2cm}{\centering SUTVA\\+ graph struc
\end{tikzpicture}
```

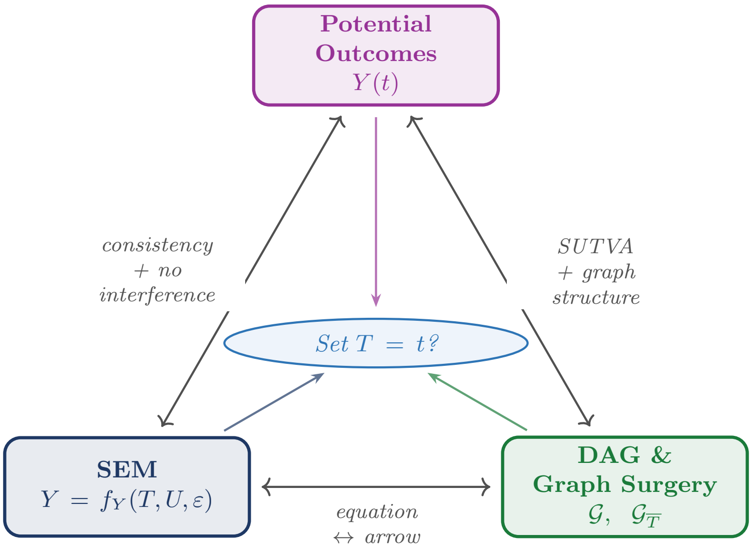

Figure 1.1: The **causal trinity**: three languages that express the same causal question from different vantage points.

1.2.1 Language 1: Structural Equation Models

A SEM represents the causal data-generating mechanism as a system of equations, one per variable, with a joint distribution over exogenous disturbances. A variable is *exogenous* if it is determined outside the system; a variable is *endogenous* if its value is produced by a structural equation within the model.

The running setup for this chapter is a three-variable confounded SEM over (U, T, Y):

$$T = f_T(U, \delta), \qquad Y = f_Y(T, U, \varepsilon), \tag{1.1}$$

where δ and ε are mutually independent exogenous disturbances and U is an unobserved exogenous confounder.

Remark: Error Independence in the NPSEM-IE

The assumption that the disturbances δ and ε are *mutually independent* is a substantive restriction. Because U is unobserved and enters both f_T and f_Y, error independence does not remove the T–$Y(t)$ dependence induced by U; it restricts only the additional channel through the equation-specific errors (δ, ε). Richardson and Robins (2014) call SEMs with this structure *Nonparametric Structural Equation Models with Independent Errors* (NPSEM-IEs). These notes adopt the NPSEM-IE framework throughout: it ensures the Markov condition holds and integrates cleanly with the graphical identification theory of later chapters.

Observational regime. Because U enters both equations, conditioning on T changes the distribution of U, so $P(y \mid T{=}t) \neq P(y \mid \mathrm{do}(T{=}t))$. An analyst who regresses Y on T without observing U faces a residual correlated with T — even though the structural error ε is independent of T by NPSEM-IE.

Interventional regime. The do-operator replaces the equation for T with a constant t, yielding the *mutilated system*: $U \sim p(u)$, $T = t$ (fixed), $Y = f_Y(t, U, \varepsilon)$. We define the *potential outcome* as $Y(t) := f_Y(t, U, \varepsilon)$; by construction, $Y(t) \sim P(y \mid \mathrm{do}(T{=}t))$.

1.2.2 Language 2: Directed Acyclic Graphs

Definition: Graph Surgery

In the observational DAG, arrows into T encode the mechanisms that determine T. Under $\mathrm{do}(T{=}t)$, all arrows into T are *deleted*, producing the mutilated graph $\mathcal{G}_{\overline{T}}$. This severs spurious back-door paths while preserving directed causal pathways from T to its descendants.

1.2.3 Language 3: Potential Outcomes

Definition: Potential Outcome [@rubin1974estimating]

In the SEM framework, $Y(t) = f_Y(t, U, \varepsilon)$ is the outcome that *would have been observed* had T been set to t, possibly contrary to fact.

Definition: SUTVA

The *stable unit treatment value assumption* (SUTVA) has two components: (i) *no interference* — unit i's potential outcome depends only on unit i's own treatment; and (ii) *no hidden versions of treatment* — each treatment level corresponds to a single, well-defined intervention. Under SUTVA, the *consistency* relation $Y_i = Y_i(T_i)$ holds. SUTVA is developed further in Chapter 4.

1.2.4 Roadmap: How the Trinity Organizes These Notes

Part I — Foundations: the graphical language (Chapters 1–3). The DAG and the do-operator are the primary language here because they make the interventional/observational distinction *syntactically explicit*: confounding is visible as an open back-door path, and identifying assumptions are visible as graph criteria.

Part II — Identification: designs and research strategies (Chapters 4–9). Part II introduces the potential outcomes framework (Chapter 4) and then studies the specific research designs through which observational and experimental data support identification: randomization and back-door adjustment (Chapter 5), propensity score methods (Chapter 6), instrumental variables (Chapter 7), mediation and front-door identification (Chapter 8), and sensitivity analysis (Chapter 9). Ignorability $(Y(t) \perp\!\!\!\perp T \mid X)$ is the potential-outcomes counterpart of the back-door criterion.

Part III — Estimation: semiparametric and machine-learning methods (Chapters 10–13). Once identification has been established, Part III turns to estimation. The semiparametric efficiency framework — estimating equations, influence functions, doubly robust estimators, and double machine learning — is largely language-neutral. Chapter 13 applies these tools to estimation under instrumental variables.

1.2.5 Historical Roots of the Causal Trinity

The three languages did not emerge from a single research program but grew independently in different disciplines over more than a century.

Structural Equation Models: genetics and econometrics. Sewall Wright (1921) introduced *path analysis* to decompose correlations among hereditary traits. The framework was transplanted into economics by Haavelmo (1943), who argued that economic relationships should be modeled as autonomous structural equations robust to policy interventions. The Cowles Commission formalized simultaneous equation systems throughout the 1940s–50s. Lucas's (1976) critique is essentially a statement of the interventional/observational distinction, decades before that phrase existed.

Potential Outcomes: experimental statistics and epidemiology. Neyman (1923) introduced the notation $Y_i(t)$ in the context of randomized agricultural experiments. The crucial extension to *observational* studies was made by Rubin (1974), who formalized the assignment mechanism and stated SUTVA explicitly. Holland's (1986) *JASA* paper brought the framework to mainstream statistics. Robins (1986) extended potential outcomes to longitudinal treatments via g-computation. Imbens and Angrist (1994) connected the framework to instrumental variables and defined the LATE.

DAGs and do-calculus: computer science and philosophy. Pearl developed Bayesian networks through the 1980s. The pivotal step came in Pearl (1995), where the *do-operator* was introduced. The 2000 monograph *Causality* synthesized DAGs, the do-calculus, identification theory, and connections to potential outcomes. Spirtes, Glymour, and Scheines (1993) — working from philosophy at Carnegie Mellon — developed constraint-based algorithms for learning causal structure from data.

Convergence. The three frameworks are closely related and often intertranslatable, but exact equivalence requires additional assumptions. These notes work mainly in an NPSEM-IE setting, where translation between the languages is especially clean.

1.3 The Core Distinction

The single most important inequality of this course is:

$$\underbrace{P(y \mid x,\, \mathrm{do}(T{=}t))}_{\text{interventional}} \neq \underbrace{P(y \mid x,\, T{=}t)}_{\text{observational}} \qquad \text{whenever } T \text{ is endogenous.} \tag{1.2}$$

Definition: Conditional Exchangeability

Given observed covariates X and confounders U, the treatment assignment is *conditionally exchangeable with the intervention* if:

$$Y(t) \perp\!\!\!\perp T \mid X,\, U, \qquad \text{or equivalently,} \qquad P(y \mid x,\, T{=}t,\, U{=}u) = P(y \mid x,\, \mathrm{do}(T{=}t),\, U{=}u).$$

Once we condition on all common causes (X, U), observing $T{=}t$ and intervening to set $T{=}t$ produce the same distribution of Y.

Definition: Positivity

The treatment assignment satisfies *positivity* if, for every treatment value t of interest, $p_{T|X,U}(t \mid x, u) > 0$ for almost all (x, u). This ensures that $P(y \mid x, T{=}t, U{=}u)$ is well-defined for every treatment value of interest.

Proposition: Back-Door Adjustment Formula

Under conditional exchangeability and positivity:

$$P(y \mid x,\, \mathrm{do}(T{=}t)) = \int P(y \mid x,\, T{=}t,\, U{=}u)\; p(u \mid x)\; du. \tag{1.3}$$

Proof

By positivity, $P(y \mid x, T{=}t, U{=}u)$ is well-defined. Applying the law of total probability to the interventional density:

$$P(y \mid x,\, \mathrm{do}(T{=}t)) = \int P(y \mid x,\, U{=}u,\, \mathrm{do}(T{=}t))\; p(u \mid x)\; du.$$

The weight is $p(u \mid x)$, not $p(u \mid x, T{=}t)$, because under $\mathrm{do}(T{=}t)$ the treatment is set externally and carries no information about U. By conditional exchangeability, $P(y \mid x, U{=}u, \mathrm{do}(T{=}t)) = P(y \mid x, T{=}t, U{=}u)$. Substituting completes the proof. □

Remark: Confounding Discrepancy

What OLS estimates is the observational density $P(y \mid x, T{=}t) = \int P(y \mid x, T{=}t, U{=}u)\; p(u \mid x, T{=}t)\; du$. The kernel $P(y \mid x, T{=}t, U{=}u)$ is the same as in Equation 1.3, but averaged over the *selection-distorted* weight $p(u \mid x, T{=}t)$ rather than the marginal $p(u \mid x)$. The confounding discrepancy:

$$P(y \mid x,\, T{=}t) - P(y \mid x,\, \mathrm{do}(T{=}t)) = \int P(y \mid x,\, T{=}t,\, U{=}u)\big[\, p(u \mid x,\, T{=}t) - p(u \mid x)\,\big]\, du$$

is nonzero when U and T are dependent *and* U affects the conditional distribution of Y.

1.3.1 Two Scenarios: Observed vs. Unobserved Confounder

Scenario 1: U is observed (no unmeasured confounding). When every component of U is recorded, both $P(y \mid x, T{=}t, U{=}u)$ and $p(u \mid x)$ can be estimated, and the back-door formula Equation 1.3 is a functional of the observable distribution. A special case is *unconfoundedness*: if $U \perp\!\!\!\perp T \mid X$ already holds, then $p(u \mid x, T{=}t) = p(u \mid x)$, and the formula reduces to $\mathbb{E}[Y \mid X{=}x, T{=}t]$. This is the key assumption underlying regression adjustment and propensity score methods (Chapters 5 and 6).

Scenario 2: U is unobserved (unmeasured confounding). When U is latent, the back-door formula is not usable from data. The main identification strategies are **instrumental variables** (Chapter 7), the **front-door criterion** (Chapter 3, applied in Chapter 8), and **sensitivity analysis** (Chapter 9).

	Scenario 1: U observed	**Scenario 2: U unobserved**
Confounding type	No unmeasured confounding	Unmeasured confounding
Back-door formula usable?	Yes: both terms estimable	No: U latent
Identification route	Back-door adjustment	IV, front-door, …
Key assumption	$Y(t) \perp\!\!\!\perp T \mid X, U$	IV, front-door, or partial-identification assumptions
Chapters in this course	Chs. 3–6 (back-door); Chs. 10–11 (IPW, DR)	Ch. 7 (IV); Ch. 8 (front-door); Ch. 9 (sensitivity)

1.4 The Gaussian Linear Confounded Model

Example: Gaussian Linear Confounded Model

The structural equations are:

$$Y = \beta T + \gamma U + \varepsilon, \qquad T = \alpha U + \delta, \tag{1.4}$$

where $U \sim \mathcal{N}(0, \sigma_U^2)$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, $\delta \sim \mathcal{N}(0, \sigma_\delta^2)$ are mutually independent. Here β is the causal effect of T on Y, γ is the direct effect of U on Y, and α is the direct effect of U on T.

The endogeneity of T is immediate: $\mathrm{Cov}(T, \gamma U + \varepsilon) = \gamma\alpha\sigma_U^2 \neq 0$ whenever $\alpha \neq 0$ and $\gamma \neq 0$. Consequently, OLS regressing Y on T does not estimate β.

```
\usetikzlibrary{arrows.meta, positioning}
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  mnode/.style={rectangle,rounded corners=3pt,draw=darkgrey,fill=gray!15,thick,minimum size=7mm,font=
  edge/.style={-{Stealth[length=5pt]},thick,color=isublue},
  dedge/.style={-{Stealth[length=5pt]},thick,color=darkgrey,dashed}
}
\begin{tikzpicture}[node distance=1.8cm]
  \node[node]  (T1) at (0.5,0)    {$T$};
  \node[node]  (Y1) at (2.5,0)    {$Y$};
  \node[unode] (U1) at (1.5,-1.3) {$U$};
  \draw[edge]  (T1) -- (Y1);
  \draw[dedge] (U1) -- (T1);
  \draw[dedge] (U1) -- (Y1);
  \node[font=\small\itshape,color=darkgrey] at (1.5,-2.1) {Observational DAG};
  \node[mnode] (T2) at (5.5,0)    {$t$};
  \node[node]  (Y2) at (7.5,0)    {$Y$};
  \node[unode] (U2) at (6.5,-1.3) {$U$};
```

```
  \draw[edge]  (T2) -- (Y2);
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  \draw[dedge] (U2) -- (Y2);
  \node[font=\small\itshape,color=darkgrey] at (6.5,-2.1) {Mutilated DAG: $\doop(T{=}t)$};
\end{tikzpicture}
```

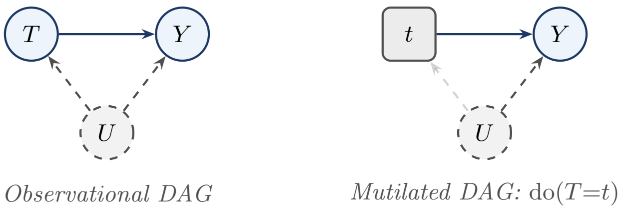

Figure 1.2: Graph surgery on the Gaussian confounded model. The faded arrow in the mutilated DAG has been severed by the intervention $\mathrm{do}(T{=}t)$. The back-door path $T \leftarrow U \rightarrow Y$ is eliminated; only the causal path $T \rightarrow Y$ remains active.

1.4.1 Two Explicit Densities

Write $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$ for the marginal variance of T.

Interventional density. Set $T = t$ externally; the back-door path through U is severed, so $U \sim \mathcal{N}(0, \sigma_U^2)$ independently:

$$P(y \mid \mathrm{do}(T{=}t)) = \mathcal{N}(\beta t,\ \ \gamma^2\sigma_U^2 + \sigma_\varepsilon^2)\,. \tag{1.5}$$

Observational density. Since (Y, T) is jointly normal, $\mathrm{Cov}(Y, T) = \beta\sigma_T^2 + \gamma\alpha\sigma_U^2$, giving:

$$P(y \mid T{=}t) = \mathcal{N}\left(\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right) t,\ \ \frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2\right). \tag{1.6}$$

1.4.2 The Endogeneity Gap

	Interventional	**Observational**
Mean	βt	$\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right) t$
Variance	$\gamma^2\sigma_U^2 + \sigma_\varepsilon^2$	$\frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2$
Residual	$(\gamma U{+}\varepsilon) \perp\!\!\!\perp T$	$(\gamma U{+}\varepsilon)$ correlated with T
Identified by	RCT (or IV, Chapter 7)	OLS
Equal when	$\alpha = 0$ or $\gamma = 0$ (no confounding)	

The *endogeneity bias* of OLS is $\gamma\alpha\sigma_U^2/\sigma_T^2$. With $\alpha = \gamma = 1$ and $\sigma_U^2 = \sigma_\delta^2 = 1$: bias $= 1/(1+1) = 0.5$.

1.4.3 Lab: Simulating the Two Densities

The analytical results were verified by simulation using $n = 10{,}000$ draws with parameters $\beta = 2$, $\alpha = 1$, $\gamma = 1$, $\sigma_U = \sigma_\varepsilon = \sigma_\delta = 1$. (R code: `chapter1_lab.R`.)

Experiment 1: OLS bias. By the omitted-variable bias formula, the OLS probability limit is $\beta + \gamma\alpha\sigma_U^2/\sigma_T^2 = 2 + 0.5 = 2.5$.

	Simulated	Theory	Formula
OLS slope	2.5147	2.5000	$\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$
True β	—	2.0000	β
Bias	0.5147	0.5000	$\gamma\alpha\sigma_U^2/\sigma_T^2$

OLS overestimates the causal effect by 25%.

Experiment 2: Two-sample comparison at $t_0 = 1$. The exact observational conditional distribution is $\mathcal{N}(2.5,\, 1.5)$; the interventional distribution is $\mathcal{N}(2,\, 2)$.

	Mean (Sim / Theory)	SD (Sim / Theory)
$\mathbb{E}[Y \mid T{=}1]$	2.497 / 2.500	1.229 / $\sqrt{1.5} = 1.225$
$\mathbb{E}[Y \mid \mathrm{do}(T{=}1)]$	2.001 / 2.000	1.428 / $\sqrt{2} = 1.414$

The variance reduction in the observational distribution reflects $\sigma^2_{\text{obs}}/\sigma^2_{\text{int}} = 1.5/2 = 0.75$: conditioning on $T{=}1$ pins down $\alpha U + \delta$, restricting the spread of Y through the correlation with γU.

Experiment 3: Density overlay for varying α. As α increases from 0 to 1.0, the observational density shifts rightward (growing bias) and narrows (reduced variance). The interventional density $\mathcal{N}(2, 2)$ is invariant to α.

1.5 The Two-Step Paradigm

Causal inference is a two-step discipline. The two steps are logically distinct and require different tools.

> Definition: Identification
>
> A causal quantity $P(y \mid \mathrm{do}(T{=}t))$ is *identified* from the observational distribution P if there exists a functional Ψ such that $P(y \mid \mathrm{do}(T{=}t)) = \Psi(P)$, and $\Psi(P)$ depends only on the observable joint distribution.

Step 1: Identification is a purely mathematical question about the causal model: can the interventional density be expressed as a function of the observed-data distribution? It does not depend on sample size. The main tools are: the back-door criterion (Chapter 3, applied in Chapters 4–6), the front-door criterion (Chapter 3), the three rules of the do-calculus (Chapter 3), and IV assumptions (Chapter 7).

Step 2: Estimation. Once $\Psi(P)$ has been identified, the statistical problem is to estimate the functional $\Psi(P)$ from n observations as efficiently as possible. The main tools are efficient influence functions (Part III), IPW (Part III), doubly robust estimators (Part III), and double machine learning (Part III).

1.5.1 Worked Example: Flu Vaccination and Infection (Simpson's Paradox)

A public health agency observes 1,000 individuals. Each person either received a flu vaccine ($T = 1$) or not ($T = 0$), and either became infected ($Y = 1$) or not. Age group X (elderly $= E$, young $= Y$) is recorded. Elderly people are both more likely to be vaccinated and more susceptible to infection, so X confounds the T–Y relationship. The causal DAG has paths: $T \to Y$ (causal) and $T \leftarrow X \to Y$ (back-door).

Observed data:

	Vaccinated ($T = 1$)	Unvaccinated ($T = 0$)	Total
Elderly ($X = E$)	360, 30% infected	40, 50% infected	400
Young ($X = Y$)	60, 5% infected	540, 10% infected	600
Total	**420, 26.4% infected**	**580, 12.8% infected**	**1,000**

Simpson's paradox:

Stratum	Vaccinated	Unvaccinated	Difference	Conclusion
Elderly	30%	50%	−20 pp	vaccine helps
Young	5%	10%	−5 pp	vaccine helps
Aggregate	**26.4%**	**12.8%**	+13.6 **pp**	**vaccine harms??**

Within every stratum the vaccine reduces infection. Yet in the aggregate the vaccinated group has a *higher* infection rate, because 86% of vaccinated are elderly vs. 7% of unvaccinated.

Step 1 (Identification). The set $\{X\}$ satisfies the back-door criterion:

$$P(Y{=}1 \mid \mathrm{do}(T{=}t)) = \sum_{x \in \{E,Y\}} P(Y{=}1 \mid T{=}t, X{=}x)\, P(X{=}x).$$

Step 2 (Estimation). With $\hat{P}(X{=}E) = 0.4$ and $\hat{P}(X{=}Y) = 0.6$:

$$\hat{P}(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.30 \times 0.4 + 0.05 \times 0.6 = 0.15,$$

$$\hat{P}(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.50 \times 0.4 + 0.10 \times 0.6 = 0.26.$$

The estimated causal risk difference is $0.15 - 0.26 = -0.11$: the vaccine causally reduces infection probability by 11 percentage points.

> The Two Steps, Concretely
>
> **Step 1 (identification)** established — from the DAG alone — that the back-door formula expresses the causal quantity as a functional of observable data. **Step 2 (estimation)** replaced the population probabilities in that formula with sample proportions.
> These two steps are logically separate: Step 1 is a mathematical argument about the causal model that does not depend on sample size; Step 2 is a statistical problem conditional on the identification formula. A researcher who skips Step 1 and reports the raw difference conflates the two and draws the wrong conclusion.

1.6 Summary

1. **Interventional vs. observational distribution.** Causal inference answers questions about interventions $P(y \mid \mathrm{do}(T{=}t))$, not about observations $P(y \mid T{=}t)$. In the Gaussian confounded model, the two densities coincide only when $\alpha = 0$ or $\gamma = 0$.

2. **The causal trinity.** The same causal question can be expressed in three languages — SEM (equation surgery), DAG (graph surgery), and potential outcomes ($Y(t)$). In the SEM framework, $Y(t)$ is defined as the solution for Y in the mutilated system.

3. **Endogeneity bias in the Gaussian confounded model.** The coefficient of T in $\mathbb{E}[Y \mid T{=}t]$ differs from the causal coefficient β by $\gamma\alpha\sigma_U^2/\sigma_T^2$. The observational variance $\gamma^2\sigma_U^2\sigma_\delta^2/\sigma_T^2 + \sigma_\varepsilon^2$ is smaller than the interventional variance $\gamma^2\sigma_U^2 + \sigma_\varepsilon^2$.

4. **Two-step paradigm.** Causal inference is: *identification* (expressing $\Psi(P)$ as a functional of observable data — a mathematical question) followed by *estimation* (constructing $\hat{\Psi}_n$ efficiently — a statistical question). The two steps are analytically separate.

Causal inference is the study of when the observational distribution contains enough information to recover the interventional distribution.

1.7 Problems

1. Do-notation fundamentals. For each statement below, decide whether it refers to an interventional or an observational distribution, and rewrite it unambiguously using do-notation where appropriate.

(a) "The probability that a patient recovers given that they took the drug."
(b) "The probability that a patient would recover if we prescribed the drug to everyone."
(c) "Among students who attended tutoring, the average exam score was 85."
(d) "If we enrolled all students in tutoring, the average exam score would be 85."

2. Deriving the observational density. In the Gaussian confounded model (Equation 1.4), verify Equation 1.6 by carrying out the following steps. Let $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$.

(a) Show that (Y, T) is jointly normal by writing both as linear combinations of the independent normals (U, ε, δ).
(b) Compute $\mathrm{Cov}(Y, T)$ and $\mathrm{Var}(T)$.
(c) Apply the conditional normal formula to derive $\mathbb{E}[Y \mid T{=}t]$ and $\mathrm{Var}[Y \mid T{=}t]$, and hence verify Equation 1.6.
(d) Show that the OLS probability limit is $\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$, and interpret this as an omitted-variable bias formula.

3. The causal trinity. Consider the following verbal causal claim: "Aspirin (T) reduces the risk of heart attack (Y) because it inhibits platelet aggregation (M), but patients with pre-existing cardiovascular disease (U, unobserved) are both more likely to take aspirin and more likely to have a heart attack."

(a) Draw the causal DAG implied by this description.
(b) Write down the recursive SEM with appropriate structural functions f_T, f_M, f_Y.
(c) State the SUTVA assumption and discuss whether it is plausible in this setting.
(d) Explain why $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$ does not identify the causal effect $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$ in this DAG.

Chapter 2

DAGs and d-Separation

Learning Objectives

By the end of this chapter, students should be able to:

1. Construct a DAG from a verbal description of a causal model and identify parents, descendants, ancestors, and non-descendants of any node.
2. Classify every intermediate node on a path as a chain, fork, or collider, and apply the blocking rule for each.
3. Apply the d-separation criterion to determine all conditional independence relationships implied by a DAG.
4. Recognize and avoid collider bias, including the case where a descendant of a collider is conditioned upon.
5. State the Markov factorization and use it to write the joint density of a DAG in terms of local conditional densities.
6. Interpret d-separation results as conditional independence assumptions on observed variables, and recognize their role as the graphical expression of causal assumptions.

Readers who want a gentler introduction to conditional independence, the three basic path motifs, and the Markov-property interpretation may consult Appendix A before or alongside this chapter.

2.1 Directed Acyclic Graphs

DAGs allow us to translate qualitative causal assumptions into quantitative statistical restrictions. Once we draw a graph encoding our causal assumptions, d-separation tells us which conditional independences are implied by the graph and helps identify candidate adjustment sets for removing confounding.

Remark

In this chapter, the word *graph* does not mean a plot of data or a graph of a function. It means a collection of nodes and edges used to represent relationships among variables. Our objective is to understand how such graphs encode statistical structure — especially conditional independence, confounding, and path blocking.

2.1.1 Basic Definition

Definition: Directed Acyclic Graph

A *directed acyclic graph* (DAG) $\mathcal{G} = (\mathcal{V}, E)$ consists of a finite set of nodes $\mathcal{V}$ and a set of directed edges $E \subseteq \mathcal{V} \times \mathcal{V}$ such that there is no directed cycle. Each node represents a variable and each directed edge $A \to B$ encodes that A is a direct cause of B relative to the variables included in the graph.

The acyclicity condition guarantees a topological ordering so that the joint density admits the recursive factorization $p(v_1, \dots, v_k) = \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i))$.

2.1.2 Structural Relationships

Definition: Structural Relationships in a DAG

For a node $v \in \mathcal{V}$ in $\mathcal{G}$:
- $\mathrm{Pa}(v) = \{w \in \mathcal{V} : w \to v \in E\}$ are the *parents* of v.
- $\mathrm{De}(v)$ = all nodes reachable from v by directed paths are the *descendants* of v.
- $\mathrm{An}(v)$ = all nodes with a directed path to v are the *ancestors* of v.
- $\mathrm{Nd}(v) = \mathcal{V} \setminus (\mathrm{De}(v) \cup \{v\})$ are the *non-descendants* of v.

Note that $\mathrm{Nd}(v)$ includes the parents of v.

Quick Terminology Summary

If $A \to B$, then A is a *parent* of B and B is a *child* of A. Two nodes are *adjacent* if connected by an edge. A *path* is any sequence of connected nodes regardless of arrow direction; a *directed path* has arrows all pointing in the same direction. A *cycle* is a directed path that returns to its starting node.

Example: Education, Earnings, and Family Background

Education (E) affects earnings (Y); family background (B) is a common cause of both; neighborhood (N) affects education but has no direct effect on earnings. The Markov factorization is $p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b)$. Reading off: $\mathrm{Pa}(E) = \{N, B\}$, $\mathrm{Pa}(Y) = \{E, B\}$, $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$.

Example: The IV DAG

The canonical IV model involves Z (instrument), T (treatment), Y (outcome), U (unobserved confounder), with $\mathrm{Pa}(T) = \{Z, U\}$, $\mathrm{Pa}(Y) = \{T, U\}$, $\mathrm{Pa}(Z) = \mathrm{Pa}(U) = \emptyset$. The descendants of Z are $\{T, Y\}$; U and Z are non-descendants of each other.

Example: The Mediation DAG

Treatment T affects outcome Y both directly and through mediator M. An unobserved confounder U creates a back-door path $T \leftarrow U \to Y$. $\mathrm{Pa}(T) = \{U\}$, $\mathrm{Pa}(M) = \{T\}$, $\mathrm{Pa}(Y) = \{M, T, U\}$. The total effect of T on Y travels along two paths: $T \to Y$ (direct) and $T \to M \to Y$ (indirect). Note: this is a standard mediation DAG, not a front-door DAG. For the front-door criterion, every directed path from T to Y must pass through M.

2.2 Paths, Blocking, and d-Separation

A DAG encodes direct causal relationships through its edges, but variables can also be statistically related through longer routes. The concept of a *path* formalizes these routes, and d-separation answers whether a path transmits or blocks statistical dependence.

2.2.1 Path Structures

Definition: Path

A *path* between nodes X and Y in $\mathcal{G}$ is any sequence of distinct nodes $X = V_0, V_1, \dots, V_k = Y$ such that each consecutive pair is connected by an edge in either direction.

Every intermediate node V_m on a path plays one of three structural roles. In a **chain** ($V_{m-1} \to V_m \to V_{m+1}$), V_m is a causal intermediary. In a **fork** ($V_{m-1} \leftarrow V_m \to V_{m+1}$), V_m is a common cause. In a **collider** ($V_{m-1} \to V_m \leftarrow V_{m+1}$), both arrows point *into* V_m; this asymmetric case has the opposite behavior under conditioning from the other two.

2.2.2 Blocking Rules

Structure	Pattern	Effect of conditioning	Key intuition
Chain	$A \to M \to B$	Blocks the path	Causal transmission
Fork	$A \leftarrow M \to B$	Blocks the path	Common cause
Collider	$A \to M \leftarrow B$	**Opens** the path	Common effect

The critical asymmetry: colliders are *closed by default* and *opened* by conditioning, while chains and forks are *open by default* and *closed* by conditioning.

Three-Node Motifs at a Glance

In a *chain* $A \to M \to B$: conditioning on M blocks the path. In a *fork* $A \leftarrow M \to B$: conditioning on M removes the spurious association. In a *collider* $A \to M \leftarrow B$: the path is blocked by default, but conditioning on M — or any descendant of M — opens it. These three motifs are the local building blocks of d-separation.

Chain: Smoking → Tar → Cancer. Marginally, smokers have elevated cancer rates (path is open). Conditioning on tar level blocks the path: once tar is fixed, the causal channel is fully accounted for.

Fork: Poverty → Poor Diet; Poverty → Lack of Exercise. Unconditionally, diet quality and exercise are correlated through poverty. Conditioning on income level removes the spurious association.

Collider: Accident → Hospitalization ← Cancer. In the general population, $A \perp\!\!\!\perp B$. Among hospitalized patients — conditioning on M — the two become negatively associated. This is Berkson's bias (Berkson 1946).

2.2.3 The d-Separation Criterion

Definition: d-Blocking and d-Separation

A path π is *d-blocked* by a set $\mathbf{S}$ if:

1. π contains a chain $A \to M \to B$ or fork $A \leftarrow M \to B$ with $M \in \mathbf{S}$; or
2. π contains a collider $A \to C \leftarrow B$ with $C \notin \mathbf{S}$ and no descendant of C is in $\mathbf{S}$.

Nodes X and Y are *d-separated* by $\mathbf{S}$, written $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$, if every path between X and Y in $\mathcal{G}$ is d-blocked by $\mathbf{S}$.

Example: Applying the Criterion to the Three Toy Settings

(i) Chain. $A \to M \to B$: $\mathbf{S} = \emptyset$ gives $A \not\perp\!\!\!\perp B$; $\mathbf{S} = \{M\}$ gives $A \perp\!\!\!\perp B \mid M$.
(ii) Fork. $A \leftarrow M \to B$: $\mathbf{S} = \emptyset$ gives $A \not\perp\!\!\!\perp B$; $\mathbf{S} = \{M\}$ gives $A \perp\!\!\!\perp B \mid M$.
(iii) Collider. $A \to M \leftarrow B$: $\mathbf{S} = \emptyset$ gives $A \perp\!\!\!\perp B$ (collider blocks the path); $\mathbf{S} = \{M\}$ gives $A \not\perp\!\!\!\perp B \mid M$ (conditioning opens the path — Berkson's bias).

Example: A Direct Probability Proof for the Chain

Consider the chain $X_1 \to X_2 \to X_3$ with factorization $p(x_1, x_2, x_3) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)$. Conditioning on X_2:

$$p(x_1, x_3 \mid x_2) = \frac{p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)}{p(x_2)} = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Hence $X_1 \perp\!\!\!\perp X_3 \mid X_2$. The analogous fork case $X_1 \leftarrow X_2 \to X_3$ is left as an exercise.

Theorem: Soundness of d-Separation [@pearl2009causality, Theorem 1.2.4]

If $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$, then $X \perp\!\!\!\perp Y \mid \mathbf{S}$ in every distribution that is Markov with respect to $\mathcal{G}$.

Remark: Soundness, Not Converse

The theorem states that d-separation *implies* conditional independence for every distribution Markov with respect to $\mathcal{G}$. The converse need not hold without additional assumptions such as *faithfulness*: a conditional independence may occur because of special parameter values even when the corresponding nodes are not d-separated in the graph.

2.2.4 Practical Ways to Check d-Separation

1. Bayes-Ball Intuition. Imagine releasing a ball from X asking whether it can reach Y, with nodes in $\mathbf{S}$ shaded. The ball: passes through chain/fork nodes not in $\mathbf{S}$; stops at chain/fork nodes in $\mathbf{S}$; stops at colliders not in $\mathbf{S}$ (with no descendant in $\mathbf{S}$); passes through colliders in $\mathbf{S}$ (or with a descendant in $\mathbf{S}$). The algorithm is formalized in Shachter (1998).

2. Moral Graph Transformation. (1) Restrict to the ancestral set $\mathrm{An}(X \cup Y \cup \mathbf{S})$. (2) Moralize: for every collider $A \to C \leftarrow B$, add an undirected edge $A - B$. (3) Remove all edge directions. (4) Delete all nodes in $\mathbf{S}$. If X and Y are disconnected, then $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$. See Lauritzen (1996, Ch. 3) for a full treatment.

Example: Checking d-Separation in a Four-Node DAG

Consider the DAG with edges $Z \to T$, $U \to T$, $T \to Y$, $U \to Y$.
Query 1. Is $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$?
Bayes-ball: The only path is $Z \to T \leftarrow U$. Node T is a collider with $T \notin \emptyset$, so the ball stops. Hence $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$.
Moral graph: Ancestral set of $\{Z, U\}$ is $\{Z, U\}$ (neither has parents); T and Y are discarded. No edges, no colliders. Z and U are disconnected $\Rightarrow (Z \perp\!\!\!\perp U)_{\mathcal{G}}$.
Query 2. Is $(Z \perp\!\!\!\perp U \mid T)_{\mathcal{G}}$?
Bayes-ball: Same path $Z \to T \leftarrow U$. Now $T \in \{T\}$: the collider is observed, so the ball passes through. Hence $Z \not\perp\!\!\!\perp U \mid T$.
Moral graph: Ancestral set of $\{Z, U, T\}$ includes T. Moralize: add edge $Z - U$ for the collider $Z \to T \leftarrow U$. After removing T, the remaining graph has edge $Z - U$. Connected $\Rightarrow Z \not\perp\!\!\!\perp U \mid T$.
Comparing: Z and U are marginally independent (instrument is exogenous) but become dependent once we condition on T — Berkson's bias.

Example: Practice DAG — Eight-Node Graph

Consider the DAG with nodes W, X_1, T, M_1, M_2, Y, C, D and edges $W \to X_1$, $W \to T$, $W \to Y$, $T \to M_1$, $M_1 \to Y$, $M_1 \to M_2$, $X_1 \to C$, $M_2 \to C$, $C \to D$.
(1) Is $X_1 \perp\!\!\!\perp Y$? No. Two open paths: $X_1 \leftarrow W \to Y$ (fork at W, unblocked) and $X_1 \leftarrow W \to T \to M_1 \to Y$ (also open). The path $X_1 \to C \leftarrow M_2$ ends at a collider C with $C \notin \emptyset$, so it is blocked.
(2) Is $X_1 \perp\!\!\!\perp Y \mid W$? Yes. Both open paths above pass through the fork W; conditioning on W

blocks them. The remaining path $X_1 \to C \leftarrow M_2 \leftarrow \cdots$ has collider $C \notin \{W\}$ — blocked. All paths blocked.
(3) Is $T \perp\!\!\!\perp M_2 \mid M_1$? Yes. The direct path $T \to M_1 \to M_2$ is a chain with $M_1 \in \{M_1\}$ — blocked. Other paths through W or Y also contain blocked colliders. All paths blocked.
(4) Does conditioning on C create collider bias? Yes. C is a collider on $X_1 \to C \leftarrow M_2$. Marginally $X_1 \perp\!\!\!\perp M_2$ (blocked by the collider). Conditioning on C opens the path, inducing spurious association.
(5) Does conditioning on D (a descendant of C) open the path $X_1 \to C \leftarrow M_2$? Yes. By clause (2) of the d-blocking definition, a collider path is unblocked whenever the collider *or any of its descendants* is in the conditioning set.

2.3 Collider Bias and the IV DAG

2.3.1 Berkson's Bias

Definition: Collider Bias

Collider bias is the spurious association induced between two variables when one conditions on their common effect (a collider), or on a descendant of that collider. In a path $A \to C \leftarrow B$, conditioning on C or any descendant of C can make A and B statistically dependent even if they are marginally independent.

Collider Bias vs. Confounding

Unlike confounding, which is *removed* by conditioning on a common cause, collider bias is *created* by conditioning on a common effect. Adding a collider or a descendant of a collider to the adjustment set introduces bias rather than reducing it.

Example: Collider Bias through Selection — Talent and Wealth

Suppose both talent (A) and wealth (B) increase the probability of admission to an elite school, so admission (S) is a collider: $A \to S \leftarrow B$. In the general population, $A \perp\!\!\!\perp B$ (the collider blocks the path). Among admitted students — conditioning on S — lower talent makes higher wealth more likely, and vice versa. This is collider bias: restricting to admitted students is precisely conditioning on the common effect.

2.3.2 Full d-Separation Analysis of the IV DAG

We work through the IV DAG with edges $Z \to T$, $T \to Y$, $U \to T$, $U \to Y$, with U *unobserved*. Because U cannot be conditioned on, the back-door path $T \leftarrow U \to Y$ cannot be blocked by adjustment, making the instrument Z the only route to identification.

Classifying paths by type:

Endpoints	Path	Type	Why
$Z \leftrightarrow Y$	$Z \to T \to Y$	*Causal*	Directed; carries the IV signal
$Z \leftrightarrow Y$	$Z \to T \leftarrow U \to Y$	*Associational*	Collider at T; activates back-door via U
$Z \leftrightarrow U$	$Z \to T \leftarrow U$	*Associational*	Collider at T; dormant unless T conditioned on

Question 1. Is $Z \perp\!\!\!\perp Y$? No. The directed path $Z \to T \to Y$ is open (chain, not conditioned on T). The path $Z \to T \leftarrow U \to Y$ has a collider at T with $T \notin \emptyset$ — blocked.

Question 2. Is $Z \perp\!\!\!\perp Y \mid T$? No. The causal path $Z \to T \to Y$ is blocked (conditioning on T in the chain). But the collider at T in $Z \to T \leftarrow U \to Y$ is now opened. Hence $Z \not\!\perp\!\!\!\perp Y \mid T$.

Question 3. Is $Z \perp\!\!\!\perp Y \mid \{T, U\}$? Yes. The causal path is blocked by conditioning on T. The path through U has the collider at T opened (by conditioning on T), but then blocked at U (by conditioning on U). Both paths blocked.

Question 4. Is $Z \perp\!\!\!\perp U$? Yes. The only path is $Z \to T \leftarrow U$ with a collider at T. Since $T \notin \emptyset$ and no descendant of T is in $\emptyset$, the path is blocked.

Interpretation. Question 1 establishes *relevance*: Z has an open causal path to Y through T. Question 4 establishes *exogeneity*: the instrument is independent of unmeasured confounding. Question 3 establishes the *exclusion restriction*: once T and U are held fixed, Z carries no residual information about Y. Note that this requires observing U, which is unavailable by assumption; the IV strategy exploits the exclusion restriction indirectly through the moment condition $\mathbb{E}[\varepsilon \cdot Z] = 0$ (Chapter 7). Question 2 is the *warning*: conditioning on T alone simultaneously closes the causal channel and opens the confounded channel — worse than no adjustment at all.

The IV DAG encodes the substantive assumptions behind instrument validity graphically, but does not make them testable in any strong sense.

2.4 The Markov Property and Factorization

Up to this point, we have used DAGs qualitatively. We now connect the graph to probability algebra: the same parent structure that governs d-separation also determines how the joint distribution factorizes.

Without structural assumptions, any joint distribution can always be written by repeated conditioning, but that generic representation is too high-dimensional to reveal much structure. A DAG becomes statistically meaningful because, together with the Markov property, it replaces the generic factorization by a sparse one involving only the parents of each node.

Proposition: Conditional Independence in the Three-Node Chain

Let $X_1 \to X_2 \to X_3$ be a chain with Markov factorization $p(x_1, x_2, x_3) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)$. Then $X_1 \perp\!\!\!\perp X_3 \mid X_2$. (Proved in the chain example above; the fork case is analogous.)

Definition: Local Markov Property

A distribution P satisfies the *local Markov property* with respect to $\mathcal{G}$ if, for every node $V_i \in \mathcal{V}$:

$$V_i \perp\!\!\!\perp \left[\mathrm{Nd}(V_i) \setminus \mathrm{Pa}(V_i)\right] \,\Big|\, \mathrm{Pa}(V_i).$$

Once we condition on the direct causes of a node, that node is independent of all variables that are neither its descendants nor its parents.

Remark

The *global Markov property* is the statement that every d-separation in $\mathcal{G}$ implies a conditional independence in P — precisely the content of the Soundness theorem. For DAGs, the local and global Markov properties are equivalent (Lauritzen 1996, Proposition 3.27): the local property implies the global one via the Markov factorization.

An immediate consequence is the **Markov factorization**:

$$p(v_1, \dots, v_k) = \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i)) \,. \tag{2.1}$$

Example: Markov Factorization in the Education DAG

For the DAG $N \to E$, $B \to E$, $B \to Y$, $E \to Y$: $p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b)$. A regression of Y on E alone is confounded by B. Conditioning on B blocks the back-door path $E \leftarrow B \to Y$ and identifies $P(y \mid \mathrm{do}(E{=}e))$ via the back-door formula (Chapter 3).

Example: Markov Factorization in the IV DAG

Including the latent U: $p(z, t, y, u) = p(z)\, p(u)\, p(t \mid z, u)\, p(y \mid t, u)$. The observable factorization is obtained by marginalizing over U:

$$p(z, t, y) = \int p(z)\, p(u)\, p(t \mid z, u)\, p(y \mid t, u)\, \mathrm{d}u.$$

This cannot be simplified to $p(z)\, p(t \mid z)\, p(y \mid t)$ when U is a common cause of T and Y — this is precisely the endogeneity problem.

2.5 Worked Example: The Education–Earnings DAG

This section is the template for how we will use DAGs throughout the course: specify the DAG, read off the factorization, use d-separation to identify the implied conditional independences, and interpret in causal terms.

Practice. Before working through the example below, return to the eight-node DAG example and rework each of the five queries from scratch, following three steps: (1) list every path; (2) classify every intermediate node as chain, fork, or collider; (3) determine which paths are blocked or open.

The causal story. Education (E) affects earnings (Y); family background (B) is a common cause of both; neighborhood (N) affects education but has no direct effect on earnings. All four variables are observed (no latent confounders).

Step 1 — Markov factorization. $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$, $\mathrm{Pa}(E) = \{N, B\}$, $\mathrm{Pa}(Y) = \{E, B\}$:

$$p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

Step 2 — d-Separation. There are two paths between N and Y:

- Path 1: $N \to E \to Y$ (a chain through E).
- Path 2: $N \to E \leftarrow B \to Y$ (a collider at E, followed by the fork leg $B \to Y$).

(i) Is $(N \perp\!\!\!\perp Y)_{\mathcal{G}}$*?* Path 1 is a chain with nothing conditioned on, so it is open. Path 2 has a collider at E (not conditioned on), so it is blocked. One open path: $N \not\!\perp\!\!\!\perp Y$.

(ii) Is $(N \perp\!\!\!\perp Y \mid E)_{\mathcal{G}}$*?* Path 1 is blocked (conditioning on E in the chain). Path 2 has a collider at E opened by conditioning, and the activated path continues through B (not conditioned on), so it is open. Hence $N \not\!\perp\!\!\!\perp Y \mid E$ — a collider trap: conditioning on E introduces bias, not removes it.

(iii) Is $(N \perp\!\!\!\perp Y \mid \{E, B\})_{\mathcal{G}}$*?* Path 1 is blocked by E. Path 2 is opened at collider E but then blocked at B. All paths blocked: $(N \perp\!\!\!\perp Y \mid E, B)_{\mathcal{G}}$.

(iv) Is $(N \perp\!\!\!\perp B)_{\mathcal{G}}$*?* The only path $N \to E \leftarrow B$ has a collider at E (not conditioned on). Path is blocked: $N \perp\!\!\!\perp B$.

Step 3 — Conditional Independence. By the Soundness theorem:

$$N \perp\!\!\!\perp B, \qquad N \perp\!\!\!\perp Y \mid E, B, \qquad N \not\!\perp\!\!\!\perp Y, \qquad N \not\!\perp\!\!\!\perp Y \mid E.$$

The statement $N \not\!\perp\!\!\!\perp Y \mid E$ is an important warning: conditioning only on education when studying the neighborhood–earnings association *introduces* bias through the activated collider path.

Step 4 — Identification. We wish to identify $P(y \mid \mathrm{do}(E{=}e))$. The only back-door path is $E \leftarrow B \to Y$ (a fork at B).

- $\mathbf{S} = \{B\}$: blocks the fork $E \leftarrow B \rightarrow Y$, and B is not a descendant of E. Valid.
- $\mathbf{S} = \{N\}$: does not block $E \leftarrow B \rightarrow Y$. Invalid.

With $\mathbf{S} = \{B\}$, the back-door formula (Chapter 3) gives:

$$P(y \mid \mathrm{do}(E{=}e)) = \sum_b P(y \mid e, b)\, P(b),$$

identifying the causal effect entirely from observational data.

2.6 The Big Picture

The central logic runs from a qualitative causal graph to an identification formula:

structural assumptions $\Longrightarrow$ DAG $\Longrightarrow$ d-separation relations $\Longrightarrow$ conditional independences (Markov) $\Longrightarrow$ identification for

Chapter 2 establishes the first four links in this chain. Chapters 3 and beyond use the same graphical machinery to derive specific identification results: back-door adjustment, front-door adjustment, and do-calculus formulas.

2.7 Summary

1. **DAGs as causal structure.** A DAG encodes which variables are causally connected, which are potential confounders, and which variables may block or open paths.
2. **Three-node motifs.** Every path is built from chains, forks, and colliders. Chains and forks are open by default and are blocked by conditioning on the middle node. Colliders are blocked by default and are opened by conditioning on the collider or on one of its descendants.
3. **d-Separation and conditional independence.** X and Y are d-separated by $\mathbf{S}$ exactly when every path between them is blocked by $\mathbf{S}$. Under the Markov property, d-separation implies the corresponding conditional independence. Some implications may be assessed empirically, but assumptions involving unobserved variables are generally not testable.
4. **Markov factorization.** The local Markov property yields $p(v_1, \ldots, v_k) = \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i))$, the bridge from graphical structure to probability calculus.
5. **Collider bias.** Conditioning on a collider — or on a descendant of a collider — can induce a spurious association between otherwise independent variables (Berkson 1946). This is the opposite of confounding adjustment.
6. **A practical workflow.** To analyze a DAG: identify the graph structure, determine which paths are open or blocked, translate d-separation statements into conditional independences under the Markov property, and interpret in light of the causal question.

2.8 Problems

1. Warm-up: a single collider. Consider the DAG $X \rightarrow Y \leftarrow Z$.

(a) Identify the structural role of Y on the path $X \rightarrow Y \leftarrow Z$.
(b) Is $X \perp\!\!\!\perp Z$? Apply the d-separation criterion and state which blocking rule applies.
(c) Is $X \perp\!\!\!\perp Z \mid Y$? Explain what happens to the path when Y is conditioned on, and describe in one sentence the real-world phenomenon this illustrates.

2. d-Separation practice. Consider the DAG: $A \rightarrow B \rightarrow D$, $A \rightarrow C \rightarrow D$, $B \rightarrow E$, $C \rightarrow E$.

(a) List all paths between A and E. (*Hint*: there are four paths; two pass through D.)
(b) For each path, identify the role (chain, fork, collider) of each intermediate node.
(c) Does $\{B, C\}$ d-separate A and E?
(d) Does $\{D\}$ d-separate B and C? What type of node is D on the path $B \rightarrow D \leftarrow C$?

3. Berkson's bias. Suppose X and Y are independent standard normal variables, and let $S = \mathbf{1}[X+Y > 0]$.

(a) Verify analytically that $\mathrm{Cov}(X, Y \mid S{=}1) < 0$.
(b) Draw the DAG for (X, Y, S) and identify S as a collider.
(c) Explain in one sentence why restricting the analysis to the subsample with $S = 1$ biases estimates of any association between X and Y.

4. Markov factorization and collider activation. Consider the DAG with edges $A \to E$, $A \to W$, $F \to E$, $E \to W$, where A = ability, F = family income, E = education, W = wages.

(a) Write down the Markov factorization $p(a, f, e, w)$.
(b) Is $(F \perp\!\!\!\perp W)_{\mathcal{G}}$? List all paths between F and W and determine which are open.
(c) Is $(F \perp\!\!\!\perp W \mid E)_{\mathcal{G}}$? Identify the role of E on each path.
(d) A researcher regresses W on E and F, omitting A. Is the coefficient on E a causal effect? Explain using the graph.

5. Soundness theorem for the collider. Consider the collider $A \to M \leftarrow B$ with factorization $p(a, m, b) = p(a)\, p(b)\, p(m \mid a, b)$.

(a) By marginalizing over M, show that $A \perp\!\!\!\perp B$ in the joint distribution. This verifies the Soundness theorem for $\mathbf{S} = \emptyset$.
(b) Show that conditioning on M breaks this independence: write out $p(a, b \mid m)$ and explain why it does *not* factorize into $p(a \mid m)\, p(b \mid m)$ in general.
(c) Explain in one sentence why parts (a) and (b) together are consistent with the Soundness theorem. (*Hint*: the theorem is a one-directional statement.)

6. Terminology check. Consider the DAG with edges $U \to X$, $U \to Z$, $X \to W$, $Z \to W$, $W \to Y$.

(a) Identify the parents, children, ancestors, descendants, and non-descendants of node W.
(b) List all pairs of adjacent nodes.
(c) Which pairs of nodes are connected by a directed path? List every such pair and the corresponding path.
(d) Write the Markov factorization $p(u, x, z, w, y)$.

7. Markov factorization and local Markov property. Consider the DAG with edges $X_1 \to X_2$, $X_1 \to X_3$, $X_2 \to X_4$, $X_3 \to X_4$.

(a) Write the joint density $p(x_1, x_2, x_3, x_4)$ implied by the Markov factorization.
(b) State the local Markov property for each of the four nodes.
(c) Is $(X_2 \perp\!\!\!\perp X_3)_{\mathcal{G}}$? Is $(X_2 \perp\!\!\!\perp X_3 \mid X_1)_{\mathcal{G}}$? Justify each answer by listing all paths.

8. Toy proof: conditional independence in the fork. Consider the fork $X_1 \leftarrow X_2 \to X_3$ with factorization $p(x_1, x_2, x_3) = p(x_2)\, p(x_1 \mid x_2)\, p(x_3 \mid x_2)$.

(a) Show directly, by conditioning on $X_2 = x_2$, that $X_1 \perp\!\!\!\perp X_3 \mid X_2$.
(b) Is $X_1 \perp\!\!\!\perp X_3$ marginally? Justify both graphically and algebraically.
(c) Explain in one sentence what the fork represents substantively and why conditioning on the common cause removes the association.

9. d-Separation in the practice DAG. Refer to the eight-node DAG in the Practice DAG example (nodes W, X_1, T, M_1, M_2, Y, C, D; edges $W \to X_1$, $W \to T$, $W \to Y$, $T \to M_1$, $M_1 \to Y$, $M_1 \to M_2$, $X_1 \to C$, $M_2 \to C$, $C \to D$). For each query, state whether it is true or false and justify by listing all relevant paths.

(a) $X_1 \perp\!\!\!\perp Y$
(b) $X_1 \perp\!\!\!\perp Y \mid W$
(c) $T \perp\!\!\!\perp M_2 \mid M_1$
(d) Does conditioning on C induce collider bias between X_1 and M_2? Identify the specific path opened.
(e) Does conditioning on D (a descendant of C) open the path $X_1 \to C \leftarrow M_2$? State which clause of the d-blocking definition applies.

Chapter 3

The Do-Calculus and Identification Criteria

Learning Objectives

By the end of this chapter, students should be able to:

1. Define the two intervention graph operations $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$, explain the causal meaning of $\mathcal{G}_{\overline{X}}$, and describe the technical role of $\mathcal{G}_{\underline{X}}$.
2. Apply the back-door criterion to determine whether a set $\mathbf{S}$ identifies a causal effect, and write the back-door adjustment formula.
3. Apply the front-door criterion in graphs where the confounder is unobserved but a suitable mediator exists.
4. State all three rules of do-calculus, identify the graph condition that licenses each rule, and use them to prove the back-door and front-door formulas.
5. State the completeness theorem of Shpitser and Pearl (2006) and explain its practical implication: if the do-calculus cannot identify a quantity from the observational distribution relative to the assumed graph, then no purely observational method can do so without additional assumptions or new data.

How to Read This Chapter

This chapter has two pedagogical layers. Section 3.1–Section 8.8 develop the two most important concrete identification criteria — back-door and front-door adjustment — using intervention graphs and direct graphical reasoning. These sections are the core material for a first reading.
Section 3.4–Section 3.6 then place these criteria inside the more general theory of the do-calculus and identifiability. These later sections are conceptually important but more abstract and may be read as a second pass after the concrete criteria are understood.

3.1 From d-Separation to Intervention: Intervention Graphs

In Chapter 2 we learned to read conditional independence from a DAG using d-separation. This chapter takes the next step: translating the graphical language into *identification formulas* — expressions that write the interventional distribution $P(y \mid \mathrm{do}(T{=}t))$ entirely in terms of quantities observable from data.

3.1.1 What Does Intervention Mean?

Conditioning versus intervening. The conditional distribution $P(Y \mid T{=}t)$ describes the subpopulation of units for whom T was *observed* to equal t. The interventional distribution $P(Y \mid \mathrm{do}(T{=}t))$, by contrast, describes the population that would result if T were *externally set* to t for everyone. The two coincide only when T has no unmeasured common causes with Y.

A simple illustration: temperature Z causes both ice-cream sales T and crime Y. $P(Y \mid T{=}t)$ is the crime rate on days when sales happen to equal t — confounded by Z. $P(Y \mid \mathrm{do}(T{=}t))$ is the crime rate we would observe if we *fixed* sales at level t by decree. If T has no direct causal path to Y, then $P(Y \mid \mathrm{do}(T{=}t)) = P(Y)$ for all t.

The structural basis of the do-operator. In the SEM framework, every variable is generated by a structural equation. For the treatment node: $T = f_T(\mathrm{Pa}(T), U_T)$. An intervention $\mathrm{do}(T{=}t)$ *replaces this entire equation* with the constant $T = t$, with two consequences: (1) T is no longer influenced by its parents; (2) all other structural equations remain unchanged.

Graph surgery as the graphical realization. Because $\mathrm{Pa}(T)$ no longer affects T after the intervention, every arrow pointing into T in the DAG should be removed. The resulting graph — the *intervention graph* $\mathcal{G}_{\overline{T}}$ — represents the post-intervention world. D-separation in $\mathcal{G}_{\overline{T}}$ encodes conditional independence in $P(\cdot \mid \mathrm{do}(T{=}t))$, not in the original P. This is the key link that allows graphical reasoning to answer causal questions.

3.1.2 The Two Graph Operations

Definition: Intervention Graphs $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a DAG and $X \subseteq \mathcal{V}$.

- $\mathcal{G}_{\overline{X}}$ (*intervention graph*, or *mutilated graph*) is obtained by **deleting all arrows into** X. This represents $\mathrm{do}(X{=}x)$, severing X's dependence on its former parents.
- $\mathcal{G}_{\underline{X}}$ (*auxiliary observation graph*) is obtained by **deleting all arrows out of** X. This is a *technical device* used in graphical conditions for certain do-calculus steps. It does *not* represent conditioning on X.

3.2 The Back-Door Criterion

Definition: Back-Door Criterion [@pearl1993comment]

A set $\mathbf{S}$ of observed variables satisfies the **back-door criterion** for the effect of T on Y in $\mathcal{G}$ if:

1. No node in $\mathbf{S}$ is a descendant of T.
2. $\mathbf{S}$ blocks every *back-door path* from T to Y — every path that begins with an arrow pointing *into* T.

Remark

The term “back-door” reflects the geometry: a back-door path enters T from behind — it begins with an arrow pointing into T — and represents a confounding route. By contrast, directed causal paths $T \to \cdots \to Y$ leave T through the front and carry the genuine effect.
Condition 2 is equivalent to requiring $(Y \perp\!\!\!\perp T \mid \mathbf{S})_{\mathcal{G}_{\underline{T}}}$: $\mathbf{S}$ d-separates T and Y in the graph obtained by deleting all arrows *out of* T (which eliminates causal paths, leaving only back-door paths). The graph $\mathcal{G}_{\overline{T}}$, which deletes arrows into T, is the wrong graph for this check.
Condition 1 adds an important constraint absent from Chapter 2: $\mathbf{S}$ must contain no descendant of T. Every valid back-door set is a d-separator in $\mathcal{G}_{\underline{T}}$, but not vice versa.

Example: A Single Confounder

Consider the DAG $T \leftarrow C \to Y$ with $T \to Y$, where C is observed. Does $\mathbf{S} = \{C\}$ satisfy the back-door criterion?

1. C is not a descendant of T.
2. $\{C\}$ blocks $T \leftarrow C \to Y$ (fork at C).

The back-door formula gives: $P(y \mid \mathrm{do}(T{=}t)) = \sum_c P(y \mid t, c)\, P(c)$.

Example: Two Back-Door Paths — Both Must Be Blocked

Consider treatment T, outcome Y, observed C_1 and C_2, unobserved U, with edges $C_1 \to T$, $C_1 \to Y$, $U \to T$, $U \to C_2$, $C_2 \to Y$, $T \to Y$. There are exactly two back-door paths:

1. $T \leftarrow C_1 \to Y$ (direct confounding by C_1)
2. $T \leftarrow U \to C_2 \to Y$ (indirect confounding via U routing through proxy C_2)

$\mathbf{S} = \{C_1\}$ fails: blocks path 1 but leaves path 2 open. $\mathbf{S} = \{C_2\}$ fails: blocks path 2 but leaves path 1 open. $\mathbf{S} = \{C_1, C_2\}$ succeeds:

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_{c_1,c_2} P(y \mid t, c_1, c_2)\, P(c_1, c_2).$$

The unobserved U never appears: conditioning on C_2 blocks the path at the chain node, exploiting the *position* of the observed variables, not direct measurement of the confounders.

Why Condition 1 Is Essential: Conditioning on a Mediator

Consider a drug T that affects recovery Y via two routes: direct ($T \to Y$) and indirect through blood pressure M ($T \to M \to Y$), with unobserved confounder $U \to T$, $U \to Y$.
Naively trying $\mathbf{S} = \{M\}$ fails for two reasons: (1) M is a descendant of T, violating Condition 1; (2) conditioning on M blocks the causal pathway $T \to M \to Y$ while leaving the back-door path $T \leftarrow U \to Y$ untouched. The adjusted quantity $\sum_m P(y \mid t, m)P(m)$ is a causally ambiguous mixture of attenuated causal signal and uncontrolled confounding — neither the total causal effect nor the direct effect.

M-Bias: Conditioning on a Collider Opens a Closed Back-Door Path

Consider $T \to Y$ with $U_1 \to T$, $U_1 \to C$, $U_2 \to C$, $U_2 \to Y$ (U_1, U_2 unobserved, C observed). The only back-door path is $T \leftarrow U_1 \to C \leftarrow U_2 \to Y$, which has a collider at C. With $\mathbf{S} = \emptyset$, the path is blocked — the causal effect is already identified: $P(y \mid \mathrm{do}(t)) = P(y \mid t)$.
Including C in the adjustment set *activates* the collider, opening the previously dormant path. The set $\{C\}$ fails Condition 2: it does not block the back-door path; it creates one. This is called *M-bias* (from the M-shaped skeleton). The practical lesson: Condition 1 alone ("include all pretreatment variables") is not sufficient.

Theorem: Back-Door Adjustment Formula [@pearl1993comment]

If $\mathbf{S}$ satisfies the back-door criterion for the effect of T on Y, and $P(T{=}t \mid \mathbf{S}{=}\mathbf{s}) > 0$ for all $\mathbf{s}$ with $P(\mathbf{S}{=}\mathbf{s}) > 0$ (positivity), then:

$$P(y \mid \mathrm{do}(T{=}t)) = \int P(y \mid T{=}t,\ \mathbf{S}{=}\mathbf{s})\, dP(\mathbf{s}). \tag{3.1}$$

Equivalently: $\mathbb{E}[h(Y) \mid \mathrm{do}(T{=}t)] = \mathbb{E}_{\mathbf{S}}[\mathbb{E}[h(Y) \mid T{=}t,\ \mathbf{S}]]$.

Remark: Positivity Is a Data-Support Condition

The positivity condition requires every value of T to be observable within each stratum of $\mathbf{S}$. It is a property of the joint distribution P, not of the graph. Both the back-door criterion (graphical) and positivity (distributional) must hold simultaneously. When positivity fails on a set of measure zero, identification still holds in theory but sparse or empty cells pose practical difficulties.

3.2.1 The Adjustment Formula as Standardization

Equation 5.5 is also known as the *standardization formula* or *g-formula* (Robins 1986). The structure is important:

- $P(y \mid t, \mathbf{s})$ is the stratum-specific conditional distribution. Within each stratum, all back-door paths are blocked by $\mathbf{S}$, so this conditional distribution is causal.
- The outer integration $\int \cdots dP(\mathbf{s})$ weights strata by the *population marginal* distribution of $\mathbf{S}$, not the treatment-conditional $dP(\mathbf{s} \mid t)$. In a randomized experiment, $\mathbf{S} = \emptyset$ and Equation 5.5 reduces to $P(y \mid \mathrm{do}(t)) = P(y \mid t)$.

Example: Back-Door Adjustment — A Numerical Walkthrough

Setup. Single confounder graph ($T \leftarrow C \rightarrow Y$, $T \rightarrow Y$), binary variables. C = disease severity (0 = mild, 1 = severe); T = treatment; Y = recovery. $P(C{=}1) = 0.5$, $P(T{=}1 \mid C{=}0) = 0.20$, $P(T{=}1 \mid C{=}1) = 0.80$.
Outcome probabilities. $P(Y{=}1 \mid T{=}1, C{=}0) = 0.55$, $P(Y{=}1 \mid T{=}0, C{=}0) = 0.45$, $P(Y{=}1 \mid T{=}1, C{=}1) = 0.45$, $P(Y{=}1 \mid T{=}0, C{=}1) = 0.35$.
Naïve (unadjusted) estimates. $P(Y{=}1 \mid T{=}1) = 0.50$, $P(Y{=}1 \mid T{=}0) = 0.39$, difference $= +0.11$.
Back-door adjusted estimates.

$$P(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.55 \times 0.5 + 0.45 \times 0.5 = 0.50.$$

$$P(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.45 \times 0.5 + 0.35 \times 0.5 = 0.40.$$

Causal risk difference $= 0.50 - 0.40 = 0.10$. The naïve estimate ($+0.11$) is close here because the confounding is mild; in general it can differ substantially.

3.3 The Front-Door Criterion

The back-door criterion requires observed variables to block every confounding path. When the confounder U is unobserved and no observed variable lies on the back-door path, a different strategy is needed: the *front-door criterion* intercepts the causal pathway at an observed mediator.

Definition: Front-Door Criterion

A set M of observed variables satisfies the **front-door criterion** for the effect of T on Y in $\mathcal{G}$ if:

1. *M intercepts all directed paths* from T to Y: every directed path from T to Y passes through some node in M.
2. There are *no unblocked back-door paths from T to M*: the path set $\{T \leftarrow \cdots \rightarrow M\}$ is empty or blocked.
3. *All back-door paths from M to Y are blocked by T*: conditioning on T blocks every path that begins with an arrow into M and ends at Y.

The front-door strategy avoids unobserved confounding by splitting the total effect into two identifiable pieces: (1) the effect of T on M (unconfounded by Condition 2), and (2) the effect of M on Y (made identifiable after conditioning on T by Condition 3).

Example: Smoking, Tar, and Cancer [@pearl1995causal]

T = smoking, M = tar deposits, Y = lung cancer, U = genetic predisposition (unobserved). Graph: $T \rightarrow M \rightarrow Y$ with $U \rightarrow T$ and $U \rightarrow Y$, no direct $T \rightarrow Y$ edge.
Conditions 1–3 hold: (1) the only directed path $T \rightarrow M \rightarrow Y$ passes through M; (2) no back-door path from T to M exists (no $U \rightarrow M$ edge); (3) the only back-door path from M to Y is $M \leftarrow T \leftarrow U \rightarrow Y$, which is blocked by conditioning on T.
The front-door formula applies:

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m P(m \mid t) \sum_{t'} P(y \mid t', m)\, P(t').$$

Example: The Prototypical Front-Door Example

The simplest front-door graph is $T \to M \to Y$ with $U \to T$, $U \to Y$ (no direct $T \to Y$ edge). Checking the criterion:

1. The only directed path is $T \to M \to Y$, which passes through M.
2. No back-door path from T to M exists (no $U \to M$ edge).
3. The only back-door path from M to Y is $M \leftarrow T \leftarrow U \to Y$; conditioning on T blocks it.

Example: When the Front-Door Criterion Fails

Case (a): Condition 2 fails — the $T \to M$ link is confounded. Adding the edge $U \to M$: now the path $T \leftarrow U \to M$ is an unblocked back-door path from T to M. Stage 1 uses $P(M{=}m \mid T{=}t)$ as if the $T \to M$ link were unconfounded, but $U \to M$ means the association mixes cause and confounding.
Case (b): Condition 3 fails — the $M \to Y$ link has an extra confounder. Adding $V \to M$ and $V \to Y$ (V unobserved): the path $M \leftarrow V \to Y$ is not blocked by conditioning on T. Stage 2 can no longer use T as a valid back-door adjustment for $M \to Y$.

Modification	Condition violated	Stage broken
Add $U \to M$	Cond. 2: $T \to M$ confounded	Stage 1
Add $V \to M$, $V \to Y$	Cond. 3: $M \to Y$ confounded by V	Stage 2

In both cases the criterion detects the failure before any formula is written down.

3.3.1 The Front-Door Formula

Theorem: Front-Door Adjustment Formula [@pearl1995causal]

If M satisfies the front-door criterion for the effect of T on Y, and $P(t) > 0$ for all t, then:

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m \underbrace{P(m \mid T{=}t)}_{\text{Stage 1}} \underbrace{\sum_{t'} P(y \mid T{=}t',\, M{=}m)\, P(t')}_{\text{Stage 2}}. \tag{3.2}$$

Equivalently, writing $\mu(t', m) = \mathbb{E}[Y \mid T{=}t', M{=}m]$:

$$\mathbb{E}[Y \mid \mathrm{do}(T{=}t)] = \mathbb{E}_{M|T=t}[\mathbb{E}_T[\mu(T,\, M)]].$$

Reading the formula. Stage 1: $P(m \mid T{=}t)$ is the distribution of M given $T{=}t$. Condition 2 guarantees no confounding between T and M, so this is directly identifiable. Stage 2: $\sum_{t'} P(y \mid T{=}t', M{=}m)\, P(t')$ is the back-door-adjusted effect of M on Y, with T as the adjustment variable. Condition 3 guarantees that conditioning on T blocks all back-door paths from M to Y.

Heuristic derivation. The formula is the composition of two back-door applications. The role of each condition: Condition 1 together with Condition 2 allow decomposing through M; Condition 2 licenses replacing $\mathrm{do}(T{=}t)$ with conditioning in the M-marginal; Condition 3 licenses back-door adjustment for $M \to Y$.

3.4 The Three Rules of Do-Calculus

Both the back-door and front-door formulas are derivable from a single algebraic engine: the three rules of the do-calculus. Each rule is a conditional independence statement in a specific intervention graph.

Graph subscripts as bookkeeping. Each rule checks d-separation in a graph formed by specific edge deletions:

Subscript	Appears in	Arrows deleted
$\mathcal{G}_{\overline{X}}$	Rule 1	all arrows *into* X
$\mathcal{G}_{\overline{X}\,\underline{Z}}$	Rule 2	all into X; all out of Z
$\mathcal{G}_{\overline{X}\,\overline{Z(W)}}$	Rule 3	all into X; into Z-nodes not ancestors of W in $\mathcal{G}_{\overline{X}}$

When $W = \emptyset$, Rule 3 uses $\mathcal{G}_{\overline{X}\,\overline{Z}}$ (into both X and Z deleted). D-separation in the modified graph is checked by the same algorithm as Chapter 2.

Theorem: The Three Rules of Do-Calculus [@pearl1995causal]

Let $\mathcal{G}$ be a DAG over $\mathcal{V}$, and let $X, Y, Z, W \subseteq \mathcal{V}$ be disjoint. The rules hold for any distribution P Markov with respect to $\mathcal{G}$.

Rule 1 (Insertion/Deletion of Observations). *Remove or insert an observation when it becomes irrelevant in the post-intervention graph.*

$$P(y \mid \mathrm{do}(x), z, w) = P(y \mid \mathrm{do}(x), w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}}}. \tag{3.3}$$

Rule 2 (Action/Observation Exchange). *Replace an intervention by an observation, or vice versa, when the modified graph makes them equivalent.*

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), z, w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\underline{Z}}}. \tag{3.4}$$

Rule 3 (Insertion/Deletion of Actions). *Remove or insert an intervention when it becomes irrelevant in the modified graph.*

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\overline{Z(W)}}}. \tag{3.5}$$

Remark: Soundness of the Rules

Each rule is *sound*: it holds for every distribution Markov with respect to $\mathcal{G}$. Rule 2's intuition is instructive: deleting Z's outgoing arrows in $\mathcal{G}_{\overline{X}\,\underline{Z}}$ simulates what intervening on Z would remove from Y's perspective. If Y is d-separated from Z after this deletion, then there is no causal channel whose behavior depends on whether Z was intervened on or merely observed.

3.4.1 Intuition for Each Rule

Rule 1 — Adding/Removing Observations. If Y and Z are independent given W in the post-do(x) world, then Z carries no additional information about Y and can be added or dropped from the conditioning set.

Rule 2 — Swapping Action for Observation. The most frequently used rule. Deleting Z's outgoing arrows cuts Z's causal paths to its descendants. If Y and Z are d-separated after this deletion, then intervention and observation on Z produce the same distribution of Y — because the only paths from Z to Y were Z's causal paths, which both operations sever (intervention) or leave intact (observation) equally.

Rule 3 — Deleting an Intervention. After deleting arrows into X and into the relevant Z-nodes, Z has no remaining path to Y. Setting Z to any value has no effect on Y, so do(z) can be dropped.

Example: Rule 1 — Deleting a Redundant Observation

Graph: $Z \to T \to Y$, no other edges. Intervene on T. In $\mathcal{G}_{\overline{T}}$: delete $Z \to T$; node Z is disconnected from Y. Hence $(Y \perp\!\!\!\perp Z)_{\mathcal{G}_{\overline{T}}}$. Rule 1 gives $P(y \mid \mathrm{do}(t), z) = P(y \mid \mathrm{do}(t))$: once T is fixed externally, its former cause Z carries no information about Y.

Example: Rule 2 — Action/Observation Exchange

Graph: $X \to Z \to Y$, no other edges. Form $\mathcal{G}_{\overline{X}\underline{Z}}$ (delete arrows into X — none — and arrows out of Z — removes $Z \to Y$). Node Z has no outgoing arrows; no path from Z to Y exists. Hence $(Y \perp\!\!\!\perp Z)_{\mathcal{G}_{\overline{X}\underline{Z}}}$. Rule 2 gives $P(y \mid \mathrm{do}(x), \mathrm{do}(z)) = P(y \mid \mathrm{do}(x), z)$.

Example: Rule 3 — Deleting an Irrelevant Intervention

Graph: $Z \to X \to Y$, no other edges. Form $\mathcal{G}_{\overline{X}\overline{Z}}$ (delete into X — removes $Z \to X$ — and into Z — none). Node Z is isolated. Hence $(Y \perp\!\!\!\perp Z)_{\mathcal{G}_{\overline{X}\overline{Z}}}$. Rule 3 gives $P(y \mid \mathrm{do}(x), \mathrm{do}(z)) = P(y \mid \mathrm{do}(x))$: once X is fixed, Z's only route to Y is severed.

Example: A Two-Rule Derivation — The Proxy Variable Graph

Graph. $T \to Y$, $U \to T$ (unobserved), $U \to S$ (observed), $S \to Y$.
Goal. Simplify $P(y \mid \mathrm{do}(t))$ to an observational expression. S lies on the back-door path $T \leftarrow U \to S \to Y$ and is observed.
Step 1. Introduce S by total probability: $P(y \mid \mathrm{do}(t)) = \sum_s P(y \mid \mathrm{do}(t), s)\, P(s \mid \mathrm{do}(t))$.
Step 2. Simplify $P(s \mid \mathrm{do}(t))$ by Rule 3. The required graph is $\mathcal{G}_{\overline{T}}$. Because S is not a descendant of T, no path from T to S exists in $\mathcal{G}_{\overline{T}}$. Hence $(S \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$, and Rule 3 gives $P(s \mid \mathrm{do}(t)) = P(s)$.
Step 3. Simplify $P(y \mid \mathrm{do}(t), s)$ by Rule 2. The required graph is $\mathcal{G}_{\underline{T}}$. In $\mathcal{G}_{\underline{T}}$, the only paths between T and Y are back-door paths. The set $\{S\}$ d-separates T and Y in $\mathcal{G}_{\underline{T}}$ (blocking $T \leftarrow U \to S \to Y$ at S). Rule 2 gives $P(y \mid \mathrm{do}(t), s) = P(y \mid t, s)$.
Combining: $P(y \mid \mathrm{do}(t)) = \sum_s P(y \mid t, s)\, P(s)$. $\square$

3.5 Do-Calculus Proofs of the Main Theorems

Proof of the Back-Door Formula via Do-Calculus

Graph: $T \to Y$, $\mathbf{S} \to T$, $\mathbf{S} \to Y$ (with $\mathbf{S}$ satisfying the back-door criterion).
Step 1. Introduce $\mathbf{S}$: $P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t), \mathbf{s})\, dP(\mathbf{s} \mid \mathrm{do}(t))$.
Step 2. Rule 3 with $X = \emptyset$, $Z = T$, $W = \emptyset$: required graph $\mathcal{G}_{\overline{T}}$. Since $\mathbf{S}$ contains no descendants of T (back-door condition 1), $(\mathbf{S} \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$, so $P(\mathbf{s} \mid \mathrm{do}(t)) = P(\mathbf{s})$.
Step 3. Rule 2 with $X = \emptyset$, $Z = T$, $W = \mathbf{S}$: required graph $\mathcal{G}_{\underline{T}}$. Back-door condition 2 states $(Y \perp\!\!\!\perp T \mid \mathbf{S})_{\mathcal{G}_{\underline{T}}}$, so $P(y \mid \mathrm{do}(t), \mathbf{s}) = P(y \mid t, \mathbf{s})$.
Combining: $P(y \mid \mathrm{do}(t)) = \int P(y \mid t, \mathbf{s})\, dP(\mathbf{s})$. $\square$

Proof of the Front-Door Formula via Do-Calculus

Graph: $T \to M \to Y$, $U \to T$ (unobserved), $U \to Y$.
Step 1. Introduce M: $P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t), m)\, dP(m \mid \mathrm{do}(t))$.
Step 2. Rule 2 to simplify $P(m \mid \mathrm{do}(t))$: required graph $\mathcal{G}_{\underline{T}}$ (delete $T \to M$). In this graph no path from T to M remains (front-door condition 2 ensures no $U \to M$ edge), so $(M \perp\!\!\!\perp T)_{\mathcal{G}_{\underline{T}}}$. Rule 2 gives $P(m \mid \mathrm{do}(t)) = P(m \mid t)$.
Step 3a. Convert conditioning on m to intervention. Rule 2 with $X = T$, $Z = M$, $W = \emptyset$: form $\mathcal{G}_{\overline{T}\underline{M}}$ (delete $U \to T$ and $M \to Y$). In this graph no path from M to Y exists, so $(Y \perp\!\!\!\perp M \mid T)_{\mathcal{G}_{\overline{T}\underline{M}}}$. Rule 2 gives $P(y \mid \mathrm{do}(t), m) = P(y \mid \mathrm{do}(t), \mathrm{do}(m))$.
Step 3b. Drop the redundant $\mathrm{do}(t)$. Rule 3 with $X = M$, $Z = T$, $W = \emptyset$: form $\mathcal{G}_{\overline{M}\overline{T}}$ (delete $T \to M$ and $U \to T$). Node T is isolated, so $(Y \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{M}\overline{T}}}$. Rule 3 gives $P(y \mid \mathrm{do}(t), \mathrm{do}(m)) = P(y \mid \mathrm{do}(m))$.
Step 3c. Back-door formula for $M \to Y$. By front-door condition 3, $\{T\}$ satisfies the back-door criterion for $M \to Y$: $P(y \mid \mathrm{do}(m)) = \int P(y \mid t', m)\, dP(t')$.
Assembling: $P(y \mid \mathrm{do}(t)) = \int [\int P(y \mid t', m)\, dP(t')] dP(m \mid t)$. $\square$

3.6 The Do-Calculus Is Complete

A natural question: are the three rules enough? Could there be a graph where the causal effect is identifiable in principle, but no sequence of rules can derive it? Shpitser and Pearl (2006) answered this definitively.

Semi-Markovian DAGs encode latent common causes compactly as bidirected edges: $X \leftrightarrow Y$ stands for an unobserved U with $U \to X$ and $U \to Y$.

Theorem: Completeness of the Do-Calculus [@shpitser2006identification]

Let $\mathcal{G}$ be a semi-Markovian DAG. A causal quantity $P(y \mid \mathrm{do}(t))$ is identifiable from P relative to $\mathcal{G}$ if and only if the do-calculus can derive a purely observational expression for it.

Example: The Bow Graph — The Simplest Non-Identifiable Structure

The *bow graph* has one directed edge $T \to Y$ and one bidirected edge $T \leftrightarrow Y$ (representing unobserved U with $U \to T$, $U \to Y$). No observed variable can block the back-door path $T \leftarrow U \to Y$; no observed mediator exists for front-door. We show directly that $P(y \mid \mathrm{do}(t))$ is not identified by constructing two models that agree on $P(T, Y)$ but disagree on $P(Y \mid \mathrm{do}(T{=}1))$.
Let $T, Y, U \in \{0, 1\}$ with $U \sim \mathrm{Bernoulli}(1/2)$.
Model $\mathcal{M}_1$ (pure confounding; no causal effect): $T = U$, $Y = U$. Both T and Y are driven by U. The arrow $T \to Y$ carries no causal influence. Observed distribution: $P(T{=}0, Y{=}0) = P(T{=}1, Y{=}1) = 1/2$. Under $\mathrm{do}(T{=}1)$: $P_1(Y{=}1 \mid \mathrm{do}(T{=}1)) = P(U{=}1) = 1/2$.
Model $\mathcal{M}_2$ (full causal effect): $T = U$, $Y = T$. Y is determined entirely by T. Since $T = U$, the observed distribution is again $P(T{=}0, Y{=}0) = P(T{=}1, Y{=}1) = 1/2$ — *identical* to $\mathcal{M}_1$. Under $\mathrm{do}(T{=}1)$: $P_2(Y{=}1 \mid \mathrm{do}(T{=}1)) = 1$.
The two models produce identical observational distributions but assign different values (1/2 vs 1) to $P(Y{=}1 \mid \mathrm{do}(T{=}1))$. No data set, however large, can distinguish them. The causal effect is *not identified*.

Proof Sketch and the ID Algorithm

Shpitser and Pearl (2006) introduced the *ID algorithm*: it takes a semi-Markovian DAG $\mathcal{G}$ and either returns an observational formula or reports non-identification.
Sufficiency (do-calculus identifies whenever possible): proved constructively. The key concept is the *c-component* — a maximal set of nodes connected by bidirected edges. The ID algorithm: (1) removes non-ancestors of Y; (2) decomposes over c-components of $\mathcal{G}_{\overline{T}}$; (3) recursively identifies each factor; (4) reports FAIL if any subproblem cannot be reduced.
Necessity (do-calculus cannot identify non-identifiable quantities): proved via the *hedge*. A hedge for $P(y \mid \mathrm{do}(t))$ is a pair of subgraphs (F, F') encoding a "loop" of confounding the do-calculus cannot break. When a hedge exists, one can construct two models $\mathcal{M}_1$ and $\mathcal{M}_2$ with identical observed P but different $P(y \mid \mathrm{do}(t))$. Combining: the do-calculus succeeds if and only if no hedge exists. □

Remark: Practical Implication of Completeness

If the ID algorithm reports non-identification, then *no estimation method — however clever — can recover the causal effect from observational data alone* given the assumed graph structure. Non-identification is not a limitation of a particular technique; it is a fundamental property of the causal model. Three remedies: impose additional structural assumptions (e.g., linearity, monotonicity); collect additional data (e.g., an instrument or a randomized experiment); or target a different, identified causal quantity (e.g., partial identification bounds, or the effect among the treated).

3.7 The Big Picture

This chapter completes the *theoretical* identification machinery of the course. The full causal inference pipeline is:

$$\text{SEM/DAG} \xrightarrow{\text{d-separation}} \text{conditional independences} \xrightarrow{\text{back-door, front-door, do-calculus}} P(y \mid \mathrm{do}(t)) \xrightarrow{\text{estimation}} \hat{\tau}.$$

The three identification criteria relate as follows. The *back-door criterion* is applicable when observed covariates can block all confounding paths; it is the workhorse of regression adjustment and propensity score methods (Chapters 5–6). The *front-door criterion* is applicable when an observed mediator intercepts all causal paths while remaining unconfounded from treatment; it is the mechanism behind the front-door formula (Chapter 8). The *do-calculus* subsumes both: it is complete in the sense of the Completeness Theorem, and its three rules constitute a universal derivation engine for identification.

3.8 Summary

Symbol	**Meaning**
$\mathcal{G}_{\overline{X}}$	Mutilated graph: all arrows into X deleted
$\mathcal{G}_{\underline{X}}$	Auxiliary graph: all arrows out of X deleted
Back-door criterion	Observed $\mathbf{S}$ not descending from T; blocks all back-door paths
Front-door criterion	Observed M intercepting all causal paths; satisfying three conditions
Rule 1	Delete/insert observations if d-separated in $\mathcal{G}_{\overline{X}}$
Rule 2	Swap action for observation if d-separated in $\mathcal{G}_{\overline{X}\,\underline{Z}}$
Rule 3	Delete action if d-separated in $\mathcal{G}_{\overline{X}\,\overline{Z(W)}}$

1. Graph surgery formalizes intervention: $\mathcal{G}_{\overline{T}}$ (delete arrows into T) represents do($T{=}t$); d-separation in $\mathcal{G}_{\overline{T}}$ encodes conditional independence in $P(\cdot \mid \mathrm{do}(t))$.
2. The back-door criterion identifies causal effects when observed variables block all confounding paths. Condition 1 (no descendants of T) and Condition 2 (d-separation in $\mathcal{G}_{\underline{T}}$) are both essential: the M-bias example shows that including a collider can introduce confounding.
3. The front-door criterion identifies causal effects via an observed mediator when direct confounding is unblockable. The two-stage structure of the formula corresponds to two unconfounded identification steps chained together.
4. The three rules of do-calculus — insertion/deletion of observations, action/observation exchange, and insertion/deletion of actions — are a sound and complete derivation engine for identification. The back-door and front-door formulas are special cases.
5. The completeness theorem of Shpitser and Pearl (2006) implies that non-identification is a property of the graph, not of the method: when the ID algorithm fails, no observational method can succeed.

3.9 Problems

1. Graph surgery and d-separation. Consider the DAG with edges $Z \to T$, $U \to T$, $U \to Y$, $T \to Y$ (U unobserved).

(a) Draw $\mathcal{G}_{\overline{T}}$ and $\mathcal{G}_{\underline{T}}$. For each graph, state which edges were deleted and why.
(b) In $\mathcal{G}_{\overline{T}}$, is $(Y \perp\!\!\!\perp Z)_{\mathcal{G}_{\overline{T}}}$? Justify by listing all paths and determining whether each is blocked.
(c) In the original $\mathcal{G}$, is $(Y \perp\!\!\!\perp Z \mid T)_{\mathcal{G}}$? How does this relate to the instruction "never condition on a mediator to remove confounding"?
(d) How does the absence of a direct edge $Z \to Y$ in the DAG relate to the exclusion restriction (the assumption that Z affects Y only through T)?

2. Back-door practice. Consider the DAG: $X \to T$, $X \to Y$, $T \to M$, $M \to Y$, $T \to Y$, where X is observed.

(a) List all back-door paths from T to Y.
(b) Does $\{X\}$ satisfy the back-door criterion for the effect of T on Y? Write the resulting adjustment formula.
(c) Does $\{M\}$ satisfy the back-door criterion? Explain why or why not.
(d) Does $\{X, M\}$ satisfy the back-door criterion? Identify which condition of the criterion M violates.
(e) Even if the formal criterion issue in (d) were set aside, explain the substantive bias that arises from conditioning on a mediator M that lies on a causal path $T \to M \to Y$.

3. Front-door identification. Consider the DAG: $T \to M \to Y$, with $U \to T$ and $U \to Y$ (unobserved U, no direct $T \to Y$ edge).

(a) Verify that M satisfies all three front-door conditions.
(b) Derive the front-door formula Equation 8.13 step by step, citing the do-calculus rule used at each step.
(c) Now add a direct edge $T \to Y$. Does M still satisfy the front-door criterion? Explain which condition fails.

4. Identification or non-identification? For each DAG, determine whether $P(y \mid \mathrm{do}(t))$ is identified non-parametrically. If identified, state the formula; if not, explain the obstruction.

(a) $T \to Y$, $U \to T$, $U \to Y$, U unobserved.
(b) Same as (a), with instrument $Z \to T$ and $Z \perp\!\!\!\perp U$ (no direct $Z \to Y$ edge). Show $P(y \mid \mathrm{do}(t))$ is *not* identified by constructing two binary SEMs $\mathcal{M}_1$ and $\mathcal{M}_2$, each compatible with the IV graph, such that $P_{\mathcal{M}_1}(Z, T, Y) = P_{\mathcal{M}_2}(Z, T, Y)$ but $P_{\mathcal{M}_1}(Y{=}1 \mid \mathrm{do}(T{=}1)) \neq P_{\mathcal{M}_2}(Y{=}1 \mid \mathrm{do}(T{=}1))$.
(c) $T \to M \to Y$, $U \to T$, $U \to Y$, $U \to M$, U unobserved.
(d) $T \to Y$, $X \to T$, $X \to Y$, X observed.

5. Back-door formula: proof and uniqueness.

(a) Under the conditions of **?@thm-backdoor**, prove Equation 5.5 using the three rules of do-calculus, following the proof sketch in Section 3.5.
(b) Suppose $\mathbf{S}_1$ and $\mathbf{S}_2$ both satisfy the back-door criterion (with positivity). Deduce from (a) that $\int P(y \mid t, \mathbf{s}_1)\, dP(\mathbf{s}_1) = \int P(y \mid t, \mathbf{s}_2)\, dP(\mathbf{s}_2)$. Interpret: the identification target $P(y \mid \mathrm{do}(t))$ is unique even when the adjustment set is not.

6. Rule sequencing on a graph with two observed confounders. Consider the DAG: $C_1 \to T$, $C_1 \to Y$, $U \to T$, $U \to C_2$, $C_2 \to Y$, $T \to Y$, with C_1, C_2 observed and U unobserved. Problem 2 established that $\{C_1, C_2\}$ is a valid back-door set. Derive $P(y \mid \mathrm{do}(t)) = \sum_{c_1, c_2} P(y \mid t, c_1, c_2)\, P(c_1, c_2)$ from first principles using the three rules. For each step: (i) name the rule applied, (ii) state which arrows are deleted and what edges remain, (iii) verify the required d-separation condition.

7. Non-identifiability by construction. Consider the graph from Problem 4(c): $T \to M \to Y$, $U \to T$, $U \to M$, $U \to Y$ (front-door condition 2 violated). Let $T, M, Y, U \in \{0, 1\}$ with $U \sim \mathrm{Bernoulli}(1/2)$.

(a) Construct two SEMs $\mathcal{M}_1$ and $\mathcal{M}_2$, each compatible with the graph, such that $P_{\mathcal{M}_1}(T, M, Y) = P_{\mathcal{M}_2}(T, M, Y)$ but $P_{\mathcal{M}_1}(Y{=}1 \mid \mathrm{do}(T{=}1)) \neq P_{\mathcal{M}_2}(Y{=}1 \mid \mathrm{do}(T{=}1))$.
(b) Explain which identification strategy fails: (i) back-door criterion, (ii) front-door criterion (which condition fails), (iii) completeness theorem (what does the existence of your two models imply?).

Part II

Identification

Chapter 4

Potential Outcomes and Adjustment

Learning Objectives

By the end of this chapter, students should be able to:

1. Define the potential outcome $Y(t)$, state SUTVA precisely, and derive the consistency equation $Y = Y(T)$.
2. Define the ATE and ATT, explain how they relate to each other, and re-express them as functionals of the structural equation for Y.
3. Show that the linear SEM structural coefficient β equals the ATE under homogeneous effects, and explain why ATE and ATT diverge under treatment effect heterogeneity.
4. State the ignorability assumption, interpret it graphically as a back-door condition, and explain why overlap is a necessary companion assumption.
5. Derive the adjustment formula for the ATE under strong ignorability and positivity, and explain how the back-door criterion provides the graphical justification for conditional exchangeability.

4.1 Motivation: A Third Language for Causality

The first three chapters developed causal inference primarily in the languages of structural equations and directed acyclic graphs. Those languages are especially effective for expressing interventions syntactically: the SEM shows how an intervention replaces an assignment mechanism, and the DAG shows how it deletes incoming arrows. This chapter introduces a third language: the *potential outcomes framework* of Neyman (1923) and Rubin (1974).

Potential outcomes provide a direct language for defining causal estimands. Quantities such as the average treatment effect, the average treatment effect on the treated, and related contrasts are most naturally written in terms of counterfactual outcomes $Y(1)$, $Y(0)$, and more generally $Y(t)$. For this reason, the potential outcomes framework has become standard in statistics, biostatistics, epidemiology, and much of econometrics.

Assumptions such as ignorability, positivity, and exclusion can be stated in this language, but DAGs and the do-operator often make their structural content more transparent. The two frameworks should be viewed as complementary rather than competing: *potential outcomes define the causal quantities of interest, while DAGs and the do-calculus clarify identification.*

4.2 The Neyman–Rubin Potential Outcomes Framework

4.2.1 The Potential Outcome

Definition: Potential Outcome [@neyman1923application; @rubin1974estimating]

For each unit i and each possible treatment value $t \in \mathcal{T}$, the *potential outcome* $Y_i(t)$ is the value of the outcome that unit i *would have exhibited* had its treatment been set to t, possibly contrary to fact.
The collection $\{Y_i(t) : t \in \mathcal{T}\}$ is called the *schedule of potential outcomes* for unit i. In the binary case $\mathcal{T} = \{0, 1\}$, the schedule is $(Y_i(0), Y_i(1))$.

$Y_i(1)$ is the outcome unit i would achieve under treatment; $Y_i(0)$ is the outcome under control. Only one of these is ever observed for any given unit. The unobserved potential outcome is called the *counterfactual*.

The Fundamental Problem of Causal Inference [@holland1986statistics]

For any unit i, at most one potential outcome $Y_i(t)$ is observed. The individual-level causal effect $Y_i(1) - Y_i(0)$ is therefore *never directly observable*. Causal inference is an inference problem precisely because this quantity must be recovered from a population of units rather than a single unit.

4.2.2 SUTVA and Consistency

Definition: SUTVA [@rubin1980randomization]

The *stable unit treatment value assumption* has two components:

1. **No interference.** The potential outcome $Y_i(t)$ depends only on unit i's own treatment: $Y_i(t_1, \dots, t_n) = Y_i(t_i)$.
2. **No hidden versions.** For each treatment level t, there is a single well-defined version. If two units both receive $T = 1$, the treatment is the same in both cases.

Under SUTVA, the observed outcome equals the potential outcome at the treatment actually received:

$$Y_i = Y_i(T_i). \tag{4.1}$$

This *consistency equation* is the bridge between the potential outcome world and the observed data world. It fails whenever SUTVA fails: if treatment spills over between units (e.g., vaccination provides herd immunity), or if the treatment label $T = 1$ covers multiple distinct interventions.

4.2.3 Connection to the Do-Operator

Proposition: Potential Outcomes and the Do-Operator

Under the structural causal model of Chapters 1–3, together with consistency and no interference (SUTVA):

$$Y(t) \overset{d}{=} Y \mid \mathrm{do}(T{=}t), \tag{4.2}$$

so that $P(Y(t) \le y) = P(Y \le y \mid \mathrm{do}(T{=}t))$ for every y. Equivalently, $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}t)]$.

Proof sketch. In the mutilated SEM $\mathcal{G}_{\overline{T}}$, the structural equation for T is replaced by $T := t$ for every unit. The structural model then determines Y from t and unit i's own background variables — which is precisely what the potential outcomes framework defines as $Y_i(t)$. SUTVA's no-interference condition ensures $Y_i(t)$ does not depend on other units' treatment values, so the marginal distribution of Y in $\mathcal{G}_{\overline{T}}$ equals the marginal distribution of $Y(t)$. □

This equivalence lets us move freely between two notational traditions. When defining estimands such as $\mathbb{E}[Y(1) - Y(0)]$, potential-outcome notation is most natural. When proving identification results from a graph, the do-operator is often more transparent: $P(y \mid \mathrm{do}(t)) \neq P(y \mid T{=}t)$ whenever T is endogenous (established in Chapter 1), whereas $Y(t)$ carries no syntactic marker for this gap.

> Remark
>
> The SEM representation $Y_i(t) = g(t, \mathbf{X}_i, U_{Y,i})$ clarifies why the two frameworks are equivalent in content but different in emphasis. The do-calculus works with the interventional distribution $P(y \mid \mathrm{do}(t))$ as a population-level object and asks when it can be recovered from observational data. The potential outcomes framework works with unit-level quantities $Y_i(t)$ and asks what population summaries (ATE, ATT) are scientifically meaningful. The SEM ties the two together.

4.3 Causal Estimands

4.3.1 The Average Treatment Effect and Its Relatives

> Definition: ATE and ATT
>
> For a binary treatment $T \in \{0, 1\}$:
> - The *average treatment effect* (ATE): $\tau_{\mathrm{ATE}} = \mathbb{E}[Y(1) - Y(0)]$.
> - The *average treatment effect on the treated* (ATT): $\tau_{\mathrm{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid T{=}1]$.

The ATE and ATT coincide only when the treatment effect does not depend on who selected into treatment — i.e., when $\mathbb{E}[Y(t) \mid T] = \mathbb{E}[Y(t)]$ for $t \in \{0, 1\}$. In a randomized experiment this holds by design.

Neither quantity is directly observable. The naïve estimator $\hat{\tau}_{\mathrm{naive}} = \bar{Y}_{T=1} - \bar{Y}_{T=0}$ estimates $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0] \neq \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$, with the gap being the selection bias induced by endogeneity.

4.3.2 Causal Estimands as Functionals of the Structural Equation

In the SEM framework, the outcome is determined by a structural equation $Y = g(T, \mathbf{X}, U_Y)$, where U_Y collects all sources of variation not accounted for by $(T, \mathbf{X})$. The potential outcome under $\mathrm{do}(T{=}t)$ is:

$$Y_i(t) = g(t, \mathbf{X}_i, U_{Y,i}). \tag{4.3}$$

Substituting into the definitions:

$$\tau_{\mathrm{ATE}} = \mathbb{E}[g(1, \mathbf{X}, U_Y) - g(0, \mathbf{X}, U_Y)], \qquad \tau_{\mathrm{ATT}} = \mathbb{E}[g(1, \mathbf{X}, U_Y) - g(0, \mathbf{X}, U_Y) \mid T{=}1]. \tag{4.4}$$

The linear SEM as a special case. In the Gaussian linear SEM $Y = \beta T + \gamma^\top \mathbf{X} + \varepsilon$, the unit-level effect is $Y_i(1) - Y_i(0) = \beta$ for every unit. Consequently, $\tau_{\mathrm{ATE}} = \tau_{\mathrm{ATT}} = \beta$. The structural coefficient β *is* the average treatment effect. This homogeneity is a special property of the linear additive model, not a general feature.

Heterogeneous effects. In the nonparametric SEM, the unit-level effect $g(1, \mathbf{X}_i, U_{Y,i}) - g(0, \mathbf{X}_i, U_{Y,i})$ varies across units. The ATE and ATT differ whenever treatment selection correlates with individual effect size — i.e., whenever units who benefit more also tend to self-select into treatment.

4.4 Ignorability, Positivity, and Adjustment

4.4.1 Strong Ignorability

The central identifying assumption in observational studies is that treatment assignment is as good as random after conditioning on observed covariates X.

> Definition: Strong Ignorability [@rosenbaum1983central]
>
> The treatment assignment T is *strongly ignorable* given X if:
> 1. **Unconfoundedness:** $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$.

2. **Overlap (positivity):** $0 < P(T{=}1 \mid X{=}x) < 1$ for all x in the support of X.

4.4.2 Adjustment Formula under Ignorability

Under strong ignorability, the ATE is identified by the standardization formula:

$$\tau_{\text{ATE}} = \mathbb{E}_X[\mathbb{E}[Y \mid T{=}1, X] - \mathbb{E}[Y \mid T{=}0, X]] . \tag{4.5}$$

Derivation. For each treatment level t, by the law of iterated expectations:

$$\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}[Y(t) \mid X]] .$$

Under unconfoundedness, $\mathbb{E}[Y(t) \mid X] = \mathbb{E}[Y(t) \mid T{=}t, X]$. By consistency, $\mathbb{E}[Y(t) \mid T{=}t, X] = \mathbb{E}[Y \mid T{=}t, X]$. Therefore $\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}[Y \mid T{=}t, X]]$. Applying once for $t = 1$ and once for $t = 0$ and taking the difference yields Equation 4.5. □

Example: A Two-Stratum Adjustment Calculation

Let $X \in \{0, 1\}$ with $P(X{=}1) = 0.4$, $P(X{=}0) = 0.6$, and observed conditional means:

	$T = 1$	$T = 0$
$X = 1$	$\mathbb{E}[Y \mid T = 1, X = 1] = 8$	$\mathbb{E}[Y \mid T = 0, X = 1] = 6$
$X = 0$	$\mathbb{E}[Y \mid T = 1, X = 0] = 5$	$\mathbb{E}[Y \mid T = 0, X = 0] = 4$

Under ignorability and positivity:

$$\mathbb{E}[Y(1)] = 0.4 \times 8 + 0.6 \times 5 = 6.2, \qquad \mathbb{E}[Y(0)] = 0.4 \times 6 + 0.6 \times 4 = 4.8.$$

Hence $\tau_{\text{ATE}} = 6.2 - 4.8 = 1.4$. The formula works by comparing treated and control outcomes within each stratum and averaging those within-stratum comparisons over the marginal distribution of X.

4.4.3 Back-Door Interpretation

The potential-outcomes assumption $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ is the counterfactual expression of the idea that, after conditioning on X, treatment assignment carries no residual information about the outcome that would be observed under intervention $T{=}t$. In DAG language, the closely related condition is that X blocks all back-door paths from T to Y.

Proposition: Back-Door Criterion Implies Ignorability

Under the NPSEM/SWIG semantics adopted in these notes, if X satisfies the back-door criterion for the effect of T on Y — i.e., (1) no node in X is a descendant of T, and (2) X blocks every back-door path from T to Y — then unconfoundedness ($Y(t) \perp\!\!\!\perp T \mid X$) holds for all t.

Proof sketch. The target statement $Y(t) \perp\!\!\!\perp T \mid X$ is *cross-world*: it mixes the counterfactual $Y(t)$ with the factual treatment T, and is not a d-separation statement in $\mathcal{G}$ itself. The rigorous translation proceeds through the single-world intervention graph $\mathcal{G}(t)$ (Appendix B). Under the NPSEM semantics, $Y(t)$ and T depend on disjoint exogenous errors together with their own ancestors, and the back-door criterion is precisely the d-separation condition on $\mathcal{G}(t)$ that makes these conditionally independent given X. See Appendix B for the explicit SWIG derivation. □

Remark

This is the direction most important for practice: a graphical adjustment set justifies the counterfactual independence needed for standardization, regression adjustment, and propensity-score methods.

The converse — whether every ignorability statement corresponds to a back-door condition — is more delicate; see Appendix B.

Example: Labor Training Program

Following LaLonde (1986) and Dehejia and Wahba (1999), let T = receipt of job training, Y = earnings two years later, X = (age, education, prior earnings, race, marital status).
The back-door path $T \leftarrow X \rightarrow Y$ is blocked by conditioning on X, leaving only the causal path $T \rightarrow Y$. Ignorability holds if this DAG is correctly specified. If there is an unobserved variable U (motivation, ability) that affects both training participation and earnings, X fails the back-door criterion and ignorability fails.

4.4.4 Overlap and Positivity

Why Overlap Is Non-Negotiable

Without overlap, some covariate strata contain only treated or only control units. In those strata, $\mathbb{E}[Y \mid T{=}0, X{=}x]$ or $\mathbb{E}[Y \mid T{=}1, X{=}x]$ is unobservable, and Equation 4.5 cannot be evaluated at those values of x. Identification of the ATE fails not because the causal structure is wrong, but because the data do not span the support needed.
The ATT can remain identified under a *weaker, one-sided* overlap condition: $P(T{=}0 \mid X{=}x) > 0$ almost surely on the support of X in the treated subpopulation. Full two-sided overlap is not required for ATT, because ATT averages only over the treated population.

4.5 Where the Frameworks Agree and Diverge

Task	Potential Outcomes	Do-Calculus / DAG	SEM
Define ATE, ATT	Most natural notation	Via $\mathbb{E}[Y \mid \mathrm{do}(t)]$	Via structural equations
Encode causal assumptions	Typically informal; graph implicit	Explicit directed edges	Structural equations
Read conditional independence	Requires auxiliary graph	d-separation	With graph only
Identification from data	Via ignorability	Back-door, front-door, do-calculus	Via exclusion restrictions
Likelihood construction	Estimating equations / semiparametric	Observational functionals only	Structural model
Cross-world assumptions	Natural to state	See Appendix B	Implicit in model
Standard in statistics	Dominant	Growing rapidly	Econometrics

Our Position in This Course

Both frameworks are indispensable. We use *potential outcomes* to *define* causal estimands. We use the *do-calculus and DAGs* for *identification*. The back-door criterion is the graphical condition that justifies conditional ignorability and hence the adjustment formula. The do-operator language is preferred throughout this course because it makes the interventional/observational distinction *syntactically enforced*: $P(y \mid \mathrm{do}(t))$ cannot be confused with $P(y \mid T{=}t)$, whereas $\mathbb{E}[Y(t)]$ carries no such syntactic marker.

4.6 Summary

Causal inference = counterfactual questions + graphical assumptions + statistical estimation.

1. The potential outcome $Y(t)$ is the outcome that would be observed under intervention $T = t$. Under consistency, no interference, and the structural causal semantics adopted in these notes, the observed outcome satisfies $Y_i = Y_i(T_i)$, and $Y(t)$ has the same distribution as $Y \mid \mathrm{do}(T{=}t)$.
2. The ATE and ATT are averages of the unit-level causal effect $g(1, \mathbf{X}_i, U_{Y,i}) - g(0, \mathbf{X}_i, U_{Y,i})$ over the full population and the treated subpopulation respectively. In the linear SEM, both equal β. They diverge under heterogeneous effects when treatment selection correlates with individual effect size.
3. Strong ignorability $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ with overlap identifies the ATE via Equation 4.5. The back-door criterion provides a sufficient graphical condition for the conditional exchangeability assumption.
4. Single world intervention graphs (Richardson and Robins 2014) provide a formal graphical representation of counterfactual variables, making ignorability a d-separation statement (Appendix B).
5. The frameworks are complementary: potential outcomes define estimands; the do-calculus identifies them. The back-door criterion connects the two by providing the graphical condition under which adjustment recovers the causal effect.

From ignorability to randomization. In observational studies, ignorability must be justified by substantive knowledge encoded in a causal graph. Randomized experiments provide a different solution: when treatment is assigned randomly, $(Y(0), Y(1)) \perp\!\!\!\perp T$ holds by design, guaranteeing ignorability *without any covariate adjustment*. Chapter 5 studies this design.

Identification vs. estimation. Chapters 1–4 have focused on *identification*: whether $\tau = \mathbb{E}[Y(1) - Y(0)]$ can be written as a functional $\Phi(P(Y, T, X))$ of the observed distribution. Estimation — constructing $\hat{\tau}$ from a finite sample to approximate $\Phi(P)$ — is studied in Part III, covering regression adjustment, IPW, doubly robust estimators, and instrumental variables.

4.7 Problems

1. SUTVA and consistency. Suppose $n = 3$ units receive binary treatments (T_1, T_2, T_3) and unit i's outcome may depend on all three treatments: $Y_i = Y_i(T_1, T_2, T_3)$.

(a) How many potential outcomes does unit 1 have? Write them out explicitly.
(b) SUTVA imposes no interference: $Y_i(T_1, T_2, T_3) = Y_i(T_i)$. How many distinct potential outcomes remain?
(c) Give a real-world example where no-interference plausibly holds and one where it plausibly fails. In the latter case, explain what identification strategy (if any) remains available.

2. ATE, ATT, and selection bias. Let $(Y(0), Y(1), T) \sim P$ with $P(T{=}1) = 0.5$, $\mathbb{E}[Y(1)] = 3$, $\mathbb{E}[Y(0)] = 1$, $\mathbb{E}[Y(1) \mid T{=}1] = 4$, $\mathbb{E}[Y(0) \mid T{=}1] = 2$, $\mathbb{E}[Y(1) \mid T{=}0] = 2$, $\mathbb{E}[Y(0) \mid T{=}0] = 0$.

(a) Compute the ATE and ATT. Are they equal?
(b) Compute the naïve estimator $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$. Decompose the gap between this and the ATE into a selection bias term and an ATT–ATE difference.
(c) Under what graphical condition on the DAG would the naïve estimator equal the ATE? State this condition in both potential outcome language (ignorability) and do-calculus language (back-door criterion).

3. Causal estimands from the structural equation. Consider the nonparametric SEM $Y = g(T, X, U_Y)$ with binary $T \in \{0, 1\}$, observed covariate X, and U_Y independent of (T, X) in the mutilated graph.

(a) Write τ_{ATE} and τ_{ATT} as expectations of $g(1, X, U_Y) - g(0, X, U_Y)$ over the appropriate distribution. Under what condition do they coincide?
(b) Specialize to the linear SEM $g(t, x, u) = \alpha + \beta t + \gamma x + u$. Show that the unit-level causal effect $Y_i(1) - Y_i(0)$ is constant across all units, and hence $\tau_{\mathrm{ATE}} = \tau_{\mathrm{ATT}} = \beta$.
(c) Now consider the heterogeneous SEM $g(t, x, u) = (\alpha + u)\, t + \gamma x$, where $U_Y \sim \mathcal{N}(0, \sigma^2)$, $\mathrm{Cov}(U_Y, T) = \rho$, and $P(T{=}1) = p \in (0, 1)$. Compute τ_{ATE} and τ_{ATT}. Show that they differ when $\rho \neq 0$, and interpret this difference.
(d) In part (c), show that the population OLS slope on T in the regression of Y on $(1, T, X)$ equals $\alpha + \rho/p = \tau_{\mathrm{ATT}}$, not $\tau_{\mathrm{ATE}} = \alpha$. What additional structure would be needed to identify τ_{ATE}?

4. SWIGs and ignorability. *(Requires Appendix B.)* Consider the DAG: $X \to T$, $X \to Y$, $T \to Y$, with X fully observed.

(a) Construct the SWIG $\mathcal{G}(t)$ by splitting T into its random and fixed halves. Draw the result, labeling the random half, fixed half, and the potential outcome $Y(t)$.

(b) In $\mathcal{G}(t)$, identify all paths between the random half T and $Y(t)$. Determine which are blocked and which are open before conditioning.

(c) Use d-separation in $\mathcal{G}(t)$ to verify that $(Y(t) \perp\!\!\!\perp T \mid X)_{\mathcal{G}(t)}$ holds. Conclude that X satisfies the back-door criterion, confirming (**prop-ign-bd?**).

(d) Now add a hidden common cause $U \to T$, $U \to Y$. Draw the revised SWIG. Does the ignorability argument still hold? State the correct conclusion and identify what additional structure would be needed for identification.

Chapter 5

Randomization and Back-Door Adjustment

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain in do-calculus terms why a randomized experiment makes $f(y \mid \mathrm{do}(T{=}t)) = f(y \mid T{=}t)$, and identify the graphical feature that produces this equality.
2. State the ignorability assumption in both potential-outcomes and graphical language, and distinguish weak from strong ignorability.
3. Apply the regression-adjusted estimator to reduce variance in a randomized experiment, and explain why consistency holds even under model misspecification.
4. Derive and compute the standardization (g-formula) estimator of the ATE from a contingency table or regression output.
5. Explain the method of stratification (subclassification) and describe when and why it removes confounding bias.
6. Identify Simpson's paradox in a numerical example and resolve it using the back-door criterion.
7. Articulate why propensity score methods are needed when X is high-dimensional, motivating Chapter 6.

5.1 Randomized Experiments

5.1.1 The Do-Calculus of Randomization

The defining feature of a randomized experiment is that the analyst *sets* the treatment for each unit, independently of all background variables. In do-calculus terms, the data come from the mutilated distribution $f(y \mid \mathrm{do}(T{=}t))$ rather than the observational distribution $f(y \mid T{=}t)$.

The Fundamental Equality of Randomized Experiments

In a completely randomized experiment, T is assigned independently of all pre-treatment variables. In the DAG, this means there are *no arrows into* T: the treatment node has no parents. By Rule 2 of the do-calculus, with $X = \emptyset$, $Z = T$, $W = \emptyset$, the graphical condition $(Y \perp\!\!\!\perp T)_{\mathcal{G}_{\underline{T}}}$ holds because T is isolated in $\mathcal{G}_{\underline{T}}$ (no parents in $\mathcal{G}$, and the outgoing edge $T \to Y$ is deleted). Rule 2 therefore gives:

$$f(y \mid \mathrm{do}(T{=}t)) = f(y \mid T{=}t). \tag{5.1}$$

In a randomized experiment, observing $T = t$ is the same as intervening to set $T = t$.

What the equality assumes beyond the graph. The fundamental equality Equation 5.1 relies on more than deletion of arrows into T. Two further conditions are needed. First, SUTVA (no interference and no hidden treatment versions), which licenses the consistency equation $Y_i = Y_i(T_i)$. Second, the realized treatment must equal the assigned treatment for every unit (full compliance); when this fails, $f(y \mid T{=}t)$ describes outcomes among those who *actually* received t rather than those who were *assigned* t. When compliance fails, the intent-to-treat versus per-protocol distinction becomes substantive (Chapters 7 and 13).

Graphical argument. In the observational DAG, arrows into T from observed covariates X and unobserved confounders U create back-door paths from T to Y. Randomization *physically* severs these arrows: assignment is determined by a coin flip, not by X or U. The mutilated graph $\mathcal{G}_{\overline{T}}$ is, in a randomized experiment, the *actual* data-generating graph. There are no back-door paths to block, because none exist.

Potential outcomes statement. In the potential outcomes language, randomization implies $(Y(0), Y(1)) \perp\!\!\!\perp T$ unconditionally — no covariate adjustment is needed. This is the strongest possible version of ignorability.

5.1.2 Estimation of the ATE in a Randomized Experiment

Lemma: Identification under Complete Randomization

Under complete randomization, $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid T{=}t]$, $t \in \{0, 1\}$.

Proof. By the PO–do equivalence, $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}t)]$. The treatment node has no parents, so Equation 5.1 gives $f(y \mid \mathrm{do}(T{=}t)) = f(y \mid T{=}t)$. □

Under complete randomization, the ATE is estimated by the *difference-in-means* (DIM) estimator:

$$\hat{\tau}_{\mathrm{DIM}} = \bar{Y}_1 - \bar{Y}_0 = \frac{1}{n_1} \sum_{i:T_i=1} Y_i - \frac{1}{n_0} \sum_{i:T_i=0} Y_i.$$

Theorem: Neyman's Theorem for the Difference-in-Means Estimator

Under complete randomization, let $\mathcal{F}_N = \{(Y_i(0), Y_i(1))\}$ be the potential outcomes (fixed), and let expectations be over the randomization distribution. Define the finite-population variances $S_t^2 = \frac{1}{n-1} \sum_i (Y_i(t) - \bar{Y}(t))^2$, cross-variance $S_{01} = \frac{1}{n-1} \sum_i (Y_i(0) - \bar{Y}(0))(Y_i(1) - \bar{Y}(1))$, and effect variance $S_\tau^2 = \frac{1}{n-1} \sum_i (\tau_i - \bar{\tau}_n)^2$. Then:

1. **Unbiasedness.** $\mathbb{E}[\hat{\tau}_{\mathrm{DIM}} \mid \mathcal{F}_N] = \bar{Y}(1) - \bar{Y}(0) = \bar{\tau}_n$.
2. **Design variance.**
$$\mathrm{Var}(\hat{\tau}_{\mathrm{DIM}} \mid \mathcal{F}_N) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{n}. \tag{5.2}$$
3. **Conservative variance estimator.** The within-arm sample variance estimator $\hat{V} = \hat{S}_1^2/n_1 + \hat{S}_0^2/n_0$ satisfies $\mathbb{E}[\hat{V} \mid \mathcal{F}_N] - \mathrm{Var}(\hat{\tau}_{\mathrm{DIM}} \mid \mathcal{F}_N) = S_\tau^2/n \geq 0$. It is exact when the unit-level treatment effects τ_i are constant.

Proof of Neyman's Theorem

Throughout we condition on $\mathcal{F}_N$. Under CRE, $\mathbb{E}[T_i \mid \mathcal{F}_N] = n_1/n$ for all i, and:

$$\mathrm{Cov}(T_i, T_j \mid \mathcal{F}_N) = \begin{cases} f_1(1 - f_1) & i = j, \\ -f_1(1 - f_1)/(n-1) & i \neq j, \end{cases}$$

where $f_1 = n_1/n$. The $i \neq j$ expression follows from sampling n_1 units without replacement.
(i) Unbiasedness. $\mathbb{E}[n_1^{-1} \sum_i T_i Y_i(1) \mid \mathcal{F}_N] = \bar{Y}(1)$ by $\mathbb{E}[T_i \mid \mathcal{F}_N] = n_1/n$; the control arm is symmetric.

(ii) Variance. Writing $Z_i(t) = Y_i(t) - \bar{Y}(t)$ and using the covariance formula:

$$\mathrm{Var}\left(\frac{1}{n_1}\sum_i T_i Y_i(1)\right) = \frac{f_1(1-f_1)}{n_1^2} \cdot \frac{n}{n-1}\sum_i Z_i(1)^2 = \frac{n_0}{n_1 n} S_1^2,$$

and the cross-arm covariance yields $-S_{01}/n$. Combining gives the first form of Equation 5.2. Rearranging using $n_0/(n_1 n) = 1/n_1 - 1/n$ gives the second form.
(iii) Conservatism. By finite-population sampling theory, $\mathbb{E}[\hat{S}_t^2 \mid \mathcal{F}_N] = S_t^2$. Therefore $\mathbb{E}[\hat{V} \mid \mathcal{F}_N] = S_1^2/n_1 + S_0^2/n_0$, and subtracting Equation 5.2 gives $S_\tau^2/n \geq 0$. □

Remark: Why the Conservatism Cannot Be Removed

The correction term S_τ^2/n depends on the unit-level effect variance. Computing S_τ^2 requires both $Y_i(0)$ and $Y_i(1)$ for every unit — precisely what the fundamental problem of causal inference forbids. The estimator $\hat{V}$ is not merely pragmatic; it is the sharpest variance estimator based on observed data alone without further modelling assumptions.

Remark: Superpopulation Variance as a Corollary

Under i.i.d. superpopulation draws with arm variances (σ_0^2, σ_1^2), the total variance has two phases: randomization variance (given $\mathcal{F}_N$) and sampling variance (over draws of $\mathcal{F}_N$). The law of total variance gives:

$$\mathrm{Var}(\hat{\tau}_{\mathrm{DIM}}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}. \tag{5.3}$$

The Neyman correction S_τ^2/n in Equation 5.2 is exactly cancelled by the sampling variance of $\bar{\tau}_n$ when averaging over superpopulation draws. See Ding (2024) for a complete treatment.

5.1.3 Fisher's Randomization Inference vs. Neyman's Repeated Sampling

Fisher's framework tests the *sharp null* $Y_i(1) = Y_i(0)$ for all i: under this null, every missing potential outcome is known, making the exact randomization distribution of any test statistic computable over all $\binom{n}{n_1}$ assignments. Neyman's framework studies the repeated-sampling behavior of estimators like $\hat{\tau}_{\mathrm{DIM}}$ for average causal effects.

Fisher's approach is aligned with exact hypothesis testing under a sharp null; Neyman's is aligned with point estimation and uncertainty quantification. Both rely on the treatment assignment mechanism but answer different questions.

Remark: Randomization Licenses Inference Without Population Assumptions

Randomization licenses causal inference *without any assumption about a population model.* In Fisher's design-based framework, the potential outcomes $\{Y_i(0), Y_i(1)\}$ are fixed numbers; the only source of randomness is the assignment vector $\mathbf{T}$, drawn uniformly from all $\binom{n}{n_1}$ possible assignments. Probability statements refer to this mechanism — not to repeated draws from a population. Units need not be a random sample; they can be a convenience sample, a census, or a targeted group. Observational methods, by contrast, must invoke either an i.i.d. sampling assumption or a superpopulation framework, because no analogous physical mechanism anchors probability statements.

Remark: Fisher's Sharp Null — Unidentifiable Effects, Exactly Testable Hypothesis

Individual causal effects $\tau_i = Y_i(1) - Y_i(0)$ are unidentifiable for any unit. Yet the sharp null $H_0 : Y_i(1) = Y_i(0)$ for all i is exactly testable. The resolution: under H_0, the sharp null completely specifies the joint potential outcome table — $Y_i(1) = Y_i(0) = Y_i^{\mathrm{obs}}$ regardless of assignment — so the full $2n$-vector of potential outcomes is known, and the permutation distribution of any test statistic is computable from observed data with no estimation step.

Remark: Further Reading

Ding (2024) provides a comprehensive graduate-level treatment of CRE, stratified experiments, and regression adjustment. Li and Ding (2017) works out the asymptotic theory underlying confidence intervals for $\hat{\tau}_{\text{DIM}}$ via a finite-population CLT requiring neither i.i.d. observations nor a superpopulation model. Ding et al. (2016) develop randomization-based tests for treatment effect heterogeneity.

5.2 Ignorability

5.2.1 Two Routes to the Same Estimand

Section Section 5.1 established that randomization achieves Equation 5.1 and that DIM is unbiased for the ATE without covariate adjustment. In observational studies, no such physical severance occurs: treatment is selected based on characteristics that may include unmeasured variables U affecting the outcome.

Both settings target the same estimand — the ATE $\mathbb{E}[Y(1) - Y(0)]$ — but reach it by fundamentally different routes:

	Randomized experiment	**Observational study**
How ignorability arises	By design: researcher severs all arrows into T	By assumption: analyst asserts no unmeasured confounders
$(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$	*Guaranteed* — holds unconditionally	*Assumed* — requires X to capture every confounder
Credibility	As strong as the randomization protocol	As strong as substantive knowledge of the DGP
Testability	Can audit the assignment mechanism	Cannot be verified from data alone
Failure mode	Protocol violations, non-compliance	Any unmeasured variable affecting both T and Y

Statistical adjustment can *implement* ignorability once assumed, but cannot *create* it. No covariate adjustment — however flexible — can close a back-door path through an unmeasured variable. This is why a well-conducted RCT is considered more credible than even the most carefully adjusted observational study.

5.2.2 Terminology

Strong ignorability was defined in Chapter 4: joint unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ together with overlap $0 < P(T{=}1 \mid X) < 1$ a.s. (Rosenbaum and Rubin 1983). The pointwise (weak) form $Y(t) \perp\!\!\!\perp T \mid X$ for each t separately is what identification of $\mathbb{E}[Y(t)]$ actually uses. Equivalent names in the literature: *unconfoundedness*, *selection on observables*, *no unmeasured confounders*, *conditional exchangeability*.

5.2.3 Three Languages for Ignorability

Language	**Statement of ignorability**
SEM	In $Y = f_Y(T, X, U_Y)$ with $T = f_T(X, U_T)$, the exogenous factor U_Y is independent of T given X — no unobserved common cause of T and Y remains after conditioning on X.

Language	Statement of ignorability
DAG / do-calculus	X satisfies the back-door criterion for (T, Y): X blocks all back-door paths and contains no descendant of T.
Potential outcomes	$Y(t) \perp\!\!\!\perp T \mid X$ for $t \in \{0, 1\}$ (weak ignorability).

These three forms are closely aligned under NPSEM-IE semantics together with consistency and no interference.

5.2.4 What Ignorability Requires

Ignorability Is Untestable from Observational Data

No statistical test can confirm or refute unconfoundedness using observed data alone. If an unobserved variable U affects both T and Y, the back-door path $T \leftarrow U \rightarrow Y$ is open, and conditioning on any set of observed variables cannot close it. Sensitivity analysis (Rosenbaum 2002) can quantify how large such an unobserved confounder would have to be to overturn a conclusion, but cannot confirm the assumption itself.

5.2.5 From Ignorability to Identification

Lemma: Identification of $\mathbb{E}[Y(t)]$

Suppose: (i) *weak ignorability* $Y(t) \perp\!\!\!\perp T \mid X$; (ii) *overlap* $P(T{=}t \mid X{=}x) > 0$ a.s.; (iii) *consistency* $Y = Y(T)$. Then:

$$\mathbb{E}[Y(t)] = \mathbb{E}[\mathbb{E}[Y \mid T{=}t, X]] = \int \mathbb{E}[Y \mid T{=}t, X{=}x]\, p(x)\, dx. \tag{5.4}$$

Proof. Apply the law of iterated expectations, then use each assumption in turn:

$$\mathbb{E}[Y(t)] \overset{\text{LIE}}{=} \mathbb{E}[\mathbb{E}[Y(t) \mid X]] \overset{(i)}{=} \mathbb{E}[\mathbb{E}[Y(t) \mid T{=}t, X]] \overset{(iii)}{=} \mathbb{E}[\mathbb{E}[Y \mid T{=}t, X]]. \quad \square$$

Overlap (ii) guarantees $\mathbb{E}[Y \mid T{=}t, X{=}x]$ is well-defined everywhere in the support of X.

Remark: Two Languages, One Formula

?@lem-ident and the back-door theorem (Chapter 3) are two proofs of the same formula in different languages. The graphical proof applies Rules 2 and 3 to the mutilated graph; the potential-outcomes proof applies the law of iterated expectations to counterfactual variables. The bridge is the structural causal semantics: under NPSEM-IE, $Y(t)$ has the same distribution as Y under $\mathrm{do}(T{=}t)$ (Chapter 4). Consistency connects $\mathbb{E}[Y(t) \mid T{=}t, X]$ to the observed $\mathbb{E}[Y \mid T{=}t, X]$.

Theorem: Identification of the ATE and ATT

Under the same three assumptions:

$$\tau_{\text{ATE}} = \int [\mathbb{E}(Y \mid T{=}1, X{=}x) - \mathbb{E}(Y \mid T{=}0, X{=}x)]\, p(x)\, dx, \tag{5.5}$$

$$\tau_{\text{ATT}} = \int [\mathbb{E}(Y \mid T{=}1, X{=}x) - \mathbb{E}(Y \mid T{=}0, X{=}x)]\, p(x \mid T{=}1)\, dx. \tag{5.6}$$

Proof. Apply **?@lem-ident** separately to $t = 1$ and $t = 0$ and subtract. For the ATT, replace $p(x)$ by $p(x \mid T{=}1)$. $\square$

Assumption traceability. Each step in the proof invokes exactly one assumption:

Proof step	Assumption invoked	Failure mode
$\mathbb{E}[Y(t) \mid X] = \mathbb{E}[Y(t) \mid T{=}t, X]$	Weak ignorability	Unmeasured confounder: $Y(t)$ depends on T within X-strata
$\mathbb{E}[Y \mid T{=}t, X{=}x]$ is well-defined	Overlap	Empty stratum: no units with $T{=}t$ at x
$\mathbb{E}[Y(t) \mid T{=}t, X] = \mathbb{E}[Y \mid T{=}t, X]$	Consistency	Interference or hidden treatment versions

Overlap is *testable* from data (check support of $X \mid T{=}1$ vs. $X \mid T{=}0$). Weak ignorability and consistency are not testable from data alone.

5.2.6 Which Variables to Condition On: The Pre-Treatment Requirement

Before estimating anything, there is a prior graphical question: *which variables should be in X?* The answer is not "all available variables." Conditioning on the wrong variable introduces bias.

Never Condition on a Post-Treatment Variable

A variable L is *post-treatment* if $T \to L$ in the DAG. Including L in X can bias the estimated treatment effect through two mechanisms:

1. **Blocking a causal pathway.** If L is a mediator ($T \to L \to Y$), conditioning on L blocks part of the causal effect of T. The result estimates a direct effect, not the total effect.
2. **Opening a collider path.** If L is a collider ($T \to L \leftarrow U \to Y$), conditioning on L opens a previously blocked path, creating a spurious T–Y association through the unobserved U.

In both cases, including L violates Condition 1 of the back-door criterion: *X must contain no descendant of T*. The back-door criterion is not merely a sufficiency condition for identification — it is the correct filter for deciding which variables to include.

The practical rule: before running any regression, classify every candidate covariate as pre-treatment or post-treatment using the causal graph. Only pre-treatment variables satisfying the back-door criterion belong in X.

5.3 Regression Adjustment and Standardization

5.3.1 The Common Three-Step Logic

Both regression adjustment and standardization implement the same back-door formula Equation 5.5. They differ only in how they estimate $\mu(t, x) = \mathbb{E}[Y \mid T{=}t, X{=}x]$.

The Three-Step Recipe: Outcome Regression / G-Formula

1. **Estimate the outcome model.** Fit a model for $\mu(t, x) = \mathbb{E}[Y \mid T{=}t, X{=}x]$ using observed data. Any regression method may be used: OLS, logistic regression, or a flexible nonparametric estimator.
2. **Predict both potential outcomes for every unit.** For each unit i — *regardless of treatment actually received* — compute:
$$\hat{Y}_i(1) = \hat{\mu}(1, X_i), \qquad \hat{Y}_i(0) = \hat{\mu}(0, X_i).$$
For a treated unit, $\hat{Y}_i(0)$ is the predicted outcome had that unit been assigned to control; for a control unit, $\hat{Y}_i(1)$ is the predicted outcome had that unit been treated.
3. **Average the individual treatment effect estimates.**
$$\hat{\tau}_{\mathrm{OR}} = \frac{1}{n}\sum_{i=1}^{n}[\hat{\mu}(1, X_i) - \hat{\mu}(0, X_i)]. \tag{5.7}$$
To estimate the ATT, average only over the n_1 treated units.

This is the *outcome regression* (OR) or *G-computation* estimator (Robins 1986). Step 3 averages out X using the empirical distribution — exactly what Equation 5.5 requires: $\tau_{\text{ATE}} = \int [\mu(1,x) - \mu(0,x)] p(x)\,dx$.

5.3.2 A Worked Example

Binary covariate X (e.g., sex), continuous outcome Y (e.g., earnings). High-X units are overrepresented in the treated arm (70% treated vs. 30% control).

	Treated ($T = 1$)	Control ($T = 0$)	
$X = 0$	$n = 30$, $\bar{Y} = 42$	$n = 70$, $\bar{Y} = 35$	
$X = 1$	$n = 70$, $\bar{Y} = 58$	$n = 30$, $\bar{Y} = 50$	
All	$n = 100$, $\bar{Y} = 53.2$	$n = 100$, $\bar{Y} = 38.5$	Unadjusted diff $= 14.7$

Step 1. $\hat{\mu}(1,0) = 42$, $\hat{\mu}(1,1) = 58$, $\hat{\mu}(0,0) = 35$, $\hat{\mu}(0,1) = 50$.

Step 2. Within-stratum effects: $42 - 35 = 7$ (for X=0), $58 - 50 = 8$ (for X=1).

Step 3. Averaging over the *marginal* distribution of X: 50% of the full sample has X=0, 50% has X=1:

$$\hat{\tau}_{\text{ATE}} = 7 \times 0.5 + 8 \times 0.5 = 7.5.$$

After adjusting for X, the estimated treatment effect is 7.5, not 14.7. The unadjusted comparison conflates the treatment effect with the higher baseline earnings of X=1 units who happen to be treated more often.

5.3.3 Standardization as a Special Case

When X is categorical, standardization computes $\hat{\mu}(t,x)$ as the within-cell sample mean — a fully saturated model:

$$\hat{\tau}_{\text{STD}} = \sum_{x \in \mathcal{X}} [\hat{\mu}(1,x) - \hat{\mu}(0,x)]\,\hat{p}(x). \tag{5.8}$$

This is *algebraically identical* to Equation 5.7 when X is categorical: standardization is regression adjustment with a saturated outcome model. The formula is also known as the *G-formula* (Robins 1986) and as *direct standardization* in epidemiology.

	Regression adjustment	**Standardization**
How $\hat{\mu}(t,x)$ is estimated	Parametric model: OLS, logistic, or flexible learner	Cell means: $\bar{Y}$ within (T=t, X=x)
Works when X is	Continuous or high-dimensional	Discrete and low-dimensional
Bias if wrong	Model misspecification	Sparse cells (some (t,x) strata empty)
Steps 2–3	Identical: predict both POs, average	Identical: predict both POs, average

5.3.4 Model Specification and What Can Go Wrong

The OR estimator is consistent if and only if $\hat{\mu}(t,x) \to \mu(t,x)$ as $n \to \infty$ — i.e., if the outcome model is correctly specified.

Misspecification bias. If the true $\mu(t,x)$ is nonlinear but a linear model is used, the estimated treatment effect absorbs the functional form error. Flexible machine learning estimators reduce this risk, at the cost of requiring cross-fitting to avoid overfitting bias (Chapter 11).

Extrapolation. Step 2 predicts $\hat{Y}_i(0)$ for treated units whose covariate values may lie outside the support of the control group. The propensity score overlap condition (Chapter 6) formalizes when extrapolation is unavoidable.

5.3.5 Regression Adjustment in Randomized Experiments: Lin (2013)

In a *randomized* experiment, outcome regression can reduce variance even though adjustment is not needed for unbiasedness. The *regression-adjusted estimator* of Lin (2013) fits the fully interacted OLS model:

$$Y_i = \alpha + \beta T_i + \gamma^\top \tilde{X}_i + \delta^\top (T_i \cdot \tilde{X}_i) + \varepsilon_i, \tag{5.9}$$

where $\tilde{X}_i = X_i - \bar{X}$ are mean-centered covariates, and takes $\hat{\beta}$ as the ATE estimate.

Why $\hat{\beta}$ equals the three-step OR estimator. The interacted regression Equation 5.9 fits arm-specific linear models with centered covariates. Under centering, the OLS intercept equals the regression-adjusted estimator of $\mathbb{E}[Y(t)]$, so $\hat{\beta} = \hat{\mu}_{1,\mathrm{reg}} - \hat{\mu}_{0,\mathrm{reg}} = \hat{\tau}_{\mathrm{reg}}$.

Theorem: Design-Consistency of the Regression-Adjusted Estimator [@lin2013agnostic]

Under complete randomization, $\hat{\beta}$ from the interacted regression Equation 5.9 is consistent for τ_{ATE} regardless of whether the linear model is correctly specified.

Proof

Work in the design-based framework of **?@thm-dim**: potential outcomes fixed, only $\mathbf{T}$ random. Let $\mathbf{B}_1 = (\sum_i \mathbf{x}_i \mathbf{x}_i^\top)^{-1} \sum_i \mathbf{x}_i Y_i(1)$ be the full-population OLS coefficient (fixed under randomization), and let $e_i(1) = Y_i(1) - \mathbf{x}_i^\top \mathbf{B}_1$. By normal equations, $\sum_i e_i(1) = 0$ and $\frac{1}{N}\sum_i \mathbf{x}_i^\top \mathbf{B}_1 = \bar{Y}(1)$.
Linearization. A short algebraic manipulation using normal equations gives:

$$\hat{\mu}_{1,\mathrm{reg}} = \bar{Y}(1) + \frac{1}{N_1}\sum_{i=1}^{N} T_i e_i(1) + O_p(N^{-1}). \tag{5.10}$$

The remainder term R_N is $O_p(N^{-1})$ because the covariate-balance gap is $O_p(N^{-1/2})$ by design, and $\hat{\beta}_1 - \mathbf{B}_1 = O_p(N^{-1/2})$ by standard within-arm OLS theory.
Consistency. Under CRE, each unit has $\Pr(T_i = 1 \mid \mathcal{F}_N) = N_1/N$, so $\mathbb{E}[N_1^{-1}\sum_i T_i e_i(1) \mid \mathcal{F}_N] = 0$ by $\sum_i e_i(1) = 0$. The conditional variance is $O(N^{-1})$ by finite-population sampling theory; by Chebyshev, $N_1^{-1}\sum_i T_i e_i(1) \to_p 0$. By Equation 5.10, $\hat{\mu}_{1,\mathrm{reg}} - \bar{Y}(1) \to_p 0$. Symmetrically, $\hat{\mu}_{0,\mathrm{reg}} - \bar{Y}(0) \to_p 0$. Therefore $\hat{\beta} \to_p \bar{Y}(1) - \bar{Y}(0) = \tau_{\mathrm{ATE}}$.
Crucially, the probability limit of $\hat{\beta}_1$ never enters: consistency comes from the randomization distribution, not from correctness of the linear model. □

Byproduct: variance reduction. Equation Equation 5.10 shows that the regression estimator is approximately $\bar{\tau} + N_1^{-1}\sum_i T_i e_i(1) - N_0^{-1}\sum_i (1 - T_i) e_i(0)$. The variance reduction relative to DIM comes from replacing raw outcomes by residuals: when the linear model explains a meaningful share of $Y_i(t)$'s variation, $S^2_{e(t)} < S^2_{Y(t)}$ and the design variance shrinks.

Remark: The Role of the Regression Model in a Randomized Experiment

In a randomized experiment, the regression model serves *one purpose only*: variance reduction. It does not contribute to unbiasedness. DIM is already design-consistent; adding covariates reduces $\mathrm{Var}(\hat{\tau})$ by absorbing residual variation in Y unrelated to T. If the model fits well, the residual variance shrinks and the estimator becomes more precise. If misspecified or covariates are weakly predictive, the variance reduction is small or zero, but *no bias is introduced*. This stands in sharp contrast to the observational setting, where the regression model carries a double burden: it must both adjust for confounding (unbiasedness) and fit the outcome surface (efficiency). Misspecification in an observational study threatens *both* properties simultaneously.

Remark: Precedence in Survey Sampling

The design-consistency result in **?@thm-lin** — model-agnostic consistency under the randomization distribution — was established in the survey sampling literature decades before Lin (2013). Isaki and Fuller (1982) proved this for the generalized regression (GREG) estimator under general probability sampling designs. A CRE is structurally equivalent to simple random sampling from the finite population of potential outcomes, so their result directly implies **?@thm-lin**. Deville and Särndal (1992) extended this to calibration estimators. Lin (2013)'s contribution was restating and proving this within the Neyman–Rubin framework for the experimental causal inference audience.

5.4 Stratification

Stratification (subclassification) is a non-parametric implementation of the back-door formula. Rather than modeling $\mathbb{E}[Y \mid T, X]$, the analyst divides the sample into *strata* — subgroups with similar values of X — and estimates the treatment effect within each stratum.

Within stratum $\mathcal{S}_k$, the treated and control units have similar covariate distributions. If the stratum is narrow enough, ignorability holds approximately, and the within-stratum DIM:

$$\hat{\tau}_k = \bar{Y}_{1,k} - \bar{Y}_{0,k}$$

is approximately unbiased for the stratum-specific ATE. The overall ATE is estimated as:

$$\hat{\tau}_{\text{STRAT}} = \sum_{k=1}^{K} \hat{\tau}_k \cdot \frac{n_k}{n}. \tag{5.11}$$

Cochran (1968) showed that with $K = 5$ equal-size strata, roughly 90% of the bias from a single continuous confounder is removed; with $K = 10$, over 95%.

Remark: Stratification, Standardization, and Survey Sampling Terminology

When X is categorical, the stratified estimator Equation 5.11 and the standardization estimator Equation 5.8 are algebraically identical: both compute within-cell treatment effect estimates and aggregate with marginal weights n_k/n. The names reflect disciplinary tradition.
When X is continuous, stratification estimates $\hat{\mu}(t, x)$ by a piecewise-constant step function; regression fits a smooth model. In the survey sampling literature, *stratification* (design-stage: divide before sampling) and *post-stratification* (analysis-stage: reweight after sampling) are distinct. Standardization in causal inference corresponds to post-stratification.

Limitations. Stratification faces the *curse of dimensionality*: with p binary covariates, there are 2^p cells, many empty in practice. Two solutions: regression adjustment, which imposes parametric structure on $\hat{\mu}(t, x)$; and propensity score stratification (Chapter 6), which collapses multivariate X to a scalar $\pi(X) = P(T{=}1 \mid X)$.

5.5 Simpson's Paradox

Simpson's paradox — the reversal of an association when conditioning on a third variable — was introduced in Chapter 1. This chapter's development gives it a second layer of interpretation. The back-door formula used in Chapter 1 is precisely the standardization estimator Equation 5.8: within-stratum conditional means averaged over the *marginal* distribution of the confounder rather than its treatment-conditional distribution. The pooled association does not do this — it weights strata by $P(X \mid T)$ instead of $P(X)$.

5.5.1 When Should You *Not* Condition on X?

Simpson's paradox has a mirror image: conditioning can *create* a spurious association or make a true causal effect disappear. This occurs when X is a mediator or a collider. The back-door criterion gives the correct prescription: include X only if it blocks back-door paths *without* opening collider paths.

Conditioning on a Mediator or Collider

Adjusting for a post-treatment variable X that lies on a causal pathway $T \to X \to Y$ blocks part of the causal effect, producing a downward-biased estimate of the total effect. Adjusting for a collider X with $T \to X \leftarrow Y$ opens a spurious association that does not exist in the population. Neither mistake is detectable from the data alone; the DAG is essential.

5.6 Lab: Simulation Study of the Outcome Regression Estimator

This simulation compares the linear and local-linear OR estimators across two designs and illustrates the bias-variance tradeoff.

Data-generating process. $X \sim \text{Uniform}(0,1)$, $\varepsilon \sim \mathcal{N}(0,1)$ independent of X. Potential outcomes:

$$Y(t) = t + 3(1+t)X^2 + \varepsilon, \qquad t \in \{0,1\}. \tag{5.12}$$

The conditional ATE is $\tau(x) = 1 + 3x^2$, so $\tau_{\text{ATE}} = 1 + 3\mathbb{E}[X^2] = 1 + 1 = 2$.

Two assignment mechanisms:

- **CRD**: $T \sim \text{Bern}(0.5)$, independent of X.
- **Observational**: $T \mid X \sim \text{Bern}(\text{expit}(-2+5X))$, giving $\pi(0) \approx 0.12$, $\pi(0.5) \approx 0.62$, $\pi(1) \approx 0.95$. Strong confounding: high-X units are nearly always treated.

Strong ignorability holds in both: under CRD by design; under the observational mechanism because T depends only on X through a known stochastic function, leaving no unmeasured variable affecting both T and Y.

Estimators. *Linear OR*: within each arm, fit OLS of Y on X (misspecified: true conditional mean is quadratic with a TX^2 interaction). *Local-linear OR*: within each arm, fit local linear regression with Gaussian kernel, bandwidth $h = 0.5 \cdot n^{-1/5}$.

Results ($n = 1000$, $B = 2000$ replications, seed 2024):

Design	Estimator	Mean	Bias	SD	RMSE
CRD	Linear OR	2.0013	+0.0013	0.0679	0.0678
CRD	Local-linear OR	2.0333	+0.0333	0.0659	0.0738
Observational	Linear OR	1.9602	−0.0398	0.0851	0.0939
Observational	Local-linear OR	2.0240	+0.0240	0.0923	0.0953

Lesson 1: Under CRD, the linear OR is unbiased regardless of model specification. This confirms **?@thm-lin**: under complete randomization, the linear OR converges to τ_{ATE} even with a wrong outcome model. The randomization distribution, not the model, does the identification work.

Lesson 2: Under an observational design, the misspecified linear OR is biased. The bias grows from $+0.0013$ under CRD to -0.0398 under the observational design. The omitted X^2 and TX^2 terms cause $\hat{\mu}(t,x)$ to misrepresent the outcome surface; because high-X units are predominantly treated, the misfit is systematically amplified in the direction of confounding. Under CRD, the balanced assignment averages out the same misfit.

Lesson 3: The local-linear OR nearly eliminates bias at the cost of higher variance. Under the observational design, local-linear OR reduces bias from -0.0398 to $+0.0240$ but its SD is 0.0923 vs. 0.0851 for the linear model. The RMSE comparison (0.0939 vs. 0.0953) favors the linear estimator in MSE, even though it is biased — the classic bias-variance tradeoff.

Model Misspecification and Confounding Interact

The linear OR bias is $+0.0013$ under CRD and -0.0398 under the observational design. The misspecification is identical in both cases — the difference comes entirely from the assignment mechanism. Under CRD, $T \perp\!\!\!\perp X$, so X within each arm is $\text{Uniform}(0,1)$. The population

OLS pointwise error $b(x) = -\frac{1}{2} + 3x - 3x^2$ integrates to exactly zero over $U(0,1)$: $\mathbb{E}[b(X)] = \int_0^1 (-\frac{1}{2} + 3x - 3x^2)\,dx = 0$. This cancellation is structural: the population OLS line is the linear projection of $\tau(X)$ onto $\{1, X\}$ under $U(0,1)$, and a linear projection always preserves the mean. Under the observational design, the within-arm distributions of X are distorted by the propensity score, so the same cancellation fails.

5.7 Chapter Summary

Estimand	Identification	Key assumption	Method
ATE under CRD	$\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid T{=}t]$	Randomization protocol	DIM, regression-adjusted DIM
ATE under observability	Equation 5.5	Ignorability + overlap	OR, standardization, stratification
ATT	Equation 5.6	Same, one-sided overlap for $t = 0$	Same methods, weighted over treated
Propensity score (Ch. 6)	Reduces X to scalar $\pi(X)$	Correctly specified PS model	IPW, matching, PS stratification

1. **Randomization as graph surgery.** A randomized experiment removes all arrows into T, so $f(y \mid \mathrm{do}(T{=}t)) = f(y \mid T{=}t)$ and DIM is unbiased for the ATE without covariate adjustment.
2. **Ignorability is the key assumption.** Under unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$, the back-door adjustment formula Equation 5.5 identifies the ATE. This assumption is substantive, design-determined in RCTs, and untestable from observational data.
3. **Three estimation strategies.** Regression adjustment (G-formula), standardization, and stratification all implement Equation 5.5 via different modeling choices. Under a randomized experiment, regression adjustment is design-consistent for the ATE even when the outcome model is misspecified (**?@thm-lin**); in observational studies, it is consistent only when the outcome model is.
4. **Simpson's paradox.** Pooled associations can reverse within subgroups when a confounder determines treatment selection. The correct causal estimate requires standardization over the *marginal* covariate distribution, guided by the back-door criterion.
5. **The propensity score dimension reduction.** When X is high-dimensional, direct stratification or cell-by-cell standardization breaks down. The propensity score $\pi(X)$ provides a scalar sufficient statistic for adjustment. Chapter 6 develops the theory and estimation methods.

5.8 Problems

1. Randomization and the do-calculus. Let the observational DAG be $\{U \to T, U \to Y, X \to T, T \to Y\}$ with U unobserved.

(a) Identify all back-door paths from T to Y.
(b) Does X satisfy the back-door criterion? Does U? Explain.
(c) Now suppose treatment is randomized. Draw the modified DAG and identify all back-door paths. Show that $f(y \mid \mathrm{do}(T{=}t)) = f(y \mid T{=}t)$ holds using Rule 2 of the do-calculus. (*Hint*: identify the appropriate (X, Z, W) instantiation, construct $\mathcal{G}_{\underline{T}}$, and verify $(Y \perp\!\!\!\perp T)_{\mathcal{G}_{\underline{T}}}$.)
(d) Under randomization, is covariate adjustment on X necessary for unbiasedness? Is it ever beneficial? Explain.

2. Ignorability and the back-door criterion. Consider the DAG $\{X \to T, X \to Y, T \to Y, U \to Y\}$ with U unobserved.

(a) Does $\{X\}$ satisfy the back-door criterion? Write out the identifying formula for τ_{ATE}.
(b) Now add the edge $U \to T$ to the DAG. Does $\{X\}$ still satisfy the back-door criterion? What does the criterion require when both X and U affect T?
(c) Explain in words what "unconfoundedness given X" means about the role of U in the data-generating process.

3. Standardization. A study of job training (T) and earnings (Y, in thousands) produces the following cell means, with $P(X{=}0) = 0.4$ and $P(X{=}1) = 0.6$:

X	$\hat{\mu}(1, x)$	$\hat{\mu}(0, x)$	n_x/n
0	28	22	0.4
1	35	31	0.6

(a) Compute $\hat{\tau}_{\text{ATE}}$ using the standardization formula Equation 5.8.
(b) Compute $\hat{\tau}_{\text{ATT}}$, given that all treated units come from $X{=}1$ — i.e., $P(X{=}1 \mid T{=}1) = 1$.
(c) The unadjusted difference in means is $33 - 25 = 8$. Compare to your answers in (a) and (b) and explain the discrepancy.

4. Stratification and Cochran's rule. You have a binary confounder $X \in \{0, 1\}$ and form two strata. Within stratum $X{=}0$: $n_0 = 600$, $\bar{Y}_1 = 10$, $\bar{Y}_0 = 8$. Within stratum $X{=}1$: $n_1 = 400$, $\bar{Y}_1 = 15$, $\bar{Y}_0 = 12$.

(a) Compute the stratified ATE estimator $\hat{\tau}_{\text{STRAT}}$ via Equation 5.11.
(b) Suppose an unadjusted DIM gives $\hat{\tau}_{\text{DIM}} = 5.5$. Explain why the two estimates differ and which is the appropriate causal estimate.
(c) Describe one limitation that would arise if X were a continuous variable with 15 dimensions.

5. Simpson's paradox. A hospital reports that ICU patients ($T{=}1$) have higher mortality than others ($T{=}0$): $P(Y{=}1 \mid T{=}1) = 0.30$ vs. $P(Y{=}1 \mid T{=}0) = 0.10$.

(a) Construct a numerical example (a $2 \times 2 \times 2$ table with disease severity X as the confounder) consistent with the pooled numbers yet showing ICU admission *reduces* mortality within both severity strata.
(b) Compute the standardized ATE using the marginal distribution of X.
(c) Draw the DAG. Identify the back-door path that the pooled comparison fails to block.
(d) A colleague argues the pooled statistic is the right answer since the hospital treats patients with both mild and severe illness. Explain, using potential outcomes notation, why this argument is incorrect.

Chapter 6

Propensity Score Methods

Learning Objectives

By the end of this chapter, students should be able to:

1. Define the propensity score $\pi(X) = P(T{=}1 \mid X)$ and explain why it is a balancing score.
2. State and prove the Balancing Property ($T \perp\!\!\!\perp X \mid \pi(X)$), the Coarsest Balancing Score Lemma, and the Propensity Score Theorem.
3. Describe standard methods for estimating the propensity score and the practical pitfalls of each.
4. Derive the IPW identification formula for the ATE and explain the role of the Horvitz–Thompson representation.
5. Distinguish the IPW estimator (targets ATE) from the nearest-neighbor matching estimator (targets ATT).
6. Distinguish positivity (weak overlap) from strong overlap, describe the practical consequences of near-violations, and apply trimming strategies.
7. Articulate the fundamental limitation of propensity score methods — the unconfoundedness assumption — and explain why this motivates instrumental variables (Chapter 7).

6.1 Motivation: The Curse of Dimensionality

Chapter 5 established that under strong ignorability, the ATE is identified by the back-door adjustment formula, and developed three estimation strategies: regression adjustment, standardization, and stratification. All three require conditioning on a covariate vector X. When X is low-dimensional, this is feasible. When X has many components, it is not.

The problem. Stratification requires cells $\{X = x\}$ to contain both treated and control units. With p binary covariates, there are 2^p cells. With $p = 10$, that is 1024 cells — far more than most datasets can support. Regression adjustment avoids the cell-count problem but requires a correctly specified model for $\mathbb{E}[Y \mid T, X]$, which becomes harder to specify reliably as p grows.

The solution. Rosenbaum and Rubin (1983) showed that it is sufficient to condition on a single scalar function of X: the *propensity score* $\pi(X) = P(T{=}1 \mid X)$. The propensity score reduces the adjustment dimension without changing the identified estimand, provided strong ignorability and positivity hold.

6.2 The Propensity Score and Its Balancing Properties

6.2.1 Balancing Scores and the Propensity Score

Definition: Balancing Score [@rosenbaum1983central]

A function $b(X)$ is called a **balancing score** if $T \perp\!\!\!\perp X \mid b(X)$.

The full covariate vector X is trivially a balancing score. The propensity score is the *coarsest* balancing score, providing the greatest dimension reduction without sacrificing identification.

Definition: Propensity Score [@rosenbaum1983central]

For a binary treatment $T \in \{0, 1\}$ and observed covariates X, the **propensity score** is $\pi(X) = P(T{=}1 \mid X)$.

In a randomized experiment, $\pi(X) = 0.5$ for all units. In an observational study, $\pi(X)$ varies with X and must be estimated from data.

Connection to strong ignorability. Under strong ignorability $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$, the potential outcomes carry no further information about treatment assignment once X is given: $P(T{=}1 \mid X, Y(0), Y(1)) = P(T{=}1 \mid X) = \pi(X)$. This equality is a *consequence* of the ignorability assumption, not part of the definition of $\pi(X)$.

Theorem: The Propensity Score Is a Balancing Score [@rosenbaum1983central]

$T \perp\!\!\!\perp X \mid \pi(X)$.

Proof

It suffices to show $P(T{=}1 \mid X, \pi(X)) = P(T{=}1 \mid \pi(X))$. Since $\pi(X)$ is a deterministic function of X, $P(T{=}1 \mid X, \pi(X)) = P(T{=}1 \mid X) = \pi(X)$. For the right-hand side, the law of iterated expectations gives:

$$P(T{=}1 \mid \pi(X)) = \mathbb{E}[P(T{=}1 \mid X) \mid \pi(X)] = \mathbb{E}[\pi(X) \mid \pi(X)] = \pi(X). \quad \square$$

Remark: Covariate Balance and Design-Stage Analysis

The balancing property implies $f(X \mid T{=}1, \pi(X)) = f(X \mid T{=}0, \pi(X))$ a.s. Imbens and Rubin (2015) call this *design before analysis*: covariate balance can be assessed by stratifying on a discretized $\hat{\pi}(X)$ and comparing covariate distributions within strata — without consulting the outcome Y, making the diagnostic immune to outcome-fishing.

Remark: Observed Covariates Only

The Balancing Property is a statement about the *observed* covariate distribution. It does *not* say anything about unobserved confounders U: if U affects both T and Y and is not captured by X, the balancing property fails to close the back-door path through U.

6.2.2 The Propensity Score Theorem

Lemma: The Propensity Score Is the Coarsest Balancing Score [@rosenbaum1983central]

If $b(X)$ is any balancing score — that is, $T \perp\!\!\!\perp X \mid b(X)$ — then $\pi(X)$ is a function of $b(X)$: $\pi(X) = g(b(X))$ for some measurable g.

Proof

Since $b(X)$ is a balancing score, $P(T{=}1 \mid X, b(X)) = P(T{=}1 \mid b(X))$. Since $b(X)$ is a function of X: $\pi(X) = P(T{=}1 \mid X) = P(T{=}1 \mid X, b(X)) = P(T{=}1 \mid b(X))$. Hence $\pi(X)$ is a measurable function of $b(X)$ alone. □

The lemma states that $\pi(X)$ is the smallest sufficient statistic for T among functions of X: any other balancing score retains at least as much information about X.

Theorem: Propensity Score Theorem [@rosenbaum1983central]

Suppose strong ignorability holds: $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ and $0 < \pi(X) < 1$ a.s. Then for any balancing score $b(X)$:

$$(Y(0), Y(1)) \perp\!\!\!\perp T \mid b(X).$$

In particular, $(Y(0), Y(1)) \perp\!\!\!\perp T \mid \pi(X)$.

Proof

We prove the result for $b(X) = \pi(X)$. For any $t \in \{0, 1\}$:

$$P(T{=}1 \mid Y(t), \pi(X)) = \mathbb{E}[P(T{=}1 \mid Y(t), X) \mid Y(t), \pi(X)] = \mathbb{E}[P(T{=}1 \mid X) \mid Y(t), \pi(X)] = \mathbb{E}[\pi(X) \mid Y(t), \pi(X)] = \pi(X$$

where the second equality uses ignorability ($T \perp\!\!\!\perp Y(t) \mid X$). Hence $T \perp\!\!\!\perp Y(t) \mid \pi(X)$. By the Coarsest Balancing Score Lemma, the result extends to any $b(X)$. □

Remark: Identification via the Propensity Score

?@thm-ps gives an alternative identification formula: $\mathbb{E}[Y(t)] = \mathbb{E}[\mathbb{E}[Y \mid \pi(X), T{=}t]]$. A common misconception is that $\mathbb{E}[Y \mid \pi(X), T{=}t]$ and $\mathbb{E}[Y \mid X, T{=}t]$ are equal pointwise — they are not. The theorem guarantees only that their marginal expectations agree: $\mathbb{E}[\mathbb{E}[Y \mid \pi(X), T{=}t]] = \mathbb{E}[\mathbb{E}[Y \mid X, T{=}t]] = \mathbb{E}[Y(t)]$.

The Dimension Reduction

?@thm-ps reduces the curse of dimensionality from a $\dim(X)$-dimensional problem to a one-dimensional one:

$$\tau_{\text{ATE}} = \mathbb{E}[\mathbb{E}[Y \mid T{=}1, \pi(X)] - \mathbb{E}[Y \mid T{=}0, \pi(X)]].$$

6.3 Estimation of the Propensity Score

Logistic regression. The classical approach models $\text{logit}(\pi(X)) = X^\top \beta$ and estimates β by maximum likelihood. The fitted values $\hat{\pi}(X_i) = \sigma(X_i^\top \hat{\beta})$ are used in place of the true propensity score.

Machine learning estimators. When X is high-dimensional or the true propensity score is nonlinear, machine learning methods — gradient boosted trees, random forests, regularized logistic regression — can estimate $\pi(X)$ more flexibly. A critical issue is *overfitting*: an estimator that perfectly separates treated and control units in-sample will produce estimated propensity scores near 0 or 1 everywhere, destroying overlap. Cross-fitting (Chapter 11) addresses this by estimating nuisance functions on held-out folds.

Propensity Score Estimation Is a Nuisance, Not the Goal

The propensity score is estimated to construct a good estimator of τ_{ATE}. Balancing tests diagnose whether the estimated score has achieved covariate balance, but they do *not* test whether unobserved confounders are balanced.

6.4 Matching on the Propensity Score

The propensity score motivates *matching*: for each treated unit i, find a control unit j with $\hat{\pi}(X_j) \approx \hat{\pi}(X_i)$, and use Y_j as a local estimate of $\mathbb{E}[Y(0) \mid \pi(X) = \pi(X_i)]$ (Abadie and Imbens 2006, 2016). The matched control outcome is not a substitute for the individual counterfactual $Y_i(0)$; it approximates the conditional mean:

$$\mathbb{E}[Y(0) \mid \pi(X) = \pi(X_i)] = \mathbb{E}[Y(0) \mid \pi(X) = \pi(X_i), T{=}0] = \mathbb{E}[Y \mid \pi(X) = \pi(X_i), T{=}0],$$

where the first equality uses **?@thm-ps** and the second uses consistency. Averaging across treated units recovers the ATT:

$$\hat{\tau}_{\text{ATT}} = \frac{1}{n_1} \sum_{i:\,T_i=1} [Y_i - Y_{\hat{\jmath}(i)}], \tag{6.1}$$

where $\hat{\jmath}(i)$ is the matched control index.

One-to-one nearest-neighbor matching. For each treated unit i:

$$\hat{\jmath}(i) = \arg\min_{j:\,T_j=0} |\hat{\pi}(X_i) - \hat{\pi}(X_j)|.$$

Matching *with replacement* reduces bias but increases variance.

Caliper matching. Restricts matches to pairs within a maximum distance δ, discarding treated units with no close match. This restricts the estimand to a subpopulation with good overlap.

Remark: Matching as Nonparametric Regression on $\pi(X)$

Propensity score matching implicitly fits a nonparametric regression of Y on $\pi(X)$ separately within each treatment arm. By **?@thm-ps**, $\mathbb{E}[Y(t) \mid \pi(X) = p]$ is identified from observed data for each t. Nearest-neighbor matching approximates this by averaging outcomes of close matches — local constant regression on the scalar propensity score. The dimension reduction from **?@thm-ps** is what makes nonparametric estimation feasible.

6.5 Inverse Probability Weighting

6.5.1 The IPW Identification Formula

Under strong ignorability, the ATE has an *inverse probability weighting* (IPW) representation:

$$\tau_{\text{ATE}} = \mathbb{E}\left[\frac{T \cdot Y}{\pi(X)}\right] - \mathbb{E}\left[\frac{(1-T) \cdot Y}{1-\pi(X)}\right].$$

Theorem: IPW Identification

Under strong ignorability ($(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ and $0 < \pi(X) < 1$): $\mathbb{E}[TY/\pi(X)] = \mathbb{E}[Y(1)]$.

Proof

$$\mathbb{E}\left[\frac{TY}{\pi(X)}\right] = \mathbb{E}\left[\frac{TY(1)}{\pi(X)}\right] = \mathbb{E}\left[\mathbb{E}\left[\frac{TY(1)}{\pi(X)}\,\middle|\, X\right]\right] = \mathbb{E}\left[\frac{Y(1)}{\pi(X)}\,\mathbb{E}[T \mid X]\right] = \mathbb{E}\left[\frac{Y(1)}{\pi(X)} \cdot \pi(X)\right] = \mathbb{E}[Y(1)].$$

The first equality uses consistency ($Y = Y(1)$ when $T{=}1$); the third uses ignorability ($Y(1) \perp\!\!\!\perp T \mid X$). □

Remark: Scope of the Ignorability Assumption

The proof invokes only the marginal independence $Y(1) \perp\!\!\!\perp T \mid X$, not the full joint $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$. An analogous proof for the control term uses only $Y(0) \perp\!\!\!\perp T \mid X$. The joint independence is used in this chapter for uniformity with Chapter 5 and because it is needed for estimands involving the joint distribution of potential outcomes.

6.5.2 Horvitz–Thompson and Hájek Estimators

The **Horvitz–Thompson (HT)** IPW estimator:

$$\hat{\tau}_{\mathrm{HT}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{T_i Y_i}{\hat{\pi}(X_i)} - \frac{(1-T_i)Y_i}{1-\hat{\pi}(X_i)}\right]. \tag{6.2}$$

The **Hájek (self-normalized)** estimator:

$$\hat{\tau}_{\mathrm{HJ}} = \frac{\sum_i T_i Y_i/\hat{\pi}(X_i)}{\sum_i T_i/\hat{\pi}(X_i)} - \frac{\sum_i (1-T_i)Y_i/(1-\hat{\pi}(X_i))}{\sum_i (1-T_i)/(1-\hat{\pi}(X_i))}. \tag{6.3}$$

The Hájek estimator caps each unit's effective weight at its share of the total within its arm, suppressing the influence of extreme weights. The efficient doubly robust AIPW estimator of Chapter 10 combines the outcome model and the propensity score; under correct nuisance specification it achieves higher efficiency than either pure IPW or pure outcome regression.

ATT estimation. The ATT re-weights the *control arm* to look like the treated population:

$$\hat{\tau}_{\mathrm{HT,ATT}} = \frac{1}{n_1}\sum_{i:\,T_i=1} Y_i - \frac{1}{n_1}\sum_{i:\,T_i=0}\frac{\hat{\pi}(X_i)}{1-\hat{\pi}(X_i)}Y_i, \tag{6.4}$$

$$\hat{\tau}_{\mathrm{HJ,ATT}} = \frac{1}{n_1}\sum_{i:\,T_i=1} Y_i - \frac{\sum_{i:\,T_i=0}\frac{\hat{\pi}(X_i)}{1-\hat{\pi}(X_i)}Y_i}{\sum_{i:\,T_i=0}\frac{\hat{\pi}(X_i)}{1-\hat{\pi}(X_i)}}. \tag{6.5}$$

Each control unit receives weight proportional to its *odds of treatment* $\hat{\pi}/(1-\hat{\pi})$. For a control unit with $\hat{\pi}(X_i) = 0.95$ the weight is 19; combined with a large baseline outcome this produces extreme variance.

6.6 Overlap and Positivity

6.6.1 Positivity and Strong Overlap

Definition: Positivity (Weak Overlap)

The **positivity** condition requires $0 < \pi(X) < 1$ almost surely. Every unit has a positive probability of receiving either treatment or control, regardless of its covariate values.

Together with unconfoundedness, positivity constitutes the strong ignorability of Rosenbaum and Rubin (1983). Positivity is the condition needed for *identification*.

Definition: Strong Overlap

The **strong overlap** condition requires $c \le \pi(X) \le 1-c$ a.s. for some $c \in (0, 1/2)$.

Strong overlap bounds the inverse weights uniformly away from infinity. It is the condition under which $\sqrt{n}$-consistent, asymptotically normal inference for the ATE goes through (Chapter 11). Positivity alone permits identification but does not guarantee stable inference: when $\pi(X)$ approaches 0 or 1, IPW weights blow up even though the parameter is technically identified (Khan and Tamer 2010).

6.6.2 Practical Consequences of Near-Violations

Near-Overlap Problems

When $\hat{\pi}(X_i)$ is close to 0 or 1, the IPW weight is very large. A small number of units with extreme weights can dominate the estimator, increasing variance dramatically.

6.6.3 Trimming Strategies

Trimming by propensity score. Restrict to units with $\eta \le \pi(X) \le 1-\eta$ for small $\eta > 0$. The estimand changes to:

$$\tau_{\text{trim}} = \mathbb{E}[Y(1) - Y(0) \mid \eta \le \pi(X) \le 1 - \eta].$$

Crump et al. (2009) rule. Crump et al. (2009) derive the optimal trimming threshold minimizing the asymptotic variance of the trimmed ATE estimator. The rule $\eta = 0.1$ is a common default.

6.6.4 Re-targeting the Estimand

When overlap fails, an alternative to trimming is to change the *estimand*. The ATT requires only $P(T{=}0 \mid X{=}x) > 0$ for x in the support of $X \mid T{=}1$, i.e., $\pi(X) < 1$ a.s. on the treated support. The ATT tolerates regions with $\pi(x) = 0$ (no treated units there) but not regions inside the treated support where $\pi(x) = 1$. The ATC is symmetric. When overlap fails where $\pi(X) \approx 0$, re-target to the ATT; when overlap fails where $\pi(X) \approx 1$, re-target to the ATC.

6.7 Lab: Simulation Study of IPW and Matching Estimators

This lab compares five estimators on a five-covariate DGP with treatment-effect heterogeneity. With five continuous confounders, direct stratification is infeasible, making propensity-score methods the natural choice.

6.7.1 Part 1: Correctly Specified Propensity Score

DGP for Lab 6

$X_j \overset{\text{i.i.d.}}{\sim} N(0,1)$ for $j = 1, \dots, 5$. All five covariates are genuine confounders. True propensity score:

$$\pi(X) = \text{expit}(-0.5 + 0.8X_1 + 0.5X_2 - 0.3X_3 + 0.2X_4 + 0.1X_5), \tag{6.6}$$

giving $P(T{=}1) \approx 0.40$. Potential outcomes:

$$Y(t) = (1 + 0.5X_1) \cdot t + 8 + 0.8X_1 + 0.5X_2 + 0.4X_3 + 0.3X_4 + 0.2X_5 + \varepsilon, \quad \varepsilon \sim N(0,1). \tag{6.7}$$

The CATE is $\tau(X) = 1 + 0.5X_1$, so $\tau_{\text{ATE}} = 1.000$ (exact). Because $\pi(X)$ is increasing in X_1, treated units have $\mathbb{E}[X_1 \mid T{=}1] > 0$, giving $\tau_{\text{ATT}} \approx 1.199$ (oracle, $n = 10^7$). The gap ≈ 0.199 reflects selection: units most likely to be treated also tend to benefit more.

Estimators. Five estimators are compared under both the true and estimated propensity score. The estimated score uses logistic regression of T on $(X_1, \dots, X_5)$ — correctly specified since the true logit is linear. The five estimators are: HT-ATE Equation 6.2, HJ-ATE Equation 6.3 (both targeting ATE); HT-ATT Equation 6.4, HJ-ATT Equation 6.5, and NNM Equation 6.1 (all targeting ATT). The large baseline mean $\mathbb{E}[Y(0)] = 8$ amplifies the consequences of extreme IPW weights, making Hájek normalization strongly recommended.

Results ($n = 1{,}000$, $B = 2{,}000$ replications, seed 42). Bias relative to each estimator's own target.

PS	Estimator	Estimand	Mean	Bias	SD	RMSE
Known	HT-ATE	ATE	1.008	+0.008	0.626	0.626

PS	Estimator	Estimand	Mean	Bias	SD	RMSE
Known	HJ-ATE	ATE	1.003	+0.003	0.145	0.145
Known	HT-ATT	ATT	1.184	−0.016	0.725	0.725
Known	HJ-ATT	ATT	1.203	+0.003	0.131	0.131
Known	NNM	ATT	1.207	+0.008	0.130	0.130
Estimated	HT-ATE	ATE	0.998	−0.002	0.194	0.194
Estimated	HJ-ATE	ATE	1.002	+0.002	0.102	0.103
Estimated	HT-ATT	ATT	1.202	+0.002	0.251	0.251
Estimated	HJ-ATT	ATT	1.202	+0.002	0.101	0.101
Estimated	NNM	ATT	1.205	+0.006	0.121	0.121

Lesson 1: Hájek normalization is strongly recommended when outcomes are non-centered. With the known PS, HT-ATE achieves SD = 0.626, while HJ-ATE achieves SD = 0.145 — a **4.3-fold reduction**. A unit with $\pi(X_i) = 0.05$ receives raw weight 20; multiplied by an outcome near $\mathbb{E}[Y(0)] = 8$, its contribution is of order 160.

Lesson 2: HT-ATT is even more unstable; Hájek normalization is equally beneficial. HT-ATT SD = 0.725. A control unit with $\hat{\pi}(X_i) = 0.95$ receives odds-ratio weight 19. HJ-ATT reduces SD to 0.131 — a **5.5-fold reduction**.

Lesson 3: HJ-ATT and NNM converge to the same target with nearly identical efficiency. Under the known PS, HJ-ATT (SD = 0.131) and NNM (SD = 0.130) are virtually indistinguishable — a clear "two routes, one estimand" demonstration.

Lesson 4: The estimand gap between ATE and ATT is large and clearly revealed. The ATE estimators converge to 1.000; the ATT estimators converge to ≈ 1.199. The 0.199 gap is not bias. A researcher who applies NNM and reports against the ATE benchmark would conclude the estimator has 20% bias; it is actually unbiased for the correct target.

Lesson 5: Estimated PS collapses HT variance. HT-ATE SD falls from 0.626 to 0.194 (69% reduction); shrinkage of fitted logistic probabilities toward the sample mean trims extreme weights automatically. HJ-ATT with estimated PS achieves SD = 0.101 — beating NNM (SD = 0.121).

The Estimand Must Be Chosen Before the Estimator

In this DGP, $\tau_{\text{ATE}} = 1.000$ and $\tau_{\text{ATT}} \approx 1.199$. The gap arises because treatment selection is correlated with the individual treatment effect: high-X_1 units are simultaneously more likely to be treated and more likely to benefit (Imbens and Rubin 2015). Choosing among estimators should be driven by the policy question (ATE or ATT?), not by which produces the preferred point estimate.

6.7.2 Part 2: Robustness to PS Model Misspecification

Modified DGP. Outcome model unchanged. True PS now contains a quadratic term:

$$\pi^*(X) = \text{expit}(-0.5 + 0.8X_1 + 0.5X_1^2 + 0.2X_4 + 0.1X_5), \tag{6.8}$$

giving a U-shaped propensity surface. $\tau_{\text{ATE}} = 1.000$ (unchanged); $\tau^*_{\text{ATT}} \approx 1.152$ (oracle). The *estimated* PS is still a linear logistic regression on $(X_1, \dots, X_5)$ — misspecified by omitting X_1^2.

Results (n = 1,000, B = 2,000, seed 42):

PS	Estimator	Estimand	Mean	Bias	SD	RMSE
True	HT-ATE	ATE	1.028	+0.028	0.824	0.825
True	HJ-ATE	ATE	1.013	+0.013	0.152	0.152
True	HT-ATT	ATT	1.186	+0.034	1.333	1.333
True	HJ-ATT	ATT	1.183	+0.031	0.212	0.215
True	NNM	ATT	1.178	+0.026	0.175	0.177
Misspecified	HT-ATE	ATE	1.478	+0.478	0.147	0.501
Misspecified	HJ-ATE	ATE	0.861	−0.139	0.087	0.164

PS	Estimator	Estimand	Mean	Bias	SD	RMSE
Misspecified	HT-ATT	ATT	1.780	+0.628	0.162	0.649
Misspecified	HJ-ATT	ATT	1.306	+0.154	0.081	0.174
Misspecified	NNM	ATT	1.172	+0.020	0.138	0.139

Lesson 6: PS misspecification makes all IPW estimators inconsistent; Hájek cannot fix this. HT-ATE bias is +0.478 and HJ-ATE bias is −0.139 — *opposite signs.* Hájek removes the *level error* (weight sums drifting from 1) but not the *shape error* (misweighting of the population). With Monte Carlo means $w_T \approx 1.045$ and $w_C \approx 0.972$, the extra HT bias is approximately $(w_T - 1)\mathbb{E}[Y(1)] - (w_C - 1)\mathbb{E}[Y(0)] \approx 0.045 \times 9 + 0.028 \times 8 \approx +0.62$, accounting for the entire gap $+0.478 - (-0.139) = +0.617$.

Lesson 7: In this simulation, matching is less sensitive to PS misspecification. NNM bias barely changes: +0.026 under the true PS versus +0.020 under misspecification. NNM does not require the PS model to be correct — it only requires that the estimated score roughly *orders* units so matched pairs are approximately balanced on true confounders. The linear logistic model, though wrong, still captures the dominant linear effects. See Yang et al. (2016) and Yang and Zhang (2023) for theoretical analyses.

Lesson 8: The relative advantage of matching over IPW reverses with model misspecification.

	Part 1: Correct PS model		Part 2: Misspecified PS model	
Estimator	Bias	RMSE	Bias	RMSE
HJ-ATT	+0.002	0.101	+0.154	0.174
NNM	+0.006	0.121	+0.020	0.139

IPW Needs a Correct PS Model; Matching Needs Covariate Balance

IPW is a *model-dependent* estimator: its consistency relies on the PS model being correctly specified. Matching is a *design-dependent* estimator: its consistency relies on matched pairs being approximately balanced on the true confounders. This condition can hold even when the PS model is wrong, provided the estimated score still separates units well enough. Neither dominates unconditionally. The recommendation is to assess PS model plausibility via balance diagnostics, or to use both as a sensitivity check (Yang et al. 2016; Yang and Zhang 2023).

6.8 The Limits of Propensity Score Methods

6.8.1 The Untestable Assumption

Every result in this chapter rests on unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$: all confounders of the T–Y relationship are observed and included in X. In graphical terms: X blocks every back-door path from T to Y.

Design-Based vs. Model-Based Identification

Randomization *guarantees* ignorability: by construction, no back-door paths exist. Propensity scores *assume* ignorability: the analyst hopes all confounders have been measured and included in X, but this can never be verified from data alone. This distinction — between identification secured by the study design and identification secured by a modeling assumption — is one of the most important dividing lines in causal inference.

Two complementary responses. *Sensitivity analysis* (Chapter 9) keeps the back-door framework but quantifies how strong an unmeasured confounder would have to be to overturn the conclusion. The *instrumental variable* approach (Chapter 7) abandons the unconfoundedness assumption entirely by exploiting an external source of variation in treatment independent of the unobserved confounders.

6.8.2 The Road to Instrumental Variables

Strategy	Key assumption	Identification mechanism
Back-door adjustment (PS)	All confounders are observed	Condition on X to block $T \leftarrow U \rightarrow Y$
Instrumental variables	Some confounders may be unobserved	Exploit exogenous variation $Z \rightarrow T$

Under IV assumptions, the ratio of the Z-induced change in Y to the Z-induced change in T — the Wald estimator — identifies a causal parameter, typically the LATE among compliers. This is derived formally in Chapter 7.

6.9 Chapter Summary

Symbol	Meaning
$\pi(X)$	Propensity score $P(T{=}1 \mid X)$
$b(X)$	Generic balancing score; $\pi(X)$ is the coarsest
$\hat{\tau}_{\mathrm{HT}}$	Horvitz–Thompson IPW estimator Equation 6.2
$\hat{\tau}_{\mathrm{HJ}}$	Hájek (self-normalized) IPW estimator Equation 6.3
NNM	Nearest-neighbor matching estimator

1. **The propensity score reduces dimension.** Under unconfoundedness, $\pi(X)$ is a balancing score ($T \perp\!\!\!\perp X \mid \pi(X)$). By **?@thm-ps**, $(Y(0), Y(1)) \perp\!\!\!\perp T \mid \pi(X)$, so identification requires adjustment for the scalar $\pi(X)$ alone.
2. **IPW identification.** The ATE is identified as $\mathbb{E}[TY/\pi(X)] - \mathbb{E}[(1-T)Y/(1-\pi(X))]$ under strong ignorability. The HT estimator is consistent but sensitive to extreme weights; the Hájek variant is recommended in practice.
3. **Matching targets the ATT.** Nearest-neighbor matching finds control units with similar propensity scores to each treated unit. It provides transparent covariate balance diagnostics but differs from IPW in estimand and methodology.
4. **Positivity is necessary; strong overlap is needed for stable estimation.** Positivity ($0 < \pi(X) < 1$ a.s.) is necessary for ATE identification. Strong overlap ($c \leq \pi(X) \leq 1-c$) ensures stable $\sqrt{n}$-inference. Trimming or re-targeting addresses near-violations.
5. **The fundamental limitation.** Unconfoundedness is untestable. When unobserved confounders are present, either sensitivity analysis or an instrumental variable is needed.

Design	Key assumption	Identified estimand
Randomized experiment	$(Y(0), Y(1)) \perp\!\!\!\perp T$ (by design)	ATE
Propensity-score adjustment	$(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$	ATE or ATT
Instrumental variables	Relevance, exogeneity, exclusion	LATE (compliers)

6.10 Problems

1. Propensity score and balancing. Suppose $X = (X_1, X_2)$ with $\operatorname{logit}(\pi(X)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$.

(a) State and prove the Balancing Property $T \perp\!\!\!\perp X \mid \pi(X)$.
(b) Two units i and j have $X_i = (1, 2.3)$ and $X_j = (0, 3.5)$ but $\hat{\pi}(X_i) = \hat{\pi}(X_j) = 0.4$. If i is treated and j is control, explain why comparing Y_i and Y_j is a valid approximation to the counterfactual comparison, and what assumption is required.
(c) Explain why matching on $\hat{\pi}(X)$ is *not* the same as matching on X directly. Under what conditions do they give the same answer?

2. ATE, ATT, and propensity score weighting. A dataset has $n = 1000$ observations with $n_1 = 400$ treated units.

(a) Write the Horvitz–Thompson IPW estimator of the ATE as a weighted sum of observed outcomes Y_i. What weights do treated units receive? What weights do control units receive?
(b) Derive an analogous IPW estimator for the ATT. (*Hint*: the ATT averages over the treated distribution; the weight for control units should reflect the treatment odds.)
(c) Show that when $\pi(X) = p$ for all units, the IPW estimator of the ATE reduces to the difference-in-means estimator. Under what experimental design does $\pi(X) = 0.5$ exactly?

3. Overlap. Let $\pi(X)$ be the true propensity score and suppose $\pi(x_0) = 1$ for some x_0.

(a) For each of τ_{ATE} and τ_{ATT}, identify the formula that fails to be identified when $\pi(x_0) = 1$, and explain why.
(b) Suppose overlap fails only on a set $\mathcal{S}$ with $P(X \in \mathcal{S}) = 0.15$. Define the trimmed ATE estimand. How does it differ from the ATE?
(c) A researcher applies the Crump et al. (2009) rule with $\eta = 0.1$, removing 8% of the sample. List two reasons the trimmed estimator may have lower variance, and state the cost in terms of external validity.

4. Hidden confounding (preview of Chapter 9). Suppose you estimate $\hat{\tau}_{\text{ATE}} = 3.2$ using propensity-score methods, and a referee questions whether unconfoundedness holds.

(a) Define what it means for U to be a "hidden confounder" in the context of the DAG, and explain why its presence invalidates the IPW identification formula.
(b) Explain in words what it means for an estimated effect to be "robust to hidden confounding," and why such an assessment depends on a quantitative yardstick. (Chapter 9 develops three formal yardsticks: Rosenbaum's Γ, the E-value, and the marginal sensitivity model.)
(c) Why does the IV strategy of Chapter 7 avoid the hidden-confounding problem entirely, and what assumption replaces unconfoundedness?

5. Matching vs. IPW. You have $n = 500$ observations comparing Hájek IPW targeting the ATE (estimator A) and 1:1 nearest-neighbor matching targeting the ATT (estimator B).

(a) Explain in one sentence why estimators A and B target different estimands even though both use $\hat{\pi}(X)$.
(b) After matching, the SMD for covariate X_1 is 0.05 (vs. 0.45 before matching). What does this tell you about the success of matching, and what assumption does balance on observed covariates not verify?
(c) Under what condition on treatment effect heterogeneity are the ATE and ATT equal?

Chapter 7

Instrumental Variables

Learning Objectives

By the end of this chapter, students should be able to:

1. Diagnose the endogeneity problem: explain why an unobserved confounder makes back-door adjustment fail, and describe how an instrumental variable provides an alternative identification route.
2. State the three IV assumptions — relevance, exogeneity, and exclusion — in the DAG/do-calculus, structural-equation, and potential-outcomes languages, and explain which are testable and which require institutional justification.
3. Follow how the IV assumptions identify the Wald estimand, and interpret the reduced form and first stage as its two observable components.
4. Explain what the Wald estimand identifies when treatment effects are heterogeneous — the Local Average Treatment Effect for compliers — why that estimand depends on the choice of instrument, and when it coincides with the ATE.
5. Compare IV and back-door adjustment on the assumptions each requires, the estimand each identifies, and the ways each can fail.

7.1 Why Instrumental Variables?

Chapter 6 showed how causal effects can be identified when all confounders are observed and can be blocked by conditioning on X. When some confounders are unobserved, back-door adjustment fails. This chapter develops instrumental variables (IV) as an alternative identification strategy: rather than blocking the confounding path $T \leftarrow U \rightarrow Y$, IV exploits an external variable Z whose effect on T is free of confounding by U. This chapter asks what IV identifies and under what assumptions; Chapter 13 asks how that estimand is computed and tested in practice.

7.1.1 The Endogeneity Problem

Unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ requires that every variable affecting both treatment and outcome is observed and included in X. In many empirical settings this is implausible: in labor economics, unobserved ability or motivation affects both schooling decisions and wages; in epidemiology, unobserved health behaviors affect both treatment uptake and outcomes. Whenever an unobserved confounder U creates a back-door path $T \leftarrow U \rightarrow Y$, the adjustment formula fails:

$$\int f(y \mid t, x)\, p(x)\, dx \;\neq\; f(y \mid \mathrm{do}(T{=}t)).$$

The gap is the *endogeneity bias.* We need a different identification strategy.

7.1.2 The IV Idea

An *instrumental variable* Z is an observed variable that: (1) *moves* treatment T (relevance); (2) *does so exogenously* — Z is unrelated to the unobserved confounder U (exogeneity); (3) *affects* Y *only through* T — Z has no direct path to Y (exclusion). Under these three conditions, the variation in T induced by Z is free of confounding by U, so the ratio of the Z-induced variation in Y to the Z-induced variation in T becomes a meaningful target for identification.

> Example: Returns to Schooling
>
> A researcher wants to estimate the causal effect of years of schooling T on log wages Y. Unobserved ability U raises both schooling and wages, so OLS is biased upward. No set of observed covariates X fully captures ability. An instrumental variable Z that shifts schooling for reasons unrelated to ability — such as proximity to a school or a policy change in compulsory attendance laws — provides a way to isolate the causal effect of schooling on wages.

IV does not eliminate the confounding path $T \leftarrow U \rightarrow Y$, nor does it make treatment as-if randomly assigned for the full population. Instead, IV *avoids* confounding rather than controlling for it — a distinction that matters both for interpreting what is identified (the LATE for compliers, not the ATE) and for understanding which assumption does the heaviest lifting (exclusion, not unconfoundedness).

7.2 Graphical Setup and Core Assumptions

7.2.1 The IV DAG

The causal structure for a basic IV model with covariates X is:

```
\usetikzlibrary{arrows.meta, positioning}
\definecolor{accent}{RGB}{46,117,182}
\definecolor{defbg}{RGB}{238,244,251}
\definecolor{backdoorred}{RGB}{192,0,0}
\definecolor{causalgreen}{RGB}{26,122,58}
\definecolor{warnbg}{RGB}{255,240,240}
\tikzset{
  w7obs/.style={circle,draw=accent,line width=1pt,minimum size=10mm,font=\small,fill=defbg},
  w7unobs/.style={circle,draw=backdoorred,line width=1pt,dashed,minimum size=10mm,font=\small,fill=wa
  w7arr/.style={-{Stealth[length=5pt]},line width=1pt,color=accent},
  w7redarr/.style={-{Stealth[length=5pt]},line width=1pt,color=backdoorred,dashed},
  w7grnarr/.style={-{Stealth[length=5pt]},line width=1pt,color=causalgreen}
}
\begin{tikzpicture}[node distance=14mm and 20mm]
  \node[w7obs] (Z) {$Z$};
  \node[w7obs,right=of Z] (T) {$T$};
  \node[w7obs,right=of T] (Y) {$Y$};
  \node[w7obs,below=12mm of T] (X) {$X$};
  \node[w7unobs,above right=10mm and 10mm of T] (U) {$U$};
  \draw[w7arr] (Z) -- node[above,font=\scriptsize\color{accent}]{(1)} (T);
  \draw[w7arr] (T) -- (Y);
  \draw[w7grnarr] (X) -- (T);
  \draw[w7grnarr] (X) -- (Y);
  \draw[w7grnarr] (X) -- (Z);
  \draw[w7redarr] (U) -- (T);
  \draw[w7redarr] (U) -- (Y);
\end{tikzpicture}
```

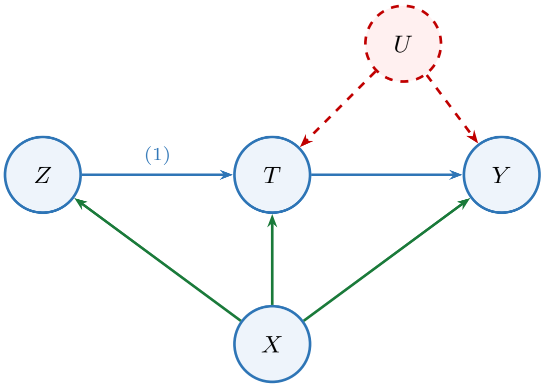

Figure 7.1: The basic IV DAG. Blue: causal effects; dashed red: confounding paths; green: covariate effects. The label (1) marks relevance; the absence of $Z \to Y$ encodes exclusion.

The three IV assumptions correspond to three distinct features of this DAG: (1) **Relevance**: there is a directed path $Z \to T$; (2) **Exogeneity**: conditional on X, the DAG implies $Z \perp\!\!\!\perp U \mid X$ by d-separation; (3) **Exclusion**: every directed path from Z to Y in $\mathcal{G}$ passes through T — there is no direct edge $Z \to Y$.

The distinction between exogeneity and exclusion is fundamental. Exogeneity says the instrument is not confounded with latent causes of the outcome. Exclusion says the instrument has no causal channel to the outcome except through treatment. A randomized encouragement may be exogenous by design, yet still violate exclusion if the encouragement changes outcomes through information, motivation, or stigma apart from the treatment itself.

7.2.2 The Three Assumptions in Three Languages

The same three assumptions can be expressed in three causal languages. These are parallel formulations, not literally identical statements: each highlights a different aspect of the design. The graphical formulation is most useful for causal design; the structural formulation is most useful for deriving moment restrictions; the potential-outcomes formulation prepares the ground for the LATE framework. The assumptions are ordered *relevance* $\to$ *exogeneity* $\to$ *exclusion*, reflecting the natural sequence in which a researcher assesses them.

Assumption	**Potential outcomes**	**Structural / econometric**	**Do-calculus / DAG**
Relevance	$P(T_i(1) \neq T_i(0)) > 0$	$\pi \neq 0$ in $T = \pi Z + \delta^\top X + \eta$	$Z \to T$ in $\mathcal{G}$ (no d-separation)
Exogeneity	$Z \perp\!\!\!\perp (Y(0), Y(1), T(0), T(1)) \mid X$	$\mathbb{E}[\varepsilon \mid Z, X] = 0$	$Z \perp\!\!\!\perp U \mid X$
Exclusion	$Y_i(t, z) = Y_i(t, z')$ for all z, z'	Z absent from structural equation for Y	$f(y \mid x, \mathrm{do}(T{=}t), z) = f(y \mid x, \mathrm{do}(T{=}t))$

Remark: Three Languages, One Identification Argument

These three formulations are aligned but not literally identical. The graphical statement encodes causal structure. The potential-outcomes statement encodes counterfactual independence. The structural formulation encodes moment orthogonality. In a well-specified IV model they support the same identification argument, but they are not interchangeable symbols.
Relevance is about the $Z \to T$ link — Z must genuinely move T. *Exogeneity* is about the absence of omitted common causes linking Z to Y. *Exclusion* is about the absence of any direct causal path $Z \to Y$ that bypasses T.

Why the do-calculus formulation is preferred. The exclusion restriction in the do-calculus column reads:

$$f(y \mid x, \mathrm{do}(T{=}t), z) = f(y \mid x, \mathrm{do}(T{=}t)).$$

This is a statement about the *interventional* density — the distribution of Y after we have *set* $T = t$ by do-surgery. The do-operator makes it impossible to confuse this with the observational statement $f(y \mid x, T{=}t, z) = f(y \mid x, T{=}t)$, which is a much weaker condition.

7.2.3 Relevance

Definition: Relevance

The instrument Z is **relevant** if it has a non-zero causal effect on the treatment T within at least some stratum of covariates X: $P(T \mid \mathrm{do}(Z{=}z), X{=}x)$ varies with z for some x. In the linear first-stage model $T = \pi Z + \delta^\top X + \eta$, this reduces to $\pi \neq 0$.

Graphically, relevance means Z and T are not d-separated in $\mathcal{G}$. The conditional covariance $\mathrm{Cov}(Z, T \mid X)$ and the first-stage F-statistic are empirical diagnostics for the observable association; the causal claim that Z shifts T is design-based.

7.2.4 Exogeneity

Definition: Exogeneity

The instrument Z is **exogenous** given covariates X if there is no open back-door path from Z to Y through unobserved causes: $Z \perp\!\!\!\perp U \mid X$ in $\mathcal{G}$ by d-separation.

In the structural linear model, the same requirement appears as $\mathbb{E}[Z\varepsilon \mid X] = 0$, because the structural residual ε is a function of U. This moment condition is a *consequence* of the graphical assumption, not a definition of exogeneity. A researcher who adopts the moment condition as a primitive has no guarantee that Z is free of back-door paths to the outcome.

Example: Quarter of Birth [@angrist1991does]

Angrist and Krueger (1991) used quarter of birth as an instrument for schooling to estimate the return to education. Exogeneity requires no open back-door path from quarter of birth to wages through unobserved variables such as ability or family background; birth timing is largely outside parental control. Bound et al. (1995) later raised concerns about weak first stages in some specifications.

Exogeneity is untestable: the unobserved confounder U is, by definition, unobserved.

7.2.5 Exclusion

Definition: Exclusion Restriction

The instrument Z satisfies the **exclusion restriction** if it affects the outcome Y *only* through its effect on the treatment T. In do-calculus terms:

$$f(y \mid x, \mathrm{do}(T{=}t), z) = f(y \mid x, \mathrm{do}(T{=}t)) \quad \text{for all } z.$$

In potential outcomes terms: $Y_i(t, z) = Y_i(t, z')$ for all $z \neq z'$.

Exclusion is a *causal* restriction, not a conditional-correlation restriction. Adding a direct arrow $Z \to Y$ to the DAG is exactly the formal counterpart of exclusion failing. In the mutilated graph $\mathcal{G}_{\overline{T}}$, the path $Z \to Y$ would remain open. Rule 1 of the do-calculus, which would allow removing Z from the density of Y given $\mathrm{do}(T{=}t)$, no longer applies.

Why Exclusion Is the Hardest Assumption

Relevance can be tested. Exogeneity can sometimes be defended by design. But exclusion — that Z has *no direct effect* on Y whatsoever — is:

- **Untestable** in just-identified models.
- **Easy to violate in practice.** An instrument that "moves treatment" often also moves other inputs. If Z is distance to a hospital (instrument for treatment uptake), it may also directly affect health outcomes through travel time in emergencies.
- **Consequential.** Even a small direct effect of Z on Y can cause substantial bias in the Wald estimand, especially when the first stage is weak (derived in Section 7.4).

Remark: Three Assumptions Are Necessary but Not Sufficient

The three IV assumptions are necessary but not sufficient for *point* identification of any causal estimand (Hernán and Robins 2006; Levis et al. 2024). A fourth *structural assumption* on the counterfactual distribution of treatment response or treatment effect heterogeneity is required. The choice of fourth assumption determines which causal quantity the Wald estimand identifies: constant treatment effects (Section 8.7), under which the Wald estimand identifies the ATE; or monotonicity (Section 7.7), under which it identifies the LATE for compliers.

Strikingly, across these different identification schemes the same conditional Wald formula $\mathrm{Cov}(Z, Y \mid X)/\mathrm{Cov}(Z, T \mid X)$ appears as the identifying expression in many cases — but its causal target changes. The three core IV assumptions are all expressible within the graphical language; the fourth assumption, by contrast, lies *outside* the graphical language in every case.

Example: Same Observables, Different ATEs

Let $Z, T \in \{0, 1\}$ with $\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] = 0.2$ and $\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0] = 0.1$, giving Wald ratio $= 0.5$. Both DGPs share the same compliance structure: $P(\mathrm{co}) = 0.2$, $P(\mathrm{at}) = 0.4$, $P(\mathrm{nt}) = 0.4$.

DGP A (constant effects): $\tau_i = 0.5$ for all units. ATE $= 0.5$. Wald ratio $= 0.5$.

DGP B (heterogeneous effects + monotonicity): $\tau_{\mathrm{co}} = 0.5$, $\tau_{\mathrm{at}} = \tau_{\mathrm{nt}} = 0$. ATE $= 0.2 \times 0.5 = 0.1$. Wald ratio $= 0.5$.

Both DGPs produce the same first stage, reduced form, and Wald ratio. Yet the ATE is 0.5 under DGP A and 0.1 under DGP B — a fivefold difference. No amount of data can distinguish them without a fourth structural assumption.

7.3 Identification in the Linear Homogeneous-Effect Model

Framework 1: Constant Treatment Effect and Linear Structure

This section works within a linear structural model in which the treatment effect is the *same* for every unit: $Y_i(t) - Y_i(t') = \beta(t - t')$ for all i, t, t'. Under this assumption, IV identifies the single parameter β, which equals the ATE, the ATT, and the LATE simultaneously: $\beta = \mathrm{ATE} = \mathrm{ATT} = \mathrm{LATE}$.

7.3.1 The Linear Structural Model

Consider the linear structural model:

$$Y = \alpha + \beta T + \gamma^\top X + \varepsilon, \tag{7.1}$$

$$T = \pi Z + \delta^\top X + \eta, \tag{7.2}$$

where ε and η are structural errors with $\mathrm{Cov}(\varepsilon, \eta) = \rho\sigma\tau \neq 0$. The non-zero covariance is the source of endogeneity: OLS applied to Equation 7.1 gives a biased estimator of β. The *reduced form* substitutes Equation 7.2 into Equation 7.1:

$$Y = \alpha + \beta\pi Z + (\beta\delta + \gamma)^\top X + (\beta\eta + \varepsilon).$$

The reduced-form coefficient on Z is $\beta\pi$: the total effect of the instrument on the outcome. Dividing by the first-stage coefficient π recovers β, provided $\pi \neq 0$.

7.3.2 The OLS Bias

The probability limit of the OLS estimator is:

$$\operatorname{plim} \hat{\beta}_{\text{OLS}} = \beta + \frac{\operatorname{Cov}(T, \varepsilon)}{\operatorname{Var}(T)} = \beta + \frac{\rho\sigma\tau}{\pi^2 \operatorname{Var}(Z) + \tau^2}.$$

The bias is zero only if $\rho = 0$ (no unobserved confounding) or if the instrument perfectly determines T. The direction of the bias depends on the sign of ρ.

7.3.3 Derivation of the Wald Estimand

The derivation has three ingredients: the first stage (how much Z moves T), the reduced form (how much Z moves Y), and the exclusion restriction (any effect of Z on Y must operate through T).

1. **Exogeneity** ($Z \perp\!\!\!\perp U \mid X$) implies $\mathbb{E}[\varepsilon \mid Z, X] = 0$.
2. **Exclusion** (no Z term in the outcome equation) combined with exogeneity yields the moment condition $\mathbb{E}[\varepsilon \cdot Z \mid X] = 0$.
3. Multiplying Equation 7.1 by $(Z - \mathbb{E}[Z \mid X])$ and applying step 2: $\operatorname{Cov}(Y, Z \mid X) = \beta \cdot \operatorname{Cov}(T, Z \mid X)$, which is the **reduced-form decomposition**: the Z–Y covariance is entirely attributable to the causal path $Z \to T \to Y$.
4. **Relevance** ($\operatorname{Cov}(Z, T \mid X) \neq 0$) ensures the denominator is non-zero:

$$\beta = \frac{\operatorname{Cov}(Y,\, Z \mid X)}{\operatorname{Cov}(T,\, Z \mid X)}. \tag{7.3}$$

In the binary instrument case ($Z \in \{0, 1\}$) without covariates, this simplifies to the **Wald estimand**:

The Wald Estimand

$$\beta = \frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]}. \tag{7.4}$$

The numerator is the *reduced form*: the total effect of Z on Y. The denominator is the *first stage*: the effect of Z on T. The exclusion restriction guarantees the entire reduced form operates through T; dividing by the first stage strips out the $Z \to T$ piece.

Remark: Estimation Deferred

The Wald estimand is an identification result: it expresses β as a ratio of observable quantities. Estimation — how to consistently estimate this ratio from a finite sample, including the correct treatment of standard errors — is taken up in Chapter 13.

Example: Returns to Schooling [@angrist1991does]

In the Mincer earnings equation $\log W = \alpha + \beta S + \gamma^\top X + \varepsilon$, where S is years of schooling and W is wages, OLS is biased upward because unobserved ability U raises both S and W. Angrist and Krueger (1991) use quarter of birth as Z: proximity to mandatory school-leaving age at different birth quarters generates exogenous variation in completed schooling. The IV estimate of the return to schooling is approximately 0.08–0.10 per year.

7.4 Why the IV Assumptions Matter

Each assumption is load-bearing, and each failure mode produces a distinct, quantifiable distortion of the Wald estimand.

When relevance fails. If $\pi = 0$, the Wald estimand is undefined. When π is small but nonzero, the instrument is *weak*. The estimator's variance diverges as $\pi \to 0$, and finite-sample bias pulls the IV estimate toward the OLS estimate at a rate proportional to $1/F$, where F is the first-stage F-statistic.

When exclusion fails. If $Y = \alpha + \beta T + \delta Z + \gamma^\top X + \varepsilon$ with $\delta \neq 0$, the Wald estimand converges to $\beta + \delta/\pi$. The bias δ/π is amplified by a weak first stage: a small direct effect combined with a weak instrument can produce large bias. This is why a weak instrument with a plausible exclusion violation is not "nearly valid" — it may be severely misleading.

When exogeneity fails. If $\mathrm{Cov}(Z, \varepsilon \mid X) \neq 0$, the Wald estimand converges to $\beta + \mathrm{Cov}(\varepsilon, Z)/\mathrm{Cov}(T, Z)$. Again the bias is amplified by weak instruments.

Assumption	Directly testable?	Basis for assessment
Relevance	Association testable; causal claim design-based	First-stage F-statistic; the causal $Z \to T$ link rests on the design
Exogeneity	No	Institutional knowledge; randomization (if available); placebo regressions on pre-determined outcomes
Exclusion	No in just-identified case; partially in overidentified case	Institutional argument; overidentification test (J-test, Chapter 13) checks mutual consistency but cannot confirm all instruments are valid (Kitagawa 2015)

7.5 Lab: OLS vs. IV Across Instrument Strengths

This lab verifies the bias formulas numerically and traces the bias–variance tradeoff across the full range of instrument strength. It studies estimator behavior *conditional* on the IV assumptions being true; it does not address whether a proposed instrument is valid in an applied study.

DGP for Lab 7

The simulation implements the linear structural model of Section 8.7 with no covariates X. Each replication draws $n = 500$ observations from:

$$Y_i = \beta T_i + \varepsilon_i, \qquad T_i = \pi Z_i + \eta_i,$$

with $Z_i \overset{\text{i.i.d.}}{\sim} N(0,1)$ and structural errors:

$$\eta_i = U_i, \qquad \varepsilon_i = \rho U_i + \sqrt{1-\rho^2}\,\xi_i, \qquad U_i, \xi_i \overset{\text{i.i.d.}}{\sim} N(0,1).$$

Fixed parameters: $\beta = 1$ (true causal effect), $\rho = 0.8$ (strong positive endogeneity). The first-stage coefficient π is varied across eight values $\pi \in \{0, 0.10, 0.15, 0.20, 0.30, 0.50, 1.00, 2.00\}$. The expected first-stage F-statistic is approximately $n\pi^2 = 500\pi^2$.

Potential Outcomes Interpretation

The potential outcome is $Y_i(t) = \beta t + \varepsilon_i$, so the individual treatment effect is constant: $Y_i(t) - Y_i(t') = \beta(t - t')$ for all i. This is Framework 1: β is simultaneously the ATE, ATT, and LATE.
The shared factor U_i creates confounding: $\mathrm{Cov}(Y_i(t), T_i) = \mathrm{Cov}(\varepsilon_i, \eta_i) = \rho \neq 0$, so unconfoundedness fails. The instrument Z_i is drawn independently of U_i, so $Z_i \perp\!\!\!\perp \varepsilon_i$ (exogeneity); Z_i does not appear in $Y_i(t)$ (exclusion); and Z_i moves T_i through π (relevance). All three IV assumptions hold exactly by construction.

Estimators. OLS regresses Y on T: $\hat{\beta}_{\mathrm{OLS}} = \mathrm{Cov}(Y, T)/\mathrm{Var}(T)$, converging to $1 + 0.8/(\pi^2 + 1)$. IV uses the Wald estimator: $\hat{\beta}_{\mathrm{IV}} = \mathrm{Cov}(Y, Z)/\mathrm{Cov}(T, Z)$, consistent for β for any $\pi \neq 0$.

Results ($n = 500$, $B = 2{,}000$ replications, seed 2024):

π	F	Theory bias	OLS mean	OLS RMSE	IV mean	IV RMSE
0.00	1	+0.800	1.801	0.802	—	—
0.10	6	+0.792	1.791	0.792	0.872	4.393
0.15	13	+0.782	1.782	0.783	0.764	6.491
0.20	21	+0.769	1.769	0.770	0.940	0.385
0.30	46	+0.734	1.735	0.735	0.975	0.165
0.50	127	+0.640	1.639	0.640	0.996	0.091
1.00	500	+0.400	1.401	0.402	0.999	0.045
2.00	2006	+0.160	1.160	0.161	1.000	0.022

Lesson 1: The OLS bias formula is exact. The theory bias $0.8/(\pi^2 + 1)$ matches the simulated OLS bias to four decimal places across all eight values of π. OLS is biased in the direction of ρ at every value of π, including $\pi = 0$.

Lesson 2: IV is consistent but has catastrophically heavy tails when the instrument is weak. At $\pi = 0.10$ ($F \approx 6$) and $\pi = 0.15$ ($F \approx 13$), the IV *mean* is far from the true value. The IV *median* is ≈ 1.01 at both values, confirming consistency — the mean is dragged off by a small fraction of replications in which the first stage is near zero. Standard deviations of 4.4 and 6.5 make these estimates useless.

Lesson 3: The RMSE crossover occurs near $F \approx 20$. IV first beats OLS on RMSE at $\pi = 0.20$ ($F \approx 21$): $0.385 < 0.770$. The Staiger–Stock rule of thumb ($F \geq 10$) is slightly too lenient: at $F \approx 13$, IV RMSE is still 6.5, nearly 8× OLS. A more conservative threshold of $F \geq$ 20–25 is needed.

Lesson 4: A strong instrument eliminates both problems. At $\pi = 1.00$ ($F \approx 500$), IV RMSE $= 0.045$ while OLS RMSE $= 0.402$ — a **9-fold** improvement from IV. OLS efficiency is illusory: its small variance is offset by a large, persistent bias.

RMSE Cannot Be the Only Criterion

OLS has lower RMSE than IV for $\pi < 0.20$ in this DGP. This does not vindicate OLS. An estimator with RMSE $= 0.78$ because it is biased by 0.80 is useless for causal inference: the bias is systematic and does not shrink with sample size. As $n \to \infty$, IV RMSE shrinks to zero while OLS RMSE stays at 0.80. The RMSE comparison is only meaningful in finite samples — it quantifies when a weak instrument is so unreliable that IV is not yet practically useful, but that is an argument for finding a stronger instrument, not for using OLS.

Remark: The First-Stage F-Statistic as a Diagnostic

The formula $F \approx n\pi^2$ gives first-stage F-statistics matching the simulation throughout. The Bound et al. (1995) threshold of $F \geq 10$ is widely used in practice and corresponds roughly to IV RMSE within a factor of two of the strong-instrument limit. This simulation suggests that threshold may understate the problem when endogeneity is strong ($\rho = 0.8$): at $F = 13$ the IV RMSE is 6.5, far above any useful threshold. A researcher should report the first-stage F alongside any IV estimate, and treat values below 20–25 with particular caution.

7.6 Multiple Instruments and Overidentification

When there are exactly as many instruments as endogenous variables ($q = p$), the model is *just-identified*. When $q > p$, the model is *overidentified*: the extra instruments impose additional moment restrictions that are testable. Under the homogeneous-effect linear model, every valid instrument must imply the same structural coefficient β, so those extra restrictions are testable. Under heterogeneous treatment effects, valid instruments can legitimately identify *different* LATEs because they shift treatment for different complier populations.

The **order condition** ($q \geq p$, counting requirement) and the **rank condition** (instruments are linearly independent in the first stage) are both necessary for identification. The Sargan–Hansen J-test (Sargan

1958; Hansen 1982) formalizes the mutual-consistency check; rejection indicates at least one instrument either violates exogeneity or exclusion, or identifies a different LATE, but does not localize which. The test statistic and asymptotic distribution are derived in Chapter 13.

What Overidentification Does Not Do

Passing the J-test does not confirm that any instrument is valid. It only shows that the sample moment conditions are mutually compatible — a weaker conclusion than validity. Multiple invalid instruments can agree with one another if they share the same violation; consistent instruments are not necessarily valid ones. Overidentification is an opportunity for a consistency check, not a substitute for the institutional argument that makes a design credible.

7.7 Heterogeneous Treatment Effects and the LATE Framework

Framework 2: Heterogeneous Treatment Effects

We now drop homogeneity and allow $\tau_i = Y_i(1) - Y_i(0)$ to vary across units. Throughout this section we work in the binary instrument, binary treatment setting ($Z, T \in \{0, 1\}$). In this setting, the Wald ratio generally no longer identifies the ATE. Instead, under an additional assumption of monotonicity, it identifies the *Local Average Treatment Effect* (LATE): the average treatment effect for the subpopulation whose treatment status is changed by the instrument.

7.7.1 Compliance Types

Definition: Compliance Types [@angrist1996identification]

For binary $Z, T \in \{0, 1\}$, the four compliance types are defined by the pair of potential treatment decisions $(T_i(0), T_i(1))$:

Compliance type	$T_i(0)$	$T_i(1)$
Complier	0	1
Always-taker	1	1
Never-taker	0	0
Defier	1	0

A **complier** takes treatment if and only if the instrument is switched on. An **always-taker** takes treatment regardless of Z. A **never-taker** never takes treatment. A **defier** does the opposite of what the instrument suggests.

The instrument Z only shifts treatment for *compliers*: always-takers and never-takers have the same treatment status regardless of Z, so they contribute nothing to the denominator $\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]$.

7.7.2 The Monotonicity Assumption

Definition: Monotonicity [@imbens1994identification]

The treatment assignment is **monotone in** Z if $T_i(1) \geq T_i(0)$ for all i. Equivalently: there are no defiers in the population.

Monotonicity is not a generic law of causal inference. It is a design-specific claim about how this particular instrument changes treatment behavior. Switching the instrument from 0 to 1 may induce some units to take treatment (compliers) and leave others unaffected (always-takers or never-takers), but it should not reverse anyone's treatment decision.

7.7.3 The LATE Theorem

Theorem: LATE Theorem [@imbens1994identification]

Suppose: (1) *Exogeneity (PO form):* $Z \perp\!\!\!\perp (Y(0), Y(1), T(0), T(1))$; (2) *Exclusion:* $Y_i(t, z) = Y_i(t)$ for all z; (3) *Relevance:* $\mathbb{E}[T(1) - T(0)] \neq 0$; (4) *Monotonicity:* $T_i(1) \geq T_i(0)$ for all i. Then the Wald estimand identifies the **Local Average Treatment Effect** (LATE):

$$\frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]} = \mathbb{E}[Y(1) - Y(0) \mid T_i(1) > T_i(0)] \equiv \tau_{\text{LATE}}.$$

Proof

Denominator. By consistency for T and exogeneity ($Z \perp\!\!\!\perp (T(0), T(1))$):

$$\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] = \mathbb{E}[T(1)] - \mathbb{E}[T(0)] = \mathbb{E}[T(1) - T(0)].$$

Monotonicity (no defiers) gives $\mathbb{E}[T(1){-}T(0)] = P(\text{complier})$, since always-takers contribute $1{-}1 = 0$ and never-takers contribute $0 - 0 = 0$: $\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] = P(\text{complier})$.
Numerator. By consistency, exclusion, and exogeneity:

$$\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0] = \mathbb{E}[Y(T(1))] - \mathbb{E}[Y(T(0))] = \sum_c P(c)\, \mathbb{E}[Y(T_c(1)) - Y(T_c(0)) \mid \text{type} = c].$$

For always-takers: $T(1) = T(0) = 1$, contribution $= 0$. For never-takers: $T(1) = T(0) = 0$, contribution $= 0$. For compliers: $T(1) = 1$, $T(0) = 0$, so $Y(T(1)) - Y(T(0)) = Y(1) - Y(0)$. No defiers by monotonicity. Therefore:

$$\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0] = P(\text{complier})\, \mathbb{E}[Y(1) - Y(0) \mid \text{complier}].$$

Ratio. Dividing numerator by denominator: $\tau_{\text{LATE}} = \mathbb{E}[Y(1) - Y(0) \mid \text{complier}]$. $\square$

Remark: The Two Senses of "Local"

The LATE is local in two senses: it is local to the complier group defined by *this* instrument, and local to the *particular instrument* that defines that group. A different instrument, even for the same treatment, will generally select a different complier population and identify a different LATE. This is developed in Section 7.8.

7.8 Interpreting IV Estimands

7.8.1 What the Two Frameworks Say

	Framework 1 (linear, homogeneous)	**Framework 2** (heterogeneous effects)
Key assumption	$\tau_i = \beta$ for all i	Monotonicity; no defiers
What IV identifies	$\beta = \text{ATE} = \text{ATT} = \text{LATE}$	$\tau_{\text{LATE}} = \mathbb{E}[\tau_i \mid \text{complier}]$
Estimand depends on instrument?	No (same β regardless of Z)	Yes (different $Z \Rightarrow$ different compliers $\Rightarrow$ different LATE)
Required for ATE?	Yes, automatically	Only if all units are compliers or effects homogeneous

Framework 1 is a special case of Framework 2: when $\tau_i = \beta$ for all i, the LATE equals the ATE equals β. In applied work, the default interpretation of the Wald estimand is the LATE; the ATE interpretation requires the additional homogeneity argument of Framework 1.

Remark: Same Formula, Different Estimand

Across a range of fourth assumptions, the conditional Wald formula $\text{Cov}(Z, Y \mid X)/\text{Cov}(Z, T \mid X)$ serves as the identifying expression in many cases (Levis et al. 2024). Two researchers using the same instrument and computing the same Wald ratio may be consistently estimating different causal quantities if they maintain different structural assumptions. The data alone cannot resolve which estimand the Wald ratio identifies; that determination requires the researcher to commit to a structural assumption about treatment response.

7.8.2 When Does LATE Equal ATE?

LATE equals ATE only under additional structure, most notably treatment-effect homogeneity. To see this, decompose the ATE by compliance type:

$$\text{ATE} = P(\text{co})\,\mathbb{E}[\tau_i \mid \text{co}] + P(\text{at})\,\mathbb{E}[\tau_i \mid \text{at}] + P(\text{nt})\,\mathbb{E}[\tau_i \mid \text{nt}].$$

The LATE equals only the first term divided by its probability weight. ATE and LATE coincide if and only if: (1) mean treatment effects are equal across compliance types, or (2) everyone is a complier, or (3) the average effects for always-takers and never-takers happen to equal the LATE — an untestable coincidence.

7.8.3 Different Instruments, Different Estimands

Because the LATE is specific to the complier population, and different instruments select different complier populations, two valid instruments for the same treatment can legitimately identify different LATEs. This is informative about treatment effect heterogeneity, not a contradiction.

Example: Compulsory Schooling Laws vs. Distance to College [@angrist1991does; @card1995using]

Compulsory schooling laws (Angrist and Krueger 1991) identify the return to schooling for students at the margin of dropping out — typically lower-income students. Distance to the nearest college (Card 1995) identifies the return for students deterred by geographic distance — again, disproportionately lower-income. Both instruments are valid; the LATEs can legitimately differ even if both are valid.

7.8.4 The Policy Relevance of LATE

For many policy questions, the LATE is exactly the right estimand. If a policy is designed to encourage a subset of the population to take treatment, then the effect on compliers (those who respond to the encouragement) is precisely what the policy-maker wants to know. When the ATE over the full population is required, IV alone is insufficient under heterogeneous effects; additional assumptions or a second instrument are needed to extrapolate from the LATE to the ATE.

7.9 Practical Guidance on Defending an IV Design

A researcher proposing an instrument should be able to answer five questions explicitly:

1. **What exactly is the instrument?** Specify Z precisely: its source of variation, the level at which it varies, and the population to which it applies.
2. **Why does it shift treatment?** Articulate the causal mechanism by which Z moves T. The first-stage F-statistic is a diagnostic for instrument strength, not a substitute for a causal account of the $Z \to T$ link.
3. **Why is it as-if random relative to latent outcome determinants?** The most credible sources are designed randomization (lotteries, randomized encouragement), natural experiments, and shift-share designs (Bartik 1991; Goldsmith-Pinkham et al. 2020). Placebo regressions on pre-determined outcomes provide partial evidence.
4. **Why can it affect the outcome only through treatment?** Exclusion is untestable in just-identified models. A useful diagnostic: how large would the direct effect δ have to be, relative to π, to overturn the estimated causal effect? When the first stage is weak, the answer is: not very large.

5. **What population margin does it shift?** Identify the complier population. This determines the LATE that is being identified and governs the external validity of the estimates.

7.10 Applied Example: Charter School Lotteries and the KIPP Lynn Study

This example illustrates a canonical randomized-encouragement IV design. The lottery randomizes *offer status* Z, not actual treatment S (years of KIPP attendance). Winning the lottery does not force a student to attend KIPP, and losing does not make later attendance impossible. Thus the lottery offer is the instrument and actual attendance is the endogenous treatment.

Setting. KIPP (Knowledge Is Power Program) schools follow a "No Excuses" model: extended school days, longer academic year, selective teacher hiring, and strict behavioral norms. KIPP Academy Lynn was substantially oversubscribed beginning in 2005. Massachusetts law requires oversubscribed charter schools to select students by lottery, so the school conducted randomized admissions lotteries from 2005 through 2008. The outcome Y_{igt} is the student's standardized score on the Massachusetts MCAS, normalized to mean zero and standard deviation one within each subject–grade–year cell statewide.

Mapping the three assumptions (Angrist et al. 2012).

Relevance. Lottery winners were offered a seat and about 80% accepted; losers rarely enrolled elsewhere at KIPP. The first-stage regression yields a coefficient of approximately 1.2: lottery winners had spent about 1.2 more years at KIPP than lottery losers at the time of each MCAS exam. The first-stage F-statistic is far above conventional thresholds.

Exogeneity. Offer status was determined by randomly drawn lottery-sequence numbers, so independence holds by design. A joint test of covariate balance yields $p = 0.615$, consistent with no pre-lottery differences.

Exclusion. The offer merely provides access to a school; it does not itself deliver instruction. One potential violation is a discouragement effect: losing the lottery might demoralize students. The authors address this by noting that scores of lottery losers are typical of demographically comparable students in Lynn, inconsistent with large discouragement effects. Random assignment of the instrument does not, by itself, imply exclusion — it only guarantees exogeneity.

First stage, reduced form, and 2SLS. The model is just-identified (one excluded instrument per endogenous variable), so the 2SLS estimator of θ (effect per year at KIPP) equals the ratio of the reduced-form coefficient on Z_i to the first-stage coefficient π.

Subject	**First stage**	**Reduced form**	**2SLS**
Math	1.221 (0.068)	0.430 (0.067)	0.352 (0.053)
ELA	1.228 (0.068)	0.164 (0.073)	0.133 (0.059)

Standard errors clustered at the student level. $N = 833$ *student-by-test observations.*

Each year at KIPP raises math scores by approximately 0.35σ and ELA scores by approximately 0.13σ. The reduced-form estimate for math (0.43σ) is larger than the 2SLS estimate (0.35σ) because the first stage exceeds 1: lottery winners accumulated somewhat more than one additional year at KIPP per unit of follow-up time.

LATE interpretation. The 2SLS estimand θ is not the effect of KIPP on all students in Lynn, nor on all applicants — it is the average per-year treatment effect for the *lottery compliers*. Compliance is partial in both directions: some lottery winners do not enroll and some lottery losers eventually find entry.

Treatment effect heterogeneity. Reading gains ($\approx 0.13\sigma$ overall) are driven almost entirely by students classified as having limited English proficiency (LEP, $\approx 0.43\sigma$) and special education needs (SPED, $\approx 0.27\sigma$); non-LEP, non-SPED students show negligible ELA gains. Math effects are large and positive across all subgroups but are largest for LEP and lower-achieving students. This connects directly to Section 7.8: different instruments would identify distinct subpopulation LATEs; the overall 2SLS estimate is a weighted average of subgroup LATEs with weights proportional to each subgroup's share of the complier population.

Remark: Lottery Design and Assumption Credibility [@angrist2009mostly]

Instruments derived from *designed* randomization — lotteries, random assignment, randomized encouragement — provide the most transparent basis for the exogeneity assumption. Exogeneity within the applicant sample, conditional on application cohort, is not merely plausible — it follows from the randomization protocol. This is why lottery-based IV designs occupy a privileged position in the program evaluation literature. The exclusion restriction still requires institutional argument.

7.11 IV versus Back-Door Adjustment

Dimension	**Back-door / propensity score**	**Instrumental variables**
Core assumption	All confounders observed: $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$	Valid instrument: relevance, exogeneity, exclusion
Unobserved confounders	Fatal: back-door adjustment fails	Permitted: IV routes around U
Estimand	ATE, ATT, or ATC; all coincide under homogeneity	LATE (compliers only); reduces to common β under homogeneous effects
Testability	Unconfoundedness is untestable; overlap is testable	Relevance testable; exogeneity and exclusion untestable (just-identified)
Main threat	Unmeasured confounder	Exclusion restriction violation
Identifies ATE?	Yes, under strong ignorability	Only under homogeneous effects

Complementary failure modes. Back-door adjustment fails when X does not capture all confounders. IV fails when the exclusion restriction is violated: the bias in the Wald estimand is δ/π, amplified by weak instruments. The two failures are orthogonal: back-door adjustment requires many observed covariates but tolerates no unobserved ones, while IV tolerates unobserved confounders but requires an instrument with no direct effect on the outcome.

When both strategies are available. The Hausman (1978) endogeneity test compares OLS and IV: under the null that T is exogenous given X, both are consistent, and a large discrepancy is evidence of endogeneity. Under the null, an appropriately scaled quadratic form in $\hat{\beta}_{\text{IV}} - \hat{\beta}_{\text{OLS}}$ is asymptotically χ^2_p. Disagreement does not by itself tell us which method is wrong: the two strategies typically target different estimands (ATE vs. LATE), and disagreement can arise from legitimate effect heterogeneity rather than failure of either assumption.

7.12 Chapter Summary

Symbol	**Meaning**
Z	Instrument
π	First-stage coefficient: $\mathbb{E}[\partial T/\partial Z]$
ρ	Endogeneity: $\text{Cov}(\varepsilon, \eta)/(\sigma\tau)$
τ_{LATE}	$\mathbb{E}[Y(1) - Y(0) \mid T_i(1) > T_i(0)]$
Wald estimand	$(\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0])/(\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0])$
Reduced form	Total effect of Z on Y
First stage	Effect of Z on T

1. **IV identifies effects from exogenous treatment variation.** When back-door adjustment fails because an unobserved U creates a path $T \leftarrow U \rightarrow Y$, a valid instrument Z identifies the causal effect by exploiting only the component of treatment variation that Z induces. IV does not block the confounding path — it avoids it.

2. **Three assumptions, ordered by testability.** Relevance can be assessed with the first-stage F-statistic (though the F-statistic is a sample diagnostic). Exogeneity and exclusion must be defended by institutional knowledge, design logic, and causal structure. Each violated assumption produces a distinct, quantifiable bias, amplified by weak instruments.
3. **Framework 1: homogeneous-effect SEM ⇒ Wald identifies β.** Under constant treatment effects and the linear structural model, the Wald estimand identifies $\beta = \text{ATE} = \text{ATT} = \text{LATE}$.
4. **Framework 2: heterogeneity + monotonicity ⇒ Wald identifies LATE.** Under heterogeneous treatment effects and monotonicity, the Wald estimand identifies the average treatment effect for *compliers* only — those whose treatment status changes with the instrument, a latent subgroup defined by $(T(0), T(1))$.
5. **Different instruments identify different effects.** The LATE depends on the instrument through the complier population it selects. This is informative about treatment effect heterogeneity, not a contradiction. LATE equals ATE only under additional structure.
6. **IV versus back-door adjustment.** Complementary failure modes: back-door fails when confounders are unobserved; IV fails when the exclusion restriction is violated or the instrument is not truly exogenous.
7. **Estimation deferred to Chapter 13.** This chapter establishes what IV identifies and under what assumptions. How the Wald ratio is estimated from finite data — reduced-form regression, two-stage least squares, asymptotic inference, and overidentification tests — is the subject of Chapter 13.

7.13 Problems

1. The three IV assumptions in three languages. Consider the DAG: $\{Z \to T,\ T \to Y,\ U \to T,\ U \to Y,\ X \to T,\ X \to Y,\ X \to Z\}$ with U unobserved.

(a) List all back-door paths from T to Y. Does X alone satisfy the back-door criterion? Explain.
(b) Verify the three IV assumptions using d-separation: (i) Relevance: show Z and T are not d-separated in $\mathcal{G}$. (ii) Exogeneity: show $Z \perp\!\!\!\perp U \mid X$ in $\mathcal{G}$. (iii) Exclusion: show $Y \perp\!\!\!\perp Z \mid T, X$ in $\mathcal{G}_{\overline{T}}$.
(c) Now add the arrow $Z \to Y$ to the DAG. Which IV assumption is violated? Show explicitly which step of the Wald derivation in Section 8.7 breaks down.
(d) Translate each of the three IV assumptions into the structural language: write the equations for T and Y and identify which coefficient restriction corresponds to each assumption.

2. Bias under assumption violations. Let $Y = \beta T + \varepsilon$ and $T = \pi Z + \eta$ with $\mathbb{E}[\varepsilon \mid Z] = 0$ and $\pi \neq 0$.

(a) Starting from $\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]$, substitute the structural equation for Y and simplify. What role does exogeneity play?
(b) Show that $\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] = \pi$ in the linear first-stage model.
(c) Derive the Wald estimand and confirm it equals β.
(d) Now suppose the exclusion restriction fails and $Y = \beta T + \delta Z + \varepsilon$ with $\delta \neq 0$. Derive the probability limit of the Wald estimator and confirm the bias formula from Section 7.4.
(e) Suppose instead that exogeneity fails: $\mathbb{E}[\varepsilon \mid Z] = cZ$ for some constant $c \neq 0$. Derive the probability limit of the Wald estimator and express the bias in terms of c and π. Compare the structure of this bias with the exclusion violation bias.

3. Order, rank, and the limits of overidentification. Consider a model with one endogenous variable T and two instruments Z_1 and Z_2, both satisfying exogeneity and exclusion.

(a) State the order condition and verify it is satisfied.
(b) State the rank condition. What would it mean geometrically if the rank condition failed — i.e., if Z_1 and Z_2 were perfectly collinear in the first-stage regression?
(c) Explain intuitively why having two valid instruments rather than one should improve estimation precision.
(d) Now suppose Z_1 is valid but Z_2 violates the exclusion restriction. Under what conditions does the Sargan–Hansen J-test have power to detect Z_2's invalidity? Under what conditions does the test fail?
(e) Why does passing the J-test not confirm that both Z_1 and Z_2 are valid? Give a concrete example in which both instruments are invalid and the J-test has no power.

4. Compliance types and the LATE. In a binary instrument, binary treatment study, suppose the population has: 30% compliers with average treatment effect $\tau_c = 6$; 25% always-takers with $\tau_a = 3$; 45%

never-takers with $\tau_n = 1$; no defiers.

(a) Compute $P(\text{complier}) = \mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]$.
(b) Compute the ATE as a weighted average of τ_c, τ_a, τ_n with appropriate weights.
(c) The Wald estimand equals $\tau_c = 6$. By how much does this overstate the ATE, and why?
(d) A second study uses a different binary instrument Z' with a complier population of 50% and a LATE of 2. Is this contradictory? What can you infer about the relative treatment effect in the two complier populations?
(e) Explain, using compliance type language, why the denominator of the Wald estimand equals $P(\text{complier})$.

5. The exclusion restriction: plausibility and violations. Evaluate the exclusion restriction for each proposed instrument. For each, state (i) whether the restriction is plausible and why; (ii) a specific mechanism by which it could be violated; and (iii) whether the violation would bias the IV estimate upward or downward.

(a) **Instrument**: rainfall in the home region of a politician, used as an instrument for government infrastructure spending. **Outcome**: local economic growth.
(b) **Instrument**: distance to the nearest hospital, used as an instrument for hospital admission. **Outcome**: 30-day mortality.
(c) **Instrument**: a randomly assigned financial incentive to enroll in a health screening program. **Outcome**: health status two years later.
(d) **Instrument**: lottery number in the Vietnam-era draft lottery, used as an instrument for military service. **Outcome**: lifetime earnings. [*This is the Angrist (1990) study; discuss why this instrument is widely regarded as satisfying the exclusion restriction.*]

6. IV versus back-door adjustment. A researcher studies the effect of job training (T) on earnings (Y). Two strategies are available: (A) a rich set of pre-treatment covariates X and a propensity-score estimator; (B) a lottery that randomly selected units to be *offered* training (not required to attend), used as instrument Z.

(a) Under what assumption does strategy (A) identify the ATE? What specific unobserved variable would most plausibly violate this assumption?
(b) Strategy (B) identifies a LATE. Describe the complier population in words. Is the LATE likely to be larger or smaller than the ATE in this setting? Explain.
(c) Both strategies are implemented and yield estimates of \$1,800 and \$2,400 per year, respectively. Describe a Hausman-type test that uses both estimates. Under what null hypothesis does the test have an approximate χ^2 distribution?
(d) If the two estimates differ significantly, which strategy would you trust more and why? What additional evidence would help distinguish the two explanations (endogeneity bias in (A) versus LATE $\neq$ ATE in (B))?

Chapter 8

Mediation and Front-Door Identification

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain the distinction between the total causal effect of T on Y and a pathway-specific effect that operates through a mediator M, and describe why this distinction matters scientifically.
2. Draw the prototype mediation DAG, write its structural equations, and identify the direct and indirect pathways.
3. Define the controlled direct effect (CDE) using the do-operator, identify it via the back-door formula for the joint intervention (T, M), and explain why it depends on the fixed level m.
4. Define the natural direct and indirect effects (NDE, NIE) using the potential outcomes notation $Y(t, M(t'))$, state the NDE+NIE=TE decomposition, and explain why these quantities involve cross-world counterfactuals.
5. State the four sequential ignorability assumptions for identification of natural effects, write the mediation formula, and identify which assumption is violated when there is an unmeasured treatment-induced mediator–outcome confounder.
6. Set up the Baron–Kenny three-equation system, derive the product and difference formulas for the indirect effect, and explain why the decomposition fails in nonlinear or interaction models.
7. State the three front-door conditions, derive the front-door formula using the do-calculus, and explain why the front-door graph enables identification despite unobserved T–Y confounding.
8. Contrast mediation analysis and instrumental variables on the dimensions of variable position, identification goal, and key assumption.

8.1 Motivation: Mechanisms

The identification results of Chapters 5–7 all answer the same question: *what is the total causal effect of T on Y?* Mediation analysis asks a finer question: *through what mechanism does that effect operate?*

More concretely, the total effect of T on Y may flow along multiple causal pathways. Some of this effect passes through an intermediate variable M — the *mediator* — along the path $T \to M \to Y$. The remainder flows directly along $T \to Y$, bypassing the mediator entirely. Decomposing the total effect into these two components is the goal of mediation analysis.

This decomposition matters for scientific and policy reasons. In a clinical trial of a behavioral intervention (T) on depression (Y), a researcher may want to know how much of the benefit operates through improved sleep quality (M) versus other pathways — because if sleep is the main channel, targeting sleep directly may be a more efficient intervention. In an economics study of education (T) on wages (Y), how much operates through occupation (M) versus cognitive skills?

Running example. Throughout this chapter we use the depression intervention as the primary example: T is a randomized behavioral intervention, M is self-reported sleep quality measured mid-trial, and Y is a depression score (e.g., PHQ-9 scale).

The challenge is that mediators are *post-treatment variables*: they are affected by the treatment, and may themselves be confounded with the outcome. Conditioning on a post-treatment variable creates exactly the collider and selection-bias problems studied in Chapters 2 and 3. A naive approach — simply including M as a covariate in a regression of Y on T — conflates adjustment with mediation and can introduce bias even in a randomized experiment.

This chapter also develops the *front-door criterion*, a distinct identification strategy that uses the mediation structure of the DAG to identify causal effects even when treatment and outcome are confounded by an unobserved variable.

8.2 The Mediation DAG

8.2.1 The Prototype Graph

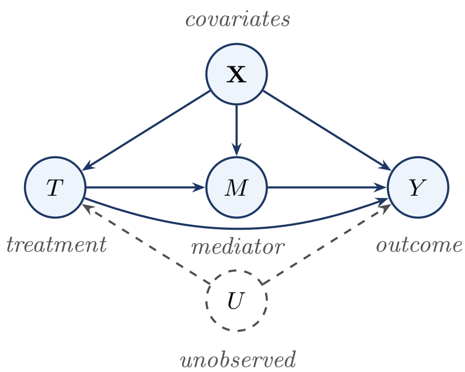

Figure 8.1: The prototype mediation DAG. The causal effect of T on Y operates through two pathways: the direct path $T \to Y$ and the indirect path $T \to M \to Y$ via the mediator. The unobserved variable U confounds the treatment–outcome relationship; $\mathbf{X}$ denotes observed pre-treatment covariates.

The graph encodes two causal pathways: the **direct pathway** $T \to Y$ (treatment affects the outcome without passing through the mediator) and the **indirect pathway** $T \to M \to Y$ (treatment first shifts the mediator, which shifts the outcome).

The prototype graph corresponds to the nonparametric SEM:

$$T = f_T(\mathbf{X}, U, \varepsilon_T), \qquad M = f_M(T, \mathbf{X}, \varepsilon_M), \qquad Y = f_Y(T, M, \mathbf{X}, U, \varepsilon_Y).$$

8.2.2 What Makes Mediation Harder Than Total Effect Estimation

The key difficulty is that M is a *post-treatment* variable. This creates two interrelated problems.

Collider bias. Conditioning on M can open collider paths. If $V \to M$ and $V \to Y$ with V unobserved, the path $T \to M \leftarrow V \to Y$ is blocked when M is not conditioned on, but opens as soon as M is included as a covariate. This is precisely why naively regressing Y on (T, M) does not isolate the direct effect.

Mediator–outcome confounding. Even when treatment is randomized, the mediator M is never randomized. An unobserved V with $V \to M$ and $V \to Y$ creates a back-door path from M to Y that randomization of T does not close.

Post-Treatment Variables Are Dangerous to Condition On

The mediator M is caused by the treatment T. This single fact makes every operation involving M structurally different from operations involving pre-treatment covariates $\mathbf{X}$.
Conditioning on a post-treatment variable in a regression or matching procedure is *not* a neutral act. It can open collider paths that were previously blocked, introduce selection bias, and produce estimates of neither the total effect nor any well-defined direct effect. This chapter studies three regimes in which conditioning on or marginalizing over M is justified: (1) **CDE**: both T and M

are intervened on simultaneously via $\mathrm{do}(T, M)$; (2) **NDE/NIE**: M is marginalized over using a distribution from a different treatment arm, justified by sequential ignorability; (3) **Front-door**: M is summed over in the front-door formula, justified by graphical conditions. Outside these three regimes, conditioning on a post-treatment variable should be treated as an error until proven otherwise.

8.2.3 Working Assumption for §§ 3–6

Throughout §§ Section 8.3–Section 8.7, we work with the *reduced prototype graph* (obtained from Figure 8.1 by assuming all T–Y and T–M confounding is captured by the observed $\mathbf{X}$). The unobserved confounder U is restored in Section 8.8.

8.3 Total Causal Effect

Definition: Total Effect

The **total effect** (TE) of T on Y is:

$$\mathrm{TE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}1)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0)].$$

The total effect captures the combined impact of all causal pathways — both $T \to Y$ and $T \to M \to Y$. It is identified by the back-door formula whenever $\mathbf{X}$ satisfies the back-door criterion for the effect of T on Y.

Running example. For the depression trial, the TE is the expected change in depression score if the entire study population were assigned to the intervention versus control. It combines the effect operating through sleep improvement with every other pathway. Mediation analysis asks: of this total, how much is due to sleep?

8.4 Controlled Direct Effect

8.4.1 Definition

The do-calculus provides the cleanest definition of the direct effect: intervene on both T *and* M simultaneously, fixing M at a specified level m. Any remaining effect of T on Y must then flow exclusively through the direct edge $T \to Y$.

Definition: Controlled Direct Effect (CDE)

The **controlled direct effect** of changing T from 0 to 1 while fixing $M = m$ by intervention is:

$$\mathrm{CDE}(m) \;=\; \mathbb{E}[Y \mid \mathrm{do}(T{=}1), \mathrm{do}(M{=}m)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0), \mathrm{do}(M{=}m)]\,. \tag{8.1}$$

This corresponds to the mutilated graph $\mathcal{G}_{\overline{TM}}$ in which all edges into both T and M are deleted. Fixing $M = m$ for everyone shuts down the indirect pathway $T \to M \to Y$.

Remark

The CDE depends on the level m at which M is fixed. In a linear additive model, $\mathrm{CDE}(m) = \tau'$ for all m, a special property of linearity. When T and M interact in their effect on Y, the CDEs at different values of m differ, and no single number summarizes the direct effect.

8.4.2 Identification of the CDE

Theorem: Identification of the CDE

Suppose $\mathbf{Z}$ satisfies the back-door criterion for the joint intervention $\mathrm{do}(T, M)$ on Y: (i) $\mathbf{Z}$ contains no descendant of T or M, and (ii) $\mathbf{Z}$ blocks every back-door path from $\{T, M\}$ to Y. Suppose further (iii) joint positivity: $P(T{=}t,\ M{=}m \mid \mathbf{Z}{=}\mathbf{z}) > 0$ for P-almost every $\mathbf{z}$. Then:

$$\mathbb{E}[Y \mid \mathrm{do}(T{=}t),\ \mathrm{do}(M{=}m)] = \sum_{\mathbf{z}} \mathbb{E}[Y \mid t,\ m,\ \mathbf{z}]\, P(\mathbf{z}). \tag{8.2}$$

Under the working assumption, $\mathbf{Z} = \mathbf{X}$ blocks every back-door path from $\{T, M\}$ to Y, so:

$$\mathbb{E}[Y \mid \mathrm{do}(T{=}t),\ \mathrm{do}(M{=}m)] = \sum_{\mathbf{x}} \mathbb{E}[Y \mid t,\ m,\ \mathbf{x}]\, P(\mathbf{x}).$$

8.4.3 The CDE Does Not Have a Complementary Indirect Effect

A common misconception is that the CDE and some "controlled indirect effect" sum to the total effect. This is not generally true. The residual quantity $\mathrm{TE} - \mathrm{CDE}(m)$ depends on m, has no clean do-calculus expression corresponding to "the indirect pathway," and cannot be interpreted as the effect that flows through M. The correct complementary decomposition uses the natural effects of Section 8.5.

Natural Effects Are a Strictly Harder Estimand Than the CDE

The CDE is a single-world estimand. The quantity $\mathbb{E}[Y \mid \mathrm{do}(T{=}t), \mathrm{do}(M{=}m)]$ involves a single joint intervention. Identification reduces to the standard back-door criterion for the pair (T, M).
The NDE and NIE are cross-world estimands. The nested counterfactual $Y(t, M(t'))$ requires an individual to simultaneously inhabit two worlds: the world where $T = t$ (which determines Y) and the world where $T = t'$ (which determines what M would *naturally* be). When $t \neq t'$, these are logically distinct states of the world that no single experiment can realize. The do-operator alone cannot express this quantity.
The identification price is steep. The CDE requires only the back-door criterion. The NDE and NIE require four sequential ignorability assumptions, including Assumption 4 (no treatment-induced mediator–outcome confounder), which cannot be satisfied by randomizing T and cannot be verified from data.

8.5 Natural Direct and Indirect Effects

8.5.1 Motivation: Cross-World Counterfactuals

The CDE fixes the mediator by external intervention. A more scientifically natural question is: what is the effect of T on Y that bypasses M when M is held at the value it *would naturally take* under the reference treatment $T = 0$?

This requires the *nested potential outcomes* notation $Y(t, M(t'))$, which denotes the outcome that would be observed if T were set to t and M were simultaneously set to the value it would naturally take if T were t'. These are called *cross-world counterfactuals* because t and t' may differ, referring to two different hypothetical states of the world simultaneously.

8.5.2 Definitions

Definition: Natural Direct and Indirect Effects

(Pearl 2001) For binary $T \in \{0, 1\}$:

$$\mathrm{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))]\,, \tag{8.3}$$

$$\mathrm{NIE} = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))]\,. \tag{8.4}$$

The **natural direct effect** (NDE) is the expected change in the outcome when T shifts from 0 to 1, holding the mediator at the value it would naturally take under $T = 0$. The **natural indirect**

effect (NIE) is the expected change in the outcome due to the shift in the mediator from $M(0)$ to $M(1)$, holding $T = 1$ fixed.

Running example. The NDE is the expected reduction in depression score when the intervention is delivered but each participant's sleep is held at the value it *would have had* without the intervention — the part of the benefit from cognitive, behavioral, or therapeutic-alliance pathways, not from sleep. The NIE is the complementary piece: how much of the depression reduction is due to the sleep improvement that the intervention itself causes.

Decomposition. The total effect decomposes as:

$$\text{TE} \;=\; \text{NDE} \;+\; \text{NIE}. \tag{8.5}$$

Proof

$$\begin{aligned}\text{NDE} + \text{NIE} &= \mathbb{E}[Y(1, M(0)) - Y(0, M(0))] + \mathbb{E}[Y(1, M(1)) - Y(1, M(0))]\\ &= \mathbb{E}[Y(1, M(1)) - Y(0, M(0))] = \mathbb{E}[Y(1) - Y(0)] = \text{TE}. \quad \square\end{aligned}$$

Remark

In the linear additive Baron–Kenny SEM with no $T \times M$ interaction, the CDE and NDE coincide: NDE $= \tau'$ and NIE $= ab$. When there is a $T \times M$ interaction, NDE $\neq$ CDE and the linear decomposition breaks down.

8.6 Identification of Natural Effects

8.6.1 Sequential Ignorability

Identification of the NDE and NIE requires four conditions, known collectively as *sequential ignorability* (Imai et al. 2010). These conditions are substantially stronger than the ignorability assumptions used for the total effect or the CDE.

Sequential Ignorability Assumptions

1. **No unmeasured treatment–outcome confounding.** $\{Y(t', m),\, M(t)\} \perp\!\!\!\perp T \mid \mathbf{X}$ for all t, t', m. Graphically: $\mathbf{X}$ blocks all back-door paths from T to Y and from T to M.
2. **No unmeasured treatment–mediator confounding.** Implied by Assumption 1 for the $T \to M$ sub-problem.
3. **No unmeasured mediator–outcome confounding given T.** $Y(t', m) \perp\!\!\!\perp M \mid T, \mathbf{X}$ for all t', m. Graphically: $(T, \mathbf{X})$ blocks all back-door paths from M to Y.
4. **No treatment-induced mediator–outcome confounder.** There is no post-treatment variable L such that $T \to L$, $L \to M$, and $L \to Y$.

Positivity (Overlap) Conditions

- **(P1) Treatment overlap.** $P(T = t \mid \mathbf{X} = \mathbf{x}) > 0$ for all $t \in \{0, 1\}$ and for P-almost every $\mathbf{x}$.
- **(P2) Mediator overlap across treatment arms.** $P(M = m \mid T = t, \mathbf{X} = \mathbf{x}) > 0$ whenever $P(M = m \mid T = t', \mathbf{X} = \mathbf{x}) > 0$.

What randomization of T does and does not provide. Randomizing T satisfies Assumptions 1 and 2 by design. It does *not* satisfy Assumptions 3 or 4. The mediator M is a post-treatment variable that is never randomized; any unobserved V with $V \to M$ and $V \to Y$ violates Assumption 3 regardless of how T was assigned. Assumption 4 can be violated by a variable L that is itself *caused* by the treatment, so randomizing T actually creates the conditions under which Assumption 4 violations can arise.

Randomization of T Does Not Secure Assumption 4

Unlike unmeasured T–Y confounders (which randomization of T can eliminate), a treatment-induced mediator–outcome confounder L is itself caused by the treatment: the path $T \to L$ is set in motion by the intervention. The absence of such an L must be argued from design, timing, measurement, or subject-matter knowledge. This is why identification of natural effects is considered strictly harder than identification of the total effect or the CDE.

8.6.2 The Mediation Formula

Theorem: Mediation Formula

(Pearl 2001) Under sequential ignorability, positivity (P1)–(P2), and with discrete M and $\mathbf{X}$:

$$\mathbb{E}[Y(t, M(t'))] = \sum_m \sum_{\mathbf{x}} \mathbb{E}[Y \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}]\, P(M{=}m \mid T{=}t',\ \mathbf{X}{=}\mathbf{x})\, P(\mathbf{X}{=}\mathbf{x}). \tag{8.6}$$

Proof Sketch (Five Steps)

Step 1 (law of total expectation): $\mathbb{E}[Y(t, M(t'))] = \sum_{m,\mathbf{x}} \mathbb{E}[Y(t,m) \mid M(t'){=}m,\ \mathbf{X}{=}\mathbf{x}]\, P(M(t'){=}m \mid \mathbf{X}{=}\mathbf{x})\, P(\mathbf{X}{=}\mathbf{x})$.
Step 2 (Assumption 4: cross-world independence): Within the NPSEM-IE framework, no-treatment-induced-confounder implies $Y(t,m) \perp\!\!\!\perp M(t') \mid \mathbf{X}$, so conditioning on $M(t')$ can be dropped: $\mathbb{E}[Y(t,m) \mid M(t'){=}m,\ \mathbf{X}{=}\mathbf{x}] = \mathbb{E}[Y(t,m) \mid \mathbf{X}{=}\mathbf{x}]$.
Step 3 (Assumptions 1 and 3): By $Y(t,m) \perp\!\!\!\perp T \mid \mathbf{X}$ and $Y(t,m) \perp\!\!\!\perp M \mid T, \mathbf{X}$: $\mathbb{E}[Y(t,m) \mid \mathbf{X}{=}\mathbf{x}] = \mathbb{E}[Y(t,m) \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}]$.
Step 4 (consistency for Y): $\mathbb{E}[Y(t,m) \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}] = \mathbb{E}[Y \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}]$.
Step 5 (Assumption 2 and consistency for M): $P(M(t'){=}m \mid \mathbf{X}{=}\mathbf{x}) = P(M{=}m \mid T{=}t',\ \mathbf{X}{=}\mathbf{x})$.
Substituting Steps 2–5 into Step 1 yields Equation 8.6. □

Interpretation. The mediation formula "mixes" the outcome regression under $T = t$ with the mediator distribution under $T = t'$. To compute the NDE, set $t = 1$ and $t' = 0$: take the conditional mean of Y evaluated at treatment 1, but weight the mediator by its distribution under treatment 0. The presence of the second treatment index t' on the right-hand side is the visible trace of the cross-world step.

The NDE and NIE from the formula:

$$\text{NDE} = \sum_{m,\mathbf{x}} \big[\mathbb{E}[Y \mid T{=}1, M{=}m, \mathbf{X}{=}\mathbf{x}] - \mathbb{E}[Y \mid T{=}0, M{=}m, \mathbf{X}{=}\mathbf{x}]\big] P(M{=}m \mid T{=}0, \mathbf{X}{=}\mathbf{x})\, P(\mathbf{X}{=}\mathbf{x}),$$

$$\text{NIE} = \sum_{m,\mathbf{x}} \mathbb{E}[Y \mid T{=}1, M{=}m, \mathbf{X}{=}\mathbf{x}] \big[P(M{=}m \mid T{=}1, \mathbf{X}{=}\mathbf{x}) - P(M{=}m \mid T{=}0, \mathbf{X}{=}\mathbf{x})\big] P(\mathbf{X}{=}\mathbf{x}).$$

Estimation Recipe: Plug-In for the Mediation Formula

Given data $\{(Y_i, T_i, M_i, \mathbf{X}_i)\}_{i=1}^n$, a plug-in estimator requires:

1. **Fit an outcome model.** Regress Y on $(T, M, \mathbf{X})$ to obtain $\hat{\mu}(t, m, \mathbf{x})$.
2. **Fit a mediator model.** Regress M on $(T, \mathbf{X})$ to obtain $\hat{p}(m \mid t, \mathbf{x})$.
3. **Predict outcomes at the evaluation arm.** For each unit i, compute $\hat{\mu}(t, m, \mathbf{X}_i)$.
4. **Reweight the mediator at the reference arm.** For discrete M: weight $\hat{\mu}(t, m, \mathbf{X}_i)$ by $\hat{p}(m \mid t', \mathbf{X}_i)$.
5. **Average over the empirical distribution of X:**

$$\frac{1}{n} \sum_{i=1}^n \sum_m \hat{\mu}(t, m, \mathbf{X}_i)\, \hat{p}(m \mid t', \mathbf{X}_i). \tag{8.7}$$

6. **Quantify uncertainty.** Nonparametric bootstrap or influence-function-based estimators

(Chapters 9–10).
The plug-in estimator is consistent only when both $\hat{\mu}$ and $\hat{p}$ are correctly specified.

8.7 The Linear Mediation Model: Baron–Kenny

8.7.1 The Three-Equation System

The regression-based approach of Baron and Kenny (1986) restricts the reduced prototype graph to a linear SEM:

$$Y = \alpha_1 + \tau T + \gamma_1^\top \mathbf{X} + \varepsilon_1, \tag{8.8}$$

$$M = \alpha_2 + aT + \gamma_2^\top \mathbf{X} + \varepsilon_2, \tag{8.9}$$

$$Y = \alpha_3 + \tau' T + bM + \gamma_3^\top \mathbf{X} + \varepsilon_3. \tag{8.10}$$

The four coefficients of interest: τ (total effect of T on Y), a (first-stage: T on M), τ' (direct: T on Y controlling for M), and b (second-stage: M on Y controlling for T).

8.7.2 The Product and Difference Formulas

Proposition: Mediation Decomposition in the Linear Model

Under Equation 8.8–Equation 8.10:

$$\tau \;=\; \tau' \;+\; ab. \tag{8.11}$$

The indirect effect is $\tau_{\text{ind}} = ab$ (product method) and the direct effect is $\tau_{\text{dir}} = \tau - ab = \tau'$ (difference method).

Proof

Substituting Equation 8.9 into Equation 8.10: $Y = (\alpha_3 + b\alpha_2) + (\tau' + ab)\,T + (\gamma_3 + b\gamma_2)^\top \mathbf{X} + (b\varepsilon_2 + \varepsilon_3)$. Comparing with Equation 8.8 gives $\tau = \tau' + ab$. $\square$

Effect	**Formula**	**Path(s)**
Total	τ	$T \to Y$ and $T \to M \to Y$ combined
Direct	$\tau' = \tau - ab$	$T \to Y$ only
Indirect	ab	$T \to M \to Y$ only
Proportion mediated	ab/τ	Share of total effect via M

Running example. Suppose the fitted coefficients are $\hat{\tau} = 0.50$, $\hat{a} = 0.40$, $\hat{b} = 0.60$, $\hat{\tau}' = 0.26$. Then: indirect effect via sleep: $\hat{a}\hat{b} = 0.40 \times 0.60 = 0.24$; direct effect: $\hat{\tau} - \hat{a}\hat{b} = 0.50 - 0.24 = 0.26 = \hat{\tau}'$; proportion mediated: $0.24/0.50 = 0.48$.

The Decomposition $\tau = \tau' + ab$ Is an Algebraic Identity, Not a Causal Theorem

The equality $\tau - \tau' = ab$ is a purely algebraic consequence of substituting one linear equation into another. It holds because linearity makes the indirect effect separable and additive. It does *not* hold in general: in **nonlinear models**, the product and difference methods yield numerically different estimates; when **T and M interact**, the CDE depends on m and the indirect effect cannot be summarized by a single number ab; for **non-continuous mediators**, the product ab has no simple causal interpretation.
The correct generalization is the mediation formula (Theorem above), which reduces to ab and τ' only in the linear, no-interaction special case.

8.7.3 Inference: The Sobel Test and Bootstrap

The delta method gives an approximate variance for the product $\hat{a}\hat{b}$. The *Sobel test* (Sobel 1982) uses the simplified approximation:

$$\widehat{\mathrm{Var}}_{\mathrm{Sobel}}(\hat{a}\hat{b}) = \hat{b}^2\,\widehat{\mathrm{Var}}(\hat{a}) + \hat{a}^2\,\widehat{\mathrm{Var}}(\hat{b}), \tag{8.12}$$

yielding $z = \hat{a}\hat{b}/\sqrt{\widehat{\mathrm{Var}}_{\mathrm{Sobel}}(\hat{a}\hat{b})}$.

Statistical validity versus causal interpretation. The Sobel test is a statistical inference procedure for the product of two fitted regression coefficients. Interpreting the tested product as an *indirect causal effect* additionally requires the causal identification conditions and the structural restrictions (linearity and no $T \times M$ interaction). In practice, bootstrap confidence intervals for ab are preferred over the Sobel test because the distribution of a product of estimates is skewed in finite samples.

The "Significance of Both Paths" Criterion Is Not a Test for Mediation

A common misuse of the Baron–Kenny framework declares mediation when $T \to Y$, $T \to M$, and $M \to Y$ are each individually significant. This has three serious defects: (1) statistical significance $\neq$ mediation; (2) the Sobel test tests a regression product, not a causal quantity; (3) zero total effect does not preclude indirect effects — direct and indirect effects of opposite sign can cancel. The modern alternative is to estimate ab (or the NIE via the mediation formula) and construct bootstrap confidence intervals.

8.7.4 The Baron–Kenny Assumptions: Two Distinct Categories

Causal ignorability conditions (identification assumptions with graphical interpretations): (1) No unmeasured T–Y confounding; (2) No unmeasured T–M confounding; (3) No unmeasured M–Y confounding given T — this is *not* implied by randomization of T.

Structural modeling restrictions (parametric assumptions, not identification conditions): (4) Linearity and additivity; (5) No $T \times M$ interaction. Restrictions 4 and 5 can in principle be tested; Conditions 1–3 cannot.

Under Conditions 1–3 and Restrictions 4–5, Baron–Kenny is a valid estimator of the NDE and NIE. Outside those conditions, τ' and ab estimate regression coefficients in misspecified structural equations, not causal effects.

Estimand	**Framework needed**	**Section**
Total effect $P(y \mid \mathrm{do}(t))$	Do-calculus	Ch. 5 (back-door)
Controlled direct effect (CDE)	Do-calculus	Section 8.4
Natural direct effect (NDE)	Potential outcomes	Section 8.5
Natural indirect effect (NIE)	Potential outcomes	Section 8.5
Front-door identification	Do-calculus	Section 8.8

8.8 Front-Door Identification

8.8.1 The Front-Door DAG

Every identification strategy in §§ Section 8.4–Section 8.7 assumed that an observed covariate set $\mathbf{X}$ blocks the back-door paths from T to Y through U. What if U is wholly unobserved and no such adjustment set exists?

The front-door criterion turns this obstacle into an opportunity: under two additional restrictions on the prototype mediation graph, the mediation structure itself provides identification of the total effect without conditioning on U. The two restrictions are: (1) remove the direct $T \to Y$ edge, so that M fully mediates the effect of T on Y; and (2) require that U has no arrow into M, so that the $T \to M$ sub-effect is unconfounded.

Why the running example does not apply. The depression/sleep scenario fails Condition 1 (full mediation): a behavioral intervention on depression plausibly operates through several non-sleep channels, so the direct edge $T \to Y$ is present. The canonical front-door example is Pearl's smoking–tar–cancer graph: T = smoking, M = tar deposited in lungs, Y = lung cancer, U = unobserved genetic susceptibility.

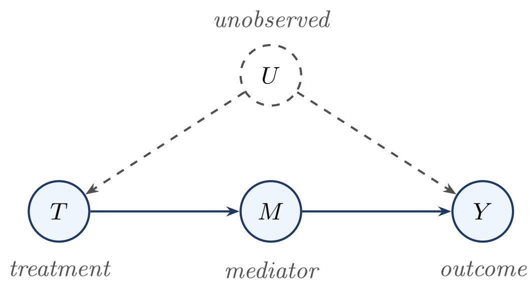

Figure 8.2: The front-door graph. Compared with the prototype mediation DAG, two edges are absent: the direct $T \to Y$ edge is removed, and U has no arrow into M. The front-door formula identifies $P(y \mid \mathrm{do}(t))$ without observing or conditioning on U.

Two Different Questions About a Mediator-Like Variable

	Ordinary mediation	**Front-door identification**
Question	How much of the total effect of T on Y flows through M?	Can M identify the total effect of T on Y despite unmeasured T–Y confounding?
Role of M	Pathway: carries part of the causal effect	Instrument-like relay: routes around confounding
Target estimand	NDE, NIE (decomposition)	$P(y \mid \mathrm{do}(t))$ (total effect itself)
Direct $T \to Y$ edge	Present	Absent (required)

8.8.2 The Three Front-Door Conditions

Front-Door Conditions

A set $\{M\}$ satisfies the **front-door criterion** for the effect of T on Y if:

1. **Full mediation.** All directed paths from T to Y pass through M (no direct $T \to Y$ edge).
2. **No unblocked back-door path from T to M.** All back-door paths from T to M are blocked (no unobserved variables affect both T and M).
3. **No unblocked back-door path from M to Y given T.** All back-door paths from M to Y are blocked by conditioning on T.

In Figure 8.2: Condition 1 holds (no $T \to Y$ edge). Condition 2 holds (U has no arrow into M). Condition 3 holds (the only back-door path from M to Y is $M \leftarrow T \leftarrow U \to Y$, blocked by conditioning on T).

8.8.3 Derivation of the Front-Door Formula

Theorem: Front-Door Formula

(Pearl 1995) Suppose M satisfies the front-door criterion for the effect of T on Y, and positivity conditions (F1)–(F2) hold. Then:

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m P(m \mid t) \sum_{t'} P(y \mid m, t')\, P(t'). \tag{8.13}$$

Proof via Three Do-Calculus Steps

Step 1: Identify the effect of T on M. By Condition 2, there are no unblocked back-door paths from T to M, so T satisfies the back-door criterion for M with the empty adjustment set:

$$P(m \mid \mathrm{do}(T{=}t)) = P(m \mid t).$$

Step 2: Identify the effect of M on Y. By Condition 3, conditioning on T blocks all back-door paths from M to Y, so $\{T\}$ is a valid back-door adjustment set:

$$P(y \mid \mathrm{do}(M{=}m)) = \sum_{t'} P(y \mid m, t')\, P(t').$$

Step 3: Combine via full mediation. By the law of total probability under $\mathrm{do}(T{=}t)$:

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_{m} P(m \mid \mathrm{do}(T{=}t))\ P(y \mid \mathrm{do}(T{=}t), \mathrm{do}(M{=}m))\,.$$

By Condition 1 (no direct $T \to Y$ edge), once M is set to m by intervention, T is d-separated from Y in $\mathcal{G}_{\overline{T}\,\overline{M}}$; Rule 3 of the do-calculus removes $\mathrm{do}(T{=}t)$ from the second factor: $P(y \mid \mathrm{do}(T{=}t), \mathrm{do}(M{=}m)) = P(y \mid \mathrm{do}(M{=}m))$.
Substituting Steps 1 and 2 yields Equation 8.13. □

8.8.4 Intuition: Routing Around Confounding

The front-door formula achieves identification in two steps that each use only unconfounded variation.

1. **T to M:** There is no confounding on the $T \to M$ edge (U does not affect M), so $P(m \mid t)$ is the causal effect of T on M.
2. **M to Y:** There is back-door confounding on $M \to Y$ through $T \leftarrow U \to Y$, but it is blocked by summing over T: $\sum_{t'} P(y \mid m, t')\, P(t')$ marginalizes out T, removing the confounding path.

The key insight: U confounds T and Y but *not* the $T \to M$ edge. The front-door formula exploits this asymmetry.

Feature	**Prototype mediation DAG**	**Front-door DAG**
Direct $T \to Y$ edge	Present	Absent
U confounds T–Y	Yes	Yes
Goal	Decompose total effect into direct/indirect	Identify total effect despite T–Y confounding
Requires Assumption 3	Yes	Satisfied by Condition 3, not by randomization
Identified by front-door formula	No	Yes

8.9 Mediation vs. Instrumental Variables

Mediation analysis and instrumental variables both involve a third variable connected to the treatment–outcome relationship, but the causal role of that variable is fundamentally different.

Feature	**Instrumental Variables**	**Mediation Analysis**
Position of third variable	Pre-treatment (Z precedes T)	Post-treatment (M follows T)
Causal role	Exogenous source of variation in T	Pathway through which T affects Y
Primary goal	Identification of $T \to Y$ effect under confounding	Decomposition of $T \to Y$ effect into pathways
Key assumption	Exclusion: Z affects Y only through T	Sequential ignorability: no unmeasured M–Y confounding given T

Feature	Instrumental Variables	Mediation Analysis
Estimand	LATE (Wald) or ATE (2SLS under homogeneity)	NDE, NIE, or CDE
Unobserved T–Y confounders	Permitted	Must be addressed separately (Assumption 1)

The Key Conceptual Distinction

	IV	Mediation
What the third variable does	Generates clean variation in T (exogenous source)	Carries part of T's causal effect (pathway)
Exclusion vs. inclusion	Z excluded from Y's structural equation	M included in Y's structural equation
Question answered	Does T cause Y?	How does T cause Y?

Can the same variable be both? Not in the same analysis. As a mediator, M is on the causal path from T to Y (inclusion required). As an instrument, M would require the exclusion restriction — that it affects Y only through T — which contradicts M being a mediator. The two frameworks make mutually incompatible structural assumptions about the intermediate variable. However, the front-door identification formula is IV's closest relative within mediation analysis: it uses the mediator M to achieve identification of the total effect of T on Y even when T is confounded.

8.10 Summary

1. **Mediation studies mechanisms.** Mediation analysis decomposes the total effect of T on Y into a direct component ($T \to Y$) and an indirect component ($T \to M \to Y$). The challenge is that the mediator is a post-treatment variable that may be confounded.

2. **The total effect is the baseline.** The TE $= \mathbb{E}[Y(1) - Y(0)]$ captures all pathways combined. Mediation analysis asks for a finer decomposition.

3. **The CDE uses do-calculus.** The controlled direct effect fixes $M = m$ by joint intervention $\mathrm{do}(T, M)$ and is identified by the back-door formula applied to the pair (T, M). No assumptions beyond the back-door criterion are needed, but the CDE depends on m and does not have a natural "indirect" complement.

4. **Natural effects require potential outcomes.** The NDE and NIE involve cross-world counterfactuals $Y(t, M(t'))$ that cannot be expressed with the do-operator alone. They decompose the total effect as TE = NDE + NIE, and are identified by the mediation formula under sequential ignorability. The critical Assumption 4 — no treatment-induced mediator–outcome confounder — is not secured by randomization of T.

5. **The linear model simplifies but restricts.** The Baron–Kenny three-equation system gives the clean formulas $\tau_{\text{ind}} = ab$ and $\tau_{\text{dir}} = \tau'$, with $\tau = \tau' + ab$. This decomposition is purely algebraic and holds only under linearity and no interaction.

6. **Front-door identification uses mediation structure.** When M fully mediates $T \to Y$, no $T \to M$ confounding exists, and T blocks $M \to Y$ confounding, the front-door formula identifies the total effect despite unobserved T–Y confounding.

7. **Mediation and IV are complementary, not equivalent.** IV uses a pre-treatment variable to generate exogenous variation in T; mediation uses a post-treatment variable to decompose the causal effect. The exclusion restriction of IV and the sequential ignorability of mediation are mutually incompatible for the same intermediate variable.

8.11 Problems

1. Identifying the CDE. Consider the DAG: $T \to M$, $T \to Y$, $M \to Y$, $X \to T$, $X \to M$, $X \to Y$, with all variables observed.

(a) Write the identification formula for $\mathbb{E}[Y \mid \mathrm{do}(T{=}1), \mathrm{do}(M{=}m)]$ using the back-door criterion for the joint intervention (T, M).
(b) Add an unobserved U with $U \to T$ and $U \to Y$. Does the back-door formula still identify the CDE? Explain which condition, if any, fails.
(c) Instead add U with $U \to M$ and $U \to Y$. Does the back-door formula still identify the CDE? Explain.

2. CDE vs. total effect. In the reduced prototype graph, let $\mathbf{Z}$ satisfy the back-door criterion for both the total effect and the joint intervention (T, M).

(a) Write expressions for the total effect and the CDE(m) using the back-door formula.
(b) Under what graphical condition does CDE(m) equal the total effect for all m? Interpret this condition.

3. Natural direct and indirect effects. Verify the NDE + NIE = TE decomposition algebraically for the linear SEM $M = \alpha T + \eta$, $Y = \beta T + \gamma M + \varepsilon$ (no interaction).

(a) Compute $Y(t, M(t'))$ in the linear model. Show that $Y(t, M(t')) = \beta t + \gamma(\alpha t') + \text{noise}$.
(b) Derive $\text{NDE} = \beta$ and $\text{NIE} = \alpha\gamma$ from the definitions Equation 8.3–Equation 8.4.
(c) Confirm $\text{NDE} + \text{NIE} = \beta + \alpha\gamma = \text{TE}$.
(d) Now suppose a $T \times M$ interaction is added: $Y = \beta T + \gamma M + \delta(T \cdot M) + \varepsilon$. Compute CDE($m$) and NDE. Show that $\text{NDE} \neq \text{CDE}(m)$ when $\delta \neq 0$.

4. The Baron–Kenny three-equation system. In the reduced prototype graph with the linear SEM Equation 8.8–Equation 8.10:

(a) State the three identification assumptions. For each, give the graphical condition in terms of back-door paths.
(b) Derive the equality $\tau = \tau' + ab$ algebraically.
(c) Suppose estimated coefficients are $\hat{\tau} = 0.50$, $\hat{a} = 0.40$, $\hat{b} = 0.60$, $\hat{\tau}' = 0.26$. Compute the indirect effect by both the product and difference methods. Do they agree? Compute the proportion mediated.
(d) An analyst reports $\hat{a} = 0.40$ with $\widehat{\text{SE}}(\hat{a}) = 0.08$, and $\hat{b} = 0.60$ with $\widehat{\text{SE}}(\hat{b}) = 0.10$. Compute the Sobel standard error for $\hat{a}\hat{b}$ using Equation 8.12 and construct an approximate 95% confidence interval.

5. The critical role of Assumption 3. Consider the graph with an unobserved V with $V \to M$ and $V \to Y$. T is randomized.

(a) Identify all back-door paths from M to Y in this graph.
(b) Can any combination of observed variables $(T, \mathbf{X})$ block all of these paths? Explain using d-separation.
(c) Suppose an analyst fits Equation 8.10 ignoring V and obtains $\hat{b} = 0.80$. In which direction is $\hat{b}$ biased if V has positive effects on both M and Y?
(d) State the additional data structure that would be needed to identify the second-stage effect non-parametrically.

6. Front-door identification. Consider the front-door graph of Figure 8.2.

(a) Verify that the three front-door conditions hold.
(b) Walk through the three-step proof of the Front-Door Formula theorem for this graph: identify which do-calculus rule justifies each step.
(c) Add a direct edge $T \to Y$ to the graph. Which front-door condition is violated? Does formula Equation 8.13 still hold?
(d) Explain why the front-door graph does not require Assumption 3 of the Baron–Kenny framework. Which back-door path — present in the graph with unobserved V — is absent here?

7. Mediation vs. instrumental variables. A researcher studies the effect of a job training program (T) on wages (Y). She proposes two intermediate variables: (A) motivation (M_A), measured after the program starts; (B) a lottery that randomly selects applicants for admission (Z), measured before the program.

(a) For variable (A): draw the mediation DAG including M_A, T, Y, and an unobserved ability variable U. State the sequential ignorability assumption needed to identify the NIE through M_A. Explain why randomization of T does not automatically satisfy this assumption.

(b) For variable (B): draw the IV DAG with Z, T, Y, and U. State the three IV assumptions. Explain why the exclusion restriction and the "mediator inclusion" of mediation analysis are mutually incompatible conditions for the same intermediate variable.

(c) The researcher argues that M_A (motivation) and Z (lottery) are both "intermediate" variables and that the analyses are interchangeable. Write a one-paragraph critique of this argument, using the distinctions from Section 8.9.

(d) Can the front-door formula be applied if motivation M_A fully mediates the effect of T on Y and U (unobserved ability) does not directly affect M_A? State the three conditions and assess whether they hold.

Chapter 9

Estimating Equations and Influence Functions

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain why identification of a causal parameter does not automatically yield a statistically reliable estimator, and articulate the distinct questions that an estimation theory must address.
2. Define an estimating equation and verify the population moment condition for standard examples (sample mean, OLS, IPW).
3. State the definition of an asymptotically linear estimator and derive the influence function from a generic first-order expansion of an estimating equation.
4. Compute the influence function for simple causal estimators when nuisance functions are known, and interpret it as the infinitesimal contribution of a single observation.
5. Describe how estimation error in nuisance functions propagates into the target estimator, and explain why this is the central statistical challenge in causal inference.
6. Use the influence function to construct a consistent variance estimator and an approximate Wald confidence interval.
7. Define efficiency in the class of regular estimators and explain the role of the efficient influence function as the foundation for doubly robust estimation in Chapter 10.

9.1 Why Estimation Needs Its Own Theory

So far the focus has been on *identification*: under what assumptions can a causal parameter be written as a functional of the observed-data distribution? Identification does not by itself provide a statistically reliable estimator. Once a causal parameter is identified, several distinct questions remain: How should the estimator be constructed from the observed sample? How does estimation of nuisance functions affect the target estimator? What is the large-sample distribution of the estimator? How can we compute valid standard errors and confidence intervals?

These questions motivate a separate theory of *estimation and inference*. In causal inference this issue is especially important because identified parameters often depend on auxiliary, or *nuisance*, functions such as the outcome regression $\mu_t(x) = \mathbb{E}(Y \mid T = t, X = x)$ or the propensity score $\pi(x) = P(T = 1 \mid X = x)$. Even when the causal parameter is identified, different estimation strategies may behave quite differently in terms of robustness, efficiency, and sensitivity to model misspecification.

The goal of this chapter is to introduce a general framework for estimation based on *estimating equations* and *influence functions*. This framework will serve as the foundation for the doubly robust and semiparametric methods developed in Chapter 10; see also Imbens and Rubin (2015) and Hernán and Robins (2020) for complementary treatments.

9.2 A Running Example: The Average Treatment Effect

Throughout this chapter we use the ATE $\psi = \mathbb{E}\{Y(1) - Y(0)\}$ as a running example. Under consistency, conditional exchangeability, and positivity, the ATE is identified by the back-door formula (Chapter 5):

$$\psi = \mathbb{E}[\mu_1(X) - \mu_0(X)], \qquad \mu_t(x) = \mathbb{E}(Y \mid T = t,\, X = x), \quad t \in \{0, 1\}.$$

This identity tells us what the target parameter is, but it does not uniquely determine how to estimate it. One may consider: a *regression-based* estimator using models for $\mu_1(x)$ and $\mu_0(x)$; a *weighting* estimator based on the propensity score $\pi(x)$ (Chapter 6); or an *augmented* estimator combining both. All of these estimators may target the same causal parameter yet differ in their statistical properties.

9.3 Estimating Equations

A broad class of estimators can be defined as solutions to *estimating equations*. Let $O_1, \dots, O_n$ be i.i.d. observations from a distribution P, and let θ denote a finite-dimensional target parameter. Write $\mathbb{P}_n f = \frac{1}{n}\sum_{i=1}^n f(O_i)$ for the empirical average.

Definition: Estimating Equation

An **estimating equation** for a parameter θ is an equation of the form $\mathbb{P}_n\{U(O;\theta)\} = 0$, where the **population moment condition** $\mathbb{E}\{U(O;\theta_0)\} = 0$ holds at the true parameter value θ_0. The function $U(O;\theta)$ is called the **estimating function**.

Example: Sample Mean

Let $\theta = \mathbb{E}(Y)$. The sample mean $\hat{\theta} = \bar{Y}$ solves $\mathbb{P}_n(Y - \theta) = 0$. The estimating function is $U(O;\theta) = Y - \theta$.

Example: Ordinary Least Squares

Let $O = (X, Y)$ and suppose β is defined by the linear moment condition $\mathbb{E}[X\{Y - X^\top\beta\}] = 0$. Then the OLS estimator solves $\mathbb{P}_n[X\{Y - X^\top\beta\}] = 0$.

Example: A Simple IPW Estimating Equation

Suppose $\pi(X)$ is known and $\psi = \mathbb{E}\left\{\frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)}\right\}$. Then ψ solves $\mathbb{E}\left[\frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)} - \psi\right] = 0$.

9.4 From Estimating Equations to Asymptotic Linearity

Under suitable regularity conditions, an estimator solving an estimating equation can typically be approximated as:

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^n \varphi(O_i) + o_p(1)$$

for some mean-zero function $\varphi(O)$.

Definition: Asymptotic Linearity and Influence Function

An estimator $\hat{\theta}$ is **asymptotically linear** with **influence function** $\varphi(O)$ if:

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^n \varphi(O_i) + o_p(1), \qquad \mathbb{E}\{\varphi(O)\} = 0.$$

This representation immediately implies asymptotic normality. When $\theta \in \mathbb{R}^p$:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(\mathbf{0},\ \Sigma)\,, \qquad \Sigma = \mathbb{E}[\varphi(O)\varphi(O)^\top].$$

Proposition: Generic First-Order Expansion

Under smoothness and regularity conditions, an estimator $\hat{\theta}$ solving $\mathbb{P}_n\{U(O;\theta)\} = 0$ admits a first-order expansion:

$$\sqrt{n}(\hat{\theta} - \theta_0) = -A^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} U(O_i;\theta_0) + o_p(1),$$

where $A = \mathbb{E}\left\{\frac{\partial}{\partial\theta^\top} U(O;\theta_0)\right\}$. Hence the influence function is $\varphi(O) = -A^{-1}U(O;\theta_0)$.

Proof

A Taylor expansion of the sample moment condition around θ_0 gives:

$$0 = \mathbb{P}_n\{U(O;\hat{\theta})\} \approx \mathbb{P}_n\{U(O;\theta_0)\} + \mathbb{P}_n\left\{\frac{\partial}{\partial\theta^\top} U(O;\theta_0)\right\}(\hat{\theta} - \theta_0).$$

By the law of large numbers, $\mathbb{P}_n\{\partial_{\theta^\top} U(O;\theta_0)\} \to A$ in probability. Rearranging and multiplying by $\sqrt{n}$ yields the stated expansion. $\square$

9.5 Influence Functions: Intuition

9.5.1 Statistical Functionals

Definition: Statistical Functional

A **statistical functional** is a map $\Psi : \mathcal{P} \longrightarrow \mathbb{R}^p$ that assigns a parameter value $\psi = \Psi(P)$ to each distribution $P \in \mathcal{P}$. The **plug-in estimator** of ψ is $\hat{\psi} = \Psi(\mathbb{P}_n)$.

The mean, OLS coefficient, and ATE are all statistical functionals. A functional Ψ is *linear* if $\Psi((1-\epsilon)P + \epsilon Q) = (1-\epsilon)\Psi(P) + \epsilon\Psi(Q)$. The mean is linear; the variance, OLS coefficient, and ATE are nonlinear, requiring local linearization — precisely what the influence function provides.

9.5.2 Influence Functions

Influence functions describe the first-order sensitivity of a statistical functional to small perturbations of the underlying distribution.

Definition: Influence Function

Let $\psi = \Psi(P)$ be a scalar parameter. A function $\varphi(O)$ is called an **influence function** for Ψ if it has mean zero under P and:

$$\hat{\psi} - \psi = \frac{1}{n} \sum_{i=1}^{n} \varphi(O_i) + o_p(n^{-1/2}).$$

The influence function is the infinitesimal contribution of a single observation to the first-order behavior of the estimator.

Remark: Formal Derivation via Pathwise Derivatives

The influence function can be defined formally through the *pathwise (Gateaux) derivative*: if $P_\epsilon = (1-\epsilon)P + \epsilon\delta_O$ is the ϵ-mixture of P with a point mass at O, then:

$$\varphi(O) = \frac{d}{d\epsilon}\Psi(P_\epsilon)\Big|_{\epsilon=0}.$$

This derivative-based definition is the starting point for semiparametric efficiency theory, developed formally in Appendix C.

Example: Influence Function of the OLS Slope

For the OLS coefficient $\beta = [\mathbb{E}(XX^\top)]^{-1}\mathbb{E}(XY)$:
Step 1. Construct the perturbed distribution $P_\epsilon = (1-\epsilon)P + \epsilon\delta_O$:

$$\Psi(P_\epsilon) = \big[(1-\epsilon)\mathbb{E}(XX^\top) + \epsilon XX^\top\big]^{-1}\big[(1-\epsilon)\mathbb{E}(XY) + \epsilon XY\big].$$

Step 2. Differentiate with respect to ϵ at $\epsilon = 0$, using $\frac{d}{d\epsilon}M(\epsilon)^{-1}\big|_0 = -[\mathbb{E}(XX^\top)]^{-1}(XX^\top - \mathbb{E}(XX^\top))[\mathbb{E}(XX^\top)]^{-1}$ and the product rule:

$$\varphi(O) = [\mathbb{E}(XX^\top)]^{-1}X(Y - X^\top\beta).$$

Verification. $\mathbb{E}[\varphi(O)] = [\mathbb{E}(XX^\top)]^{-1}\mathbb{E}[X(Y - X^\top\beta)] = 0$ by definition of β.
Connection to estimating equations: The OLS estimating function is $U(O;\beta) = X(Y - X^\top\beta)$, $A = -\mathbb{E}(XX^\top)$, and Proposition 1 gives the same $\varphi(O) = [\mathbb{E}(XX^\top)]^{-1}X(Y - X^\top\beta)$.

9.6 Influence Functions for Simple Causal Estimators

9.6.1 Known Outcome Regression

Suppose $\psi = \mathbb{E}[\mu_1(X) - \mu_0(X)]$ and the functions $\mu_1(\cdot)$ and $\mu_0(\cdot)$ are known. The natural plug-in estimator is $\hat{\psi} = \mathbb{P}_n\{\mu_1(X) - \mu_0(X)\}$. Its influence function is:

$$\varphi(O) = \mu_1(X) - \mu_0(X) - \psi.$$

9.6.2 Known Propensity Score

Suppose $\pi(X)$ is known and $\psi = \mathbb{E}\Big\{\frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)}\Big\}$. The empirical analogue is the Horvitz–Thompson estimator, and its influence function is:

$$\varphi(O) = \frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)} - \psi.$$

Remark: Nuisance Functions and the Influence Function

When nuisance functions are *known*, the influence function is simply the centered estimating function. This observation becomes much more consequential in the next section, where nuisance functions are unknown and must be estimated from data.

Remark: Estimating Function, Influence Function, and Efficient Influence Function

These three objects are closely related but not the same.

1. An *estimating function* defines an estimator through a sample moment equation $\mathbb{P}_n\{\phi(O;\theta)\} = 0$.
2. An *influence function* describes the first-order asymptotic behavior of an estimator: $\sqrt{n}(\hat{\theta} -$

$\theta_0) = n^{-1/2}\sum_i \varphi(O_i) + o_p(1)$.

3. The *efficient influence function* is the influence function with the smallest possible asymptotic variance among regular estimators in the statistical model.

In simple examples with known nuisance functions, the centered estimating function often coincides with the influence function. This is not automatic when nuisance parameters are estimated.

9.7 Z-Estimation with Nuisance Parameters

In real applications, nuisance functions must be estimated from the same data used to estimate the target parameter. Once this happens, the first-order expansion of $\hat\psi$ is no longer obtained by simply centering the estimating function with the nuisance held fixed. Z-estimation provides the correct framework: it treats both the target parameter and the nuisance parameter as components of a single stacked estimating equation.

9.7.1 The Stacked Estimating Equation Framework

Suppose the target parameter $\psi \in \mathbb{R}$ depends on an unknown nuisance parameter $\alpha \in \mathbb{R}^q$. Both are estimated by solving a stacked system:

$$\mathbb{P}_n\{U_1(O;\psi,\alpha)\} = 0, \qquad \mathbb{P}_n\{U_2(O;\alpha)\} = 0. \tag{9.1}$$

Write $\theta = (\psi^\top, \alpha^\top)^\top$ and $U(O;\theta) = (U_1(O;\psi,\alpha),\, U_2(O;\alpha))^\top$.

Theorem: Z-Estimation Theorem (Stacked Equations)

Suppose the true parameter value is $\theta_0 = (\psi_0, \alpha_0^\top)^\top$ and the population moment condition $\mathbb{E}\{U(O;\theta_0)\} = 0$ holds, the Jacobian $\mathcal{A} = \mathbb{E}\{\partial_{\theta^\top} U(O;\theta_0)\}$ is invertible, and regularity conditions hold. Then $\hat\theta$ is consistent for θ_0 and:

$$\sqrt{n}(\hat\theta - \theta_0) = -\mathcal{A}^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n U(O_i;\theta_0) + o_p(1).$$

The influence function of $\hat\psi$ is:

$$\varphi(O) = -A_{11}^{-1}\, U_1(O;\psi_0,\alpha_0) + A_{11}^{-1}A_{12}A_{22}^{-1}\, U_2(O;\alpha_0), \tag{9.2}$$

where $A_{11} = \mathbb{E}\{\partial_\psi U_1\}$, $A_{12} = \mathbb{E}\{\partial_{\alpha^\top} U_1\}$, $A_{22} = \mathbb{E}\{\partial_{\alpha^\top} U_2\}$.

Proof

Partitioning $\mathcal{A}$ conformably with (ψ,α):

$$\mathcal{A} = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}, \qquad \mathcal{A}^{-1} = \begin{pmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{pmatrix}.$$

The lower-left block is zero because U_2 does not depend on ψ. Reading off the first row of $-\mathcal{A}^{-1}U(O;\theta_0)$ gives Equation 9.2. □

Remark: Interpreting the Correction Term

Equation 9.2 decomposes the influence function into two parts. The first term, $-A_{11}^{-1}U_1(O;\psi_0,\alpha_0)$, is exactly what we would obtain if α_0 were known. The second term, $A_{11}^{-1}A_{12}A_{22}^{-1}U_2(O;\alpha_0)$, is a *nuisance correction* that accounts for the additional variability introduced by estimating α. If $A_{12} = 0$, the target estimating function U_1 does not depend on α at α_0, and the correction vanishes. This is the key condition exploited by *doubly robust* estimators in Chapter 10.

9.7.2 Working Example: IPW with Estimated Propensity Score

Assume a logistic regression model $\pi(X;\alpha) = \operatorname{expit}(X^\top\alpha)$. The IPW estimating equation for ψ is $U_1(O;\psi,\alpha) = \frac{TY}{\pi(X;\alpha)} - \frac{(1-T)Y}{1-\pi(X;\alpha)} - \psi$, and the score for the logistic regression is $U_2(O;\alpha) = X\{T - \pi(X;\alpha)\}$.

Computing the Jacobian blocks: $A_{11} = -1$, $A_{22} = -\Sigma_\pi := -\mathbb{E}\{\pi(X;\alpha_0)(1-\pi(X;\alpha_0))XX^\top\}$. After computing A_{12}, the corrected influence function is:

$$\varphi(O) = U_1(O;\psi_0,\alpha_0) + A_{12}\,\Sigma_\pi^{-1}\,U_2(O;\alpha_0). \tag{9.3}$$

Example: Variance Reduction from ML Estimation of Propensity Score

We show that estimating α by maximum likelihood can only *reduce* the asymptotic variance relative to using the true α_0.
Write $\sigma_1^2 = \mathbb{E}[U_1^2]$ for the variance of the naive IPW estimating function. Two Bartlett-type identities are the key:
Identity 1 (Bartlett for ML score): $\mathbb{E}[U_2U_2^\top] = \Sigma_\pi$.
Identity 2 (cross-identity): $A_{12} = -\mathbb{E}[U_1U_2^\top]$.
Substituting into $\mathbb{E}[\varphi(O)^2] = \sigma_1^2 + 2A_{12}\Sigma_\pi^{-1}\mathbb{E}[U_1U_2] + A_{12}\Sigma_\pi^{-1}\mathbb{E}[U_2U_2^\top]\Sigma_\pi^{-1}A_{12}^\top$:

$$\mathbb{E}[\varphi(O)^2] = \sigma_1^2 - A_{12}\Sigma_\pi^{-1}A_{12}^\top \le \sigma_1^2.$$

Since Σ_π is positive definite, the variance is reduced. *ML estimation of the propensity score parameters reduces the asymptotic variance of the IPW estimator* — the *estimated propensity score paradox* (Robins 1986).

Remark: Sandwich Variance Estimator

Equation 9.2 provides the basis for the *sandwich variance estimator*: $\hat V = n^{-2}\sum_{i=1}^n \hat\varphi_i^2$, where $\hat\varphi_i$ is $\varphi(O_i)$ evaluated at $(\hat\psi,\hat\alpha)$. This estimator automatically accounts for nuisance estimation uncertainty. Standard errors from a logistic regression routine that ignores the link between $\hat\alpha$ and $\hat\psi$ will in general be *incorrect*.

The Central Challenge in Causal Estimation

The primary difficulty in practice is often not identifying the target parameter, but *correctly accounting for the effect of nuisance estimation* on the asymptotic distribution of the final estimator. Treating $\hat\alpha$ as if it were the true α_0 yields correct point estimates under a correctly specified propensity score model but *incorrect standard errors*. The Z-estimation framework resolves this variance-accounting issue whenever the nuisance model is correctly specified and the nuisance estimator converges at rate $n^{-1/2}$.

Remark: Machine Learning for Nuisance Estimation

The Z-estimation theorem as stated requires $\hat\alpha$ to converge at rate $n^{-1/2}$, which holds for finite-dimensional parametric models but fails for nonparametric or machine-learning estimators. When flexible methods are used for $\pi(x)$ or $\mu_t(x)$, the correction term in Equation 9.2 no longer has the simple form derived here. This is the starting point for the doubly robust and debiased machine-learning estimators introduced in Chapter 11.

9.8 Variance Estimation and Confidence Intervals

Once an estimator admits the asymptotic linear representation, its asymptotic variance is $\operatorname{Var}(\hat\psi) \approx n^{-1}\operatorname{Var}\{\varphi(O)\}$. A natural estimator is:

$$\hat V = \frac{1}{n^2}\sum_{i=1}^n \hat\varphi_i^2 = \frac{1}{n(n-1)}\sum_{i=1}^n(\hat\varphi_i - \bar\varphi)^2. \tag{9.4}$$

This yields the approximate Wald confidence interval $\hat{\psi} \pm z_{1-\alpha/2}\sqrt{\hat{V}}$.

Proposition: Variance from the Influence Function

If $\sqrt{n}(\hat{\psi} - \psi) = n^{-1/2}\sum_{i=1}^{n}\varphi(O_i) + o_p(1)$ with $\mathbb{E}\{\varphi(O)\} = 0$ and $\mathbb{E}\{\varphi(O)^2\} < \infty$, then:

$$\sqrt{n}(\hat{\psi} - \psi) \xrightarrow{d} N(0, \mathbb{E}[\varphi(O)^2]), \qquad \mathrm{Var}(\hat{\psi}) = \frac{1}{n}\mathbb{E}[\varphi(O)^2] + o(n^{-1}).$$

9.9 Toward Efficiency: Semiparametric Models and the EIF

Different regular estimators of the same parameter may have different large-sample variances. What is the smallest achievable asymptotic variance among all regular estimators? Answering that question leads to semiparametric efficiency theory. The formal machinery — regular parametric submodels, scores, tangent spaces, and the characterization of the EIF as a canonical gradient — is developed in Appendix C.

9.9.1 Semiparametric Models

Definition: Semiparametric Model

A **semiparametric model** is a statistical model $\mathcal{P}$ for the distribution P of the observed data in which (i) the **parameter of interest** $\psi = \Psi(P) \in \mathbb{R}^p$ is finite-dimensional, and (ii) the remaining aspects of P — collectively called the **nuisance parameter** — are infinite-dimensional.

The canonical example is the nonparametric model $\mathcal{P} = \{P : 0 < \pi(x) < 1 \ \forall x\}$ with target parameter $\psi = \mathbb{E}[Y(1) - Y(0)]$. The nuisance consists of $\mu_t(x)$, $\pi(x)$, and the marginal distribution of X.

9.9.2 Regular Estimators

Definition: Regular Estimator (informal)

An estimator $\hat{\psi}$ is called **regular** at P_0 if, along every smooth parametric submodel $\{P_t\}$ with $P_0 = P_{t=0}$, the limiting distribution of $\sqrt{n}(\hat{\psi} - \psi_t)$ under $P_t = P_{1/\sqrt{n}}$ does not depend on the submodel.

Regularity rules out pathological "superefficient" estimators. All estimators in this course — regression, IPW, and their augmented variants — are regular.

9.9.3 The Semiparametric Efficiency Bound and the EIF

Theorem: Semiparametric Convolution Theorem

(Bickel et al. 1993) In a semiparametric model $\mathcal{P}$, the asymptotic distribution of any regular estimator $\hat{\psi}$ of ψ_0 satisfies:

$$\sqrt{n}(\hat{\psi} - \psi_0) \xrightarrow{d} N(0, V^*) * M$$

for some distribution M, where $V^* = \mathbb{E}_P[\varphi^*(O)\,\varphi^*(O)^\top]$ is the **semiparametric efficiency bound**. The function $\varphi^*(O)$ with $\mathbb{E}[\varphi^*(O)] = 0$ and $\mathbb{E}[\varphi^*(O)^2] = V^*$ is called the **efficient influence function** (EIF). An estimator achieves V^* if and only if it is asymptotically linear with influence function $\varphi^*(O)$.

Definition: Semiparametrically Efficient Estimator

A regular estimator $\hat{\psi}$ is **semiparametrically efficient** at $P_0 \in \mathcal{P}$ if $\sqrt{n}(\hat{\psi} - \psi_0) \xrightarrow{d} N(0, V^*)$. Equivalently, $\hat{\psi}$ is asymptotically linear with influence function $\varphi^*(O)$.

Remark: How the EIF is Computed

The EIF $\varphi^*(O)$ can be derived as the pathwise derivative of $\Psi(P)$ in the *least favorable* direction. In practice: (i) compute the pathwise derivative $\varphi(O; P_\epsilon)$ for an arbitrary parametric submodel; (ii) project onto the tangent space of $\mathcal{P}$ at P_0. For nonparametric or large semiparametric models, the tangent space is rich enough that this projection is often the identity, and the EIF equals the ordinary pathwise derivative. See Appendix C for the formal treatment.

9.9.4 The Efficient Influence Function for the ATE

Remark: EIF for the ATE

For the ATE $\psi = \mathbb{E}\{Y(1) - Y(0)\}$ in the nonparametric observational-data model under strong ignorability and positivity, the efficient influence function is:

$$\varphi^*(O) = \frac{T\{Y - \mu_1(X)\}}{\pi(X)} - \frac{(1-T)\{Y - \mu_0(X)\}}{1 - \pi(X)} + \mu_1(X) - \mu_0(X) - \psi. \tag{9.5}$$

This expression has a natural decomposition: the first two terms are an IPW-style residual correction, and the last two are the regression estimator centered at ψ. The EIF depends on *both* nuisance functions $\pi(X)$ and $\mu_t(X)$, and an estimator that plugs in consistent estimates of both achieves the efficiency bound $V^* = \mathbb{E}[\varphi^*(O)^2]$.

Remark: Bridge to Chapter 10

Two important properties of $\varphi^*(O)$ will be central in Chapter 10:
(i) **Double robustness.** The estimator obtained by replacing $\pi(X)$ and $\mu_t(X)$ by estimates and averaging $\varphi^*(O_i)$ is consistent for ψ if *either* $\hat{\pi}$ or $\hat{\mu}_t$ is consistent — not necessarily both.
(ii) **Semiparametric efficiency.** When both nuisance estimators are consistent and converge at suitable rates, the resulting estimator achieves the efficiency bound V^*.
The *augmented IPW* (AIPW) estimator is the principal tool for exploiting these two properties simultaneously, and will be derived and analyzed in Chapter 10.

9.10 Chapter Summary

Object	**Role**
Estimating function $U(O;\theta)$	Defines the estimator via $\mathbb{P}_n\{U(O;\hat{\theta})\} = 0$
Influence function $\varphi(O)$	First-order term in the expansion of $\hat{\psi} - \psi$; used for CLT and variance estimation
Asymptotic variance $\mathbb{E}[\varphi(O)^2]/n$	Governs precision; the target for efficiency comparisons
Efficient influence function $\varphi^*(O)$	Achieves the semiparametric efficiency bound; basis for doubly robust estimators (Chapter 10)

1. **Identification is not estimation.** Identification provides a population formula, but not automatically a satisfactory estimator.
2. **Estimating equations.** Many estimators are defined as solutions to sample moment conditions $\mathbb{P}_n\{U(O;\theta)\} = 0$, with the population condition $\mathbb{E}\{U(O;\theta_0)\} = 0$ ensuring consistency.

3. **Asymptotic linearity.** Estimating equations often yield a first-order expansion, immediately implying asymptotic normality via the CLT.
4. **Influence functions.** The function $\varphi(O)$ describes the first-order sensitivity of the estimator to a single observation. It is the key object for both asymptotic theory and variance estimation.
5. **Nuisance estimation.** In causal inference, nuisance functions must be estimated, and this estimation error propagates into the target estimator. The Z-estimation framework corrects for this propagation.
6. **Variance from the influence function.** The asymptotic variance is $n^{-1}\mathbb{E}[\varphi(O)^2]$, consistently estimated by $n^{-2}\sum_i \hat{\varphi}_i^2$.
7. **Efficiency.** Among regular estimators, the one with the smallest asymptotic variance is efficient. The EIF characterizes this lower bound and will guide estimator construction in Chapter 10.

9.11 Problems

1. Estimating equations and moment conditions.

(a) Let $O = (X, Y)$ and define $\theta = \mathrm{Cov}(X, Y)/\mathrm{Var}(X)$. Write down an estimating equation for θ and verify that the population moment condition holds at θ_0.

(b) Let $\theta = F^{-1}(0.5)$ be the population median. Propose an estimating function $U(O; \theta)$ and verify the population moment condition. (*Hint*: consider $U(O; \theta) = \mathbf{1}(Y \le \theta) - 0.5$.)

(c) Show that the OLS estimator and the IPW estimator of the ATE are both special cases of the general M-estimator framework.

2. Asymptotic linearity and the CLT.

(a) Let $\hat{\psi} = \bar{Y}$ be the sample mean with $\mathbb{E}Y = \psi$ and $\mathrm{Var}(Y) = \sigma^2 < \infty$. Write down the influence function, state its asymptotic distribution, and give a consistent estimator of $\mathrm{Var}(\hat{\psi})$.

(b) Suppose $\hat{\psi}_1$ and $\hat{\psi}_2$ are two asymptotically linear estimators with influence functions φ_1 and φ_2. Show that $\hat{\psi}_\lambda = \lambda\hat{\psi}_1 + (1-\lambda)\hat{\psi}_2$ is also asymptotically linear for any fixed $\lambda \in [0, 1]$, and find its influence function.

(c) Use part (b) to explain why, among linear combinations of two estimators, the one minimizing asymptotic variance is generally not the simple average ($\lambda = 1/2$).

3. Influence functions for causal estimators. Consider the ATE $\psi = \mathbb{E}\{Y(1) - Y(0)\}$ identified under strong ignorability.

(a) Assume $\pi(X)$ and $\mu_t(X)$ are both known. Compute the influence functions of: (i) the regression estimator $\hat{\psi}_{\text{reg}} = \mathbb{P}_n\{\mu_1(X) - \mu_0(X)\}$; (ii) the IPW estimator $\hat{\psi}_{\text{IPW}} = \mathbb{P}_n\{TY/\pi(X) - (1-T)Y/(1-\pi(X))\}$.

(b) Under what conditions on the data-generating process will $\hat{\psi}_{\text{reg}}$ have a smaller asymptotic variance than $\hat{\psi}_{\text{IPW}}$?

(c) Using the EIF Equation 10.7, verify that it has mean zero under P by computing $\mathbb{E}[\varphi^*(O)]$.

4. Variance estimation.

(a) Let $\hat{\varphi}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) - \hat{\psi}_{\text{reg}}$ be the estimated influence values for the regression estimator. Write down the plug-in variance estimator $\hat{V}$ and simplify. Under what conditions is $\hat{V}$ consistent for $\mathrm{Var}(\hat{\psi}_{\text{reg}})$?

(b) A researcher reports $\hat{\psi} = 2.4$ and $\hat{V} = 0.09$ based on $n = 400$ observations. Compute a 95% Wald confidence interval. Is the treatment effect statistically distinguishable from zero at the 5% level?

(c) Explain why simply plugging in $\hat{\mu}_t$ and $\hat{\pi}$ into the influence function formula may yield an inconsistent variance estimator when these nuisance estimators converge at a slower rate than $n^{-1/2}$.

5. Efficiency comparisons.

(a) Suppose φ_{reg} and φ_{IPW} denote the influence functions of the regression and IPW estimators. Without calculation, explain why neither estimator is always more efficient than the other.

(b) The semiparametric efficiency bound for the ATE is $\mathbb{E}[\varphi^*(O)^2]$, where φ^* is given in Equation 10.7. Show that $\mathbb{E}[\varphi^*(O)^2] \le \mathbb{E}[\varphi_{\text{IPW}}(O)^2]$ by expanding the squared EIF. (*Hint*: use the law of iterated expectations to show that the cross-terms cancel.)

(c) Give one practical reason why an efficient estimator based on the EIF may not always be preferred over a simpler, less efficient estimator.

Part III

Estimation

Chapter 10

Doubly Robust Estimation and Semiparametric Efficiency

Learning Objectives

By the end of this chapter, students should be able to:

1. Derive the AIPW estimator as a bias-corrected prediction estimator, and explain the role of each term.
2. Prove the double robustness property using the law of iterated expectations.
3. Describe the class of augmented IPW estimators indexed by arbitrary control functions $b_t(x)$, verify unbiasedness for any b_t, and identify the optimal choice that minimizes total variance.
4. Interpret the augmentation step as a Hilbert-space projection onto the orthogonal complement of the augmentation space, and derive the Pythagorean variance decomposition.
5. Carry out the projection argument in the causal inference setting, starting from the Horvitz–Thompson estimator.
6. State the semiparametric efficiency bound for the ATE and explain why the AIPW estimating function is the efficient influence function.
7. Derive the two approaches to building double robustness directly into the outcome model: IPW-weighted regression and the augmented model with clever covariate.
8. Identify the product-rate condition as the key sufficient condition for nuisance estimation error to be asymptotically negligible, and construct a consistent variance estimator and Wald confidence interval.

10.1 Why Combine Outcome Regression and Weighting?

Under consistency, conditional exchangeability, and positivity, the ATE $\tau = \mathbb{E}\{Y(1) - Y(0)\}$ is identified by either the regression formula $\tau = \mathbb{E}\{\mu_1(X) - \mu_0(X)\}$ or the Horvitz–Thompson formula $\tau = \mathbb{E}\left\{\frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)}\right\}$.

Outcome regression estimates $\mu_1(x)$ and $\mu_0(x)$ and averages $\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)$ over the sample. **IPW** estimates $\pi(x)$ and reweights observed outcomes. Each has weaknesses: regression can be biased under misspecification; IPW can be unstable when estimated propensity scores are near 0 or 1 (Rosenbaum and Rubin 1983).

This chapter develops a third strategy: combine both models to construct an estimator that is consistent when *either* model is correct, and achieves the semiparametric efficiency bound when both are correct. We derive the same estimator in three complementary ways: as a bias-corrected prediction estimator (Section 10.2), as the optimal member of a class of augmented estimators (Section 10.4), and as the efficient influence function in a semiparametric model (Section 10.7).

10.2 The Prediction Estimator and Its Bias

Working with generic *working regression functions* $m_t(x)$, the **prediction estimator** is $\hat{\tau}_{\text{pred}}(m) = n^{-1}\sum_{i=1}^{n}\{m_1(X_i) - m_0(X_i)\}$. The population bias is:

$$\tau_{\text{pred}}(m) - \tau = \mathbb{E}\big[(m_1(X) - \mu_1(X)) - (m_0(X) - \mu_0(X))\big]. \tag{10.1}$$

If a propensity score estimator $\hat{\pi}$ is available, we can estimate and subtract this bias. The bias-corrected prediction estimator is:

$$\hat{\tau}_{\text{AIPW}} = \hat{\tau}_{\text{pred}} - \widehat{\text{Bias}} = \hat{\mu}_{1,\text{dr}} - \hat{\mu}_{0,\text{dr}}, \tag{10.2}$$

where:

$$\hat{\mu}_{1,\text{dr}} = \frac{1}{n}\sum_{i=1}^{n}\frac{T_i}{\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_1(X_i)\} + \frac{1}{n}\sum_{i=1}^{n}\hat{\mu}_1(X_i),$$

$$\hat{\mu}_{0,\text{dr}} = \frac{1}{n}\sum_{i=1}^{n}\frac{1-T_i}{1-\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_0(X_i)\} + \frac{1}{n}\sum_{i=1}^{n}\hat{\mu}_0(X_i).$$

Remark: Two Roles of the Augmentation Terms

From the bias-correction perspective, $T_i\{Y_i - \hat{\mu}_1(X_i)\}/\hat{\pi}(X_i)$ are IPW-weighted residuals estimating prediction error. From an estimating-equation perspective, they make the estimating function orthogonal to nuisance perturbations. The estimated bias term has zero expectation when the outcome model is correct; the bias-corrected estimator is unbiased when the propensity model is correct. Hence the name *doubly robust*.

10.3 The Augmented IPW Estimator

Notation. $\mu_t(x)$ denotes a generic *working* outcome regression; the truth is $\mu_t^*(x)$. The propensity score $\pi(x)$ is always the *true* propensity.

The AIPW estimator solves $\mathbb{P}_n\{\phi(O;\tau,\hat{\eta})\} = 0$ where:

$$\phi(O;\tau,\eta) = \left[\frac{T}{\pi(X)}\{Y - \mu_1(X)\} + \mu_1(X)\right] - \left[\frac{1-T}{1-\pi(X)}\{Y - \mu_0(X)\} + \mu_0(X)\right] - \tau. \tag{10.3}$$

Solving explicitly:

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{T_i}{\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_1(X_i)\} - \frac{1-T_i}{1-\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_0(X_i)\}\right]. \tag{10.4}$$

Theorem: Double Robustness

Let $\phi(O;\tau,\eta)$ be the estimating function in Equation 10.3. Under consistency, conditional exchangeability, and positivity, $\mathbb{E}\{\phi(O;\tau,\eta)\} = 0$ if either:

1. $\mu_t(x) = \mu_t^*(x)$ for $t = 0, 1$, regardless of whether $\pi(x)$ is correctly specified; or
2. $\pi(x) = P(T = 1 \mid X = x)$, regardless of whether μ_0 and μ_1 are correctly specified.

Proof

Case 1: outcome regression correct. If $\mu_t(X) = \mathbb{E}(Y \mid T = t, X)$, then $\mathbb{E}\Big[\frac{T}{\pi(X)}\{Y - \mu_1(X)\}\,\Big|\,X\Big] = \frac{P(T=1|X)}{\pi(X)}\mathbb{E}\{Y - \mu_1(X) \mid T = 1, X\} = 0$, and similarly for the control arm. Therefore $\mathbb{E}\{\phi\} = \mathbb{E}\{\mu_1(X) - \mu_0(X)\} - \tau = 0$.
Case 2: propensity score correct. Suppose $\pi(X) = P(T = 1 \mid X)$. For the treated bracket:

$$\mathbb{E}\left[\frac{T}{\pi(X)}\{Y - \mu_1(X)\} + \mu_1(X)\,\middle|\,X\right] = \frac{\mathbb{E}[TY \mid X]}{\pi(X)} - \mu_1(X)\cdot\frac{\mathbb{E}[T \mid X]}{\pi(X)} + \mu_1(X).$$

Using $\mathbb{E}[T \mid X] = \pi(X)$ (so the μ_1 terms cancel) and $\mathbb{E}[TY \mid X] = \pi(X)\mu_1^*(X)$, this equals $\mu_1^*(X)$. By symmetry, the control bracket gives $\mu_0^*(X)$. Hence $\mathbb{E}\{\phi \mid X\} = \mu_1^*(X) - \mu_0^*(X) - \tau$, and taking expectations gives $\mathbb{E}\{\phi\} = 0$. $\square$

Remark: The Algebra of Case 2

The key cancellation: $\mu_1(X)$ enters the bracket twice with equal and opposite conditional expectations because $\mathbb{E}[T/\pi(X) \mid X] = 1$ when π is correct. The bracket "forgets" μ_1 entirely and converges to $\mu_1^*(X)$. A wrong μ_t is subtracted out by the correct π.

10.4 A Class of Augmented Estimators

For any square-integrable functions $b_1(x)$ and $b_0(x)$, define:

$$\hat{\mu}_{1,b} = \frac{1}{n}\sum_{i=1}^{n}\frac{T_i}{\pi(X_i)}Y_i - \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{T_i}{\pi(X_i)} - 1\right\}b_1(X_i),$$

$$\hat{\mu}_{0,b} = \frac{1}{n}\sum_{i=1}^{n}\frac{1-T_i}{1-\pi(X_i)}Y_i - \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{1-T_i}{1-\pi(X_i)} - 1\right\}b_0(X_i),$$

and $\hat{\tau}_b = \hat{\mu}_{1,b} - \hat{\mu}_{0,b}$. The correction is mean-zero under the true π for any b_t, so every member is unbiased; only the variance depends on b_t. When $b_t = \mu_t$, $\hat{\mu}_{t,b}$ equals the AIPW form.

Theorem: Unbiasedness and Variance of the AIPW Class

Under conditional exchangeability, SUTVA, and positivity, with the true propensity score, for any square-integrable b_0, b_1: (1) $\mathbb{E}(\hat{\tau}_b) = \tau$; and (2):

$$\mathrm{Var}(\hat{\tau}_b) = \frac{1}{n}\,\mathbb{E}\left[\left(\frac{1}{\pi(X)} - 1\right)\{Y(1) - b_1(X)\}^2 + 2\{Y(1) - b_1(X)\}\{Y(0) - b_0(X)\} + \left(\frac{1}{1-\pi(X)} - 1\right)\{Y(0) - b_0(X)\}^2\right] \tag{10.5}$$

Theorem: Optimal Control Functions

The control functions $b_1^*(X) = \mathbb{E}\{Y(1) \mid X\}$ and $b_0^*(X) = \mathbb{E}\{Y(0) \mid X\}$ minimize the total variance in Equation 10.5. The proof shows that the b_t-dependent term equals a perfect square $\mathbb{E}\left[\left(\sqrt{\frac{1-\pi}{\pi}}u + \sqrt{\frac{\pi}{1-\pi}}v\right)^2\right] \geq 0$, where $u = \mu_1^* - b_1$ and $v = \mu_0^* - b_0$, which equals zero iff $b_t = \mu_t^*$.

10.5 The Projection Interpretation

The optimality of AIPW has an elegant geometric interpretation in terms of projections in a Hilbert space of estimators (Tsiatis 2006, Ch. 2). Define the **augmentation space** $\Lambda = \{\hat{b} : \mathbb{E}(\hat{b}) = 0\}$.

Theorem: Optimal Projection

The optimal correction is $\hat{b}^* = \Pi(\hat{\theta}_0 \mid \Lambda)$, characterized by $\hat{b}^* \in \Lambda$ and $\mathrm{Cov}(\hat{\theta}_0 - \hat{b}^*, \hat{b}) = 0$ for all $\hat{b} \in \Lambda$. The optimal estimator $\hat{\theta}_{\mathrm{opt}} = \Pi(\hat{\theta}_0 \mid \Lambda^{\perp})$ satisfies the **Pythagorean identity**:

$$\mathrm{Var}(\hat{\theta}_0) = \mathrm{Var}(\hat{\theta}_{\mathrm{opt}}) + \mathrm{Var}(\hat{b}^*). \tag{10.6}$$

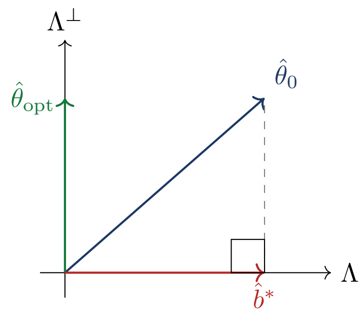

Figure 10.1: Geometric illustration of the projection. $\hat\theta_0$ decomposes into $\hat\theta_{\rm opt} \in \Lambda^\perp$ and $\hat b^* \in \Lambda$. The Pythagorean identity gives $\mathrm{Var}(\hat\theta_0) = \mathrm{Var}(\hat\theta_{\rm opt}) + \mathrm{Var}(\hat b^*)$.

10.6 Projection in the Causal Inference Setting

We now specialize to the ATE, with $\pi(X)$ known. Decompose the Horvitz–Thompson (HT) estimator as $\hat\tau_{\rm HT} = \hat\mu_{1,\rm HT} - \hat\mu_{0,\rm HT}$. Define arm-specific augmentation spaces:

$$\Lambda_t = \left\{ n^{-1} \sum_{i=1}^n \left(\frac{T_i^{(t)}}{\pi_i^{(t)}} - 1 \right) b_t(X_i) : b_t \in \mathcal{L}^2 \right\}, \quad \Lambda = \Lambda_1 \oplus \Lambda_0.$$

Theorem: Arm-wise Projections

The optimal arm-wise corrections are given by $b_t^*(x) = \mathbb{E}\{Y(t) \mid x\}$:

$$\Pi(\hat\mu_{t,\rm HT} \mid \Lambda_t) = n^{-1} \sum_{i=1}^n \left(\frac{T_i^{(t)}}{\pi_i^{(t)}} - 1 \right) \mu_t^*(X_i).$$

The orthogonality condition $\mathbb{E}[r_{t,i}\, b_i] = 0$ holds because $\mathbb{E}\{\mu_t^*(X_i) - \mu_t^*(X_i) \mid X_i\} = 0$ and $\mathbb{E}[T_i/\pi_i \cdot (T_i/\pi_i - 1) \mid X_i] = 1/\pi_i - 1$.

The arm-wise optimal estimators are:

$$\hat\mu_{t,\rm opt} = \frac{1}{n} \sum_{i=1}^n \left[\frac{T_i^{(t)}}{\pi_i^{(t)}} \{Y_i - \mu_t^*(X_i)\} + \mu_t^*(X_i) \right],$$

and $\hat\tau_{\rm opt} = \hat\mu_{1,\rm opt} - \hat\mu_{0,\rm opt}$ is precisely the AIPW estimator Equation 10.4, confirming the Optimal Control Functions theorem by a different route.

10.7 The Efficient Influence Function and Semiparametric Efficiency

Under the nonparametric model for $O = (X, T, Y)$, the efficient influence function for the ATE is:

$$\varphi_{\rm eff}(O) = \frac{T}{\pi(X)}\{Y - \mu_1(X)\} - \frac{1-T}{1-\pi(X)}\{Y - \mu_0(X)\} + \mu_1(X) - \mu_0(X) - \tau. \tag{10.7}$$

Comparing Equation 10.7 with Equation 10.3, the AIPW estimating function *is* the efficient influence function $\varphi^*(O)$ from Chapter 9. The augmentation space Λ is the nuisance tangent space of the nonparametric model.

Theorem: Semiparametric Efficiency Bound for the ATE

Under the nonparametric model for $O = (X, T, Y)$, every regular asymptotically linear estimator $\hat{\tau}$ of the ATE satisfies $\mathrm{Avar}(\sqrt{n}(\hat{\tau} - \tau)) \geq \mathbb{E}\{\varphi_{\text{eff}}(O)^2\}$, and the bound is attained by estimators whose influence function equals φ_{eff}.

This is a specialization of the Hájek–Le Cam convolution theorem; see Bickel et al. (1993) or Vaart (1998, Ch. 25) for a rigorous development.

Remark: EIF as the Semiparametric Score

The efficient influence function plays the same role in semiparametric theory that the score plays in parametric models: its variance gives the information lower bound. The three derivations in this chapter — bias correction, optimal augmentation, and semiparametric identification — converge on the same estimating function. This is the deepest explanation of why AIPW occupies a central place in the causal inference toolkit.

10.8 Doubly Robust Regression: Weighted and Augmented Approaches

The **internal bias calibration** (IBC) conditions (Firth and Bennett 1998) enforce double robustness directly within the outcome model fit: when satisfied, the prediction estimator coincides with AIPW without a separate augmentation step:

$$\sum_{i=1}^{n} \frac{T_i}{\hat{\pi}_i}\{Y_i - \hat{\mu}_1(X_i)\} = 0, \qquad \sum_{i=1}^{n} \frac{1 - T_i}{1 - \hat{\pi}_i}\{Y_i - \hat{\mu}_0(X_i)\} = 0. \tag{10.8}$$

Weighted regression. Use IPW-weighted least squares: minimize $\sum_{i=1}^{n} \frac{T_i}{\hat{\pi}_i}\{Y_i - \mu_1(X_i; \theta_1)\}^2$ for arm $t = 1$. The normal equation for the intercept is exactly the IBC condition Equation 10.8. IPW-weighted fitted values automatically satisfy IBC for any model containing a constant (Robins et al. 1994; Bang and Robins 2005).

Augmented model with clever covariate. Let $\hat{\mu}_1^{(0)}(X_i)$ be any initial fit. Include the **clever covariate** $\hat{\pi}_i^{-1}$ (Laan and Rubin 2006; Laan and Rose 2011) and run OLS among treated units:

$$Y_i = \alpha_1 + \beta_1 \hat{\mu}_1^{(0)}(X_i) + \gamma_1 \hat{\pi}_i^{-1} + e_i(1), \qquad T_i = 1. \tag{10.9}$$

The fitted model is $\hat{\mu}_1(X_i) = \hat{\alpha}_1 + \hat{\beta}_1 \hat{\mu}_1^{(0)}(X_i) + \hat{\gamma}_1 \hat{\pi}_i^{-1}$. The normal equation for $\hat{\gamma}_1$ is exactly the IBC condition, so the prediction estimator can be written in AIPW form:

$$\hat{\mu}_1^{\text{pred}} = \frac{1}{n}\sum_{i=1}^{n} \hat{\mu}_1(X_i) + \frac{1}{n}\sum_{i=1}^{n} \frac{T_i}{\hat{\pi}_i}\{Y_i - \hat{\mu}_1(X_i)\}. \tag{10.10}$$

Remark: Behavior Under Correct Outcome Model

If $\hat{\mu}_1^{(0)}$ is correctly specified, then $\hat{\alpha}_1 \xrightarrow{p} 0$, $\hat{\beta}_1 \xrightarrow{p} 1$, and $\hat{\gamma}_1 \xrightarrow{p} 0$. Including the clever covariate carries no asymptotic efficiency cost.

Remark: Connection to TMLE

This approach captures the targeting idea underlying TMLE (Laan and Rubin 2006; Laan and Rose 2011). Standard TMLE uses the fixed-offset form (coefficient on $\hat{\mu}_1^{(0)}$ fixed at 1). Model Equation 10.9 goes further by freely estimating $\hat{\beta}_1$. See Laan and Rose (2018) for a comprehensive treatment.

10.9 Lab: Simulation Study

This lab compares four estimators of the ATE: $\hat{\tau}_{\text{HT}}$, $\hat{\tau}_{\text{AIPW}}$, the fixed-offset augmented-model estimator $\hat{\tau}_{\text{aug}}$, and the improved augmented-model estimator $\hat{\tau}_{\text{aug}^+}$ of Equation 10.9. A 2×2 design over nuisance-model correctness — OR correct or misspecified, crossed with PS correct or misspecified — demonstrates double robustness directly. The lab also reports empirical 95% Wald coverage, illustrating the distinction between double-robust consistency and valid asymptotic inference.

Data-generating process. $n = 1000$ observations. Draw $X_i \sim N(0,1)$, then $T_i \mid X_i \sim \text{Bernoulli}(\pi^*(X_i))$ with:

$$\pi^*(x) = \text{expit}\{0.2\,x + 0.2\,(x^2 - 1)\}.$$

The potential outcomes are $Y_i(1) = 1 + X_i + 0.5X_i^2 + \varepsilon_i(1)$ and $Y_i(0) = X_i + 0.5X_i^2 + \varepsilon_i(0)$, with $\varepsilon_i(t) \sim N(0,1)$ i.i.d., true ATE $\tau = 1$. The quadratic term enters both the true OR and true PS; a linear-only fit omits it.

The 2×2 design. The fitted $\hat{\pi}$ is clipped to $[10^{-3}, 1 - 10^{-3}]$.

Scenario	OR fit	PS fit
S1	correct: $Y \sim (1, X, X^2)$	correct: $T \sim (1, X, X^2)$
S2	correct: $Y \sim (1, X, X^2)$	misspecified: $T \sim (1, X)$
S3	misspecified: $Y \sim (1, X)$	correct: $T \sim (1, X, X^2)$
S4	misspecified: $Y \sim (1, X)$	misspecified: $T \sim (1, X)$

Results ($B = 2000$ replications, `set.seed(2025)`). Bias, Var, MSE in units of 10^{-3}; Cov = empirical 95% Wald coverage.

Scenario	Metric	$\hat{\tau}_{\text{HT}}$	$\hat{\tau}_{\text{AIPW}}$	$\hat{\tau}_{\text{aug}}$	$\hat{\tau}_{\text{aug}^+}$
S1: OR , PS	Bias	0.55	−0.44	−0.45	−0.45
	Var	6.82	4.25	4.25	4.30
	MSE	6.82	4.25	4.25	4.30
	Cov (%)	99.9	95.0	94.9	94.5
S2: OR , PS	Bias	188.56	0.62	0.62	0.62
	Var	6.32	4.33	4.33	4.33
	MSE	41.87	4.32	4.32	4.32
	Cov (%)	67.4	94.2	94.2	94.2
S3: OR , PS	Bias	1.46	2.02	2.25	−25.26
	Var	5.74	4.71	4.41	8.02
	MSE	5.74	4.71	4.41	8.66
	Cov (%)	99.9	98.4	98.2	93.2
S4: OR , PS	Bias	188.42	188.77	188.62	−5.42
	Var	6.28	6.30	6.29	4.26
	MSE	41.78	41.93	41.87	4.29
	Cov (%)	67.5	33.1	33.1	94.7

S1 (both correct). All four estimators are approximately unbiased. $\hat{\tau}_{\text{HT}}$ has about 60% higher MSE (6.82 vs 4.25×10^{-3}) because it discards the outcome model. The three augmented estimators are essentially equivalent, confirming that $(\hat{\alpha}_t, \hat{\beta}_t, \hat{\gamma}_t) \to (0, 1, 0)$ when the OR is correct. HT's coverage is above nominal because its influence function has larger variance than φ_{eff}.

S2 (OR correct, PS misspecified). HT is badly biased (MSE $\approx 42 \times 10^{-3}$, coverage 67%). The three augmented estimators are indistinguishable: when the OR is correct they converge to the same limit regardless of PS specification. This is the **first direct demonstration of double robustness**: PS misspecification is absorbed by the correctly specified OR model.

S3 (OR misspecified, PS correct). HT, AIPW, and $\hat{\tau}_{\text{aug}}$ remain consistent through the correct PS (biases under 3×10^{-3}, coverage close to nominal). $\hat{\tau}_{\text{aug}}$ achieves the lowest MSE (4.41×10^{-3}): the targeted regression reduces residual variance without disturbing the IBC constraint. $\hat{\tau}_{\text{aug}^+}$ shows a finite-sample bias (-25×10^{-3}) from high-leverage clever-covariate values. This is the **second double-robustness demonstration**: OR misspecification is absorbed by the correctly specified PS.

S4 (both misspecified). Double robustness provides no guarantee. HT, AIPW, and $\hat{\tau}_{\text{aug}}$ are all badly biased (MSE $\approx 42\times10^{-3}$); Wald coverage collapses to 33% for AIPW and aug — the variance estimator is concentrated around the wrong value. The apparent "recovery" of $\hat{\tau}_{\text{aug}^+}$ (coverage 94.7%) is a DGP-specific artifact: the regressors $(1, \hat{\mu}_t^{(0)}(X), \hat{\pi}^{-1}(X))$ happen to capture the omitted quadratic curvature through the nonlinearity of $\hat{\pi}^{-1}$, but this does **not** indicate triple robustness.

Takeaway. Comparing S2–S3 against S4 is the crisp illustration of the Double Robustness theorem: AIPW is consistent whenever at least one nuisance model is correct, and only S4 breaks the guarantee. Wald coverage mirrors this pattern. The lab also shows that $\hat{\tau}_{\text{aug}}$ can reduce MSE in S3 by shrinking residual variance through IBC-enforced regression — the practical motivation for targeting methods such as TMLE.

10.10 Asymptotic Inference with Estimated Nuisance Functions

Under suitable regularity conditions:

$$\sqrt{n}(\hat{\tau}_{\text{AIPW}} - \tau) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varphi_{\text{eff}}(O_i) + o_p(1) \xrightarrow{d} N(0,\ \mathbb{E}\{\varphi_{\text{eff}}(O)^2\}).$$

Neyman orthogonality implies nuisance estimation error enters only through a second-order remainder. A sufficient condition for this remainder to be negligible is the **product-rate condition**:

$$\|\hat{\pi} - \pi\| \cdot \|\hat{\mu}_t - \mu_t\| = o_p(n^{-1/2}), \qquad t = 0, 1, \tag{10.11}$$

where $\|\cdot\|$ denotes the $L_2(P)$ norm (equivalently, the root-mean-squared-error under the data distribution). This allows each nuisance estimator to converge at rate as slow as $n^{-1/4}$. In modern applications, this is usually paired with **cross-fitting**; Chapter 11 develops this in detail.

> Remark: Neyman Orthogonality
>
> The first-order insensitivity of the AIPW estimating function to nuisance perturbations is an instance of *Neyman orthogonality*. Chapter 11 exploits this via cross-fitting, yielding the *double/debiased machine learning* (DML) estimator (Chernozhukov et al. 2018).

Compute plug-in influence values:

$$\hat{\varphi}_i = \frac{T_i}{\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_1(X_i)\} - \frac{1-T_i}{1-\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_0(X_i)\} + \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) - \hat{\tau}_{\text{AIPW}}, \tag{10.12}$$

and set:

$$\hat{V} = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\hat{\varphi}_i - \bar{\varphi})^2, \qquad \bar{\varphi} = \frac{1}{n}\sum_{i=1}^{n}\hat{\varphi}_i. \tag{10.13}$$

The Wald CI is $\hat{\tau}_{\text{AIPW}} \pm z_{1-\alpha/2}\sqrt{\hat{V}}$.

> Remark: Double Robustness versus Efficient Inference
>
> It is important not to conflate two distinct results. *Double-robust consistency* requires only that *one* of the two nuisance models be correctly specified; no rate conditions are imposed. *Efficient asymptotic inference* — the $\sqrt{n}$-normal expansion and the Wald interval based on $\hat{V}$ — is a strictly stronger requirement. It requires *both* nuisance functions to be estimated consistently at rates satisfying the product-rate condition Equation 11.7, together with overlap and moment conditions. Plugging estimated nuisance functions into Equation 10.13 produces valid standard errors only

when the asymptotic linear representation holds; that representation does *not* follow from double robustness alone. In particular, if the outcome regression is misspecified but the propensity score is correct, the AIPW estimator remains consistent but the influence-function variance estimator targets the variance of an estimator whose nuisance functions converge to their true values, not to misspecified limits. The simulation in Section 11.10 illustrates this: coverage is approximately nominal in S2 and S3 (where double robustness operates), but this relies on the surviving nuisance model being correctly specified, not just present. Careful rate verification — or cross-fitting as in Chapter 11 — is essential before reporting the Wald interval.

Extreme Propensity Scores

When $\hat{\pi}(X_i)$ is near 0 or 1, the IPW weights become large and can destabilize the estimator. Practical remedies: overlap diagnostics, truncation of extreme propensity scores, or restricting to a subgroup with adequate overlap. The augmented model approach of Section 10.8 provides a complementary strategy.

10.11 Comparison of Regression, IPW, and AIPW

Estimator	Uses μ_t	Uses π	Consistent if	Main weakness
Regression	Yes	No	outcome model correct	Sensitive to OR misspecification
IPW	No	Yes	propensity model correct	Unstable under weak overlap
AIPW	Yes	Yes	either model correct	Requires both nuisances and careful inference

10.12 Chapter Summary

Symbol	Meaning
τ	ATE $= \mathbb{E}\{Y(1) - Y(0)\}$
$\mu_t(x)$	Outcome regression $\mathbb{E}(Y \mid T = t, X = x)$
$\pi(x)$	Propensity score $P(T = 1 \mid X = x)$
$b_t(x)$	Control function; optimal $b_t^*(x) = \mathbb{E}\{Y(t) \mid X = x\}$
$\hat{\tau}_{\mathrm{AIPW}}$	AIPW estimator Equation 10.4
Λ	Augmentation space $\{\hat{b} : \mathbb{E}(\hat{b}) = 0\}$
$\varphi_{\mathrm{eff}}(O)$	Efficient influence function Equation 10.7
$\hat{\pi}_i^{-1}$	Clever covariate; enforces IBC via its normal equation
IBC	Internal bias calibration conditions Equation 10.8

1. The prediction estimator bias can be estimated and subtracted using the propensity score, yielding the AIPW estimator.
2. AIPW has two key properties: **double robustness** (consistent when either model is correct) and **class-optimality** ($b_t^* = \mu_t^*$ minimizes variance).
3. The optimality follows also from a Hilbert-space projection onto $\Lambda^{\perp}$; the Pythagorean identity governs the variance reduction.
4. The AIPW estimating function *is* the efficient influence function; three derivations converge on the same object.
5. Double robustness can be built directly into the outcome regression via IPW-weighted regression or including the clever covariate $\hat{\pi}_i^{-1}$ (TMLE connection).

6. The simulation (Section 11.10) demonstrates double robustness across a 2×2 design: AIPW remains consistent whenever at least one nuisance model is correct (S1–S3) and loses the guarantee when both are misspecified (S4). Wald coverage is approximately nominal in S1–S3 but collapses in S4, illustrating that double-robust consistency and valid asymptotic inference are distinct properties.
7. For asymptotic inference, the central issue is whether the effect of estimating nuisance functions is asymptotically negligible. Neyman orthogonality yields a second-order remainder, and the product-rate condition Equation 11.7 is sufficient for root-n inference. In modern machine-learning settings, cross-fitting is often used to make these conditions more plausible; Chapter 11 develops this approach.

10.13 Exercises

1. Bias of the prediction estimator.

(a) Verify the bias formula Equation 10.1 by writing $\tau_{\text{pred}}(m) - \tau$ as a difference of working-model errors and simplifying.
(b) Under what condition on the propensity score model does $\widehat{\text{Bias}}(\hat{\tau}_{\text{pred}})$ have zero expectation? Give a careful argument using iterated expectations.
(c) Construct a simple example (binary X, binary T, binary Y) in which the estimated bias is nonzero and the outcome model is misspecified, but the AIPW estimator is still consistent.

2. The AIPW class and optimal control functions.

(a) Verify that $\hat{\tau}_b$ with $b_t = \mu_t$ equals the AIPW estimator Equation 10.4.
(b) Using Equation 10.5, show that $b_t^*(x) = \mathbb{E}\{Y(t) \mid x\}$ minimizes conditional variance by completing the square in b_t.
(c) Suppose $\pi(x) = 1/2$ for all x. Simplify Equation 10.5 and interpret the result.

3. Double robustness.

(a) Verify Case 1 of the Double Robustness theorem: with $\mu_t = \mu_t^*$ and arbitrary π, show $\mathbb{E}\{\phi\} = 0$.
(b) Verify Case 2: with $\pi = \pi^*$ and arbitrary μ_t, show $\mathbb{E}\{\phi\} = 0$.
(c) Provide a counterexample showing $\mathbb{E}\{\phi\} \neq 0$ when both models are misspecified.

4. Projection and the Pythagorean identity.

(a) Verify $\text{Cov}(\hat{\theta}_{\text{opt}}, \hat{b}^*) = 0$ from the definition of $\hat{b}^*$.
(b) Deduce Equation 10.6 from the decomposition $\hat{\theta}_0 = \hat{\theta}_{\text{opt}} + \hat{b}^*$.
(c) Explain why $\hat{\theta}_{\text{opt}}$ is still unbiased for θ even though $\hat{b}^* \in \Lambda$ is subtracted.

5. Semiparametric efficiency.

(a) Show $\mathbb{E}\{\varphi_{\text{eff}}(O)^2\} \leq \mathbb{E}\{\varphi_{\text{IPW}}(O)^2\}$ by writing $\varphi_{\text{IPW}} = \varphi_{\text{eff}} + (\varphi_{\text{IPW}} - \varphi_{\text{eff}})$ and showing the cross term vanishes.
(b) Interpret the efficiency gain $\mathbb{E}\{\varphi_{\text{IPW}}^2\} - \mathbb{E}\{\varphi_{\text{eff}}^2\}$ in terms of the variance of the outcome regression.
(c) Give one reason why an efficient estimator based on φ_{eff} may still be disfavored in practice.

6. Augmented model and the clever covariate.

(a) Show that the normal equation for $\hat{\gamma}_1$ in the augmented model Equation 10.9 is exactly the IBC condition Equation 10.8, and conclude that the prediction estimator can always be written in AIPW form Equation 10.10.
(b) Explain why $\hat{\gamma}_1 \xrightarrow{p} 0$ when the initial model $\hat{\mu}_1^{(0)}$ is correctly specified.
(c) Explain in words why $\hat{\pi}_i^{-1}$ is called the clever covariate: what bias does it absorb, and why does this make the prediction estimator doubly robust?

7. Coverage of the Wald confidence interval (computational). Use the DGP of Section 11.10 with the propensity score generalized to $\pi^*(x) = \text{expit}\{0.2\,x + \gamma(x^2 - 1)\}$, so that the baseline ($\gamma = 0.2$) coincides with the lab and larger γ degrades overlap. Implement the AIPW estimator and variance estimator Equation 10.13. Following Scenario S3 of the lab, use logistic regression of T on $(1, X, X^2)$ for $\hat{\pi}$ (correct PS) and OLS of Y on $(1, X)$ separately in each arm for $\hat{\mu}_t$ (misspecified OR); truncate $\hat{\pi}$ to $[10^{-3}, 1 - 10^{-3}]$.

(a) With $\gamma = 0.2$, $n = 500$, $B = 2000$ replications, compute empirical coverage of the 95% Wald interval $\hat{\tau}_{\text{AIPW}} \pm z_{0.975}\sqrt{\hat{V}}$. Compare the average SE $\overline{\sqrt{\hat{V}}}$ with the empirical SD of $\hat{\tau}_{\text{AIPW}}$ across replications.

(b) Repeat with $\gamma \in \{0.5, 1.0\}$, producing increasingly weak overlap at large $|X|$. Report coverage, average SE, and empirical SD. What do you observe?

(c) Which assumption of Section 10.10 is stressed as γ grows, and which of the remedies in the Extreme Propensity Scores warning would you try first? (No simulation required; a paragraph suffices.)

Chapter 11

Flexible Nuisance Estimation, Orthogonal Scores, and Cross-Fitting

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain why flexible nuisance estimation can produce misleading inference for a causal parameter even when predictive accuracy is high.
2. Define Neyman orthogonality, verify it for a given estimating function, and explain how it reduces sensitivity to first-order nuisance estimation error.
3. Identify the orthogonal score for the ATE, recognize it as the efficient influence function from Chapter 10, and state the three roles it simultaneously fulfills.
4. Describe sample splitting, explain why independence between nuisance estimation and score evaluation simplifies asymptotic arguments, and articulate its efficiency cost.
5. Describe K-fold cross-fitting, write the cross-fitted moment equation explicitly, and explain how cross-fitting recovers efficiency relative to a single split.
6. State the product-rate condition for the cross-fitted AIPW estimator, connect it to Neyman orthogonality, and explain why it allows each nuisance estimator to converge at rate $n^{-1/4}$.
7. Construct a consistent variance estimator from out-of-fold estimated influence values and form a Wald confidence interval.
8. Articulate what machine learning can and cannot contribute to causal inference, and follow the practical workflow for cross-fitted orthogonal-score estimation.

11.1 Why Flexible Nuisance Estimation Is Both Attractive and Dangerous

In Chapters 9 and 10 we developed estimation and inference using estimating equations, influence functions, doubly robust scores, and semiparametric efficiency. The causal parameter of interest depends on nuisance functions such as the outcome regressions $\mu_t(x) = \mathbb{E}(Y \mid T = t, X = x)$ and the propensity score $\pi(x) = P(T = 1 \mid X = x)$. In many real applications, these may be too complex to model accurately with low-dimensional parametric forms, making flexible methods attractive: penalized regression, splines, random forests, boosting, neural networks, and ensemble learning.

These methods can substantially improve predictive accuracy, but they also create new statistical difficulties: they may converge more slowly than parametric estimators; they may overfit the observed sample; their asymptotic behavior may be difficult to characterize; and naive plug-in inference can fail even when prediction quality is good.

Better nuisance prediction therefore does not automatically imply valid inference for the target causal parameter. The goal of this chapter is to explain how orthogonal scores, sample splitting, and cross-fitting make it possible to combine flexible nuisance estimation with valid large-sample inference.

11.2 Why Naive Plug-In Estimation Can Fail

Suppose the parameter of interest is $\psi = \Psi(\eta)$, where η denotes a nuisance object such as (μ_0, μ_1, π). The *plug-in estimator* $\widehat{\psi}_{\text{plug}} = \Psi(\widehat{\eta})$ has a first-order Taylor expansion around the true nuisance η_0:

$$\widehat{\psi}_{\text{plug}} - \psi = \underbrace{D_\eta \Psi(\eta_0)[\widehat{\eta} - \eta_0]}_{\text{linear term}} + \underbrace{R(\widehat{\eta},\, \eta_0)}_{\text{second-order remainder}}, \tag{11.1}$$

where $D_\eta \Psi(\eta_0)[h]$ denotes the Gateaux derivative and $|R(\widehat{\eta}, \eta_0)| \lesssim \|\widehat{\eta} - \eta_0\|^2$.

The second-order remainder is typically manageable: if $\|\widehat{\eta} - \eta_0\| = o_p(n^{-1/4})$ then $R = o_p(n^{-1/2})$. The obstacle is the *linear term*, proportional to the nuisance estimation error.

Finite-dimensional nuisance. If $\eta \in \mathbb{R}^d$ and $\widehat{\eta} - \eta_0 = O_p(n^{-1/2})$, the linear term is $O_p(n^{-1/2})$ and contaminates the root-n limiting distribution. If the functional satisfies $D_\eta \Psi(\eta_0) = 0$ — *Neyman orthogonality* — the linear term vanishes and the expansion reduces to the second-order remainder alone, $O_p(n^{-1})$, asymptotically negligible.

Infinite-dimensional nuisance. When η belongs to a function space, orthogonality removes the *population bias* but does not control the *empirical process term*. Writing the plug-in estimating equation as $\mathbb{P}_n\{\phi(\,\cdot\,; \psi, \widehat{\eta})\} = 0$, the deviation from the population equation decomposes as:

$$\mathbb{P}_n\{\phi(\,\cdot\,; \psi, \widehat{\eta})\} - \mathbb{P}_n\{\phi(\,\cdot\,; \psi, \eta_0)\} = \underbrace{P\{\phi(\,\cdot\,; \psi, \widehat{\eta}) - \phi(\,\cdot\,; \psi, \eta_0)\}}_{\text{population bias}} + \underbrace{(\mathbb{P}_n - P)\{\phi(\,\cdot\,; \psi, \widehat{\eta}) - \phi(\,\cdot\,; \psi, \eta_0)\}}_{\text{empirical process term}}. \tag{11.2}$$

Neyman orthogonality controls the *population bias*: because the Gateaux derivative vanishes at the truth, the population bias is second order in $\|\widehat{\eta} - \eta_0\|$. For flexible machine-learning estimators, the *empirical process term* can remain non-negligible even as $\widehat{\eta}$ converges consistently. This is why orthogonality and cross-fitting are *complements* rather than substitutes.

> Remark
>
> This problem is not specific to machine learning. It arises whenever nuisance parameters are estimated in an infinite-dimensional model. Machine learning makes the issue more visible because it encourages highly adaptive nuisance estimation; see Chernozhukov et al. (2018) for a systematic treatment.

11.3 Orthogonal Scores

> Definition: Neyman Orthogonality
>
> The score function $\phi(O;\, \psi, \eta)$ is called **orthogonal** (or **Neyman orthogonal**) at the truth (ψ_0, η_0) if the Gateaux derivative of the estimating equation with respect to the nuisance, evaluated at the truth, vanishes:
>
> $$\frac{\partial}{\partial r} \mathbb{E}\{\phi(O;\, \psi_0,\, \eta_0 + rh)\}\Big|_{r=0} = 0$$
>
> for all perturbation directions h in a suitable class.

This means small local perturbations of the nuisance parameter have no first-order effect on the estimating equation at the truth. Orthogonality does not mean nuisance estimation no longer matters: it controls only the population-level bias, and a separate argument (cross-fitting) is needed to rule out the additional bias from evaluating the score on the same sample used to estimate $\widehat{\eta}$.

> Remark: Neyman's Original Insight
>
> The terminology honors Jerzy Neyman, whose work on hypothesis testing introduced the idea of constructing test statistics that are insensitive to nuisance parameters. In the modern semiparametric literature the concept was formalized by Robins et al. (1994) and others, and later made systematic

in the double machine learning framework of Chernozhukov et al. (2018).

11.4 The Orthogonal Score for the Average Treatment Effect

The efficient influence function from Chapter 10 is:

$$\varphi_{\text{eff}}(O;\,\tau,\eta) = \frac{T}{\pi(X)}\{Y-\mu_1(X)\} - \frac{1-T}{1-\pi(X)}\{Y-\mu_0(X)\} + \mu_1(X) - \mu_0(X) - \tau, \tag{11.3}$$

where $\eta = (\mu_0, \mu_1, \pi)$. We now prove this score is Neyman orthogonal.

Lemma: Neyman Orthogonality of the ATE Score

The score $\varphi_{\text{eff}}(O;\,\tau,\eta)$ in Equation 11.3 is Neyman orthogonal at every truth (τ_0, η_0) satisfying the identification assumptions. That is, for every bounded perturbation direction $h(x)$, $g(x)$, or $\ell(x)$:

$$\frac{\partial}{\partial r}\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau_0,\,\mu_0,\,\mu_1,\,\pi+rh)\}\Big|_{r=0} = 0,$$

$$\frac{\partial}{\partial r}\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau_0,\,\mu_0,\,\mu_1+rg,\,\pi)\}\Big|_{r=0} = 0,$$

$$\frac{\partial}{\partial r}\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau_0,\,\mu_0+r\ell,\,\mu_1,\,\pi)\}\Big|_{r=0} = 0.$$

Proof

Throughout, recall $\mathbb{E}(Y \mid T=1,X) = \mu_1(X)$, $\mathbb{E}(Y \mid T=0,X) = \mu_0(X)$, $\mathbb{E}(T \mid X) = \pi(X)$.
Perturbation in π. Replace π by $\pi + rh$ and differentiate. The derivative becomes $\mathbb{E}\Big[-\frac{T\,h(X)}{\pi(X)^2}\{Y-\mu_1(X)\} - \frac{(1-T)\,h(X)}{(1-\pi(X))^2}\{Y-\mu_0(X)\}\Big]$. Conditioning on X: $\mathbb{E}[T\{Y-\mu_1(X)\} \mid X] = \pi(X)\mu_1(X) - \mu_1(X)\pi(X) = 0$, and similarly $\mathbb{E}[(1-T)\{Y-\mu_0(X)\} \mid X] = 0$. Both conditional expectations vanish.
Perturbation in μ_1. Replace μ_1 by $\mu_1 + rg$. The derivative is $\mathbb{E}\Big[g(X)\left(1-\frac{T}{\pi(X)}\right)\Big]$. Conditioning on X: $\mathbb{E}[1 - T/\pi(X) \mid X] = 1 - \pi(X)/\pi(X) = 0$.
Perturbation in μ_0. Replace μ_0 by $\mu_0 + r\ell$. The derivative is $\mathbb{E}\Big[\ell(X)\left(\frac{1-T}{1-\pi(X)} - 1\right)\Big]$. Conditioning on X: $\mathbb{E}[(1-T)/(1-\pi(X)) - 1 \mid X] = (1-\pi(X))/(1-\pi(X)) - 1 = 0$. □

Remark: What the Proof Reveals

Each of the three derivative calculations reduces to a single key fact: the conditional mean of the IPW weight equals one. Specifically, $\mathbb{E}(T/\pi(X) \mid X) = 1$ and $\mathbb{E}((1-T)/(1-\pi(X)) \mid X) = 1$ are immediate consequences of $\mathbb{E}(T \mid X) = \pi(X)$, the engine behind both the balancing theorem and the IPW identification formula (Chapter 6). Neyman orthogonality of the ATE score is therefore a direct consequence of how the propensity score is defined.

The score Equation 11.3 simultaneously fulfills three purposes:

1. It *identifies* τ through the moment condition $\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau,\eta)\} = 0$.
2. It is *doubly robust*, yielding a consistent estimator whenever either the outcome model or the propensity score model is correctly specified.
3. It is the *efficient influence function*, so any estimator based on it achieves the semiparametric efficiency bound.

11.5 Why Reusing the Same Data Can Be Problematic

Define the centered empirical process $\mathbb{G}_n f = \sqrt{n}\,(\mathbb{P}_n - P)f$. With the plug-in AIPW estimating equation, the remainder R_n decomposes exactly as in Equation 11.2:

$$R_n = \underbrace{P\{\varphi_{\text{eff}}(\cdot\,;\tau,\hat{\eta}) - \varphi_{\text{eff}}(\cdot\,;\tau,\eta_0)\}}_{\text{population bias}} + \underbrace{(\mathbb{P}_n - P)\{\varphi_{\text{eff}}(\cdot\,;\tau,\hat{\eta}) - \varphi_{\text{eff}}(\cdot\,;\tau,\eta_0)\}}_{\text{empirical process term}}. \tag{11.4}$$

By Neyman orthogonality (Section 11.4), the population bias is $o_p(n^{-1/2})$ under the product-rate condition. The *empirical process term* $n^{-1/2}\mathbb{G}_n\{f_{\hat{\eta}} - f_{\eta_0}\}$ depends on $\hat{\eta}$ estimated from the same sample. Whether it is $o_p(n^{-1/2})$ depends on how complex the class $\{f_\eta : \eta \in \mathcal{H}\}$ is — a question the Donsker condition answers.

Definition: Donsker Class

A class of measurable functions $\mathcal{F}$ is a **Donsker class** if the empirical process $\{\mathbb{G}_n f : f \in \mathcal{F}\}$ converges weakly in $\ell^\infty(\mathcal{F})$ to a tight Gaussian process. Equivalently, $\sup_{f\in\mathcal{F}} |(\mathbb{P}_n - P)f| = O_p(n^{-1/2})$ and the fluctuations of $n^{1/2}(\mathbb{P}_n - P)$ over $\mathcal{F}$ are stochastically equicontinuous; see Vaart (1998), Ch. 19.

Example: Donsker and Non-Donsker Classes

- **Parametric classes** (e.g., logistic regression models indexed by a finite-dimensional parameter) are Donsker under mild moment conditions.
- **Hölder-smooth function classes** on $[0,1]^d$ with smoothness index $s > d/2$ are Donsker.
- **Indicator classes** $\{x \mapsto \mathbf{1}(x \le t) : t \in \mathbb{R}\}$ are Donsker by the classical theorem.
- **Random forests and neural networks** typically fall outside classical Donsker regimes in the forms used for modern flexible prediction.
- **High-dimensional lasso** with a growing number of selected variables also typically falls outside Donsker regimes.

If $\{f_\eta : \eta \in \mathcal{H}\}$ is Donsker and $\|\hat{\eta} - \eta_0\| \to 0$, then by stochastic equicontinuity $\mathbb{G}_n\{f_{\hat{\eta}} - f_{\eta_0}\} = o_p(1)$, making the empirical process term $o_p(n^{-1/2})$. When the Donsker condition fails, this need not hold: the empirical process term can remain non-negligible even as $\hat{\eta} \to \eta_0$.

Remark: Donsker Conditions and Machine Learning

The Donsker condition fails for random forests, neural networks, and high-dimensional lasso precisely because their function classes are too rich. Cross-fitting sidesteps this entirely: because $\hat{\eta}^{(-k)}$ is independent of the observations in $\mathcal{I}_k$, the empirical process term is replaced by a sum of conditionally independent terms, each $o_p(n^{-1/2})$ by a conditional law of large numbers, with no Donsker assumption required.

11.6 Sample Splitting

Sample splitting resolves the data-reuse problem by construction. Partition $\{1,\dots,n\}$ into a training sample $\mathcal{I}_{\text{train}}$ and an evaluation sample $\mathcal{I}_{\text{eval}}$. Fit nuisance estimators $\hat{\eta}^{(\text{train})}$ on $\mathcal{I}_{\text{train}}$, then define $\hat{\tau}$ by solving the estimating equation on the held-out sample:

$$\frac{1}{|\mathcal{I}_{\text{eval}}|}\sum_{i\in\mathcal{I}_{\text{eval}}} \varphi_{\text{eff}}\big(O_i;\,\tau,\,\hat{\eta}^{(\text{train})}\big) = 0.$$

Because $\hat{\eta}^{(\text{train})}$ is estimated on $\mathcal{I}_{\text{train}}$, it is *independent* of the observations in $\mathcal{I}_{\text{eval}}$. Conditioning on $\hat{\eta}^{(\text{train})}$, the summands in the empirical process term are independent and mean zero, so by the conditional law of large numbers:

$$(\mathbb{P}_{n_e} - P)\{\varphi_{\text{eff}}(\cdot\,;\tau,\hat{\eta}^{(\text{train})}) - \varphi_{\text{eff}}(\cdot\,;\tau,\eta_0)\} = o_p(n^{-1/2}).$$

No Donsker condition is needed: independence does the work that Donsker equicontinuity would otherwise have to do.

The main disadvantage is inefficiency: only $|\mathcal{I}_{\text{eval}}|$ observations contribute to estimating τ, and the estimate depends on the particular random partition chosen.

11.7 Cross-Fitting

Cross-fitting extends sample splitting to recover full-sample efficiency while preserving the independence argument. Partition $\{1, \dots, n\}$ into K approximately equal folds $\mathcal{I}_1, \dots, \mathcal{I}_K$. For each fold k: (1) estimate nuisance functions on $\{1, \dots, n\} \setminus \mathcal{I}_k$ to obtain $\hat{\eta}^{(-k)} = (\hat{\mu}_0^{(-k)}, \hat{\mu}_1^{(-k)}, \hat{\pi}^{(-k)})$; (2) evaluate the orthogonal score on $\mathcal{I}_k$; (3) aggregate across all folds. The cross-fitted estimator $\hat{\tau}_{\text{DML}}$ solves:

$$\frac{1}{n} \sum_{k=1}^{K} \sum_{i \in \mathcal{I}_k} \varphi_{\text{eff}}(O_i;\, \tau,\, \hat{\eta}^{(-k)}) = 0. \tag{11.5}$$

For each fold k, the nuisance estimate $\hat{\eta}^{(-k)}$ is trained on $\{1, \dots, n\} \setminus \mathcal{I}_k$, independent of the observations in $\mathcal{I}_k$. Conditioning on $\hat{\eta}^{(-k)}$, the fold-k summands are independent and mean zero, and the conditional law of large numbers gives $R_n^{(k)} = o_p(n^{-1/2})$ for each k without any Donsker assumption. Averaging over folds, $R_n = o_p(n^{-1/2})$.

Cross-fitting therefore achieves two goals simultaneously: it preserves the held-out independence structure, and it ensures every observation serves once as an evaluation point, so the full sample contributes to estimating τ.

Algorithm: Cross-Fitted Orthogonal-Score Estimator for the ATE

1. Partition the sample into K folds $\mathcal{I}_1, \dots, \mathcal{I}_K$.
2. For each $k = 1, \dots, K$, fit nuisance estimators $\hat{\eta}^{(-k)} = (\hat{\mu}_0^{(-k)}, \hat{\mu}_1^{(-k)}, \hat{\pi}^{(-k)})$ on $\{1, \dots, n\} \setminus \mathcal{I}_k$.
3. For each $i \in \mathcal{I}_k$, compute the out-of-fold score $\hat{\varphi}_i = \varphi_{\text{eff}}(O_i;\, \tau,\, \hat{\eta}^{(-k)})$.
4. Solve the cross-fitted moment equation Equation 11.5. Because the score is linear in τ, the solution is:
$$\hat{\tau}_{\text{DML}} = \hat{\mu}_{1,\text{DML}} - \hat{\mu}_{0,\text{DML}}, \tag{11.6}$$
where $\hat{\mu}_{t,\text{DML}} = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in \mathcal{I}_k} \Big[\hat{\mu}_t^{(-k)}(X_i) + \frac{T_i^{(t)}}{\hat{\pi}^{(-k)}(X_i)^{(t)}} \{Y_i - \hat{\mu}_t^{(-k)}(X_i)\} \Big]$, with $T_i^{(1)} = T_i$, $T_i^{(0)} = 1 - T_i$.

11.8 Double Machine Learning for the Average Treatment Effect

The estimator Equation 11.6 is simply the AIPW estimator from Chapter 10 with out-of-fold nuisance estimates. The term *double machine learning* (DML), introduced by Chernozhukov et al. (2018), refers to the combination of three ingredients:

- *Neyman orthogonality*, which eliminates the population bias term so nuisance estimation error enters only at second order;
- *cross-fitting*, which eliminates the empirical process term via fold-by-fold conditional independence, removing the need for Donsker conditions; and
- *flexible machine-learning estimation* of the nuisance functions, which is made valid by the first two ingredients.

The term "double" should not be read as merely meaning "two models"; it refers to the combination of nuisance learning with orthogonalization and debiasing. DML is not a new causal estimand and not a new identification strategy; it is a modern estimation protocol built around an orthogonal score. In the biostatistics literature, the same estimator is often called the *cross-fitted AIPW estimator*.

"Double" Does Not Mean "Two Models Are Enough"

It is tempting to think that estimating any two of μ_0, μ_1, and π flexibly is sufficient. What matters is that *all* nuisance components entering the orthogonal score are estimated out-of-fold, and that the product-rate condition is satisfied. Using flexible methods for only one nuisance function while misspecifying another can still produce invalid inference.

11.9 Rate Conditions and Asymptotic Normality

Theorem: Asymptotic Linearity Under Cross-Fitting

Suppose: (i) the score φ_{eff} is orthogonal at the truth; (ii) the nuisance estimators are consistent in mean squared error; (iii) *strong overlap* holds: there exists $c > 0$ such that $c \le \pi(X) \le 1 - c$ and $c \le \hat{\pi}^{(-k)}(X) \le 1 - c$ a.s. for each fold k; (iv) the *product-rate condition* holds:

$$\|\hat{\pi}^{(-k)} - \pi\| \cdot \|\hat{\mu}_t^{(-k)} - \mu_t\| = o_p(n^{-1/2}), \qquad t = 0, 1. \tag{11.7}$$

Then the cross-fitted estimator satisfies:

$$\sqrt{n}(\hat{\tau}_{\text{DML}} - \tau) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi_{\text{eff}}(O_i; \tau, \eta) + o_p(1) \xrightarrow{d} N\Big(0,\ \mathbb{E}[\varphi_{\text{eff}}(O; \tau, \eta)^2]\Big).$$

The asymptotic variance equals the semiparametric efficiency bound, so $\hat{\tau}_{\text{DML}}$ is asymptotically efficient whenever the product-rate condition holds.

Remark: Weak Overlap vs. Strong Overlap

Weak overlap ($0 < \pi(X) < 1$ a.s.) is the condition needed for *identification*: every covariate stratum contains units from both treatment arms. *Strong overlap* ($c \le \pi(X) \le 1 - c$ for some $c > 0$) is what the proof actually uses: it bounds the inverse-probability weights away from infinity. Without strong overlap, influence function values can have infinite variance, and $\sqrt{n}$-consistent inference breaks down even if the ATE is identified (Khan and Tamer 2010). Trimming extreme weights or restricting the target population to a region of stronger overlap are common practical responses.

Proof Sketch

Write $n_k = |\mathcal{I}_k|$. The cross-fitted moment equation gives $0 = K^{-1} \sum_k \mathbb{P}_{n_k}\{\varphi_{\text{eff}}(O; \tau, \hat{\eta}^{(-k)})\}$. Adding and subtracting $\varphi_{\text{eff}}(O; \tau, \eta_0)$ and rearranging:

$$\hat{\tau}_{\text{DML}} - \tau = \frac{1}{n} \sum_{i=1}^{n} \varphi_{\text{eff}}(O_i; \tau, \eta_0) + \frac{1}{K} \sum_{k=1}^{K} R_n^{(k)},$$

where each fold remainder $R_n^{(k)}$ decomposes into population bias $B_n^{(k)}$ and empirical process term $E_n^{(k)}$.
Step 1 (population bias). By iterated expectations and $\mathbb{E}(T \mid X) = \pi(X)$:

$$|B_n^{(k)}| \le C\Big(\|\hat{\mu}_1^{(-k)} - \mu_1\| \cdot \|\hat{\pi}^{(-k)} - \pi\| + \|\hat{\mu}_0^{(-k)} - \mu_0\| \cdot \|\hat{\pi}^{(-k)} - \pi\|\Big) = o_p(n^{-1/2})$$

by the product-rate condition (strong overlap ensures denominators are bounded below).
Step 2 (empirical process term). Condition on $\hat{\eta}^{(-k)}$. Since $\hat{\eta}^{(-k)}$ is trained on $\{1, \dots, n\} \setminus \mathcal{I}_k$, the evaluation-fold summands are conditionally i.i.d. with conditional mean zero. The conditional variance of $E_n^{(k)}$ satisfies $\mathrm{Var}(\sqrt{n}\, E_n^{(k)} \mid \hat{\eta}^{(-k)}) = (n/n_k)\, \|f_k\|_{L_2}^2 = o_p(1)$ under consistency. By conditional Chebyshev, $\sqrt{n}\, E_n^{(k)} = o_p(1)$.
Conclusion. $\sqrt{n}\, R_n^{(k)} = o_p(1)$ for each k, so $\sqrt{n}\, K^{-1} \sum_k R_n^{(k)} = o_p(1)$. The CLT applies to the first term, and asymptotic normality follows by Slutsky's theorem. □

Remark: Interpreting the Product-Rate Condition

The product-rate condition requires the *product* of the two nuisance estimation errors to be $o_p(n^{-1/2})$, substantially weaker than requiring each error individually to be $o_p(n^{-1/2})$. A sufficient symmetric case: each nuisance estimator converges at rate $o_p(n^{-1/4})$ in mean square, achievable by many nonparametric and machine-learning methods. This is what makes flexible nuisance estimation feasible in semiparametric causal inference.

11.10 Lab: Simulation Study of the DML Estimator

Data-generating process. $n = 2000$ observations; $X_1, X_2, X_3 \sim \text{Uniform}(-1, 1)$ i.i.d. True nuisance functions:

$$\pi(X) = \text{expit}\big(\sin(\pi X_1) + X_2^2 - 0.5\big), \quad \mu_1(X) = 2\sin(\pi X_1) + X_2^2 + X_3, \quad \mu_0(X) = \sin(\pi X_1) + X_2^2 - X_3.$$

True ATE: $\tau = \mathbb{E}\{\mu_1(X) - \mu_0(X)\} = \mathbb{E}\{\sin(\pi X_1) + 2X_3\} = 0$ (by symmetry). The nonlinearity of π and μ_t means random forests can fit them consistently, while the Donsker condition fails for the function class they trace out — exactly the setting of Section 11.5.

The three estimators. (1) *Oracle AIPW*: plug in the true nuisance functions directly — infeasible but provides the semiparametric efficiency benchmark. (2) *Naive RF AIPW*: estimate π, μ_1, μ_0 using random forests on the **full sample**, then evaluate the AIPW score on the same sample — the Donsker condition fails. (3) *DML (cross-fitted AIPW)*: same random forest learners but via $K = 5$-fold cross-fitting — the fold-by-fold independence argument eliminates the empirical process term.

Estimators (2) and (3) use identical learners; the only difference is data reuse versus cross-fitting.

Implementation. The simulation was run in Python using scikit-learn random forests with fixed hyperparameters (`n_estimators=100`, `min_samples_leaf=5`; all other settings at library defaults). No nested cross-validation is used to tune learners inside the cross-fitting loop: this keeps the comparison between Naive and DML clean — any difference is attributable to cross-fitting alone, not to different tuning paths. The propensity score is clipped to $[0.01, 0.99]$ after fitting. Replication seed is $1000 + r$ for $r = 0, \ldots, 199$.

Pseudocode: One Replication

1. **Generate data.** Draw (X_i, T_i, Y_i) for $i = 1, \ldots, n$ from the DGP above.
2. **Oracle AIPW.** Compute $\hat{\tau}_{\text{oracle}} = n^{-1} \sum_i \varphi_{\text{eff}}(O_i; \tau, \pi, \mu_0, \mu_1)$ using the true nuisance functions.
3. **Naive RF AIPW.** Fit $\hat{\pi}$, $\hat{\mu}_0$, $\hat{\mu}_1$ as random forests on the full sample; clip $\hat{\pi}$ to $[0.01, 0.99]$; compute $\hat{\tau}_{\text{naive}}$ with the same-sample fits.
4. **DML.** Partition $\{1, \ldots, n\}$ into K shuffled folds. For each fold k: fit $\hat{\pi}^{(-k)}$, $\hat{\mu}_0^{(-k)}$, $\hat{\mu}_1^{(-k)}$ on $\{1, \ldots, n\} \setminus \mathcal{I}_k$; clip $\hat{\pi}^{(-k)}$ to $[0.01, 0.99]$; predict on $\mathcal{I}_k$. Compute $\hat{\tau}_{\text{DML}}$ from Equation 11.6.

Simulation results ($n = 2000$, $K = 5$, 200 replications):

Estimator	**Bias**	**Variance**	**RMSE**
Oracle AIPW	0.009	0.0029	0.055
Naive RF AIPW	0.039	0.0031	0.068
DML	0.007	0.0036	0.061

Oracle AIPW is nearly unbiased, serving as the efficiency benchmark. **Naive RF AIPW** exhibits a clear positive bias (0.039, roughly four times the oracle bias) despite using consistent random forest estimators. The bias arises from the uncontrolled empirical process term: reusing the same sample allows overfitting in the random forests to propagate into the AIPW score, exactly as analyzed in Section 11.5. **DML** reduces the bias to 0.007 — within Monte Carlo error of zero — at a modest finite-sample variance cost (from 0.0031 to 0.0036). The small variance increase does *not* represent an asymptotic efficiency cost: every observation is used once as an evaluation point, and cross-fitting is designed to recover first-order efficiency. At $n = 2000$ with $K = 5$, the increase arises from two finite-sample sources: each out-of-fold nuisance estimator is trained on $(K-1)n/K = 1600$ observations rather than the full sample, so its predictions are slightly noisier; and fold-assignment randomness adds a small additional source of sampling variability. (The Naive estimator's lower variance is in part a consequence of overfitting, which reduces plug-in variance at the cost of the bias this lab is designed to expose.) The net effect is a **lower RMSE for DML** (0.061) than for Naive RF AIPW (0.068): the bias reduction outweighs the variance difference.

Remark

The key comparison is between Naive RF AIPW and DML, not between either and the oracle. Both use identical random forest learners; the only difference is cross-fitting. The bias reduction from 0.039 to 0.007 is therefore attributable solely to removing the same-sample data-reuse problem. The accompanying small increase in variance is a finite-sample phenomenon, not an asymptotic penalty: cross-fitting is designed to recover first-order efficiency, and the gap should narrow as n grows.

11.11 Variance Estimation and Confidence Intervals

Inference for $\hat{\tau}_{\text{DML}}$ proceeds exactly as in Chapter 10, except that nuisance estimates are now out-of-fold. Let $k(i)$ denote the fold containing observation i, and define:

$$\hat{\varphi}_i = \varphi_{\text{eff}}(O_i;\, \hat{\tau}_{\text{DML}},\, \hat{\eta}^{(-k(i))}). \tag{11.8}$$

The variance estimator is:

$$\hat{V} = \frac{1}{n(n-1)} \sum_{i=1}^{n} (\hat{\varphi}_i - \bar{\varphi})^2, \qquad \bar{\varphi} = \frac{1}{n} \sum_{i=1}^{n} \hat{\varphi}_i. \tag{11.9}$$

The Wald confidence interval at level $1-\alpha$ is $\hat{\tau}_{\text{DML}} \pm z_{1-\alpha/2}\sqrt{\hat{V}}$.

Theorem: Consistency of the Variance Estimator

Under the conditions of the Asymptotic Linearity theorem, the variance estimator $\hat{V}$ in Equation 11.9 is consistent for the asymptotic variance of $\hat{\tau}_{\text{DML}}$; equivalently, $n\hat{V} \xrightarrow{p} \mathbb{E}[\varphi_{\text{eff}}(O;\, \tau, \eta)^2]$.

The formula Equation 11.9 is the sample variance of the out-of-fold estimated influence values divided by n — identical to Chapter 10's formula, but $\hat{\varphi}_i$ now uses the out-of-fold nuisance estimate $\hat{\eta}^{(-k(i))}$.

11.12 A Practical Workflow

Workflow: Cross-Fitted Causal Estimation

Phase 1: Identification and setup

1. **Specify the estimand.** State the target causal parameter and verify the identification assumptions (conditional exchangeability, positivity, and consistency for the ATE).
2. **Choose the orthogonal score.** For the ATE under conditional exchangeability, use the efficient influence function Equation 11.3.
3. **Select nuisance learners.** Choose flexible learners for $\mu_0(x)$, $\mu_1(x)$, $\pi(x)$. Verify informally that the product-rate condition Equation 11.7 is plausible.
4. **Monitor overlap.** Examine the empirical distribution of covariates by treatment group before fitting.

Phase 2: Estimation

5. **Partition and cross-fit.** Partition into K folds ($K = 5$ is a common default). For each fold k, fit nuisance learners on the complement to obtain $\hat{\eta}^{(-k)}$.
6. **Compute out-of-fold orthogonal scores.** For each $i \in \mathcal{I}_k$, compute $\hat{\varphi}_i$.

7. **Solve the moment equation.** Average scores across all observations to obtain $\hat{\tau}_{\mathrm{DML}}$ via Equation 11.6.
8. **Estimate the variance.** Compute $\widehat{V}$ via Equation 11.9 and form the Wald confidence interval.

Phase 3: Diagnostics and interpretation

9. **Inspect estimated propensity scores.** Examine $\hat{\pi}^{(-k)}(X_i)$ across folds; extreme values near 0 or 1 inflate influence-value variance.
10. **Assess sensitivity.** Repeat with alternative learners. Substantial sensitivity signals that the product-rate condition may not be satisfied.
11. **Interpret relative to identification assumptions.** The credibility of the causal estimate rests on identification assumptions, not on predictive accuracy of the nuisance learners.

Step 11 deserves emphasis. A common error is to assess credibility by citing good out-of-sample predictive accuracy for the nuisance models. Predictive accuracy is necessary but not sufficient. The identifying assumptions are untestable from the data:

> *Modern causal estimation is not merely machine learning plus a causal estimand. It is machine learning embedded inside a carefully constructed semiparametric procedure, whose validity rests on identification assumptions that no algorithm can verify.*

11.13 What Machine Learning Does Not Solve

Flexible nuisance estimation is a genuine improvement over parametric methods, but it does not resolve the fundamental difficulties of causal inference. In particular, machine learning does not solve:

- **Unmeasured confounding.** If important confounders are absent from X, conditional exchangeability $Y(t) \perp\!\!\!\perp T \mid X$ fails. No amount of flexibility in estimating $\pi(x)$ or $\mu_t(x)$ can correct this.
- **Violations of consistency.** If the treatment version is ill-defined or SUTVA is violated, the potential outcomes framework itself is compromised.
- **Failure of positivity.** When $\pi(x) \approx 0$ or $\pi(x) \approx 1$, influence values become unstable and coverage can degrade severely.
- **Ambiguity about the estimand.** Flexible estimation cannot resolve disagreement about what causal quantity is scientifically meaningful.
- **Invalid instrumental variable assumptions.** The exclusion restriction and exogeneity conditions (Chapter 7) are not testable; machine learning cannot substitute for subject-matter justification.
- **Fragile mediation assumptions.** Sequential ignorability and no interference between mediators (Chapter 8) must be argued on substantive grounds.

Machine learning improves the estimation of nuisance functions *within* a given identification strategy. It does not create identification where none exists.

11.14 Chapter Summary

Symbol	Meaning
$\eta = (\mu_0, \mu_1, \pi)$	Nuisance tuple
$\phi(O;\, \psi, \eta)$	Generic orthogonal score; zero mean at truth
$\varphi_{\mathrm{eff}}(O;\, \tau, \eta)$	Efficient influence function for the ATE Equation 11.3
$\mathcal{I}_k$	k-th fold; $k = 1, \ldots, K$
$\hat{\eta}^{(-k)}$	Nuisance estimates trained without fold k
$\hat{\tau}_{\mathrm{DML}}$	Cross-fitted AIPW (DML) estimator Equation 11.6
$\hat{\varphi}_i$	Out-of-fold estimated influence value
$\widehat{V}$	Variance estimator Equation 11.9

1. Flexible methods are attractive for nuisance estimation when true nuisance functions are complex,

but naive plug-in estimators can fail for two reasons: orthogonality may be absent, and even when orthogonality holds, same-sample nuisance fitting can create an additional empirical-process bias.
2. A score function is Neyman orthogonal if its Gateaux derivative with respect to the nuisance vanishes at the truth, so nuisance estimation error enters only at second order.
3. The efficient influence function for the ATE Equation 11.3 is orthogonal, doubly robust, and efficiency-achieving.
4. Even with an orthogonal score, reusing the same data for nuisance estimation and score evaluation can induce overfitting bias, especially when the nuisance class falls outside classical fixed Donsker regimes.
5. Sample splitting removes this bias by separating nuisance estimation from score evaluation, but wastes data.
6. Cross-fitting recovers efficiency by rotating training and evaluation roles across K folds.
7. The cross-fitted AIPW estimator $\hat{\tau}_{\mathrm{DML}}$ is the standard double machine learning estimator for the ATE.
8. Under the product-rate condition Equation 11.7, orthogonality and cross-fitting together yield asymptotic linearity, root-n consistency, and asymptotic normality of $\hat{\tau}_{\mathrm{DML}}$, with asymptotic variance equal to the semiparametric efficiency bound. A sufficient condition: each nuisance estimator converges at rate $o_p(n^{-1/4})$ in $L_2(P)$ norm, equivalently that its squared L_2 risk is $o_p(n^{-1/2})$.
9. The asymptotic variance is estimated consistently from the empirical variance of the out-of-fold estimated influence values.
10. Machine learning improves nuisance estimation, but does not resolve identification problems: unmeasured confounding, positivity failures, and untestable structural assumptions remain the researcher's responsibility.

11.15 Problems

1. Verifying orthogonality.

(a) Write out the efficient influence function $\varphi_{\mathrm{eff}}(O;\tau,\eta)$ for the ATE and compute its expectation. Confirm it equals zero at the truth.
(b) Consider a perturbation $\pi \mapsto \pi + r \cdot h$ for a bounded function $h(x)$. Differentiate $\mathbb{E}\{\varphi_{\mathrm{eff}}(O;\tau,\mu_0,\mu_1,\pi + rh)\}$ with respect to r and evaluate at $r = 0$. Show the derivative is zero, confirming Neyman orthogonality in the propensity score direction.
(c) Repeat the calculation for a perturbation in μ_1.

2. First-order bias of the plug-in estimator. Suppose $\hat{\mu}_1 = \mu_1 + \delta$ for a deterministic function $\delta(x)$, and $\hat{\mu}_0 = \mu_0$, $\hat{\pi} = \pi$.

(a) Show that the bias of the prediction estimator $\hat{\tau}_{\mathrm{pred}} = n^{-1}\sum_i\{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}$ is $\mathbb{E}\{\delta(X)\}$.
(b) Show that the population bias of the plug-in AIPW estimator (using $\hat{\mu}_1 = \mu_1 + \delta$ with the true π and μ_0) is zero. Explain which property of the AIPW score is responsible. Does this mean the plug-in AIPW estimator is asymptotically unbiased when $\hat{\mu}_1$ is estimated flexibly from the same sample used to evaluate the score? Explain why or why not.

3. Cross-fitting with $K = 2$. Partition $n = 200$ observations into two halves $\mathcal{I}_1$ and $\mathcal{I}_2$.

(a) Write down the cross-fitted moment equation Equation 11.5 explicitly for $K = 2$.
(b) Explain why the contribution from $\mathcal{I}_1$ uses nuisance estimates trained on $\mathcal{I}_2$ and vice versa.
(c) Compare this procedure to single sample splitting with $\mathcal{I}_1$ as the training fold. What is gained and what is lost?

4. The product-rate condition. Suppose $\|\hat{\pi} - \pi\|_{L_2} = o_p(n^{-\alpha})$ and $\|\hat{\mu}_t - \mu_t\|_{L_2} = o_p(n^{-\beta})$ for $\alpha, \beta > 0$.

(a) State the condition on α and β under which the product-rate condition Equation 11.7 holds.
(b) Show that $\alpha = \beta = 1/4$ satisfies your condition.
(c) Does $\alpha = 1/2$, $\beta = 0$ satisfy it? What does this say about the case where the propensity score is estimated parametrically but the outcome model converges only at an unspecified rate?

5. Variance estimation. Using the explicit formula Equation 11.6, write out the estimated influence value $\hat{\varphi}_i$ for a generic observation $i \in \mathcal{I}_k$. Show that $n^{-1}\sum_i \hat{\varphi}_i = 0$ when $\hat{\tau}_{\mathrm{DML}}$ solves the cross-fitted moment equation, and explain why the centering in Equation 11.9 is therefore asymptotically inconsequential.

6. What machine learning cannot do. For each of the following scenarios, identify the identification assumption that is violated and explain why flexible nuisance estimation cannot remedy the problem.

(a) A study of job-training effects on earnings omits pre-program earnings, a strong predictor of both program participation and subsequent earnings.

(b) A study of a medical treatment estimates the propensity score accurately, but the treatment was assigned in clusters (hospital wards) and outcomes may be correlated within clusters in a way that depends on the fraction of patients treated.

(c) An instrument is used to estimate the effect of education on wages, but the instrument also directly affects wages through a channel not related to education.

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Appendix A

Graphical Intuition for Conditional Independence and d-Separation

This appendix provides a gentle introduction to the probabilistic and graphical ideas that underlie Chapters 2 and 3. Our goal is not to give a complete treatment of graphical models, but rather to develop enough intuition so that the formal machinery of d-separation, back-door adjustment, and intervention graphs does not appear abruptly.

A directed acyclic graph (DAG) is more than a picture. It is a compact language for expressing assumptions about how variables are related. Once those assumptions are represented graphically, the graph tells us which variables may be associated, which paths transmit dependence, and which variables should or should not be conditioned on.

The key pedagogical idea of this appendix is simple: before learning the full d-separation criterion, it is helpful to understand three elementary three-node patterns. Those local patterns explain most of what happens later in larger graphs.

A.1 Conditional Independence: The Probabilistic Language Behind Graphs

Before introducing graphs, we first recall the probabilistic notion that graphs are designed to encode.

Definition: Conditional Independence

Let X, Y, and Z be random variables. We say that X and Y are *conditionally independent given* Z, written $X \perp\!\!\!\perp Y \mid Z$, if

$$p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$$

for all z with positive probability.

Equivalently, once Z is known, learning X gives no additional information about Y. In terms of densities, conditional independence admits the following equivalent characterizations:

$$X \perp\!\!\!\perp Y \mid Z \iff f(x, y, z)\, f(z) = f(x, z)\, f(y, z) \iff \exists\, a, b\colon f(x, y, z) = a(x, z)\, b(y, z).$$

The last form is especially useful: it says that the joint density factors into one piece depending on (x, z) and another depending on (y, z), with no cross-term in x and y.

Proposition: Fundamental Properties of Conditional Independence

(Dawid 1979, 1980) For random variables X, Y, Z, and W, the following hold:

- **(C1) Symmetry.** $X \perp\!\!\!\perp Y \mid Z \;\Rightarrow\; Y \perp\!\!\!\perp X \mid Z$.
- **(C2) Decomposition.** $X \perp\!\!\!\perp (Y, W) \mid Z \;\Rightarrow\; X \perp\!\!\!\perp Y \mid Z$.

- **(C3) Weak Union.** $X \perp\!\!\!\perp (Y, W) \mid Z \Rightarrow X \perp\!\!\!\perp Y \mid (Z, W)$.
- **(C4) Contraction.** $X \perp\!\!\!\perp Y \mid Z$ and $X \perp\!\!\!\perp W \mid (Y, Z) \Rightarrow X \perp\!\!\!\perp (Y, W) \mid Z$.
- **(C5) Intersection.** If $f(x, y, z, w) > 0$ for all (x, y, z, w), then $X \perp\!\!\!\perp Y \mid (Z, W)$ and $X \perp\!\!\!\perp Z \mid (Y, W) \Rightarrow X \perp\!\!\!\perp (Y, Z) \mid W$.

Properties (C1)–(C4) hold for any probability distribution and are known as the *semigraphoid axioms*. Property (C5) additionally requires the joint density to be strictly positive; together with (C1)–(C4) it forms the *graphoid axioms*. These properties are used implicitly throughout the course whenever conditional independence statements are combined or simplified.

Conditional independence is not the same as marginal independence. Two variables may be dependent marginally but independent after conditioning on a third variable. Conversely, two variables may be independent marginally but become dependent after conditioning. Both phenomena occur repeatedly in causal inference.

Example: Ice Cream, Drowning, and Season

Let X = ice cream sales, Y = drowning incidents, and Z = season. Marginally, X and Y are positively associated because both tend to be higher in summer. This does not mean that ice cream sales cause drowning. A more plausible explanation is that season is a common cause of both variables. Once season is fixed, the association largely disappears: $X \perp\!\!\!\perp Y \mid Z$.

This example illustrates a recurring theme in causal inference: an observed association may be induced by a third variable, and conditioning on that variable can remove the spurious dependence.

Remark: The Central Question

At this stage, the main question for the reader is: *Does conditioning on a variable remove association, preserve it, or create it?* Graphs provide a systematic answer to exactly this question.

A.2 Three Basic Motifs: Chain, Fork, and Collider

Every path in a DAG is built from local three-node configurations. There are three fundamental types: a *chain*, a *fork*, and a *collider*. Their behavior under conditioning is the foundation of d-separation.

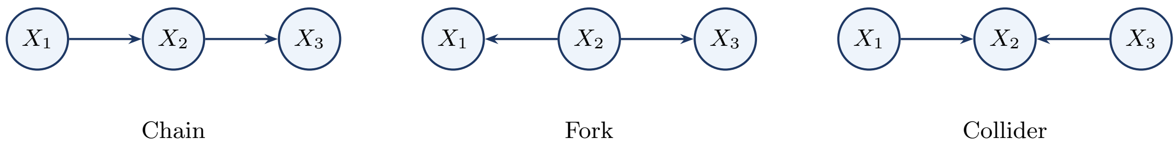


Figure A.1: The three fundamental three-node motifs. Chains and forks are blocked by conditioning on the middle node. Colliders are blocked by default but opened by conditioning on the middle node or one of its descendants.

A.2.1 Chain

Consider the pattern $X_1 \to X_2 \to X_3$, where the middle node X_2 lies on a directed pathway from X_1 to X_3.

Example: Exercise, Body Weight, and Blood Pressure

Let X_1 = exercise, X_2 = body weight, and X_3 = blood pressure. Exercise may affect blood pressure partly through its effect on body weight, so the path $X_1 \to X_2 \to X_3$ transmits association from X_1 to X_3. Marginally, exercise and blood pressure are associated; but if we condition on body weight, that particular pathway is blocked: $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

Once the middle variable is fixed, the chain no longer transmits additional information from X_1 to X_3.

A.2.2 Fork

Consider the pattern $X_1 \leftarrow X_2 \rightarrow X_3$, where the middle node X_2 is a common cause of X_1 and X_3.

Example: Ice Cream, Season, and Drowning

With X_1 = ice cream sales, X_2 = season, and X_3 = drowning incidents, season affects both variables, so X_1 and X_3 are associated even though neither causes the other. Conditioning on season blocks this path: $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

A fork is the simplest graphical form of confounding: the middle node creates association, and conditioning on it blocks the path.

A.2.3 Collider

Consider the pattern $X_1 \rightarrow X_2 \leftarrow X_3$, where the middle node X_2 is a common effect of X_1 and X_3.

Example: Talent, Legacy Status, and College Admission

Let X_1 = academic talent, X_3 = legacy status, and X_2 = admission to an elite university. Talent and legacy status may be unrelated in the general applicant pool. But among admitted students they can become statistically associated: low legacy status makes unusually high talent more likely among admitted applicants, and vice versa. Symbolically, $X_1 \perp\!\!\!\perp X_3$ but $X_1 \not\!\perp\!\!\!\perp X_3 \mid X_2$.

Unlike chains and forks, a collider *blocks* the path by default. Conditioning on the collider opens the path and may induce association that was not present marginally.

A.2.4 Summary of the Three Motifs

A chain ($X_1 \rightarrow X_2 \rightarrow X_3$) and a fork ($X_1 \leftarrow X_2 \rightarrow X_3$) are each open by default and blocked by conditioning on the middle node X_2. A collider ($X_1 \rightarrow X_2 \leftarrow X_3$) is blocked by default and opened by conditioning on the middle node or any of its descendants.

Conditioning Is Not Always Beneficial

Many adjustment mistakes arise from forgetting this asymmetry. Conditioning on a collider can create bias rather than remove it. The informal rule "control for more variables" is unsafe in causal inference; d-separation teaches a more precise lesson: condition on the *right* variables, not simply on many variables.

Chapter 2 develops these three motifs into the full d-separation criterion; see in particular Section 2.3 for the blocking rules and Section 2.4 for an extended treatment of collider bias.

A.3 d-Separation: When Does Conditioning Block a Path?

The three-node motifs explain what happens locally on a path. The next step is to extend this logic to a general DAG, where two variables may be connected by many paths.

Definition: Path

A *path* between two nodes is a sequence of distinct nodes such that each consecutive pair is connected by an edge, regardless of edge direction.

Definition: Blocked Path and d-Separation

A path is *blocked* by a conditioning set S if at least one of the following holds:

1. The path contains a chain or fork node that belongs to S, or
2. The path contains a collider such that neither the collider nor any of its descendants belongs

to S.
A path is *open* given S if it is not blocked. Two nodes X and Y are *d-separated* by a set S if every path between X and Y is blocked by S.

A.3.1 A Confounding Example

Consider the DAG with edges $X \to T$, $T \to Y$, and $X \to Y$. Here T is the treatment, Y is the outcome, and X is a pre-treatment covariate that affects both. There are two paths from T to Y: the directed causal path $T \to Y$, and the back-door path $T \leftarrow X \to Y$.

Without conditioning, the back-door path is open, so the observed association between T and Y mixes the causal effect with confounding. Conditioning on X blocks the fork $T \leftarrow X \to Y$, thereby isolating the causal path.

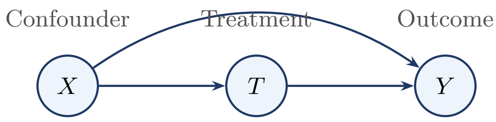


Figure A.2: A simple confounding graph. The path $T \leftarrow X \to Y$ is a back-door path from treatment to outcome. Conditioning on X blocks this noncausal path.

This confounding graph anticipates the back-door criterion of Chapter 3: the graphical condition that makes adjustment valid is precisely that all back-door paths are blocked.

A.3.2 A Collider Warning

Now consider the DAG with edges $T \to C$, $U \to C$, and $U \to Y$. Here C is a collider on the path $T \to C \leftarrow U \to Y$. Without conditioning on C, the path is blocked at the collider. Conditioning on C opens the path, creating a spurious association between T and Y through U.

Even more subtly, conditioning on a *descendant* of C can also open the path. If additionally $C \to D$, then conditioning on D may also induce association between T and Y through the collider at C.

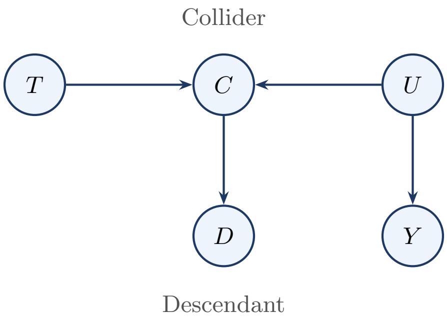


Figure A.3: Collider bias with a descendant. The path $T \to C \leftarrow U \to Y$ is blocked by default at the collider C. Conditioning on C, or on its descendant D, can open the path and induce a spurious association between T and Y.

A.3.3 A Practical Checklist

To decide whether X and Y are d-separated by S, proceed as follows. First, list all paths between X and Y. On each path, classify each interior node as part of a chain, fork, or collider. Then check whether each path is blocked by S. Finally, conclude that X and Y are d-separated if and only if every path is blocked.

For beginners, this path-by-path procedure is often more useful than starting with the most compact formal definition.

A.4 DAG Factorization and the Markov Property

Up to this point, we have used DAGs qualitatively, to decide which paths are open or blocked. We now connect the graph to probability algebra: the same parent structure that governs d-separation also determines how the joint distribution factorizes.

Definition: DAG Factorization

Let $\mathcal{G}$ be a DAG with nodes $V_1, \dots, V_p$. A joint distribution $p(v_1, \dots, v_p)$ is said to *factorize according to* $\mathcal{G}$ if

$$p(v_1, \dots, v_p) = \prod_{j=1}^{p} p\big(v_j \mid \mathrm{Pa}(V_j)\big),$$

where $\mathrm{Pa}(V_j)$ denotes the set of parents of V_j in $\mathcal{G}$.

Example: A Simple Confounding Graph

For the DAG with edges $X \to T$, $T \to Y$, and $X \to Y$, the joint density factorizes as

$$p(x, t, y) = p(x)\, p(t \mid x)\, p(y \mid t, x).$$

Definition: Local Markov Property

A distribution P satisfies the *local Markov property* with respect to $\mathcal{G}$ if, for every node $V_i \in V$,

$$V_i \perp\!\!\!\perp (\mathrm{Nd}(V_i) \setminus \mathrm{Pa}(V_i)) \mid \mathrm{Pa}(V_i).$$

That is, once we condition on the direct causes of a node, that node is independent of all variables that are neither its descendants nor its parents.

Remark: Global Markov Property

The global Markov property is the statement that every d-separation in $\mathcal{G}$ implies a conditional independence in P. Thus, if two sets of variables are d-separated by a set S in the graph, then they are conditionally independent given S in the distribution. This is studied in detail in Chapter 2.

Remark: Equivalence for DAGs

For DAGs, the local and global Markov properties are equivalent under the usual graphical-model assumptions. The local property implies the global one via the Markov factorization, which is what is actually used in identification arguments.

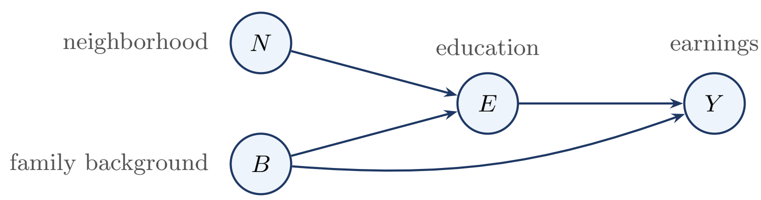


Figure A.4: Education–earnings DAG used to illustrate factorization and the local and global Markov properties.

Example: Education–Earnings Graph

Consider the DAG in Figure A.4, with edges $N \to E$, $B \to E$, $E \to Y$, and $B \to Y$. The corresponding factorization is

$$p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

The parent set of Y is $\mathrm{Pa}(Y) = \{E, B\}$, so the local Markov property gives $Y \perp\!\!\!\perp N \mid \{E, B\}$. The same conclusion follows from the global viewpoint: the only path from N to Y is $N \to E \to Y$, which is blocked by conditioning on E, and conditioning additionally on B opens no new path.

Remark: Edges as Scientific Claims

In a causal DAG, an arrow is typically drawn only when a direct dependence-generating relation is believed to be present. This explains why edges are not included gratuitously: every arrow represents a substantive scientific claim. Minimality and faithfulness are discussed in Chapter 2.

A.5 Optional: Moralization as an Alternative Criterion

Note to Reader

This section may be skipped on a first reading.

There is an alternative graph-theoretic way to check d-separation based on constructing an undirected graph called the *moral graph*. The method is elegant and useful in more advanced graphical-model theory, but it is not necessary for understanding the main causal ideas of Chapters 2 and 3. For most students, the path-by-path method using chains, forks, and colliders is more intuitive on first exposure.

The procedure is as follows. First, restrict attention to the ancestors of the variables of interest. Second, connect any two parents of a common child by an undirected edge. Third, drop all arrow directions. Finally, check ordinary graph separation in the resulting undirected graph. This procedure yields a criterion equivalent to d-separation.

Example: Moral Graph of a Collider

Suppose a DAG contains the pattern $A \to C \leftarrow B$. In the moral graph, the two parents A and B are connected by an undirected edge, producing the triangle $A - C - B$ together with $A - B$. This added edge reflects the special role of colliders in d-separation.

A.6 Summary

This appendix introduced the graphical ideas that support Chapters 2 and 3. Conditional independence is the probabilistic language that graphs are designed to encode. The three local motifs — chain, fork, and collider — determine how conditioning affects association along a path. D-separation extends these local rules to arbitrary graphs: two variables are d-separated by S if every path between them is blocked by S. The most important application in causal inference is to distinguish confounding paths from causal paths and to identify valid adjustment sets. Finally, the Markov property gives a probabilistic interpretation to the graph by linking graphical structure to a factorization of the joint distribution.

Appendix B

Single World Intervention Graphs

This appendix develops single world intervention graphs (SWIGs), a graphical device introduced by Richardson and Robins (2014) and given an accessible practical treatment by Bezuidenhout et al. (2025). A SWIG provides a formal bridge between the potential outcomes framework and the do-calculus by encoding both in a single diagram. A reader familiar with Chapters 2–4 will find that SWIGs require no new conceptual ingredients: they are ordinary DAGs in which the treatment node is split into two pieces, making potential outcomes and identifiability assumptions graphically explicit rather than leaving them implicit.

Chapter 4 used the back-door criterion and the adjustment formula to identify causal effects. SWIGs make that identification argument graphically explicit: potential outcomes appear as labeled nodes inside the graph, so ignorability becomes a routine d-separation statement rather than a free-standing probabilistic assumption. Section B.5 completes the picture by establishing the biconditional equivalence between ignorability and the back-door criterion that was only stated in one direction in Chapter 4.

B.1 The SWIG Construction

In the original DAG $\mathcal{G}$, the node T represents the treatment as it naturally occurs. Under do($T{=}t$), however, the treatment is externally fixed. The SWIG separates these two roles by *splitting* T into two pieces displayed side by side in the graph:

- a **random half** (left side, capital letter T): represents the treatment value that would naturally occur, and retains all incoming edges from T's causal parents;
- a **fixed half** (right side, lowercase letter t): represents the externally imposed intervention value, carries all outgoing edges that formerly left T, and has *no* edge to or from the random half.

> Definition: Single World Intervention Graph
>
> (Richardson and Robins 2014) The *SWIG* $\mathcal{G}(t)$ is constructed from $\mathcal{G}$ by:
> 1. Replacing T with two nodes: random half T (all incoming edges retained) and fixed half t (all outgoing edges retained).
> 2. Severing the direct connection between the two halves.
> 3. Redrawing causal arrows at the split node: *all arrows into the split node land on the random half*; *all arrows departing the split node leave from the fixed half.*
> 4. Relabeling every descendant V of T in $\mathcal{G}$ as $V(t)$, indicating it is a potential outcome under do($T{=}t$).

Figure B.1 illustrates the construction for the simple confounded-treatment graph.

B.2 The Fixed Half as a Source Node

After the SWIG $\mathcal{G}(t)$ is constructed, d-separation among its nodes is evaluated using the standard rules of Chapter 2. The key structural fact that governs all path analysis in a SWIG is:

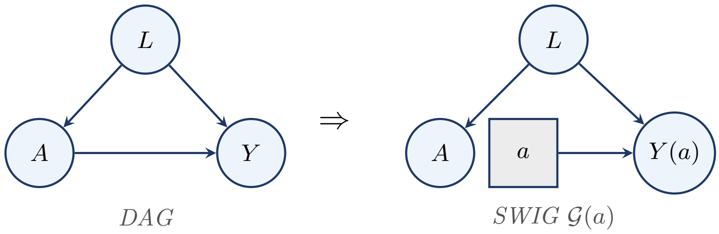


Figure B.1: Converting the confounded-treatment DAG (L is a confounder) into the SWIG for do(A=a). The treatment node splits: the random half A (circle) retains the incoming arrow from L; the fixed half a (square) carries the outgoing arrow to $Y(a)$. D-separation in the SWIG gives $A \perp\!\!\!\perp Y(a) \mid L$ (conditional ignorability).

The Fixed Half Is a Source Node

In $\mathcal{G}(t)$, the fixed half t has **no incoming edges**. Step 2 of the construction severs the connection from the random half T, and no other node in $\mathcal{G}$ has a direct arrow into t. Three consequences follow immediately.

1. **No path arrives at t from another node.** The edge $T \to t$ does not exist in $\mathcal{G}(t)$. In particular, the path $T \to t \to Y(t)$ *cannot be formed*: it is not "blocked" by a d-separation rule — it simply does not exist as a path in the graph.
2. **t cannot lie strictly between two other nodes on a path.** Because t has no parents, it is never a transit node; it can only be an *origin* of paths leading to descendants $V(t)$.
3. **The random half T and the fixed half t are d-separated by the empty set.** With no edge between them and no common ancestor, $T \perp\!\!\!\perp t$ holds unconditionally in $\mathcal{G}(t)$.

This is not an additional d-separation rule: it is a direct consequence of the SWIG construction. The fixed half t represents an externally stipulated intervention — it is not *caused* by anything within the system, so it has no parents.

Path analysis in $\mathcal{G}(t)$. With t as a source node, every path from the random half T to a potential outcome $V(t)$ must travel *away* from T through T's parents in $\mathcal{G}$ — that is, every such path is a back-door path. This is the structural fact that makes the back-door criterion readable directly from the SWIG.

Recovering ignorability from the SWIG. In Figure B.1, the only paths from the random half A to $Y(a)$ are back-door paths through A's parents. The path $A \leftarrow L \to Y(a)$ is open, but blocked by conditioning on L. No other path connects A to $Y(a)$. Conditioning on L achieves d-separation:

$$A \perp\!\!\!\perp Y(a) \,\big|\, L \quad \text{in } \mathcal{G}(a),$$

which is exactly the conditional ignorability of the Strong Ignorability definition in Chapter 4, now *derived* as a graph-structural consequence rather than asserted as an assumption.

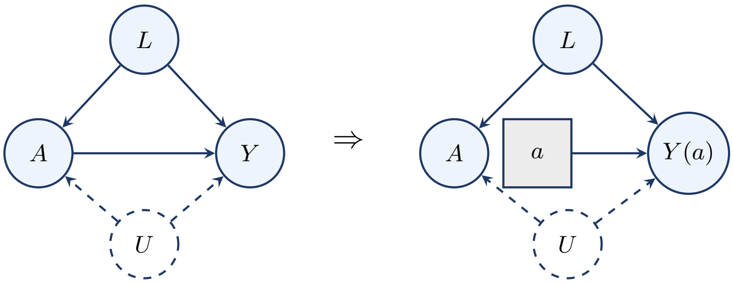
B.3 When Ignorability Fails: Hidden Confounding

The SWIG is equally useful for diagnosing *failures* of ignorability. Figure B.2 shows the SWIG when a hidden common cause U affects both A and Y.

In the SWIG $\mathcal{G}(a)$ with hidden U, the paths from A to $Y(a)$ are: (1) $A \leftarrow L \to Y(a)$: blocked by conditioning on L; (2) $A \leftarrow U \to Y(a)$: open regardless of L, since U is unobserved. Even after conditioning on L, the path through U remains open, so $A \not\perp\!\!\!\perp Y(a) \mid L$ in $\mathcal{G}(a)$, and ignorability fails. The failure is *immediately visible* from the SWIG: an unblocked dashed arrow entering the random half A signals that selection into treatment carries information about the potential outcome that no set of observed covariates can remove.

Remark

This failure motivates the instrumental variable designs of Chapter 7. When a valid instrument Z is available — one that affects A but has no direct effect on Y and is independent of U — the IV



DAG (with hidden U) *SWIG $\mathcal{G}(a)$ (with hidden U)*

Figure B.2: Adding a hidden confounder U (dashed circle) to the confounded-treatment DAG and its SWIG. In $\mathcal{G}(a)$, two back-door paths connect A to $Y(a)$: the observed path $A \leftarrow L \rightarrow Y(a)$, which can be blocked by conditioning on L, and the hidden path $A \leftarrow U \rightarrow Y(a)$, which cannot. Ignorability fails.

strategy sidesteps the unblocked U-path rather than attempting to block it by covariate adjustment.

B.4 What SWIGs Achieve

1. Potential outcomes as random variables in a DAG. In $\mathcal{G}(t)$, the variable $Y(t)$ is an ordinary node with well-defined parents: the fixed half t and any other causes of Y retained from $\mathcal{G}$. Its marginal distribution equals the interventional distribution $f(y \mid \mathrm{do}(T{=}t))$. $Y(t)$ is the counterfactual random variable, and $f(y \mid \mathrm{do}(T{=}t))$ is its induced interventional distribution; the SWIG makes both explicit in the same graph.

2. The mutilated graph as a special case. Removing the random half T from $\mathcal{G}(t)$ yields exactly the mutilated graph $\mathcal{G}_{\overline{T}}$ of Chapter 1. The SWIG is a more informative version that retains the random half alongside the fixed half, making it possible to ask — and answer by d-separation — questions about the relationship between the natural treatment T and the potential outcomes $V(t)$.

3. Unconfoundedness as a d-separation statement. The ignorability condition $(Y(t) \perp\!\!\!\perp T \mid X)$ is a standard d-separation statement *in the SWIG* $\mathcal{G}(t)$, verifiable by path-tracing. This gives the analyst a concrete graphical check: draw the SWIG, identify all paths from the random half T to $Y(t)$, and verify that X blocks all of them.

4. A single graph for both frameworks. The SWIG simultaneously hosts the natural treatment T (random half), the intervention value t (fixed half), the potential outcomes $V(t)$, and all d-separation relationships among them.

Cross-World Independence and Its Limits

Consider $Y(1) \perp\!\!\!\perp Y(0)$: $Y(1)$ lives in SWIG $\mathcal{G}(1)$ and $Y(0)$ lives in SWIG $\mathcal{G}(0)$. No unit is ever observed in both worlds simultaneously, so no data can directly test this independence. More generally, statements such as $Y(t) \perp\!\!\!\perp T(t') \mid X$ for $t \neq t'$ involve variables from *different SWIGs*. SWIGs make such assumptions explicit: the analyst must specify a joint distribution across multiple SWIGs — a commitment that potential-outcome notation alone does not force.

Remark

More than one node may be split in a single SWIG. In longitudinal settings with time-varying treatments (A_0, A_1), splitting both nodes yields the sequential-randomization SWIG, from which the two *sequential exchangeability* conditions needed to identify the g-formula follow directly by d-separation.

B.5 Biconditional Equivalence under Faithfulness

Proposition 4.1 in Chapter 4 established the practically important direction: if X satisfies the back-door criterion, then unconfoundedness holds. That proof required only the Markov property. The converse — that every ignorability statement corresponds to a back-door condition — additionally requires faithfulness and follows naturally from the SWIG construction.

Definition: Faithfulness

A distribution P is **faithful** to a DAG $\mathcal{G}$ if every conditional independence that holds in P is entailed by d-separation in $\mathcal{G}$: that is, $X \perp\!\!\!\perp Y \mid \mathbf{Z}$ in P only if X and Y are d-separated by $\mathbf{Z}$ in $\mathcal{G}$. Equivalently, faithfulness rules out *accidental* cancellations of causal paths. Together, the Markov property and faithfulness make d-separation a *complete* characterization of the conditional independence structure of P. Faithfulness fails only on a measure-zero set of parameter values, so it is considered a mild regularity condition in practice.

Theorem: Ignorability and the Back-Door Criterion Are Equivalent

Under the Markov and faithfulness assumptions for $\mathcal{G}$, unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ holds if and only if X satisfies the back-door criterion for the effect of T on Y in $\mathcal{G}$.

Proof

($\Rightarrow$) This is Proposition 4.1 in Chapter 4, which uses only the Markov property.
($\Leftarrow$) Suppose $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ holds. By faithfulness, every conditional independence in P corresponds to a d-separation in $\mathcal{G}$. Consider the SWIG $\mathcal{G}(t)$. Because the fixed half t has no incoming edges (Section B.2), every path from the random half T to a potential outcome $Y(t)$ must pass through T's parents in $\mathcal{G}$ — that is, every such path is a back-door path. The d-separation $(Y(t) \perp\!\!\!\perp T \mid X)_{\mathcal{G}(t)}$ therefore requires X to block all back-door paths from T to Y. If any member of X were a descendant of T, conditioning on it could open a previously blocked collider path, contradicting the assumed independence; hence no node in X is a descendant of T. Both conditions of the back-door criterion are therefore satisfied. $\square$

The SWIG thus provides a *unified* perspective: ignorability is neither an assumption of the potential outcomes framework alone nor of the do-calculus in isolation, but a graph-structural property that both frameworks share. The back-door criterion and conditional exchangeability are two names for the same identifying condition, seen from two different representational languages.

B.6 Problems

1. SWIG construction. Consider the DAG: $X \to T$, $X \to Y$, $T \to Y$, with X fully observed.

(a) Construct the SWIG $\mathcal{G}(t)$ by splitting T into its random and fixed halves. Draw the result, labeling the random half, fixed half, and the potential outcome $Y(t)$.
(b) In $\mathcal{G}(t)$, identify all paths between the random half T and $Y(t)$. Apply d-separation to determine which are blocked and which are open before conditioning.
(c) Use d-separation in $\mathcal{G}(t)$ to verify that $(Y(t) \perp\!\!\!\perp T \mid X)_{\mathcal{G}(t)}$ holds. Conclude that X satisfies the back-door criterion, confirming Theorem B.1 above.
(d) Now add a hidden common cause $U \to T$, $U \to Y$. Draw the revised SWIG. Does the ignorability argument still hold? State the correct conclusion and identify what additional structure (if any) would be needed for identification.

2. Sequential exchangeability in a longitudinal SWIG. Suppose the treatment is time-varying: T_0 is assigned at baseline and T_1 at a second time point. The DAG has edges $X \to T_0 \to T_1 \to Y$, $X \to Y$, and $T_0 \to Y$.

(a) Construct the SWIG $\mathcal{G}(t_0, t_1)$ by splitting both T_0 and T_1. Label all random halves, fixed halves, and potential outcomes.

(b) List all back-door paths from the random half T_0 to $Y(t_0, t_1)$ and from T_1 to $Y(t_0, t_1)$. For each, determine whether conditioning on X and the history up to that point is sufficient to achieve d-separation.

(c) State the two sequential exchangeability conditions that the SWIG makes graphically transparent, and explain what they say about treatment assignment at each time point.

Appendix C

Pathwise Differentiability and Efficient Influence Functions

Section 9.9 in Chapter 9 introduced the efficient influence function of the average treatment effect functional and stated, without proof, that this object is the *canonical gradient* of Ψ relative to the nonparametric tangent space. This appendix supplies the geometric machinery behind that statement. The treatment follows Bickel et al. (1993) and Tsiatis (2006), with notation aligned to Chapter 9.

The reader should think of this appendix as the formal counterpart of Sections 9.3–9.7: those sections showed how an influence function determines the asymptotic distribution of an estimator; here we develop the parallel notion of an influence function as a *derivative of a functional*, and characterize the unique influence function that achieves the semiparametric efficiency bound.

The applied consequences — the AIPW estimator, double robustness, Neyman orthogonality, cross-fitting, and rate conditions — are developed in Chapters 10 and 11. This appendix supplies only the foundational geometry on which those chapters rest.

C.1 Regular Parametric Submodels and Scores

Let $\mathcal{P}$ be a statistical model for the distribution of the observed data O on a sample space $\mathcal{O}$, with true distribution $P \in \mathcal{P}$. All densities below are taken with respect to a common dominating measure μ. The parameter of interest is a smooth real-valued functional $\Psi : \mathcal{P} \to \mathbb{R}$, $\psi = \Psi(P)$.

To define what it means for Ψ to be "smooth," we need a notion of a path through $\mathcal{P}$ that begins at P.

Definition: Regular Parametric Submodel

A **regular parametric submodel** of $\mathcal{P}$ passing through P is a one-parameter family $\{P_\varepsilon : \varepsilon \in (-\delta, \delta)\} \subset \mathcal{P}$ with $P_0 = P$, with densities p_ε such that $\varepsilon \mapsto \log p_\varepsilon(O)$ is differentiable at $\varepsilon = 0$ in $L_2(P)$, with **score**:

$$S(O) = \frac{\partial}{\partial \varepsilon} \log p_\varepsilon(O)\Big|_{\varepsilon=0} \in L_2(P).$$

The L_2-differentiability requirement is a mild smoothness condition that ensures $\int p_\varepsilon \, d\mu$ can be differentiated under the integral. For a rigorous formulation, see Vaart (1998, sec. 7.2).

Because $\int p_\varepsilon \, d\mu = 1$ for all ε, differentiating under the integral gives:

$$\mathbb{E}_P[S(O)] = 0, \qquad \mathbb{E}_P[S(O)^2] < \infty, \tag{C.1}$$

so every score lies in $L_2^0(P) := \{f \in L_2(P) : \mathbb{E}_P[f] = 0\}$.

Remark: Why Submodels Rather Than Point-Mass Contamination

A heuristic device sometimes used for "probing" a functional is the contamination path $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_o$, which yields the classical Hampel influence function $\Psi(\delta_o) - \Psi(P)$. Contamination is intuitive but does not yield a regular submodel: when P is continuous, the formal "score" of such a path is $\delta_o/p - 1$, which is not in $L_2(P)$. All formal results below use the definition above, while the contamination heuristic remains useful for guessing the form of an influence function in concrete examples.

C.2 Pathwise Differentiability

The functional Ψ is differentiable along a submodel if the map $\varepsilon \mapsto \Psi(P_\varepsilon)$ is differentiable at $\varepsilon = 0$. Pathwise differentiability asserts that this derivative can be represented as an inner product between a fixed function and the score, uniformly over submodels.

Definition: Pathwise Differentiability and Influence Function

The functional $\Psi : \mathcal{P} \to \mathbb{R}$ is **pathwise differentiable** at P if there exists a function $\varphi \in L_2^0(P)$ such that, for every regular parametric submodel $\{P_\varepsilon\}$ with score S:

$$\frac{\partial}{\partial\varepsilon}\Psi(P_\varepsilon)\bigg|_{\varepsilon=0} = \mathbb{E}_P[\varphi(O)\, S(O)]. \tag{C.2}$$

Any such φ is called an **influence function** (or **gradient**) of Ψ at P. The set of influence functions is denoted $\mathrm{IF}(\Psi, P)$.

Equation C.2 is the defining identity of semiparametric theory. It says the perturbation of Ψ along a submodel is fully encoded by the $L_2(P)$ inner product of a fixed function φ with the score S. Geometrically, φ is a representer for the linear functional $S \mapsto \partial_\varepsilon \Psi(P_\varepsilon)|_0$ restricted to the space of scores.

Example: Mean Functional

Let $O = Y$ with $\mathbb{E}_P|Y| < \infty$ and $\Psi(P) = \mathbb{E}_P[Y]$. For any regular submodel $\{P_\varepsilon\}$ with score S:

$$\frac{\partial}{\partial\varepsilon}\int y\, p_\varepsilon(y)\, d\mu(y)\bigg|_0 = \mathbb{E}_P[Y \cdot S(O)].$$

Since $\mathbb{E}_P[S] = 0$, subtracting $\mathbb{E}_P[Y]\cdot\mathbb{E}_P[S] = 0$ gives $\partial_\varepsilon\Psi(P_\varepsilon)|_0 = \mathbb{E}_P[(Y - \mathbb{E}_P Y)\, S(O)]$, so $\varphi(O) = Y - \Psi(P)$ is an influence function of the mean functional. Under the nonparametric model, this is in fact the *unique* influence function, and hence the efficient influence function.

Remark: Connection to Estimation

If an estimator $\hat\psi$ is asymptotically linear at P with influence function φ in the sense of Chapter 9, and if $\hat\psi$ is regular along every submodel, then φ satisfies Equation C.2 and is therefore a gradient of Ψ. Pathwise differentiability is thus a necessary condition for the existence of a regular asymptotically linear estimator of Ψ (Tsiatis 2006, Theorem 3.1).

C.3 Non-Uniqueness of Influence Functions

Definition Section C.2 does not produce a unique influence function. If φ satisfies Equation C.2 and $h \in L_2^0(P)$ is orthogonal to every score in $L_2(P)$, then $\varphi + h$ also satisfies Equation C.2, since $\mathbb{E}_P[(\varphi+h)S] = \mathbb{E}_P[\varphi S] + \mathbb{E}_P[hS] = \mathbb{E}_P[\varphi S]$. Whether φ is unique depends on how rich the collection of scores is — a point made precise through the notion of the tangent space.

The terminological distinction can now be made precise:

- An **estimating function** is any function $U(O;\theta)$ whose root defines an estimator (Chapter 9, Section 9.3).
- An **asymptotic influence function** of an estimator is the function φ in its asymptotic expansion (Chapter 9, Section 9.4).
- A **(pathwise) influence function** of a functional is a $\varphi \in L_2^0(P)$ satisfying Equation C.2.

For a regular asymptotically linear estimator of a pathwise-differentiable functional, all three notions coincide. The third depends crucially on $\mathcal{P}$ through the collection of admissible submodels, which is what makes $\mathrm{IF}(\Psi, P)$ generally non-unique in semiparametric models.

C.4 The Tangent Space

Definition: Tangent Space

The **tangent space** of $\mathcal{P}$ at P, denoted $T_P(\mathcal{P})$ or simply T, is the closed linear span in $L_2^0(P)$ of all scores of regular parametric submodels of $\mathcal{P}$ passing through P.

The tangent space is a closed subspace of the Hilbert space $L_2^0(P)$. Two extreme cases are illustrative.

Nonparametric model. If $\mathcal{P}$ contains all distributions satisfying mild regularity, then for any bounded $h \in L_2^0(P)$, the submodel $p_\varepsilon(o) \propto (1 + \varepsilon\, h(o))p(o)$ is regular with score $S = h$. Since bounded mean-zero functions are dense in $L_2^0(P)$, taking closure gives $T = L_2^0(P)$.

Fully parametric model. If $\mathcal{P} = \{P_\theta : \theta \in \Theta \subset \mathbb{R}^k\}$ with score S_θ at θ_0, then $T = \mathrm{span}\{S_{\theta_0,1}, \dots, S_{\theta_0,k}\}$ is k-dimensional.

Most causal models lie between these extremes: the parameter of interest ψ is finite-dimensional, but the rest of P is unrestricted, making T infinite-dimensional but not all of $L_2^0(P)$.

Definition: Nuisance Tangent Space

A submodel $\{P_\varepsilon\}$ is a **nuisance submodel** at P if $\Psi(P_\varepsilon) = \Psi(P)$ for all ε, i.e., the parameter of interest is held fixed along the path. The **nuisance tangent space**, denoted T_η, is the closed linear span of scores of all nuisance submodels.

The nuisance tangent space is a closed subspace of T, and hence of $L_2^0(P)$. The orthogonal complement is:

$$T_\eta^\perp := \{f \in L_2^0(P) : \mathbb{E}_P[fg] = 0 \text{ for all } g \in T_\eta\}. \tag{C.3}$$

By the projection theorem, the Hilbert-space decomposition $L_2^0(P) = T_\eta \oplus T_\eta^\perp$ holds automatically.

Proposition: Every Influence Function Lies in $T_\eta^\perp$

Let Ψ be pathwise differentiable at P. Then every influence function $\varphi \in \mathrm{IF}(\Psi, P)$ satisfies $\varphi \in T_\eta^\perp$.

Proof

For any nuisance submodel $\{P_\varepsilon\}$ with score S_η, the definition of nuisance submodel gives $\Psi(P_\varepsilon) = \Psi(P)$ for all ε, hence $\partial_\varepsilon \Psi(P_\varepsilon)|_0 = 0$. Combined with Equation C.2:

$$0 = \frac{\partial}{\partial\varepsilon}\Psi(P_\varepsilon)\bigg|_0 = \mathbb{E}_P[\varphi\, S_\eta].$$

This holds for every score S_η of a nuisance submodel; by linearity and L_2-continuity it extends to every element of the closed linear span T_η. Hence $\varphi \perp T_\eta$, i.e., $\varphi \in T_\eta^\perp$. □

Remark: Causal Example

For the ATE under the nonparametric model on $O = (X, T, Y)$, $T = L_2^0(P)$ and T_η is the closed linear span of all scores of submodels along which $\mathbb{E}_P[Y(1) - Y(0)]$ is constant. A standard calculation (Tsiatis 2006, sec. 7.4) identifies $T_\eta^\perp$ as a one-dimensional subspace spanned by the EIF φ^* of Chapter 9, Equation 10.7; this is the formal content of the claim that φ^* is uniquely determined.

C.5 The Canonical Gradient and the Efficiency Bound

Among all influence functions, there is a unique one lying in the full tangent space T, and it is the one with smallest variance.

Theorem: Canonical Gradient

Let Ψ be pathwise differentiable at P with non-empty influence function set $\mathrm{IF}(\Psi, P)$. Then:

(i) Any two influence functions differ by an element of $T^\perp$: for $\varphi_1, \varphi_2 \in \mathrm{IF}(\Psi, P)$, $\varphi_1 - \varphi_2 \in T^\perp$.

(ii) There exists a unique $\varphi^* \in \mathrm{IF}(\Psi, P)$ with $\varphi^* \in T$, namely $\varphi^* = \Pi[\varphi \mid T]$ for any $\varphi \in \mathrm{IF}(\Psi, P)$, where $\Pi[\,\cdot \mid T\,]$ denotes orthogonal projection in $L_2^0(P)$ onto T.

(iii) For every $\varphi \in \mathrm{IF}(\Psi, P)$, $\mathbb{E}_P[\varphi(O)^2] \geq \mathbb{E}_P[\varphi^*(O)^2]$, with equality iff $\varphi = \varphi^*$ in $L_2(P)$.

Proof

(i) For $\varphi_1, \varphi_2 \in \mathrm{IF}(\Psi, P)$ and every score S of a regular submodel, $\mathbb{E}_P[(\varphi_1 - \varphi_2)S] = \partial_\varepsilon \Psi|_0 - \partial_\varepsilon \Psi|_0 = 0$. By linearity and L_2-continuity, $\mathbb{E}_P[(\varphi_1 - \varphi_2)g] = 0$ for every $g \in T$, i.e., $\varphi_1 - \varphi_2 \in T^\perp$.
(ii) Fix any $\varphi \in \mathrm{IF}(\Psi, P)$ and set $\varphi^* := \Pi[\varphi \mid T]$, so $\varphi - \varphi^* \in T^\perp$. For any score $S \in T$:

$$\mathbb{E}_P[\varphi^* S] = \mathbb{E}_P[\varphi S] - \mathbb{E}_P[(\varphi - \varphi^*)S] = \mathbb{E}_P[\varphi S] = \frac{\partial}{\partial \varepsilon} \Psi(P_\varepsilon)\bigg|_0,$$

so $\varphi^* \in \mathrm{IF}(\Psi, P)$. Uniqueness: if $\tilde{\varphi}^* \in \mathrm{IF}(\Psi, P) \cap T$, then by (i), $\varphi^* - \tilde{\varphi}^* \in T^\perp \cap T = \{0\}$.
(iii) Write $\varphi = \varphi^* + (\varphi - \varphi^*)$ with the two summands orthogonal in $L_2(P)$ (since $\varphi^* \in T$ and $\varphi - \varphi^* \in T^\perp$). The Pythagorean identity gives:

$$\mathbb{E}_P[\varphi^2] = \mathbb{E}_P[(\varphi^*)^2] + \mathbb{E}_P[(\varphi - \varphi^*)^2] \geq \mathbb{E}_P[(\varphi^*)^2],$$

with equality iff $\varphi = \varphi^*$. $\square$

Definition: Efficient Influence Function and Efficiency Bound

The unique element $\varphi^* \in \mathrm{IF}(\Psi, P) \cap T$ is called the **efficient influence function** (EIF), or **canonical gradient**, of Ψ at P. Its variance:

$$V^*(\Psi, P) = \mathbb{E}_P[\varphi^*(O)^2]$$

is the **semiparametric efficiency bound** for estimating $\Psi(P)$ in the model $\mathcal{P}$.

Remark: Role of the Nuisance Tangent Space

Since every influence function already lies in $T_\eta^\perp$, the EIF can equivalently be described as the unique influence function in $T \cap T_\eta^\perp$. This intersection contains exactly the directions in T that are orthogonal to nuisance perturbations — directions along which ψ genuinely moves. In practice, the EIF is often constructed by taking a naive influence function (such as the IPW form) and subtracting its projection onto T_η to remove nuisance components; this is the projection view exploited in Chapter 10.

Theorem: Asymptotic Efficiency Bound

Let $\hat{\psi}_n$ be a regular asymptotically linear estimator of $\Psi(P)$ in the model $\mathcal{P}$. Then its asymptotic variance satisfies:

$$\mathrm{AVar}(\sqrt{n}(\hat{\psi}_n - \psi)) \geq V^*(\Psi, P),$$

and the bound is achieved if and only if the influence function of $\hat{\psi}_n$ is φ^* in $L_2(P)$.

This theorem is a corollary of the Hájek–Le Cam convolution theorem, which states the stronger result that $\sqrt{n}(\hat{\psi}_n - \psi) \rightsquigarrow Z + W$, where $Z \sim N(0, V^*)$ and W is independent of Z. A precise statement requires the local asymptotic normality framework of Vaart (1998, secs. 25.3–25.6).

Remark: Why the EIF Appears in Concrete Formulas

The Canonical Gradient Theorem explains why specific objects such as the AIPW influence function in Chapter 9 look "constructed" rather than guessed. Given a naive influence function φ (e.g., the IPW form $TY/\pi - (1-T)Y/(1-\pi) - \psi$), it may fail to lie in T; its projection $\varphi^* = \Pi[\varphi \mid T]$ is the EIF. In Chapter 10 this projection is computed explicitly for the ATE under ignorability, producing the AIPW formula. The construction in Chapter 10 should therefore be read as an instance of the Canonical Gradient Theorem applied to a specific causal functional.

Bibliographic Notes

The modern formulation of pathwise differentiability and tangent spaces is developed in the monographs of Bickel et al. (1993) and Vaart (1998, chap. 25); the latter is the standard reference for the convolution theorem and local asymptotic normality underlying the Asymptotic Efficiency Bound theorem. Tsiatis (2006) gives a treatment oriented specifically toward missing data and causal inference, and is a natural companion to the material developed here.